HCSRN Laboratory Test Types

Reference for the HCSRN VDW Laboratory Results Test Type

HCSRN VDW Lab Workgroup

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1 Laboratory Test Types

A Reference for the HCSRN VDW Laboratory Results Test Type. Version 2.1, April 2026

This reference document is not inclusive of all lab tests that have been deemed priority tests to incorporate into VDW format. The reference will be added to and updated on a regular basis as laboratory content is expanded in the VDW.

1.1 Chemistry Lab Tests

1.1.1 A_G_RATIO (Albumin to globulin ratio)

- Common Name: Albumin/globulin (A/G) ratio
- Long Name: Albumin to globulin ratio; A/G ratio in blood/serum
- Test Type: A_G_RATIO
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications: To screen for some liver and kidney disorders as well as other diseases; to investigate the cause of abnormal pooling of fluid in tissue (edema).

- Description: Proteins form the structural part of most organs and make up enzymes and hormones that regulate body functions. There are two classes of proteins found in the blood, albumin and globulin. Albumin is a carrier of many small molecules, but its main purpose is to keep fluid from leaking out of blood vessels through osmotic pressure. Globulin proteins include enzymes, antibodies, and more than 500 other proteins. The ratio of albumin to globulin (A/G ratio) is calculated from values obtained by direct measurement of total protein and albumin. It represents the relative amounts of albumin and globulins. Normally, there is a little more albumin than globulins, resulting in a normal A/G ratio of slightly over 1. Because disease states affect the relative changes in albumin and globulins in different ways, this may assist in determining the cause of the change in protein levels. A low A/G ratio may reflect overproduction of globulins such as seen in multiple myeloma or autoimmune diseases, underproduction of albumin such as occurs with cirrhosis, or selective loss of albumin from the circulation as occurs with kidney disease (nephrotic syndrome). A high A/G ratio suggests underproduction of immunoglobulins as may be seen in some genetic deficiencies and in some leukemias. More specific tests, such as albumin, liver enzyme tests, and serum protein electrophoresis must be performed to make an accurate diagnosis.
- Related Tests: Albumin; Liver panel; Protein electrophoresis
- Approximate Reference Range: Ratio of 1.7 - 2.2 to 1
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 12175-6, 1759-0, 44429-9
  - Associated CPT:

1.1.2 AFP (Alpha-1 Fetoprotein)

- Common Name: Alpha-1 Fetoprotein
- Long Name: Alpha-1 Fetoprotein in Serum or Plasma
- Test Type: AFP

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The AFP test is a screening tool used to assess the risk of certain fetal abnormalities, including neural tube defects (like spina bifida and anencephaly) and some chromosomal disorders such as Down syndrome and Edwards syndrome. In non-pregnant individuals it can be used as a tumor marker to detect and monitor certain cancers, especially liver cancer, testicular cancer, and ovarian cancer. It can also help monitor treatment effectiveness and detect potential cancer recurrence.
- Description: The Alpha-Fetoprotein (AFP) in serum or plasma test measures the level of AFP, a protein, in a blood sample. AFP is primarily produced during fetal development by the liver and yolk sac. In healthy individuals who are not pregnant, AFP levels are typically very low.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For healthy non-pregnant adults, the normal range is generally considered low, with values typically between 0 and 40 ng/mL, although some sources specify a normal range of 0 to 8 ng/mL or less than 10 ng/mL. In pregnant women, AFP levels increase from around the 14th week of gestation until about the 32nd week. For instance, between 15 and 21 weeks of gestation, a typical range for AFP is 10 to 150 ng/mL.
- Comments: These LOINCs represent AFP used as a tumor marker and not when used in maternal testing for birth defects.
- Test Method Variance:
- Other:
  - Associated LOINC: 1834-1, 19176-7, 19177-5, 53960-1, 53961-9, 53962-7, 83072-9, 83073-7
  - Associated CPT:

1.1.3 ALBUMIN (Albumin in blood)

- Common Name: Albumin
- Long Name: Albumin, total in blood/serum
- Test Type: ALBUMIN
- Panels: Liver Panel
- Equivalent Tests: N/A
- Not Equivalent Tests: Prealbumin, microalbumin, urinalysis, liver panel
- Indications:

To screen for a liver disorder or kidney disease, or to evaluate nutritional status, along with or instead of a prealbumin test.

- Description:

This test measures the level of albumin in the blood. Albumin is the most abundant protein in the fluid portion of the blood (plasma). Albumin keeps fluid from leaking out of blood vessels, nourishes tissues, and transports hormones, vitamins, drugs, and ions like calcium throughout the body. Albumin is made in the liver. The concentration of albumin drops when the liver is damaged, when a person has a kidney disease that causes nephrotic syndrome, when a person is malnourished, has inflammation, or is in shock. Albumin levels can rise when a person is dehydrated. This is a relative increase that occurs as the volume of plasma decreases.

- Related Tests:

Prealbumin, microalbumin, urinalysis, liver panel

- Approximate Reference Range: 3.5 – 5.5 g/dL.
- Comments: An albumin test may be ordered as part of a liver panel to evaluate liver function, along with a creatinine and BUN (Blood Urea Nitrogen) to evaluate kidney function, or along with a prealbumin to evaluate nutritional status. Certain drugs increase albumin in the blood, including anabolic steroids, androgens, growth hormones, and insulin. Receiving large amounts of intravenous fluids may make the test results inaccurate for a short time after the intravenous fluids are administered.

- Test Method Variance:
- Other:
  - Associated LOINC: 101198-0, 1751-7, 54347-0 , 61151-7, 61152-5, 62234-0, 62235-7, 77148-5, 103577-3, 76631-1
  - Associated CPT: 82040

1.1.4 ALP (Alkaline phosphatase)

- Common Name: ALP
- Long Name: Alkaline phosphatase, total isoenzymes in blood/serum
- Test Type: ALP
- Panels: Liver Panel
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications: Assists in detecting liver disease or bone disorders. In conditions affecting the liver, damaged liver cells release increased amounts of ALP into the blood.
- Description: ALP is an enzyme found in several body tissues, including liver, bone, kidney, bowel (intestine), and in the placenta of pregnant women. The highest concentrations of ALP are in bone and liver cells. Any condition causing excessive bone formation, including bone disorders such as Paget's disease and others such as rheumatoid arthritis and healing fractures, can cause increased ALP levels. Children and adolescents typically have higher blood ALP levels because their bones are still growing. ALP is often used to detect blocked bile ducts because ALP is especially high in the edges of the cells that join to form bile ducts. If one or more bile ducts are obstructed, for example by a tumor, blood levels of ALP can be high. Smaller increases of blood ALP levels are seen in liver cancer and cirrhosis, with use of drugs toxic to the liver, and in hepatitis.
- Related Tests: AST; ALT; GGT; Bilirubin

- Approximate Reference Range (Each site will have slight variations): Approximately 30-120 international units (IU)/L, although the normal ranges can vary slightly, e.g., some laboratories may have a normal range of 44 - 147 IU/L. Normal values vary with age and gender. High ALP levels are seen in children undergoing growth spurts and in pregnant women.
- Comments: Low levels of ALP may be present with zinc deficiency and are temporarily low after blood transfusions or heart bypass surgery. A rare genetic disorder of bone metabolism called hypophosphatasia can cause very low levels of ALP. Temporary elevations are also seen with healing fractures. Oral contraceptives may decrease ALP; antiepileptics may increase ALP.
- Test Method Variance:
- Other:
  - Associated LOINC: 1783-0, 6768-6, 6769-4, 77141-0
  - Associated CPT: 82040

1.1.5 ALT (Alanine Aminotransferase, SGPT)

- Common Name: ALT/SGPT
- Long Name: Alanine Aminotransferase/SGPT
- Test Type: ALT
- Panels: Comprehensive metabolic profile
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications: Estimate liver function. ALT is an enzyme that is produced in liver cells (hepatocytes); it is relatively specific for liver disease. High levels of ALT in the bloodstream mean that there may be liver inflammation and/or damage. Many drugs and disease states can result in minor (1.5 to 2 times above normal) elevations of ALT that are asymptomatic.

- Description: Alanine aminotransferase has a relatively long half life (37-57 hours) and may remain elevated after liver inflamation has resolved. ALT rises greater than 10-30 times the upper normal limit often indicate infectious conditions (e.g., viral hepatitis) or serious drug-induced disease while more common, minor drug induced elevations of ALT are typically < 3 times the upper normal limit
- Related Tests: SGPT is an older name for the same enzyme - Approximate Reference Range (Each site will have slight variations): approximately 10 - 65 IU/L
- Comments: - Test Method Variance: Method and reference ranges vary. Reported in international units per liter.
- Other:
  - Associated LOINC: 1742-6, 1743-4, 1744-2, 44785-4, 76625-3, 77144-4, 96586-3
  - Associated CPT: 84460

1.1.6 AMYLASE (Amylase in blood)

- Common Name: Amylase total
- Long Name: Amylase, total in blood
- Test Type: AMYLASE
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Elevated amylase levels often suggest a problem with the pancreas, such as acute or chronic pancreatitis, pancreatic tumors, or blockages in the pancreatic duct. Other conditions, including salivary gland infections (like mumps), ruptured ectopic pregnancy, intestinal blockage, or kidney problems, can also cause high amylase levels. Low amylase levels are less common but can indicate chronic pancreatitis (where the pancreas is damaged and produces less amylase), liver failure, or cystic fibrosis.

- Description: An amylase blood test, also known as a serum amylase test, measures the level of the enzyme amylase in your bloodstream. Amylase is primarily produced by the pancreas and salivary glands and plays a vital role in digesting carbohydrates into simpler sugars for absorption.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for amylase in a blood test for an adult is generally 30 to 110 units per liter (U/L)
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 1798-8, 6697-7, 76630-3, 77146-9
  - Associated CPT: 82150

1.1.7 AST (Aspartate Aminotransferase, SGOT)

- Common Name: AST/SGOT
- Long Name: Aspartate Aminotransferase/SGOT
- Test Type: AST
- Panels: Comprehensive metabolic profile
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications: Estimate liver function. AST is found in other organs besides the liver, notably skeletal muscle, myocardium, brain, kidney and red blood cells. High AST levels in the bloodstream can be a sign of liver damage or heart attack. Many drugs and disease states can result in minor (1.5 to 2 times above normal) elevations of ALT that are asymptomatic.

- Description: AST rises higher than ALT in response to alcohol-related hepatic injury, usually at a ratio of 2:1 to 8:1, with the AST at 1 - 5 times the upper limit of normal. If the AST:ALT ratio $>$ 8:1, consider other forms of hepatic damage. In general, alcohol and drug-related AST elevations are usually less than 5 times the norm, or under 150u/l.
- Related Tests: SGOT is the older name for the same enzyme
- Approximate Reference Range (Each site will have slight variations): Approximately 15-35 IU/L
- Comments: Hemolysis will cause falsely elevated values.
- Test Method Variance: Method and reference ranges vary. Reported in international units per liter
- Other:
  - Associated LOINC: 1920-8, 27344-1, 30239-8, 44786-2, 88112-8, 96587-1
  - Associated CPT: 84450

1.1.8 (BILI) Bilirubin Test Types

1. BILI_DIR (Bilirubin, Direct)
   - Common Name: Direct Bilirubin
   - Long Name: Bilirubin, direct/conjugated/glucuronidated in blood/serum
   - Test Type: BILI_DIR
   - Panels: Hepatic/Liver Function Panel
   - Equivalent Tests: Direct bilirubin; Conjugated bilirubin;
   - Not Equivalent Tests:
   - Indications: The bilirubin test is used to check liver function and watch for signs of liver disease, such as hepatitis or cirrhosis, or the effects of medicines that can damage the liver. The test is also used to find out if something is blocking the bile ducts. This may occur if gallstones, tumors of the pancreas, or other conditions are present. Bilirubin tests are also used to diagnose conditions that cause increased destruction of red blood cells, such as hemolytic anemia or hemolytic disease of the newborn.

- Description: Bilirubin testing checks for levels of bilirubin — an orange-yellow pigment — in blood. Bilirubin results from the normal breakdown of red blood cells. As a normal process, bilirubin is carried in the blood and passes through your liver. Too much bilirubin may indicate liver damage or disease. Before reaching the liver, as a breakdown product of red blood cells, the bilirubin is called indirect (unconjugated) bilirubin. Once in the liver, most bilirubin attaches to certain sugars creating direct (conjugated) bilirubin. Direct bilirubin is released into bile and stored in the gallbladder before eventually being excreted in stool. Higher than normal levels of direct or indirect bilirubin may indicate different types of liver problems.
- Related Tests: Other tests related to liver function (alanine transaminase [ALT], aspartate transaminase [AST], alkaline phosphatase [ALP], albumin and total protein, gammaglutamyltransferase [GGT], lactate dehydrogenase [LDH], prothrombin time [PT]).
- Approximate Reference Range (Each site will have slight variations): 0.1 – 0.3 mg/dL or 1.7 – 5.1 mmol/L
- Comments: Bilirubin levels tend to be slightly higher in males than females, while African Americans show lower values. Strenuous exercise may also increase bilirubin levels.
- Test Method Variance: Urine levels of bilirubin may also be clinically significant.
- Other: It is not uncommon to see high bilirubin levels in newborns, typically 1 to 3 days old. This is sometimes called physiologic jaundice of the newborn. Within the first 24 hours of life, up to 50% of full-term newborns, and an even greater percentage of pre-term babies, may have a high bilirubin level. After birth, newborns begin breaking down the excess red blood cells (RBCs) they are born with and, since the newborn's liver is not fully mature, it is unable to process the extra bilirubin, causing the infant's bilirubin levels to rise in the blood and other body tissues. This situation usually resolves itself within a few days. In other instances, newborns' red blood cells may be being destroyed because of blood incompatibilities between the baby and her mother, called hemolytic disease of the newborn.
  - Associated LOINC: 14629-0, 15152-2, 18264-2, 1968-7, 29760-6, 35191-6, 43820-0

- Associated CPT: 82248

2. BILI_INDIR (Bilirubin, Indirect)

   - Common Name: Indirect Bilirubin
   - Long Name: Bilirubin, indirect/unconjugated/nonglucuronidated in blood/serum
   - Test Type: BILI_INDIR
   - Panels: Hepatic/Liver Function Panel
   - Equivalent Tests: Indirect bilirubin; unconjugated bilirubin
   - Not Equivalent Tests:
   - Indications: The bilirubin test is used to check liver function and watch for signs of liver disease, such as hepatitis or cirrhosis, or the effects of medicines that can damage the liver. The test is also used to find out if something is blocking the bile ducts. This may occur if gallstones, tumors of the pancreas, or other conditions are present. Bilirubin tests are also used to diagnose conditions that cause increased destruction of red blood cells, such as hemolytic anemia or hemolytic disease of the newborn.
   - Description: Bilirubin testing checks for levels of bilirubin — an orange-yellow pigment — in blood. Bilirubin results from the normal breakdown of red blood cells. As a normal process, bilirubin is carried in the blood and passes through your liver. Too much bilirubin may indicate liver damage or disease. Before reaching the liver, as a break-down product of red blood cells, the bilirubin is called indirect (unconjugated) bilirubin. Once in the liver, most bilirubin attaches to certain sugars creating what's called direct (conjugated) bilirubin. Direct bilirubin is released into bile and stored in the gallbladder before eventually being excreted in stool. Higher than normal levels of direct or indirect bilirubin may indicate different types of liver problems.
   - Related Tests: Other tests related to liver function (alanine transaminase [ALT], aspartate transaminase [AST], alkaline phosphatase [ALP], albumin and total protein, gammaglutamyltransferase [GGT], lactate dehydrogenase [LDH], prothrombin time [PT])
   - Approximate Reference Range (Each site will have slight variations): 0.2 – 0.8 mg/dL or 3.4 – 12.0 mmol/L

- Comments: Bilirubin levels tend to be slightly higher in males than females, while African Americans show lower values. Strenuous exercise may also increase bilirubin levels.
- Test Method Variance:
- Other: It is not uncommon to see high bilirubin levels in newborns, typically 1 to 3 days old. This is sometimes called physiologic jaundice of the newborn. Within the first 24 hours of life, up to 50% of full-term newborns, and an even greater percentage of pre-term babies, may have a high bilirubin level. After birth, newborns begin breaking down the excess red blood cells (RBCs) they are born with and, since the newborn's liver is not fully mature, it is unable to process the extra bilirubin, causing the infant's bilirubin levels to rise in the blood and other body tissues. This situation usually resolves itself within a few days. In other instances, newborns' red blood cells may be being destroyed because of blood incompatibilities between the baby and her mother, called hemolytic disease of the newborn.
  - Associated LOINC: 14630-8, 15153-0,1970-3, 1971-1, 22665-4, 35192-4
  - Associated CPT:

3. BILI_TOT (Bilirubin, Total)

- Common Name: Total Bilirubin
- Long Name: Bilirubin, total in blood/serum
- Test Type: BILI_TOT
- Panels: Hepatic/Liver Function Panel
- Equivalent Tests: Total bilirubin; TBIL; Neonatal bilirubin
- Not Equivalent Tests: The following bilirubin tests are subsets (components) of total bilirubin but are not the same as total bilirubin and have separate TEST_TYPES: Direct bilirubin; Conjugated bilirubin; Indirect bilirubin; Unconjugated bilirubin.
- Indications: The bilirubin test is used to check liver function and watch for signs of liver disease, such as hepatitis or cirrhosis, or the effects of medicines that can damage the liver. The test is also used to find out if something is blocking the bile ducts. This may occur if gallstones, tumors of the pancreas, or other conditions are present. Bilirubin tests are also used to diagnose conditions that

cause increased destruction of red blood cells, such as hemolytic anemia or hemolytic disease of the newborn.

- Description: Bilirubin testing checks for levels of bilirubin — an orange-yellow pigment — in blood. Bilirubin results from the normal breakdown of red blood cells. As a normal process, bilirubin is carried in the blood and passes through your liver. Too much bilirubin may indicate liver damage or disease. Before reaching the liver, as a break-down product of red blood cells, the bilirubin is called indirect (unconjugated) bilirubin. Once in the liver, most bilirubin attaches to certain sugars creating what's called direct (conjugated) bilirubin. Direct bilirubin is released into bile and stored in the gallbladder before eventually being excreted in stool. Higher than normal levels of direct or indirect bilirubin may indicate different types of liver problems. Total bilirubin is indirect plus direct bilirubin plus various amounts of uncommon bilirubin subsets all combined into the total bilirubin measurement.
- Related Tests: Other tests related to liver function (alanine transaminase [ALT], aspartate transaminase [AST], alkaline phosphatase [ALP], albumin and total protein, gammaglutamyltransferase [GGT], lactate dehydrogenase [LDH], prothrombin time [PT])
- Approximate Reference Range (Each site will have slight variations): 0.3 – 1.0 mg/dL or 5.1 – 17.0 mmol/L
  - Normal Values in Newborns: Normal values in newborns depend on the age of the baby in hours and whether the baby was premature or full term. Normal values may vary from lab to lab.
    * Total bilirubin levels in newborns less than 7 days old
    * Age Premature baby Full-term baby
    * Less than 24 hours $<$ 8.0 mg/dL ($<$ 137 mmol/L) $<$ 6.0 mg/dL ($<$ 103 mmol/L)
    * Less than 48 hours $<$ 12.0 mg/dL ($<$ 205 mmol/L) $<$ 10.0 mg/dL ($<$ 170 mmol/L)
    * 3 to 5 days $<$ 15.0 mg/dL ($<$ 256 mmol/L) $<$ 12.0 mg/dL ($<$ 205 mmol/L)
    * 7 days or older $<$ 15.0 mg/dL ($<$ 256 mmol/L) $<$ 10.0 mg/dL ($<$ 170 mmol/L)
- Comments: Bilirubin levels tend to be slightly higher in males

than females, while African Americans show lower values. Strenuous exercise may also increase bilirubin levels.

- Test Method Variance:
- Other: It is not uncommon to see high bilirubin levels in newborns, typically 1 to 3 days old. This is sometimes called physiologic jaundice of the newborn. Within the first 24 hours of life, up to 50% of full-term newborns, and an even greater percentage of pre-term babies, may have a high bilirubin level. After birth, newborns begin breaking down the excess red blood cells (RBCs) they are born with and, since the newborn's liver is not fully mature, it is unable to process the extra bilirubin, causing the infant's bilirubin levels to rise in the blood and other body tissues. This situation usually resolves itself within a few days. In other instances, newborns' red blood cells may be being destroyed because of blood incompatibilities between the baby and her mother, called hemolytic disease of the newborn.
  - Associated LOINC: 14631-6, 1975-2, 33898-8, 33899-6, 35194-0, 42719-5, 54363-7, 59827-6, 59828-4, 77137-8, 89871-8, 89872-6, 97770-2
  - Associated CPT: 82247

1.1.9 BNP (Brain Natriuretic Peptide)

- Common Name: BNP
- Long Name: Natriuretic Peptide B or B-type; Brain Natriuretic Peptide
- Test Type: BNP
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: N-Terminal-ProBNP
- Indications: To help diagnose the presence and severity of heart failure. BNP levels can assist in differentiating between heart failure and other problems, such as lung disease.

- Description: This test measures the concentration of BNP in the blood. The heart normally produces low levels of a precursor protein, pro-BNP, which is cleaved to release the active hormone BNP and an inactive fragment, NT-proBNP. The purpose of BNP is to help regulate blood volume and, therefore, the work the heart must do in pumping blood throughout the body. BNP is produced mainly in the heart's left ventricle (the organ's main pumping chamber). When the left ventricle is stretched from having to work harder, the concentrations of BNP in blood can increase markedly. This situation may occur in heart failure as well as other diseases that affect the heart and circulatory system. Higher-than-normal results suggest that a person is in heart failure, and the level of BNP in the blood is related to the severity of heart failure. Higher levels of BNP also may be associated with a worse prognosis for the patient.
- Related Tests: N-Terminal-ProBNP; Cardiac biomarkers such as CK and Troponin.
- Approximate Reference Range: Normal $<$ 100pg/ml
- Comments: BNP levels decrease in most patients who have been taking drug therapies for heart failure. Levels of BNP tend to increase with age and are increased in persons with kidney disease.
- Test Method Variance:
- Other:
  - Associated LOINC: 30934-4, 35257-5, 42637-9, 47092-2
  - Associated CPT: 83880

1.1.10 BNP_PROHORMONE (N-terminal prohormone-B-type natriuretic peptide)

- Common Name: N-terminal prohormone-B-type natriuretic peptide (NT-proBNP)
- Long Name: Natriuretic peptide.B prohormone N-Terminal [Mass/volume] in Serum or Plasma
- Test Type: BNP_PROHORMONE
- Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Heart Failure Diagnosis and Severity Assessment is the primary use of this test to help diagnose or rule out heart failure in individuals experiencing symptoms like shortness of breath and swelling. This test helps monitor the effectiveness of treatment for heart failure and determine if the condition is worsening or improving. This test can also be used to assess the risk of future cardiovascular events and mortality in patients with heart failure or other cardiac conditions.
- Description: NT-proBNP is a protein fragment that the heart produces as a precursor to Brain Natriuretic Peptide (BNP), a hormone involved in regulating blood pressure and fluid balance. When the heart is under stress or working harder than normal, it releases more BNP and NT-proBNP.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): normal levels are less than 125 pg/mL for people under 75 and less than 450 pg/mL for those over 75.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 33762-6, 33763-4, 71425-3, 83107-3, 83108-1
  - Associated CPT:

1.1.11 BUN (Urea nitrogen in blood/serum)

- Common Name: BUN
- Long Name: Urea nitrogen in blood/serum
- Test Type: BUN
- Panels: Basic Metabolic Panel (BMP); Comprehensive Metabolic Panel (CMP)
- Equivalent Tests: N/A

- Not Equivalent Tests: Urine urea nitrogen
- Indications: To evaluate kidney function or monitor the effectiveness of dialysis and other treatments related to kidney disease or damage. BUN is also used to evaluate general health status when ordered as part of a basic metabolic panel or comprehensive metabolic panel.
- Description: BUN measures the amount of urea nitrogen in the blood. Nitrogen, in the form of ammonia, is produced in the liver when protein is broken into its component parts (amino acids) and metabolized. The nitrogen combines with other molecules in the liver to form the waste product urea. Urea is then released into the bloodstream and carried to the kidneys where it is filtered out of the blood and excreted in the urine. Because this is an ongoing process, there is normally a small but stable amount of urea nitrogen in the blood. Most diseases or conditions that affect the kidneys or liver have the potential to affect the amount of urea present in the blood. If increased amounts of urea are produced by the liver or decreased amounts are excreted by the kidneys, then blood urea nitrogen concentrations will rise. If significant liver damage or disease inhibits the production of urea, then BUN concentrations may fall.
- Related Tests: Creatinine; Creatinine Clearance; eGFR; CMP; BMP; Urinalysis; Microalbumin
- Approximate Reference Range (As the BUN reference range is lab specific, there is no "standard" BUN reference range): Approximately = 6-23 mg/dL.
- Comments: BUN levels increase with age. BUN levels in very young babies are about 2/3 of the levels found in healthy young adults, while levels in adults over 60 years of age are slightly higher than in younger adults. Levels are also slightly higher in men than women. Both decreased and increased BUN concentrations may be seen during a normal pregnancy. If one kidney is fully functional, BUN concentrations may be normal even when significant dysfunction is present in the other kidney.
- Test Method Variance:
- Other:

- Associated LOINC: 12961-9, 12962-7, 12963-5, 14937-7, 3094-0, 35234-4, 59570-2, 6299-2
- Associated CPT: 84520, 84525

1.1.12 (CA) Cancer Antigen Test Types

1. CA125 (Cancer Antigen 125), Quantitative in Blood

   - Common Name: CA125
   - Long Name: Cancer Antigen 125
   - Test Type: CA125
   - Panels:
   - Equivalent Tests: Blood CA125
   - Not Equivalent Tests:
   - Indications: A doctor may order a CA-125 test before a woman starts ovarian cancer treatment as a baseline to compare against future measurements. During therapy, physicians order CA-125 testing at intervals to monitor response to therapy. CA-125 may also be measured periodically after therapy is completed. An increase in CA-125 may indicate that the cancer has returned.
   - Description: CA-125 is primarily used to monitor therapy during treatment for ovarian cancer. CA-125 is also used to detect whether cancer has come back after treatment is complete. Series of CA-125 tests that show rising or falling concentrations are often more useful than a single result. This test is sometimes used to test and monitor high-risk women who have a family history of ovarian cancer but who do not yet have the disease. This test is not used to screen for ovarian cancer because it is non-specific. Levels in the blood can be elevated in other conditions such as normal menstruation, pregnancy, endometriosis, and pelvic inflammatory disease.
   - Related Tests: Tumor markers, BRCA-1, BRCA-2
   - Approximate Reference Range (Each site will have slight variations): Normal range less that 35 U/ml
   - Comments: If CA-125 levels fall during therapy, this generally indicates that the cancer is responding to treatment. If CA-125 levels rise or stay the same, then the cancer may not be responding to therapy. High CA-125 levels after treatment is complete

may indicate that the cancer has come back. If a woman who has been diagnosed with ovarian cancer has a baseline CA-125 level that is normal, then the test is not likely to be useful to monitor her ovarian cancer. In this case, the ovarian cancer may not be producing CA-125 so it is not a good marker of disease progression.

- Test Method Variance:
- Other:
  - Associated LOINC: 10334-1, 83082-8
  - Associated CPT: 86304

2. CA125_BF (Cancer Antigen 125), Quantitative in Body Fluid

   - Common Name: CA125 in body fluid
   - Long Name: CA 125 Body Fluid
   - Test Type: CA125_BF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A doctor may order a CA-125 test before a woman starts ovarian cancer treatment as a baseline to compare against future measurements. During therapy, physicians order CA-125 testing at intervals to monitor response to therapy. CA-125 may also be measured periodically after therapy is completed. An increase in CA-125 may indicate that the cancer has returned.
   - Description: CA-125 is primarily used to monitor therapy during treatment for ovarian cancer. CA-125 is also used to detect whether cancer has come back after treatment is complete. Series of CA-125 tests that show rising or falling concentrations are often more useful than a single result. This test is sometimes used to test and monitor high-risk women who have a family history of ovarian cancer but who do not yet have the disease. This test is not used to screen for ovarian cancer because it is non-specific. Levels in the blood can be elevated in other conditions such as normal menstruation, pregnancy, endometriosis, and pelvic inflammatory disease.
   - Related Tests: Tumor markers, BRCA-1, BRCA-2

- Approximate Reference Range (Each site will have slight variations):
- Comments: If CA-125 levels fall during therapy, this generally indicates that the cancer is responding to treatment. If CA-125 levels rise or stay the same, then the cancer may not be responding to therapy. High CA-125 levels after treatment is complete may indicate that the cancer has come back. If a woman who has been diagnosed with ovarian cancer has a baseline CA-125 level that is normal, then the test is not likely to be useful to monitor her ovarian cancer. In this case, the ovarian cancer may not be producing CA-125 so it is not a good marker of disease progression.
- Test Method Variance:
- Other:
  - Associated LOINC: 11210-2 (body fluid), 19165-0 (pleural) ,40618-1 (peritoneal), 50775-6 (CSF), 68923-2 (pericard)
  - Associated CPT: 86304

3. CA125_BF_T (Cancer Antigen 125) Titer in Body Fluid

   - Common Name: CA125 titer in body fluid
   - Long Name: CA 125 Titer, Body Fluid
   - Test Type: CA125_BF_T
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A doctor may order a CA-125 test before a woman starts ovarian cancer treatment as a baseline to compare against future measurements. During therapy, physicians order CA-125 testing at intervals to monitor response to therapy. CA-125 may also be measured periodically after therapy is completed. An increase in CA-125 may indicate that the cancer has returned.
   - Description: CA-125 is primarily used to monitor therapy during treatment for ovarian cancer. CA-125 is also used to detect whether cancer has come back after treatment is complete. Series of CA-125 tests that show rising or falling concentrations are often more useful than a single result. This test is sometimes used

to test and monitor high-risk women who have a family history of ovarian cancer but who do not yet have the disease. This test is not used to screen for ovarian cancer because it is non-specific. Levels in the blood can be elevated in other conditions such as normal menstruation, pregnancy, endometriosis, and pelvic inflammatory disease.

- Related Tests: Tumor markers, BRCA-1, BRCA-2
- Approximate Reference Range (Each site will have slight variations):
- Comments: If CA-125 levels fall during therapy, this generally indicates that the cancer is responding to treatment. If CA-125 levels rise or stay the same, then the cancer may not be responding to therapy. High CA-125 levels after treatment is complete may indicate that the cancer has come back. If a woman who has been diagnosed with ovarian cancer has a baseline CA-125 level that is normal, then the test is not likely to be useful to monitor her ovarian cancer. In this case, the ovarian cancer may not be producing CA-125 so it is not a good marker of disease progression.
- Test Method Variance:

Other:

- Associated LOINC: 15156-3
- Associated CPT: 86304

4. CA125_T (Cancer Antigen 125), Titer in Blood

   - Common Name: CA125 titer
   - Long Name: CA 125 titer
   - Test Type: CA125_T
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A doctor may order a CA-125 test before a woman starts ovarian cancer treatment as a baseline to compare against future measurements. During therapy, physicians order CA-125 testing at intervals to monitor response to therapy. CA-125 may

also be measured periodically after therapy is completed. An increase in CA-125 may indicate that the cancer has returned.

- Description: CA-125 is primarily used to monitor therapy during treatment for ovarian cancer. CA-125 is also used to detect whether cancer has come back after treatment is complete. Series of CA-125 tests that show rising or falling concentrations are often more useful than a single result. This test is sometimes used to test and monitor high-risk women who have a family history of ovarian cancer but who do not yet have the disease. This test is not used to screen for ovarian cancer because it is non-specific. Levels in the blood can be elevated in other conditions such as normal menstruation, pregnancy, endometriosis, and pelvic inflammatory disease.
- Related Tests: Tumor markers, BRCA-1, BRCA-2
- Approximate Reference Range (Each site will have slight variations):
- Comments: If CA-125 levels fall during therapy, this generally indicates that the cancer is responding to treatment. If CA-125 levels rise or stay the same, then the cancer may not be responding to therapy. High CA-125 levels after treatment is complete may indicate that the cancer has come back. If a woman who has been diagnosed with ovarian cancer has a baseline CA-125 level that is normal, then the test is not likely to be useful to monitor her ovarian cancer. In this case, the ovarian cancer may not be producing CA-125 so it is not a good marker of disease progression.
- Test Method Variance:
- Other:
  - Associated LOINC: 15157-1
  - Associated CPT: 86304

5. CA125_QL (Cancer Antigen 125), Qualitative in Blood

   - Common Name: CA125
   - Long Name: CA 125 Qualitative
   - Test Type: CA125_QL
   - Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A doctor may order a CA-125 test before a woman starts ovarian cancer treatment as a baseline to compare against future measurements. During therapy, physicians order CA-125 testing at intervals to monitor response to therapy. CA-125 may also be measured periodically after therapy is completed. An increase in CA-125 may indicate that the cancer has returned.
- Description: CA-125 is primarily used to monitor therapy during treatment for ovarian cancer. CA-125 is also used to detect whether cancer has come back after treatment is complete. Series of CA-125 tests that show rising or falling concentrations are often more useful than a single result. This test is sometimes used to test and monitor high-risk women who have a family history of ovarian cancer but who do not yet have the disease. This test is not used to screen for ovarian cancer because it is non-specific. Levels in the blood can be elevated in other conditions such as normal menstruation, pregnancy, endometriosis, and pelvic inflammatory disease.
- Related Tests: Tumor markers, BRCA-1, BRCA-2
- Approximate Reference Range (Each site will have slight variations): Not applicable (qualitative test)
- Comments: If CA-125 levels fall during therapy, this generally indicates that the cancer is responding to treatment. If CA-125 levels rise or stay the same, then the cancer may not be responding to therapy. High CA-125 levels after treatment is complete may indicate that the cancer has come back. If a woman who has been diagnosed with ovarian cancer has a baseline CA-125 level that is normal, then the test is not likely to be useful to monitor her ovarian cancer. In this case, the ovarian cancer may not be producing CA-125 so it is not a good marker of disease progression.
- Test Method Variance:
- Other:
  - Associated LOINC: 2006-5
  - Associated CPT: 86304

6. CA15_3 (Cancer Ag 15-3 Qn), Quantitative in Serum or Plasma

- Common Name: CA15-3
- Long Name: Cancer Antigen 15-3 Quantitative
- Test Type: CA15_3
- Panels:
- Equivalent Tests: Blood CA15-3
- Not Equivalent Tests:
- Indications: A doctor may order a CA-15-3 test to measure the amount of a protein called cancer antigen 15-3 (CA 15-3) in your blood. CA 15-3 is released into the bloodstream by some breast cancer cells. This test is primarily used to monitor breast cancer patients, assess treatment effectiveness, and detect disease recurrence.
- Description: Doctors mainly use CA15-3 tests to check if advanced breast cancer is responding to treatment or to see if it has come back (recurred) after treatment. They will use this test to measure the CA15-3 level in the blood before, during and after treatment.
- Related Tests: Tumor markers, BRCA-1, BRCA-2
- Approximate Reference Range (Each site will have slight variations): Normal range less that 30 U/ml
- Comments: If CA15-3 levels fall during therapy, this generally indicates that the cancer is responding to treatment. If CA15-3 levels rise or stay the same, then the cancer may not be responding to therapy. High CA15-3 levels after treatment is complete may indicate that the cancer has come back. If a woman who has been diagnosed with breast cancer has a baseline CA15-3 level that is normal, then the test is not likely to be useful to monitor breast cancer. In this case, the breast cancer may not be producing CA15-3 so it is not a good marker of disease progression.
- Test Method Variance:
- Other:
  - Associated LOINC: 6875-9, 83083-6
  - Associated CPT:

7. CA15_3_QL (Cancer Ag 15-3 Ql), Qualitative in Serum or Plasma
   - Common Name: CA15-3 Qualitative

- Long Name: Cancer Antigen 15-3 Qualitative
- Test Type: CA15_3_QL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A doctor may order a CA-15-3 test to measure the amount of a protein called cancer antigen 15-3 (CA 15-3) in your blood. CA 15-3 is released into the bloodstream by some breast cancer cells. This test is primarily used to monitor breast cancer patients, assess treatment effectiveness, and detect disease recurrence.
- Description: Doctors mainly use CA15-3 tests to check if advanced breast cancer is responding to treatment or to see if it has come back (recurred) after treatment. They will use this test to measure the CA15-3 level in the blood before, during and after treatment.
- Related Tests: Tumor markers, BRCA-1, BRCA-2
- Approximate Reference Range (Each site will have slight variations): Not applicable (qualitative test)
- Comments: If CA15-3 levels fall during therapy, this generally indicates that the cancer is responding to treatment. If CA15-3 levels rise or stay the same, then the cancer may not be responding to therapy. High CA15-3 levels after treatment is complete may indicate that the cancer has come back. If a woman who has been diagnosed with breast cancer has a baseline CA15-3 level that is normal, then the test is not likely to be useful to monitor breast cancer. In this case, the breast cancer may not be producing CA15-3 so it is not a good marker of disease progression.
- Test Method Variance:
- Other:
  - Associated LOINC: 2007-3
  - Associated CPT:

8. CA199 (Cancer Ag 19-9), in Serum or Plasma

- Common Name: CA19-9
- Long Name: Cancer Antigen 19-9 Quantitative

- Test Type: CA199
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: CA19-9 is usually used to monitor the progress in treating pancreatic cancers. Elevated levels of CA 19-9 can indicate pancreatic cancer, but it's also associated with other cancers like those of the stomach, bile duct, and colon, as well as gallbladder, lung, and ovarian cancers. Additionally, non-cancerous conditions like gallstones, pancreatitis, and liver disease can also cause elevated CA 19-9 levels.
- Description: CA19-9 is a protein that is normally found in small amounts in the body. It is made by cells in the pancreas, liver and gallbladder. It can also be found on certain types of cancer cells.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range less that 37 U/ml
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 2009-9, 2010-7, 24108-3, 83084-4
  - Associated CPT:

9. CA27_29 (Cancer Ag 27-29 Qn)

- Common Name: CA27_29
- Long Name: Cancer Antigen 27_29 Quantitative
- Test Type: CA27_29
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: CA 27-29 is a cancer antigen used as a tumor marker to monitor breast cancer, specifically in cases of metastasis or recurrence. It is not used for screening or diagnosing breast cancer, but rather to track how well treatment is working or if the cancer has spread or returned.

- Description: The CA 27-29 marker (similar to CA 15-3) is used to follow the course of treatment in women with advanced breast cancer. Cancers of the colon, stomach, kidney, lung, ovary, pancreas, uterus, and liver may also raise CA 27-29 levels. Noncancerous conditions that may increase CA27-29 are: first trimester pregnancy, endometriosis, ovarian cysts, benign breast disease, kidney disease, and liver disease.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range less that 38 U/ml
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 17842-6
  - Associated CPT:

10. CA27_29_QL (Cancer Ag 27-29 Ql)

    - Common Name: CA27_29 Qualitative
    - Long Name: Cancer Antigen 27_29 Qualitative
    - Test Type: CA27_29_QL
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: CA 27-29 is a cancer antigen used as a tumor marker to monitor breast cancer, specifically in cases of metastasis or recurrence. It is not used for screening or diagnosing breast cancer, but rather to track how well treatment is working or if the cancer has spread or returned.
    - Description: The CA 27-29 marker (similar to CA 15-3) is used to follow the course of treatment in women with advanced breast cancer. Cancers of the colon, stomach, kidney, lung, ovary, pancreas, uterus, and liver may also raise CA 27-29 levels. Noncancerous conditions that may increase CA27-29 are: first trimester pregnancy, endometriosis, ovarian cysts, benign breast disease, kidney disease, and liver disease.
    - Related Tests:

- Approximate Reference Range (Each site will have slight variations): Not applicable (qualitative test)
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 2012-3
  - Associated CPT:

1.1.13 CALCIUM (Total Serum Calcium)

- Common Name: Total serum calcium; Ca++
- Long Name: Calcium, total in serum/blood
- Test Type: CALCIUM
- Panels: Basic metabolic profile, comprehensive metabolic profile, general health panel, renal function panel
- Equivalent Tests:
- Not Equivalent Tests: Free calcium; ionized calcium; body fluids other than blood (i.e. urine)
- Indications: To screen for, diagnose, and monitor a range of conditions relating to the bones, heart, nerves, kidneys, and parathyroid glands; also monitored in certain types of cancer. Certain drugs can affect calcium levels.
- Description:
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): ~ 9.0-10.5 mg/dl
- Comments: Blood levels of total calcium may be affected by levels of protein in the blood, since calcium in the blood is attached to albumin; ionized calcium is not attached to albumin. Certain antacids and calcium supplements may affect levels.
- Test Method Variance:

- Other:
  - Associated LOINC: 17861-6, 1996-8, 2000-8, 35246-8, 42593-4, 42857-3, 49765-1
  - Associated CPT: 82310

1.1.14 CEA (Carcinoembryonic Ag)

- Common Name: Carcinoembryonic Ag
- Long Name: Carcinoembryonic Ag in Serum or Plasma
- Test Type: CEA
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Carcinoembryonic antigen (CEA) is increased in certain cancers, particularly colorectal cancer, but can also be elevated in other conditions and in people who smoke. CEA testing is a blood test that measures CEA levels, primarily used to monitor for cancer, assess treatment effectiveness, or detect recurrence.
- Description: Carcinoembryonic antigen (CEA) is a protein normally found in very low levels in the blood of adults. A CEA test is most commonly used for colorectal cancer.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range less that 3 ng/ml
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 19166-8, 19167-6, 2039-6, 83085-1
  - Associated CPT:

1.1.15 CHLORIDE (Chloride, total)

- Common Name: Chloride
- Long Name: Chloride in serum, plasma, blood
- Test Type: CHLORIDE
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Symptoms of potential chloride imbalances (both low and high) can include fatigue, weakness, difficulty breathing, dehydration, and muscle twitching.
- Description: A chloride blood test measures the amount of chloride in your blood. Chloride is an electrolyte that helps maintain fluid balance, blood volume, and blood pressure. The test can be used to help diagnose and monitor conditions related to electrolyte imbalances, kidney problems, or lung disorders.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal range for blood chloride is typically between 96 and 106 mEq/L.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 100745-9, 2069-3, 2075-0, 41649-5, 41650-3, 51590-8, 77138-6
  - Associated CPT:

1.1.16 (CK) Creatine Kinase Test Types

1. CK (Creatine Kinase)
   - Common Name: CK

- Long Name: Creatine Kinase or CPK - total concentration in blood of all CK subtypes
- Test Type: CK
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: BNP; Troponin
- Indications: Blood levels of CK rise when muscle or heart cells are injured. This test is used if a patient has chest pain or other signs and symptoms of a heart attack. In the first 4 to 6 hours after a heart attack, the concentration of CK in blood begins to rise. It reaches its highest level in 18 to 24 hours and returns to normal within 2 to 3 days. This test is also used to monitor skeletal muscles damage.
- Description: Creatine kinase is an enzyme found in the heart, brain, skeletal muscle, and other tissues. Enzymes are proteins that help cells to perform their normal functions. In muscle and heart cells, most of this energy is used when muscles contract. There are three different forms (or isoenzymes) of CK: CK-MM (found in skeletal muscles and heart); CK-MB (found mostly in the heart); and CK-BB (found mostly in the brain). The small amount of CK that is normally in the blood comes mainly from muscles and not the brain.
- Related Tests: CK_MB; CK_MB; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_BF, CK_CSF; CK_ISO; CK_ISO_EP; CK_MB; CK_MB_EIA; CK_MB_EP; CK_MBI_CLC; CK_MBI_EIA; CK_MBI_EP; CK_MM_EP; CK_MMI; CK_MMI_EP; CK_QN; Troponin; BNP
- Approximate Reference Range (Normal values vary from lab to lab and from one type of testing protocol to another): Male: 38 - 174 units/L; Female: 96 - 140 units/L
- Comments: People who have greater muscle mass have higher CK levels than those who do not. African-Americans may have higher CK levels than other racial/ethnic groups. Very heavy exercise can also increase CK. Other forms of muscle damage, such as from a fall, a car accident, surgery, or an injection, can also increase CK. Several drugs, including cholesterol-lowering drugs (statins), can damage muscle and increase CK. Drinking excessive amounts

of alcohol also may increase CK. Early pregnancy can decrease CK levels.

- Test Method Variance:
- Other:
  - Associated LOINC: 2157-6, 24335-2, 50756-6
  - Associated CPT: 82550

2. CK_BB_EP (Creatine Kinase BB Electrophoresis) NOTE: This single reference document contains information pertinent to the individual TEST_TYPES CK_BB_EP, and the indexes CK_BBI, or CK_BBI_EP.

   - Common Name: CK BB (also known as CPK BB)
   - Long Name: Creatine Kinase BB (Brain-type)
   - Test Type: CK_BB_EP is CK_BB by the testing method electrophoresis
     - CK_BBI is the Creatine Kinase BB/Creatine Kinase Total as a ratio or index
     - CK_BBI_EP is the Creatine Kinase BB/Creatine Kinase Total as a ratio or index by the testing method electrophoresis
   - Panels: N/A
   - Equivalent Tests: N/A
   - Not Equivalent Tests: All other CK TEST_TYPES that do not contain "BB" in name.
   - Indications: Blood levels of CK rise when muscle, brain or heart cells are injured. CK_BB is found primarily in the brain. CK_BB is sometimes used in diagnosing of cancer of the lung and stomach. However, CK_BB is expressed in all tissues at low levels and has little clinical relevance.
   - Description: CK_BB is one of three separate forms (isoenzymes) of the enzyme creatine kinase (CK). CK_BB is found mostly in the brain, lungs, intestinal tract and smooth muscle.
   - Related Tests: CK; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_CSF; CK_ISO; CK_ISO_EP
   - Approximate Reference Range:
   - Comments: $<$ 1% of total CK

- Test Method Variance:
- Other:
  - Associated LOINC: 2152-7
  - Associated CPT: 82552, 82554

3. CK_BBI (Creatine Kinase BB Total) NOTE: This single reference document contains information pertinent to the individual TEST_TYPEs CK_BB_EP, and the indexes CK_BBI, or CK_BBI_EP.

   - Common Name: CK BB (also known as CPK BB)
   - Long Name: Creatine Kinase BB/Creatine Kinase Total
   - Test Type: CK_BBI
   - Panels: N/A
   - Equivalent Tests: N/A
   - Not Equivalent Tests: All other CK TEST_TYPES that do not contain "BB" in name.
   - Indications: Blood levels of CK rise when muscle, brain or heart cells are injured. CK_BB isfound primarily in the brain. CK_BB is sometimes used in diagnosing of cancer of the lung andstomach. However, CK_BB is expressed in all tissues at low levels and has little clinicalrelevance.
   - Description: CK_BBI is the Creatine Kinase BB/Creatine Kinase Total as a ratio or index CK_BBI_EP is the Creatine Kinase BB/Creatine Kinase Total as a ratio or index by the testing method electrophoresis. CK_BB is one of three separate forms (isoenzymes) of the enzyme creatinekinase (CK). CK_BB is found mostly in the brain, lungs, intestinal tract and smooth muscle.
   - Related Tests: CK; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_CSF; CK_ISO; CK_ISO_EP
   - Approximate Reference Range:
   - Comments: $<$ 1% of total CK
   - Test Method Variance:
   - Other:
     - Associated LOINC: 9642-0
     - Associated CPT: 82552, 82554

4. CK_BBI_EP (Creatine Kinase BB Total Electrophoresis) NOTE: This single reference document contains information pertinent to the individual TEST_TYPES CK_BB_EP, and the indexes CK_BBI, or CK_BBI_EP.

    - Common Name: CK BB (also known as CPK BB)
    - Long Name: Creatine Kinase BB/Creatine Kinase Total Electrophoresis
    - Test Type: CK_BBI_EP
    - Panels: N/A
    - Equivalent Tests: N/A
    - Not Equivalent Tests: All other CK TEST_TYPES that do not contain "BB" in name.
    - Indications: Blood levels of CK rise when muscle, brain or heart cells are injured. CK_BB isfound primarily in the brain. CK_BB is sometimes used in diagnosing of cancer of the lung andstomach. However, CK_BB is expressed in all tissues at low levels and has little clinicalrelevance.
    - Description: CK_BB_EP is CK_BB by the testing method electrophoresis CK_BBI is the Creatine Kinase BB/Creatine Kinase Total as a ratio or indexCK_BBI_EP is the Creatine Kinase BB/Creatine Kinase Total as a ratio or index by the testing-method electrophoresis. CK_BB is one of three separate forms (isoenzymes) of the enzyme creatinekinase (CK). CK_BB is found mostly in the brain, lungs, intestinal tract and smooth muscle.
    - Related Tests: CK; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_CSF; CK_ISO; CK_ISO_EP
    - Approximate Reference Range:
    - Comments: $<$ 1% of total CK
    - Test Method Variance:
    - Other:
        - Associated LOINC: 15048-2, 72565-5
        - Associated CPT: 82552, 82554

5. CK_BF (Creatine Kinase in Body Fluid)

    - Common Name: Creatine Kinase in Body Fluid

- Long Name: Creatine Kinase Body Fluid Quantitative
- Test Types: CK_BF
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests:
- Indications: The CK test is primarily used to diagnose and monitor injuries and conditions that cause damage to skeletal muscles.
- Description: A creatine kinase (CK) test, also known as creatine phosphokinase (CPK) test, measures the level of the enzyme creatine kinase in a sample of blood or other body fluids. CK is primarily found in muscle tissue (skeletal muscle and heart muscle) and the brain. Under normal conditions, only small amounts of CK leak into the bloodstream as a result of normal wear and tear on the muscles. However, significant damage to muscle tissue (skeletal muscle, heart muscle), or the brain can cause larger amounts of CK to leak into the bloodstream, leading to elevated CK levels.
- Related Tests:
- Approximate Reference Range: The normal range for adults generally falls between 22 and 198 U/L.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 16688-4
  - Associated CPT: 82550

6. CK_CSF (Creatine Kinase CSF Quantitative)

- Common Name: CSF CK
- Long Name: Creatine Kinase in Cerebral Spinal Fluid
- Test Type: CK_CSF
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: All other CK TEST_TYPES.
- Indications: CK activity in CSF is indicative of cerebral injury (brain damage).

- Description: Creatine kinase is an enzyme found in the heart, brain, skeletal muscle, and other tissues. For more information on CK total, refer to CK information document.
- Related Tests: CK; CK_MB; CK_MB; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_BF, CK_ISO; CK_ISO_EP; CK_MB; CK_MB_EIA; CK_MB_EP; CK_MBI_CLC; CK_MBI_EIA; CK_MBI_EP;CK_MM_EP; CK_MMI; CK_MMI_EP; CK_QN
- Approximate Reference Range:
- Comments: Creatine kinase is not normally present in cerebrospinal fluid (CSF).
- Test Method Variance:
- Other:
  - Associated LOINC: 2151-9
  - Associated CPT: 82550

7. CK_ISO (Creatine Kinase Isoenzymes)

- Common Name: Creatine Kinase Isoenzymes
- Long Name: Creatine Kinase Isoenzymes Interpretation - analytic method not specified
- Test Type: CK_ISO
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Healthcare providers utilize the Creatine Kinase isoenzymes test to diagnose and monitor conditions involving muscle damage, including:
  - Muscle injuries: Assessing the extent of damage from trauma, strenuous exercise, or certain medications.
  - Muscle diseases: Diagnosing or tracking the progression of conditions like muscular dystrophy.
  - Heart conditions: Evaluating heart problems such as heart attacks (myocardial infarction) or inflammation of the heart muscle (myocarditis).
  - Neurological injuries: Assessing brain injury or stroke.

- Description: The Creatine Kinase (CK) isoenzymes test, also known as Creatine Phosphokinase (CPK) isoenzymes or CK fractions, measures the levels of different types of the enzyme creatine kinase in your blood. This test helps determine the source of muscle damage within the body, as different CK isoenzymes are predominantly found in specific tissues:
  - CK-MM: Primarily found in skeletal muscles, but also present in the heart according to University of Rochester Medical Center.
  - CK-MB: Predominantly found in heart muscle, but small amounts are present in skeletal muscles.
  - CK-BB: Primarily found in brain tissue and smooth muscles (like those in the intestines or uterus).
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
  - CK-MM: 96-100% of total CK is considered within the normal range.
  - CK-MB: 0% of total CK, or in quantitative terms, 0-5 ng/mL or less than 5% of total CK, is considered normal.
  - CK-BB: 0% or 0-0.001% of total CK is the expected normal finding.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 49259-5, 5912-1
  - Associated CPT: 82552, 82554

8. CK_ISO_EP (Creatine Kinase Isoenzymes Electrophoresis)
   - Common Name: Creatine Kinase Isoenzymes Electrophoresis
   - Long Name: Creatine Kinase Isoenzymes Electrophoresis Interpretation
   - Test Type: CK_ISO_EP
   - Panels:
   - Equivalent Tests:

- Not Equivalent Tests:
- Indications: High levels of CK-MM suggest damage to skeletal muscles, potentially from strenuous exercise, injuries, or muscle diseases. Elevated levels of CK-MB are often associated with heart muscle damage, such as a heart attack or inflammation of the heart muscle. Increased levels of CK-BB may suggest brain injury or stroke, or conditions affecting smooth muscles.
- Description: A creatine kinase (CK) isoenzyme electrophoresis test is a blood test used to determine the source of muscle damage by separating and measuring the levels of different forms (isoenzymes) of CK. There are three main CK isoenzymes:
  - CK-MM: Primarily found in skeletal muscles and heart muscle.
  - CK-MB: Found mostly in the heart muscle.
  - CK-BB: Primarily found in the brain and smooth muscles.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): CK-MM: 96-100% or 100%, CK-MB: 0-4% or 0%, CK-BB: 0%
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 14680-3, 33547-1, 49258-7, 72561-4
  - Associated CPT: 82552, 82554

9. CK_MB (Creatine Kinase MB - analytic method not specified)

   - Common Name: CK_MB (also known as CPK MB)
   - Long Name: Creatine Kinase-MB
   - Test Type: CK_MB
   - Panels: N/A
   - Equivalent Tests: Use this TEST_TYPE ONLY when the analytic method is not specified in the lab data.
   - Not Equivalent Tests: All other CK TEST_TYPES.

- Indications: B lood levels of CK rise when muscle or heart cells are injured. CK_MB levels (along with total CK) are tested in persons who have chest pain to diagnose if they have had a heart attack. Since a high total CK could indicate damage to either the heart or other muscles, CK_MB helps to distinguish between these two sources.
- Description: CK–MB is one of three separate forms (isoenzymes) of the enzyme creatine kinase. CK–MB is found mostly in heart muscle. It rises when there is damage to heart muscle cells.
- Related Tests: CK; CK_MB; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_BF, CK_CSF; CK_ISO; CK_ISO_EP; CK_MB; CK_MB_EIA; CK_MB_EP; CK_MBI_CLC; CK_MBI_EIA; CK_MBI_EP; CK_MM_EP; CK_MMI; CK_MMI_EP; CK_QN; Troponin; BNP
- Approximate Reference Range: Normal CK-MB is 5% or less of total CK, often 5 – 25 IU/L.
- Comments: If the value of CK_MB is elevated and the ratio of CK_MB to total CK (relativeindex) is high, it is likely that the heart was damaged. A high CK with a low relative index suggests that skeletal muscles were damaged. Although CK_MB is a good test, it has been largely replaced by troponin, which is more specific for damage to the heart. Sometimes persons who are having trouble breathing have to use their chest muscles. Chest muscles have more CK_MB than other muscles, which would raise the amount of CK_MB in the blood. Persons whose kidneys have failed can also have high CK_MB levels without having had a heart attack. Rarely, chronic muscle disease, low thyroid hormone levels, and alcohol abuse can increase CK_MB, producing changes similar to those seen in aheart attack.
- Test Method Variance: Can be measured by test methods such as enzyme immunoassay, column chromatography, or electrophoresis. If the test method is specified, that TEST_TYPE should be used rather than CK_MB.
- Other:
  - Associated LOINC: 101446-3, 13969-1, 32673-6, 49551-5, 6773-6, 83092-7
  - Associated CPT: 82553

10. CK_MB_EIA (Creatine Kinase MB – Enzyme Immunoassay)

- Common Name: CK_MB by immunoassay (also known as CPK MB)
- Long Name: Creatine Kinase MB – Enzyme Immunoassay
- Test Type: CK_MB_EIA
- Panels: N/A
- Equivalent Tests: Use the TEST_TYPE only when the test method is stated to be enzyme immunoassay
- Not Equivalent Tests: All other CK TEST_TYPES.
- Indications: Blood levels of CK rise when muscle or heart cells are injured. CK_MB levels,along with total CK, are tested in persons who have chest pain to diagnose whether they have had a heart attack. Since a high total CK could indicate damage to either the heart or other muscles, CK_MB helps to distinguish between these two sources.
- Description: CK_MB is one of three separate forms (isoenzymes) of the enzyme creatine kinase (CK). CK_MB is found mostly in heart muscle. It rises when there is damage to heartmuscle cells.
- Related Tests: CK; CK_MB; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_BF, CK_CSF; CK_ISO; CK_ISO_EP; CK_MB; CK_MB_EP; CK_MBI_CLC; CK_MBI_EIA; CK_MBI_EP; CK_MM_EP; CK_MMI; CK_MMI_EP; CK_QN; Troponin; BNP
- Approximate Reference Range: Normal CK-MB is 5% or less of CK, often 5 to 25 IU/L.
- Comments: If the value of CK_MB is elevated and the ratio of CK_MB to total CK (relative index) is high, it is likely that the heart was damaged. A high CK with a low relative index suggests that skeletal muscles were damaged.Although CK-MB is a good test, it has been largely replaced by troponin, which is more specific for damage to the heart. Persons whose kidneys have failed or who are having trouble breathing can have high CK_MB levels without having had a heart attack. Rarely, chronic muscle disease, low thyroid hormone levels, and alcohol abuse can increase CK_MB, producing changes similar to those seen in a heart attack.
- Test Method Variance: Separation of CK into isoenzymes may be accomplished by electrophoresis, column chromatography, or

immunoassay. Immunoassay methods for isoenzymes can be accomplished rapidly, are highly sensitive and specific, and do not show the interferences common to traditional electrophoresis.

- Other:
  - Associated LOINC: (No LOINCs have been identified at this time)
  - Associated CPT: 82553

11. CK_MB_EP (Creatine Kinase MB - Electrophoresis)

    - Common Name: CK_MB by electrophoresis (also known as CPK MB)
    - Long Name: Creatine Kinase MB - Electrophoresis
    - Test Type: CK_MB_EP
    - Panels: N/A
    - Equivalent Tests: Use this TEST_TYPE only when the test method is stated to beelectrophoresis.
    - Not Equivalent Tests: All other CK TEST_TYPES.
    - Indications: Blood levels of CK rise when muscle or heart cells are injured. CK_MB levels,along with total CK, are tested in persons who have chest pain to diagnose whether they have had a heart attack. Since a high total CK could indicate damage to either the heart or other muscles, CK_MB helps to distinguish between these two sources.
    - Description: CK_MB is one of three separate forms (isoenzymes) of the enzyme creatine kinase (CK). CK_MB is found mostly in heart muscle. It rises when there is damage to heart muscle cells.
    - Related Tests: CK; CK_MB; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_BF, CK_CSF; CK_ISO; CK_ISO_EP; CK_MB; CK_MB_EIA; CK_MBI_CLC; CK_MBI_EIA; CK_MBI_EP; CK_MM_EP;CK_MMI; CK_MMI_EP; CK_QN; Troponin; BNP
    - Approximate Reference Range: Normal CK-MB is 5% or less of CK, or 5 to 25 IU/L.
    - Comments: If the value of CK_MB is elevated and the ratio of CK_MB to total CK (relative index) is high, it is likely that the heart was damaged. A high CK with a low relative index suggests that skeletal muscles were damaged.Although CK-MB

is a good test, it has been largely replaced by troponin, which is more specific for damage to the heart. Persons whose kidneys have failed or who are having trouble breathing can have high CK–MB levels without having had a heart attack. Rarely, chronic muscle disease, low thyroid hormone levels, and alcohol abuse can increase CK–MB, producing changes similar to those seen in a heart attack.

- Test Method Variance: Separation of CK into isoenzymes may be accomplished by electrophoresis, column chromatography, or immunoassay. Electrophoretically, CK_BB is most mobile, CK_MB is intermediate, and CK_MM is neutral. Although electrophoresis is possiblyless sensitive than column chromatography or immunoassay, there has been extensiveexperience and it is adequate for routine clinical use.
- Other:
  - Associated LOINC: 2154-3
  - Associated CPT: 82553

12. CK_MBI_CLC (Creatine Kinase MB/Creatine Kinase Total Calculated)

    - Common Name: CK_MB / CK (calculated ratio)
    - Long Name: Creatine Kinase MB/Creatine Kinase Total Calculated
    - Test Type: CK_MBI_CLC
    - Panels: N/A
    - Equivalent Tests: Use this TEST_TYPE only when the result is a ratio of CK_MB to CK total asa calculated ratio.
    - Not Equivalent Tests: All other CK tests.
    - Indications: Blood levels of CK rise when muscle or heart cells are injured. CK_MB levels,along with total CK, are tested in persons who have chest pain to diagnose whether they have had a heart attack. Since a high total CK could indicate damage to either the heart or other muscles, the ratio of CK_MB to total CK helps to distinguish between these two sources.
    - Description: CK_MB is one of three separate forms (isoenzymes) of the enzyme creatine kinase (CK). CK_MB is found mostly in heart muscle. It rises when there is damage to heart muscle cells.

This test is the calculated ratio (or relative index) of the CK_MB to the total amount of the three CK isoenzymes.

- Related Tests: CK; CK_MB; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_BF, CK_CSF; CK_ISO; CK_ISO_EP; CK_MB; CK_MB_EIA; CK_MB_EP; CK_MBI_EIA; CK_MBI_EP; CK_MM_EP;CK_MMI; CK_MMI_EP; CK_QN; Troponin; BNP
- Approximate Reference Range: If the calculated ratio of CK_MB to the total CK (relativeindex) is more than 2.5 – 3 (some say more than 5), it is likely that the source is cardiac (and that the heart was damaged). Ratios between 3 and 5 represent a gray zone. A ratio less than 2.5 – 3 is consistent with a skeletal muscle source.
- Comments: CK_MB to total CK (often multiplied by 100) can assist in differentiating false positive elevations of CK_MB arising from skeletal muscle. No definitive diagnosis can be established without serial determinations to detect a rise.
- Test Method Variance:
- Other:
  - Associated LOINC: 12188-9, 12189-7, 20569-0, 49136-5, 72564-8
  - Associated CPT: 82550, 82553

13. CK_MBI_EIA (Creatine Kinase MB/Creatine Kinase Total Enzyme Immunoassay)

    - Common Name: CK_MB/CK (ratio determined by enzyme immunoassay)
    - Long Name: Creatine Kinase MB/Creatine Kinase Total Enzyme Immunoassay
    - Test Type: CK_MBI_EIA
    - Panels: N/A
    - Equivalent Tests: Use this TEST_TYPE only when the result is a ratio of CK_MB to CK total

    determined by enzyme immunoassay (EIA).

    - Not Equivalent Tests: All other CK tests.
    - Indications: Blood levels of CK rise when muscle or heart cells are injured. CK_MB levels,along with total CK, are tested in

persons who have chest pain to diagnose whether they have had a heart attack. Since a high total CK could indicate damage to either the heart or othermuscles, the ratio of CK_MB to total CK helps to distinguish between these two sources.

- Description: CK_MB is one of three separate forms of CK. CK_MB is found mostly in heart muscle. It rises when there is damage to heart muscle cells. Separation of CK into isoenzymes may be accomplished by electrophoresis, column chromatography, or immunoassay.Immunoassay methods for isoenzymes can be accomplished rapidly and is highly sensitive andspecific. Further, it does not show the interferences common to traditional electrophoresis.
- Related Tests: CK; CK_MB; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_BF, CK_CSF; CK_ISO; CK_ISO_EP; CK_MB; CK_MB_EIA; CK_MB_EP; CK_MBI_CLC; CK_MBI_EP; CK_MM_EP; CK_MMI; CK_MMI_EP; CK_QN; Troponin; BNP
- Approximate Reference Range: If the ratio of CK_MB to the total CK is more than 2.5 – 3 (some say more than 5), it is likely that the source is cardiac (and that the heart was damaged). Ratios between 3 and 5 represent a gray zone. A ratio less than 2.5 – 3 is consistent with a skeletal muscle source.
- Comments: CK_MB to total CK (often multiplied by 100) can assist in differentiating false positive elevations of CK_MB arising from skeletal muscle. No definitive diagnosis can be established without serial determinations to detect a rise.
- Test Method Variance:
- Other:
  - Associated LOINC: (No LOINCs have been identified at this time)
  - Associated CPT: 82550, 82553

14. CK_MBI_EP (Creatine Kinase MB/ Creatine Kinase Total Electrophoresis)

    - Common Name: CK_MB/CK (ratio determined by electrophoresis)
    - Long Name: Creatine Kinase MB/ Creatine Kinase Total Electrophoresis
    - Test Type: CK_MBI_EP

- Panels: N/A
- Equivalent Tests: Use this TEST_TYPE only when the result is a ratio of CK_MB to CK total determined by electrophoresis.
- Not Equivalent Tests: All other CK tests.
- Indications: Blood levels of CK rise when muscle or heart cells are injured. CK_MB levels, along with total CK, are tested in persons who have chest pain to diagnose whether they have had a heart attack. Since a high total CK could indicate damage to either the heart or other muscles, the ratio of CK_MB to total CK helps to distinguish between these two sources.
- Description: CK_MB is one of three separate forms of CK. CK_MB is found mostly in heart muscle. It rises when there is damage to heart muscle cells. Separation of CK into isoenzymes may be accomplished by electrophoresis, column chromatography, or immunoassay.
- Related Tests: CK; CK_MB; CK_BB_EP, CK_BBI, CK_BBI_EP; CK_BF, CK_CSF; CK_ISO; CK_ISO_EP; CK_MB; CK_MB_EIA; CK_MB_EP; CK_MBI_CLC; CK_MBI_EIA; CK_MM_EP; CK_MMI; CK_MMI_EP; CK_QN; Troponin; BNP
- Approximate Reference Range: Normal If the ratio of CK_MB to the total CK is more than 2.5–3 (some say more than 5), it is likely that the source is cardiac (and that the heart was damaged). Ratios between 3 and 5 represent a gray zone. A ratio less than 2.5 – 3 is consistent with a skeletal muscle source.
- Comments: CK_MB to total CK (often multiplied by 100) can assist in differentiating falsepositive elevations of CK_MB arising from skeletal muscle. No definitive diagnosis can be established without serial determinations to detect a rise.
- Test Method Variance:
- Other:
  - Associated LOINC: 12187-1, 72563-0
  - Associated CPT: 82550, 82553

15. CK_MM_EP (Creatine Kinase MM Electrophoresis)

- Common Name: Creatine Kinase MM Electrophoresis
- Long Name: Creatine Kinase Muscle-Muscle Electrophoresis

- Test Type: CK_MM_EP
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: To help diagnose and monitor muscle injuries and diseases, including muscular dystrophy, rhabdomyolysis, and myositis. To aid in the diagnosis of heart attack, although troponin testing is now more commonly used due to its higher sensitivity and specificity for cardiac damage.
- Description: The Creatine Kinase (CK) isoenzymes electrophoresis lab test is used to detect and differentiate between damage to different types of muscle tissue, particularly skeletal and cardiac muscle.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): results typically show that Creatine Kinase MM accounts for 96-100% of total Creatine Kinase.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 15049-0, 2155-0
  - Associated CPT: 82552, 82554

16. CK_MMI (Creatine Kinase MM)

    - Common Name: Creatine Kinase MM
    - Long Name: Creatine Kinase MM/Creatine Kinase Total
    - Test Type: CK_MMI
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: Elevated CK-MM levels suggest damage to skeletal muscle tissue. This could be due to injury, strenuous exercise, muscle diseases like muscular dystrophy or rhabdomyolysis, or certain medications.

- Description: The creatine kinase muscle-muscle total (CK-MM total) lab test measures the level of creatine kinase (CK) in your blood, specifically focusing on the CK-MM isoenzyme, which is predominantly found in skeletal muscle.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for total creatine kinase (CK) in adults typically falls between 30-170 U/L.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 9643-8
  - Associated CPT: 82552, 82554

17. CK_MMI_EP (Creatine Kinase MM Electrophoresis)

- Common Name: Creatine Kinase MM Electrophoresis
- Long Name: Creatine Kinase MM/Creatine Kinase Total Electrophoresis
- Test Type: CK_MMI_EP
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A total CK test can indicate general muscle damage, a CK isoenzyme electrophoresis test, or a CK with isoenzymes test, can help pinpoint the specific source of the damage by separating and measuring the three isoenzymes.
- Description: The creatine kinase (CK) test is a blood test that measures the levels of the CK enzyme, which plays a role in energy production for muscle contraction.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The reference range for total creatine kinase (CK) in a lab test generally falls between 30-170 U/L for adults.
- Comments:
- Test Method Variance:

- Other:
  - Associated LOINC: 72562-2
  - Associated CPT: 82552, 82554

18. CK_QN (Creatinine Kinase Macromolecular Quantitative)

- Common Name: Serum creatinine
- Long Name: Blood/Serum/Plasma Creatinine
- Test Type: CREATININE
- Panels: Basic metabolic panel, comprehensive metabolic profile
- Equivalent Tests: Plasma creatinine, whole blood creatinine
- Not Equivalent Tests: 24 hour creatinine; creatinine clearance; microalbumin/creatinine ratio;

creatine. Body fluids other than blood are commonly tested i.e., urine creatinine, peritoneal dialysis fluid, amniotic fluid.

- Indications: Estimate kidney function
- Description: Urinary excretion of creatinine is relatively constant from day to day. Creatinine

measurement can also be used to provide an internal reference for comparing other tests that are reported with creatinine, for example, the microalbumin/creatinine ratio. Creatinine tests that are reported per creatinine unit such as 24 hour urine creatinine tests relate serum values to urine values. Kidney function is approximately the inverse of the creatinine value, i.e., creatinine of 2.0 ~ 1/2 of normal kidney function.

- Related Tests: Calculated glomerular filtration rate, creatinine clearance
- Approximate Reference Range (Each site will have slight variations): Approximately 0.5-

1.5 mg/dL. Reference range for females is usually slightly lower than reference range for males.

- Comments: Some cephalosporins will react with Jaffre reagent causing falsely elevated

creatinine

- Test Method Variance: Not common
- Other:
  - Associated LOINC: (No LOINCs have been identified)
  - Associated CPT: 82550

1.1.17 (CO2) Carbon Dioxide Test Types

1. CO2_A (Carbon dioxide in Arterial blood)
   - Common Name: Carbon dioxide in Arterial blood
   - Long Name: Carbon dioxide, total [Moles/volume] in Arterial blood
   - Test Type: CO2_A
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A carbon dioxide (CO2) blood test may be indicated for several reasons, including when a person experiences symptoms of an electrolyte imbalance, like breathing difficulties, weakness, or fatigue.
   - Description: An arterial blood test, specifically measuring carbon dioxide, is a test that analyzes blood from an artery to assess lung function, oxygen and carbon dioxide levels, and acid-base balance. It measures the partial pressure of carbon dioxide (PaCO2), which indicates how well the body is removing carbon dioxide from the bloodstream.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): normal range for carbon dioxide in arterial blood, specifically the partial pressure of carbon dioxide (PaCO2), is typically 35 to 45 mmHg (or 4.7 to 6.0 kPa). Another way to express this range is 23 to 30 mEq/L or 23 to 29 mmol/L
   - Comments: Also, see HCO3_ test_types for bicarbonate
   - Test Method Variance:
   - Other:
     - Associated LOINC: 2026-3

- Associated CPT:

2. CO2_C (Carbon dioxide in Capillary blood)
    - Common Name: Carbon dioxide in Capillary blood
    - Long Name: Carbon dioxide, total [Moles/volume] in Capillary blood
    - Test Type: CO2_C
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: A carbon dioxide (CO2) blood test using capillary blood, also known as a capillary blood gas (CBG) test, is typically used when an arterial blood gas (ABG) test is not feasible or necessary, and is mainly used for monitoring acid-base balance and respiratory status. It can be a good alternative to arterial blood draw in certain situations.
    - Description: A capillary blood test to measure carbon dioxide (pCO2) is a common laboratory procedure used to assess ventilation status and acid-base balance, especially in cases where arterial blood gas (ABG) collection is difficult or less desirable, such as in neonates and children. This test provides a less invasive alternative to drawing blood from an artery.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): The normal range for total carbon dioxide (TCO2) in blood, when measured using capillary blood, is generally 22-28 mEq/L or 22-28 mmol/L.
    - Comments: Also, see HCO3_ test_types for bicarbonate
    - Test Method Variance:
    - Other:
        - Associated LOINC: 16551-4
        - Associated CPT:

3. CO2_V (Carbon dioxide in Veinous blood)
    - Common Name: Carbon dioxide in Veinous blood

- Long Name: Carbon dioxide, total [Moles/volume] in Veinous blood)
- Test Type: CO2_V
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A venous carbon dioxide blood test (CO2 test) is often done as part of an electrolyte panel or basic metabolic panel when a healthcare provider suspects an electrolyte imbalance, potential side effects of certain medications, or to check for conditions like kidney disease or Cushing's disease. It's also used to monitor the progress of diseases that affect bicarbonate levels.
- Description: The carbon dioxide (CO2) in veinous blood test helps healthcare providers determine if the body is balancing electrolytes properly.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): normal range of 20 to 29 millimoles per liter (mmol/L)
- Comments: Also, see HCO3_ test_types for bicarbonate
- Test Method Variance:
- Other:
  - Associated LOINC: 19223-7, 2027-1, 97545-8
  - Associated CPT:

4. CO2_NS (Carbon dioxide source not specified)

   - Common Name: Carbon dioxide source not specified
   - Long Name: Carbon dioxide, total [Moles/volume] in Serum, Plasma or Blood
   - Test Type: CO2_NS
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A carbon dioxide test is typically done to check for electrolyte imbalances, kidney or lung disorders, or metabolic problems.

- Description: A CO2 blood test measures the amount of carbon dioxide in your blood, primarily in the form of bicarbonate and helps assess your body's acid-base balance and kidney and lung function.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): normal range of 20 to 29 millimoles per liter (mmol/L)
- Comments: Also, see HCO3_ test_types for bicarbonate
- Test Method Variance:
- Other:
  - Associated LOINC: 2028-9, 20565-8, 77143-6
  - Associated CPT:

5. CO2_A_CLC (Carbon dioxide in Arterial blood Calculated)

   - Common Name: Carbon dioxide in Arterial blood Calculated
   - Long Name: Carbon dioxide, total [Moles/volume] in Arterial blood by calculation
   - Test Type: CO2_A_CLC
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A CO2 blood test can be performed using a calculated result, particularly in the context of an arterial blood gas (ABG) test. Carbon dioxide (CO2) is a waste product of your body's metabolism. Most of the CO2 in your blood exists in the form of bicarbonate (HCO3). Bicarbonate is a vital electrolyte that helps maintain the acid-base balance (pH) in your blood.
   - Description: While a standard venous CO2 test measures bicarbonate (which is the main form of CO2 in blood), arterial blood gas analysis can involve measuring pH and pCO2 (partial pressure of carbon dioxide).

   Using these measured values (pH and pCO2), the total CO2 content (TCO2) can be calculated using a standardized form of the Henderson-Hasselbalch equation.

   - Related Tests:

- Approximate Reference Range (Each site will have slight variations):
- Comments: Also, see HCO3_ test_types for bicarbonate
- Test Method Variance:
- Other:
  - Associated LOINC: 41647-9
  - Associated CPT:

6. CO2_C_CLC (Carbon dioxide in Capillary blood by calculated)
   - Common Name: Carbon dioxide in Capillary blood by calculated
   - Long Name: Carbon dioxide, total [Moles/volume] in Capillary blood by calculation
   - Test Type: CO2_C_CLC
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Calculated CO2 values in capillary blood (typically bicarbonate - HCO3-) are primarily used to assess and monitor acid-base balance and ventilation adequacy in patients. A direct measurement of CO2 in capillary blood may be less reliable compared to arterial blood in some cases, capillary blood gas analyzers measure pH and pCO2 (carbon dioxide tension). From these measurements, other parameters like bicarbonate (HCO3-) and base excess (BE) are then calculated.
   - Description:While CO2 levels can be measured directly in blood, in certain situations, particularly in neonates and children, a capillary blood gas (CBG) test may be performed, and some results, like bicarbonate (HCO3), are often calculated.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations):
   - Comments: Also, see HCO3_ test_types for bicarbonate
   - Test Method Variance:
   - Other:
     - Associated LOINC: 51781-3

- Associated CPT:

7. CO2_V_CLC (Carbon dioxide in Venous blood by calculation)
    - Common Name: Carbon dioxide in Venous blood by calculation
    - Long Name: Carbon dioxide, total [Moles/volume] in Venous blood by calculation
    - Test Type: CO2_V_CLC
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: A carbon dioxide test using a calculated result from venous blood might be indicated when a patient exhibits symptoms of electrolyte imbalance or acid-base imbalance, or if there are concerns about respiratory or kidney function, or if the patient is experiencing symptoms like shortness of breath, weakness, confusion, or fatigue, as these can all be associated with imbalances in the body's acid-base balance and electrolytes.
    - Description: A venous blood test that measures the amount of carbon dioxide (CO2) in the blood is often performed as part of an electrolyte or basic metabolic panel. This test, also known as a CO2 content test, total CO2, or bicarbonate test, helps evaluate the body's acid-base balance.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations):
    - Comments: Also, see HCO3_ test_types for bicarbonate
    - Test Method Variance:
    - Other:
        - Associated LOINC: 57920-1, 48391-7
        - Associated CPT:

8. CO2_NS_CLC (Carbon dioxide source not specified by calculation)
    - Common Name: Carbon dioxide source not specified by calculation
    - Long Name: Carbon dioxide, total [Moles/volume] in Serum, Plasma or Blood by calculation

- Test Type: CO2_NS_CLC
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A calculated bicarbonate (HCO3) level, which is a key indicator of the total carbon dioxide (CO2) in your blood, is typically tested in various situations to help assess your body's acid-base balance and overall health.
- Description: Most laboratories measure the total carbon dioxide content in a blood sample, which mainly comprises bicarbonate. This is often done using an autoanalyzer.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
- Comments: Also, see HCO3_ test_types for bicarbonate
- Test Method Variance:
- Other:
  - Associated LOINC: 34728-6, 57922-7
  - Associated CPT:

1.1.18 COLOGUARD (Noninvasive colorectal cancer screen)

- Common Name: Noninvasive colorectal cancer screen
- Long Name: Noninvasive colorectal cancer DNA and occult blood screening [Presence] in Stool
- Test Type: COLOGUARD
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A colorectal cancer screening is generally indicated for average-risk individuals, typically those between 45 and 75 years old, as recommended by organizations like the U.S. Preventive Services Task Force. It's also a suitable option for individuals who are unable or

unwilling to undergo more invasive screening methods like colonoscopy. Additionally, it can be used to investigate potential digestive issues if symptoms like abdominal pain, rectal bleeding, or anemia are present.

- Description: A DNA occult blood test, also known as a stool DNA test or multitarget stool DNA test (like Cologuard®), is a non-invasive screening method for colorectal cancer and precancerous polyps. This test looks for DNA changes in the cells shed into the stool by colorectal cancer or polyps, as well as for traces of hidden (occult) blood.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): No numeric results
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 77354-9, 77353-1
  - Associated CPT:

1.1.19 CREATININE (Creatinine Blood/Serum/Plasma)

- Common Name: Serum creatinine
- Long Name: Blood/Serum/Plasma Creatinine
- Test Type: CREATININE
- Panels: Basic metabolic panel, comprehensive metabolic profile
- Equivalent Tests: Plasma creatinine, whole blood creatinine
- Not Equivalent Tests: 24 hour creatinine; creatinine clearance; microalbumin/creatinine ratio; creatine. Body fluids other than blood are commonly tested i.e., urine creatinine, peritoneal dialysis fluid, amniotic fluid.
- Indications: Estimate kidney function

- Description: Urinary excretion of creatinine is relatively constant from day to day. Creatinine measurement can also be used to provide an internal reference for comparing other tests that are reported with creatinine, for example, the microalbumin/creatinine ratio. Creatinine tests that are reported per creatinine unit such as 24 hour urine creatinine tests relate serum values to urine values. Kidney function is approximately the inverse of the creatinine value, i.e., creatinine of 2.0 ~ 1/2 of normal kidney function.
- Related Tests: Calculated glomerular filtration rate, creatinine clearance
- Approximate Reference Range (Each site will have slight variations): Approximately 0.5- 1.5 mg/dL. Reference range for females is usually slightly lower than reference range for males.
- Comments: Some cephalosporins will react with Jaffre reagent causing falsely elevated creatinine
- Test Method Variance: Not common
- Other:
  - Associated LOINC: 101475-2, 11041-1 (postdialysis), 11042-9 (predialysis), 14682-9, 21232-4, 2160-0, 35203-9, 38483-4, 40248-7 (baseline), 40264-4 (baseline), 44784-7, 51619-5 (predialysis), 51620-3 (postdialysis), 59826-8, 77140-2, 96590-5
  - Associated CPT: 82565, 82575

1.1.20 CRP (C-reactive protein)

- Common Name: CRP
- Long Name: C-Reactive Protein
- Test Type: CRP
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: High-sensitivity C-reactive protein (hs-CRP); Erythrocyte sedimentation rate (ESR)

- Indications: To identify the presence of inflammation and to monitor response to treatment for patients with inflammatory disorder (arthritis; rheumatoid arthritis, lupus and other autoimmune disorders; inflammatory bowel disease) or to monitor for infection (after surgery, organ transplants, or burns).

- Description: C-reactive protein (CRP) is an acute phase reactant, a substance made by the liver and secreted into the bloodstream within a few hours after the start of an infection or inflammation. Increased levels are observed after a heart attack, in sepsis, and after a surgical procedure. Its rise in the blood can also precede pain, fever, or other clinical indicators. The level of CRP can jump a thousandfold in response to inflammation and can be valuable in monitoring disease activity. High normal levels of CRP in otherwise healthy individuals have been found to be predictive of the future risk of a heart attack, stroke, sudden cardiac death, and peripheral arterial disease, even when cholesterol levels are within an acceptable range.

- Related Tests: High-sensitivity C-reactive protein (hs-CRP); Erythrocyte sedimentation rate (ESR)

- Approximate Reference Range (Each site will have slight variations): 0 – 5 mg/L (0.0 – 0.5 mg/dL).

- Comments: CRP levels $<$ 1 mg/L are considered low relative cardiovascular risk; CRP levels 1-3 mg/L are considered average relative cardiovascular risk; CRP levels 3.1 – 10 are considered high relative cardiovascular risk; CRP levels persistently $>$ 10 mg/L may represent non-cardiovascular inflammation. NSAIDs, aspirin or statins may reduce CRP levels in blood. Recent illness, tissue injury, infection, or other acute or chronic inflammation will raise the amount of CRP. CRP levels can be elevated in the later stages of pregnancy as well as with use of birth control pills or hormone replacement therapy. Higher levels of CRP have also been observed in obese individuals.

- Test Method Variance:

- Other:

  - Associated LOINC: 11039-5, 1988-5, 48421-2, 76485-2
  - Associated CPT: 86140

1.1.21 CRP_TIT (C-reactive protein titer)

- Common Name: C-Reactive Protein
- Long Name: C-reactive protein serum/plasma titer
- Test Type: CRP_TIT
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: High-sensitivity C-reactive protein (hs-CRP); Erythrocyte sedimentation rate (ESR)
- Indications: To identify the presence of inflammation and to monitor response to treatment for patients with inflammatory disorder (arthritis; rheumatoid arthritis, lupus and other autoimmune disorders; inflammatory bowel disease) or to monitor for infection (after surgery, organ transplants, or burns).
- Description: C-reactive protein (CRP) is an acute phase reactant, a substance made by the liver and secreted into the bloodstream within a few hours after the start of an infection or inflammation. Increased levels are observed after a heart attack, in sepsis, and after a surgical procedure. Its rise in the blood can also precede pain, fever, or other clinical indicators. The level of CRP can jump a thousandfold in response to inflammation and can be valuable in monitoring disease activity. High normal levels of CRP in otherwise healthy individuals have been found to be predictive of the future risk of a heart attack, stroke, sudden cardiac death, and peripheral arterial disease, even when cholesterol levels are within an acceptable range.
- Related Tests: High-sensitivity C-reactive protein (hs-CRP); Erythrocyte sedimentation rate (ESR)
- Approximate Reference Range (Each site will have slight variations): 0 – 5 mg/L (0.0 – 0.5 mg/dL).
- Comments: CRP levels $<$ 1 mg/L are considered low relative cardiovascular risk; CRP levels 1 – 3 mg/L are considered average relative cardiovascular risk; CRP levels 3.1 – 10 are considered high relative cardiovascular risk; CRP levels persistently $>$ 10 mg/L may represent

non-cardiovascular inflammation. NSAIDs, aspirin or statins may reduce CRP levels in blood. Recent illness, tissue injury, infection, or other acute or chronic inflammation will raise the amount of CRP. CRP levels can be elevated in the later stages of pregnancy as well as with use of birth control pills or hormone replacement therapy. Higher levels of CRP have also been observed in obese individuals.

- Test Method Variance:
- Other:
  - Associated LOINC: 14634-0
  - Associated CPT:

1.1.22 CRPHS (High-sensitivity C-reactive protein in serum/plasma)

- Common Name: hsCRP
- Long Name: High-sensitivity C-Reactive Protein
- Test Type: CRPHS
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: C-reactive protein (CRP); Erythrocyte sedimentation rate (ESR)
- Indications: To assess risk of developing heart disease, cardiovascular disease (CVD), or other processes involving inflammation. Inflammation plays a major role in atherosclerosis, which is often associated with CVD. Studies have shown that measuring CRP with the improved methodology of the highly sensitive assay can identify the risk level for CVD in apparently healthy people; the hs-CRP test may be used to screen healthy people.
- Description: CRP is made by the liver and secreted into the bloodstream. It can be measured with two different tests: CRP and hs-CRP, each measuring different blood ranges. The hs-CRP test can more accurately detect lower concentrations of the protein that may be within the normal range than the standard CRP test. Individuals with hs-CRP results in the high end of the normal range have 1.5 to 4 times

the risk of having a heart attack as those with hs-CRP values at the low end of the normal range.

- Related Tests: Lipid profile, Cardiac risk assessment, CRP, Erythrocyte sedimentation rate (ESR)
- Approximate Reference Range: 0 – 5 mg/L (0.0 – 0.5 mg/dL).
- Comments: Hs-CRP usually is ordered as one of several tests in a cardiovascular risk profile, often along with tests for cholesterol and triglycerides. The American Heart Association and US Centers for Disease Control and Prevention have defined risk groups as follows: Low risk ($<$ 1.0 mg/L); Average risk (1.0 to 3.0 mg/L); High risk $>$ 3.0 mg/L) NSAIDs, aspirin or statins may reduce CRP levels in blood. Recent illness, tissue injury, infection, or other acute or chronic inflammation will raise the amount of CRP. CRP levels can be elevated in the later stages of pregnancy as well as with use of birth control pills or hormone replacement therapy. Higher levels of CRP have also been observed in the obese.
- Test Method Variance:
- Other:
  - Associated LOINC: 30522-7, 71426-1, 76486-0
  - Associated CPT: 86141

1.1.23 CRPHS_QUIN (High-sensitivity C-reactive protein quintile)

- Common Name: HsCRP quintile
- Long Name: High-sensitivity C-Reactive Protein
- Test Type: CRPHS_QUIN
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: C-reactive protein (CRP); Erythrocyte sedimentation rate (ESR)

- Indications: To assess risk of developing heart disease, cardiovascular disease (CVD), or other processes involving inflammation. Inflammation plays a major role in atherosclerosis, which is often associated with CVD. Studies have shown that measuring CRP with the improved methodology of the highly sensitive assay can identify the risk level for CVD in apparently healthy people; the hs-CRP test may be used to screen healthy people.

- Description: refers to the use of high-sensitivity C-reactive protein (hs-CRP) levels to assess cardiovascular risk, categorized into five groups (quintiles) based on their relative risk.

  - Quintile 1 (Lowest Risk): Typically, individuals with hs-CRP levels below 0.7 mg/L are considered to be at the lowest risk for cardiovascular disease.
  - Quintile 2 (Low Risk): Individuals with hs-CRP levels between 0.7 and 1.1 mg/L are considered at low risk.
  - Quintile 3 (Moderate Risk): Individuals with hs-CRP levels between 1.2 and 1.9 mg/L are considered at moderate risk.
  - Quintile 4 (High Risk): Individuals with hs-CRP levels between 2.0 and 3.8 mg/L are considered at high risk.
  - Quintile 5 (Highest Risk): Individuals with hs-CRP levels above 3.9 mg/L are considered to be at the highest risk of cardiovascular disease.

- Related Tests: Lipid profile, Cardiac risk assessment, CRP, Erythrocyte sedimentation rate (ESR)

- Approximate Reference Range: See the quintile descriptions

- Comments: Hs-CRP usually is ordered as one of several tests in a cardiovascular risk profile, often along with tests for cholesterol and triglycerides. The American Heart Association and US Centers for Disease Control and Prevention have defined risk groups as follows: Low risk ($<$ 1.0 mg/L); Average risk (1.0 to 3.0 mg/L); High risk $>$ 3.0 mg/L) NSAIDs, aspirin or statins may reduce CRP levels in blood. Recent illness, tissue injury, infection, or other acute or chronic inflammation will raise the amount of CRP. CRP levels can be elevated in the later stages of pregnancy as well as with use of birth control pills or hormone replacement therapy. Higher levels of CRP have also been observed in the obese.

- Test Method Variance:
- Other:
  - Associated LOINC: 35648-5
  - Associated CPT:

1.1.24 CYSTATIN_C (Cystatin C)

- Common Name: Cystatin C
- Long Name: Cystatin C [Mass/volume] in Serum or Plasma
- Test Type: CYSTATIN_C
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: This test helps to evaluate kidney health, especially in cases where creatinine levels (another measure of kidney function) might be influenced by factors like muscle mass, diet, age, or sex. Cystatin C levels are less affected by these factors, potentially making it a more accurate indicator in some situations.
- Description: A cystatin C blood test measures the level of the protein cystatin C in your blood. Cystatin C is a small protein produced constantly by cells in the body. When the kidneys are healthy, they filter cystatin C from the blood, which is then excreted in the urine.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): typically falls between 0.61 and 1.23 mg/L
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 33863-2
  - Associated CPT: 82610

1.1.25 (D_DIME) D-dimer fibrin derivatives Test Types

1. D_DIME_DDU (D-dimer fibrin derivatives, D-Dimer Units)

   - Common Name: D-dimer fibrin derivatives, D-Dimer Units
   - Long Name: D-dimer fibrin derivatives, quantitative, D-Dimer Units
   - Test Type: D_DIME_DDU
   - Panels: None
   - Equivalent Tests: None Related
   - Not Equivalent, Tests: FEU, and tests from Cerebral spinal fluid are not included.
   - Indications: D-dimer tests are ordered to help rule out the presence of a clot (thrombus). Some of the conditions that the d-dimer test is used to help rule out include deep vein thrombosis (DVT), pulmonary embolism (PE), strokes. The D-dimer test may be used to determine if further testing is necessary to help diagnose diseases and conditions that cause inappropriate clotting. A D-dimer level may be used to help diagnose disseminated intravascular coagulation (DIC) and to monitor the effectiveness of DIC treatment.
   - Description: D-dimer is one of the protein fragments produced when a blood clot dissolves in the body. It is a specific plasmin-mediated breakdown product of crosslinked fibrin. When a blood vessel or tissue is injured and bleeds, the body creates a blood clot to stop the bleeding. During this process, threads of a protein called fibrin are produced. These threads crosslink together to form a fibrin net, which, together with platelets, helps hold the forming blood clot in place at the site of the injury until it heals. Once the clot is no longer needed, the body uses an enzyme called plasmin to break up the clot. The fragments of the disintegrating fibrin in the clot are called fibrin degradation products (FDP). One of the fibrin degradation products produced is D-dimer. D-dimer is normally undetectable. It is produced only after a clot has formed and is in the process of being broken down. When there is significant formation and breakdown of blood clots in the body, the D-dimer blood level rises. A positive d-dimer test cannot predict whether or not a clot is present, but rather indicates that further testing is required. A negative D- dimer test means

that it is unlikely that a clot is present. A normal D-dimer result has a negative predictive value of approximately 95% for the exclusion of acute PE or DVT when there is low or moderate pretest PE probability. The DDU lab test measures D-dimer levels based on the molecular weight of D-dimer.

- Approximate Reference Range: the normal range is typically less than 250 ng/mL when using D-dimer units (DDU)
- Comments: The method of determining DDU or FEU classification will differ from site to site. Some sites will be able to use the LOINC code. Some sites may find the text 'DDU' or 'FEU' as part of the result unit or in a note associated with a test result.
- Test Method Variance: There are several different methods of testing for D-dimer. Most of the D-dimer tests that yield quantitative results are done in a hospital lab, while those that yield qualitative or semi-quantitative results are performed at the patient's bedside (point of care).
- Other:
  - Associated LOINC: 48058-2, 48066-5, 91556-1
  - Associated CPT:

2. D_DIME_FEU (D-dimer fibrin derivatives, Fibinogen Equivalent Units)

   - Common Name: D-dimer fibrin derivatives, Fibinogen Equivalent Units
   - Long Name: D-dimer fibrin derivatives, quantitative, Fibinogen Equivalent Units
   - Test Type: D_DIME_FEU
   - Panels: None

   Equivalent Tests: None Related, but Not Equivalent, Tests: Fibrin Degradation Products (FDP); Fibrin Split Products (FSP); Prothrombin Time (PT); Partial Thromboplastin Time (PTT); Fibrinogen; Platelet Count

   - Indications: D-dimer tests are ordered to help rule out the presence of a clot (thrombus). Some of the conditions that the d-dimer test is used to help rule out include deep vein thrombosis (DVT), pulmonary embolism (PE), strokes. The D-dimer test

may be used to determine if further testing is necessary to help diagnose diseases and conditions that cause inappropriate clotting. A D-dimer level may be used to help diagnose disseminated intravascular coagulation (DIC) and to monitor the effectiveness of DIC treatment.

- Description: D-dimer is one of the protein fragments produced when a blood clot dissolves in the body. It is a specific plasmin-mediated breakdown product of crosslinked fibrin. When a blood vessel or tissue is injured and bleeds, the body creates a blood clot to stop the bleeding. During this process, threads of a protein called fibrin are produced. These threads crosslink together to form a fibrin net, which, together with platelets, helps hold the forming blood clot in place at the site of the injury until it heals. Once the clot is no longer needed, the body uses an enzyme called plasmin to break up the clot. The fragments of the disintegrating fibrin in the clot are called fibrin degradation products (FDP). One of the fibrin degradation products produced is D-dimer. D-dimer is normally undetectable. It is produced only after a clot has formed and is in the process of being broken down. When there is significant formation and breakdown of blood clots in the body, the D-dimer blood level rises. A positive d-dimer test cannot predict whether or not a clot is present, but rather indicates that further testing is required. A negative D- dimer test means that it is unlikely that a clot is present. A normal D-dimer result has a negative predictive value of approximately 95% for the exclusion of acute PE or DVT when there is low or moderate pretest PE probability. In an FEU lab test D-dimer levels are reported based on the molecular weight of fibrinogen.
- Approximate Reference Range: FEU D-dimer result is typically less than 500 ng/mL.
- Comments: The method of determining DDU or FEU classification will differ from site to site. Some sites will be able to use the LOINC code. Some sites may find the text 'DDU' or 'FEU' as part of the result unit or in a note associated with a test result.
- Test Method Variance: There are several different methods of testing for D-dimer. Most of the D-dimer tests that yield quantitative results are done in a hospital lab, while those that yield qualitative or semi-quantitative results are performed at the patient's bedside (point of care).

- Other:
  - Associated LOINC: 48065-7, 48067-3, 71427-9
  - Associated CPT:

3. D_DIME_NS (D-dimer fibrin derivatives, Not Specified)

   - Common Name: D-dimer fibrin derivatives, Not Specified
   - Long Name: D-dimer fibrin derivatives, quantitative, Not Specified
   - Test Type: D_DIME_NS
   - Panels: None
   - Equivalent Tests:
   - Indications: D-dimer tests are ordered to help rule out the presence of a clot (thrombus). Some of the conditions that the d-dimer test is used to help rule out include deep vein thrombosis (DVT), pulmonary embolism (PE), strokes. The D-dimer test may be used to determine if further testing is necessary to help diagnose diseases and conditions that cause inappropriate clotting. A D-dimer level may be used to help diagnose disseminated intravascular coagulation (DIC) and to monitor the effectiveness of DIC treatment.
   - Description: D-dimer is one of the protein fragments produced when a blood clot dissolves in the body. It is a specific plasmin-mediated breakdown product of crosslinked fibrin. When a blood vessel or tissue is injured and bleeds, the body creates a blood clot to stop the bleeding. During this process, threads of a protein called fibrin are produced. These threads crosslink together to form a fibrin net, which, together with platelets, helps hold the forming blood clot in place at the site of the injury until it heals. Once the clot is no longer needed, the body uses an enzyme called plasmin to break up the clot. The fragments of the disintegrating fibrin in the clot are called fibrin degradation products (FDP). One of the fibrin degradation products produced is D-dimer. D-dimer is normally undetectable. It is produced only after a clot has formed and is in the process of being broken down. When there is significant formation and breakdown of blood clots in the body, the D-dimer blood level rises. A positive d-dimer test cannot predict whether or not a clot is present, but rather indicates

that further testing is required. A negative D- dimer test means that it is unlikely that a clot is present. A normal D-dimer result has a negative predictive value of approximately 95% for the exclusion of acute PE or DVT when there is low or moderate pretest PE probability.

- Approximate Reference Range: A normal qualitative D-dimer result is "undetectable: or "negative" or "not detected." Quantitative results of < = 250 ng/mL D-Dimer Units (DDU) OR < = 0.5 mcg/mL Fibrinogen Equivalent Units (FEU) are normal. The normal reportable range is sometimes listed as 110 - 250 ng/mL DDU OR 0.22 - 0.50 mcg/mL FEU. Results can be reported in as many as eight different combinations of types (e.g., FEU, DDU) and magnitude (e.g., ng/mL, mcg/L [g/L], mcg/ml [g/mL], mg/L) of units. Unfortunately, at times result units are only reported as the magnitude (whether FEU or DDU type is not reported). In these cases, ranges and abnormal flags can assist in guiding the interpretation, but it is not always possible to assign magnitude. Do not make assumptions if you cannot determine the magnitude of the results.
- Comments: If a test cannot be classified as 'DDU' or 'FEU' by LOINC code or by a text string in either the result_unit field or a note associated with the test, the test is Not Specified and should be placed in this test_type. Generally the LOINCs listed in this row are Not Specified. If you have reason to believe that a test associated with a LOINC code listed on this row is either DDU or FEU please allow your local site knowledge to override the LOINC classification of Not Specified. In such a case the test should be placed in either the D_DIME_DDU or D_DIME_FEU test_type.
- Test Method Variance: There are several different methods of testing for D-dimer. Most of the D-dimer tests that yield quantitative results are done in a hospital lab, while those that yield qualitative or semi-quantitative results are performed at the patient's bedside (point of care).
- Other:
  - Associated LOINC: 15129-0, 30240-6, 3246-6, 38898-3, 7799-0
  - Associated CPT:

4. D_DIMER_QL (D-dimer fibrin derivatives, qualitative)

- Common Name: D-dimer fibrin derivatives, qualitative
- Long Name: D-dimer fibrin derivatives, qualitative
- Test Type: D_DIMER_QL
- Panels: None
- Equivalent Tests: None

Related, but Not Equivalent, Tests: Fibrin Degradation Products (FDP); Fibrin Split Products (FSP); Prothrombin Time (PT); Partial Thromboplastin Time (PTT); Fibrinogen; Platelet Count

- Indications: D-dimer tests are ordered to help rule out the presence of a clot (thrombus). Some of the conditions that the d-dimer test is used to help rule out include deep vein thrombosis

(DVT), pulmonary embolism (PE), strokes. The D-dimer test may be used to determine if further testing is necessary to help diagnose diseases and conditions that cause inappropriate clotting. A D-dimer level may be used to help diagnose disseminated intravascular coagulation (DIC) and to monitor the effectiveness of DIC treatment.

- Description: D-dimer is one of the protein fragments produced when a blood clot dissolves in the body. It is a specific plasmin-mediated breakdown product of crosslinked fibrin. When a blood vessel or tissue is injured and bleeds, the body creates a blood clot to stop the bleeding. During this process, threads of a protein called fibrin are produced. These threads crosslink together to form a fibrin net, which, together with platelets, helps hold the forming blood clot in place at the site of the injury until it heals. Once the clot is no longer needed, the body uses an enzyme called plasmin to break up the clot. The fragments of the disintegrating fibrin in the clot are called fibrin degradation products (FDP). One of the fibrin degradation products produced is D-dimer. D-dimer is normally undetectable. It is produced only after a clot has formed and is in the process of being broken down. When there is significant formation and breakdown of blood clots in the body, the D-dimer blood level rises. A positive d-dimer test cannot predict whether or not a clot is present, but rather indicates that further testing is required. A negative D- dimer test means that it is unlikely that a clot is present. A normal D-dimer result

has a negative predictive value of approximately 95% for the exclusion of acute PE or DVT when there is low or moderate pretest PE probability.

- Approximate Reference Range: A normal qualitative D-dimer result is "undetectable: or "negative" or "not detected." Quantitative results of < = 250 ng/mL D-Dimer Units (DDU) OR < = 0.5 mcg/mL Fibrinogen Equivalent Units (FEU) are normal. The normal reportable range is sometimes listed as 110 - 250 ng/mL DDU OR 0.22 - 0.50 mcg/mL FEU. Results can be reported in as many as eight different combinations of types (e.g., FEU, DDU) and magnitude (e.g., ng/mL, mcg/L [g/L], mcg/ml [g/mL], mg/L) of units. Unfortunately, at times result units are only reported as the magnitude (whether FEU or DDU type is not reported). In these cases, ranges and abnormal flags can assist in guiding the interpretation, but it is not always possible to assign magnitude. Do not make assumptions if you cannot determine the magnitude of the results.
- Comments: A normal or negative D-dimer result means that it is most likely that the person tested does not have an acute condition or disease that is causing abnormal clot formation and breakdown. The test is used to help rule out clotting as the cause of symptoms. A positive Ddimer result may indicate the presence of an abnormally high level of fibrin degradation products. It tells the doctor that there may be significant blood clot formation and breakdown in the body, but it does not tell the location or cause. An elevated D-dimer does not always indicate the presence of a clot because a number of other factors can cause an increased level (e.g., recent surgery, pregnancy, trauma, infection, heart disease, some cancers). D-dimer is not used as the only test to diagnose a condition.
- Test Method Variance: There are several different methods of testing for D-dimer. Most of the D-dimer tests that yield quantitative results are done in a hospital lab, while those that yield qualitative or semi-quantitative results are performed at the patient's bedside (point of care).
- Other:
  - Associated LOINC: 15179-5, 29280-5, 3247-4
  - Associated CPT: 85378, 85379, 85362

1.1.26 ESR (Erythrocyte sedimentation rate)

- Common Name: Erythrocyte sedimentation rate
- Long Name: Erythrocyte sedimentation rate
- Test Type: ESR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Diagnose and Monitor Inflammatory Conditions: ESR is often used alongside other tests to help diagnose and monitor conditions that cause inflammation, such as arthritis, vasculitis, infections, and inflammatory bowel disease.
- Description: The Erythrocyte Sedimentation Rate (ESR), also known as the sed rate, is a blood test that helps detect inflammation in the body.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range varies by Sex and age:
  - Males younger than 50: Less than 15 mm/hr
  - Males older than 50: Less than 20 mm/hr
  - Females younger than 50: Less than 20 mm/hr
  - Females older than 50: Less than 30 mm/hr
  - Newborn babies: 0 to 2 mm/hr
  - Children before puberty: 3 to 13 mm/hr
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 4537-7, 4538-5, 30341-2,
  - Associated CPT: 85651, 85652

1.1.27 (FE) Iron Test Types

1. FE (Iron)
    - Common Name: Iron
    - Long Name: Iron in Serum or Plasma
    - Test Type: FE
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: There are different reasons that an iron blood test should be given:
        - Symptoms of Iron Deficiency: If the patient experiences symptoms like fatigue, weakness, pale skin, dizziness, shortness of breath, headache, or restless legs syndrome, your doctor might recommend an iron test to determine if iron deficiency is the cause.
        - Symptoms of Iron Overload: If the patient exhibits symptoms such as joint pain, abdominal pain, fatigue, weakness, loss of sex drive, or changes in skin color (bronze or gray), an iron blood test could be used to investigate the possibility of iron overload or hemochromatosis.
        - Low Red Blood Cell Count or Abnormalities: If a routine complete blood count (CBC) reveals a low red blood cell count, low hemoglobin, or small, pale red blood cells, an iron test may be necessary to identify the cause of the anemia.
    - Description: An iron blood test assesses various aspects of iron in your blood to determine if your levels are too high or too low. Iron is crucial for overall health as it's a key component of hemoglobin, the protein in red blood cells that transports oxygen throughout the body.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): The normal reference range for a serum iron blood test typically falls between 60 to 170 micrograms per deciliter (mcg/dL).
    - Comments:
    - Test Method Variance:

- Other:
  - Associated LOINC: 35214-6, 2498-4, 14798-3
  - Associated CPT:

2. FE_QL (Iron qualitative)

   - Common Name: Iron qualitative
   - Long Name: Iron in Serum or Plasma qualitative
   - Test Type: FE_QL
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: These tests produce qualitative results that is either "normal", "abnormal high" (this indicates excessive iron in the blood), or "abnormal low" (this indicates an iron deficiency).
     - Symptoms of Iron Deficiency: If the patient experiences symptoms like fatigue, weakness, pale skin, dizziness, shortness of breath, headache, or restless legs syndrome, your doctor might recommend an iron test to determine if iron deficiency is the cause.
     - Symptoms of Iron Overload: If the patient exhibits symptoms such as joint pain, abdominal pain, fatigue, weakness, loss of sex drive, or changes in skin color (bronze or gray), an iron blood test could be used to investigate the possibility of iron overload or hemochromatosis.
   - Description: An iron blood test assesses various aspects of iron in your blood to determine if your levels are too high or too low. Iron is crucial for overall health as it's a key component of hemoglobin, the protein in red blood cells that transports oxygen throughout the body.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): This is a qualitative result so there is no reference range
   - Comments:
   - Test Method Variance:
   - Other:

  - Associated LOINC: 14797-5, 44326-7
  - Associated CPT:

3. FE_SATURATION (Iron Saturation)

   - Common Name: Iron Saturation
   - Long Name: Iron Saturation in Serum or Plasma
   - Test Type: FE_SATURATION
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: There are different reasons that an iron saturation in blood test should be given:
     - Symptoms of Iron Deficiency: If the patient experiences symptoms like fatigue, weakness, pale skin, dizziness, shortness of breath, headache, or restless legs syndrome, your doctor might recommend an iron test to determine if iron deficiency is the cause.
     - Symptoms of Iron Overload: If the patient exhibits symptoms such as joint pain, abdominal pain, fatigue, weakness, loss of sex drive, or changes in skin color (bronze or gray), an iron blood test could be used to investigate the possibility of iron overload or hemochromatosis.
     - Low Red Blood Cell Count or Abnormalities: If a routine complete blood count (CBC) reveals a low red blood cell count, low hemoglobin, or small, pale red blood cells, an iron test may be necessary to identify the cause of the anemia.
   - Description: Iron is transported through the bloodstream bound to a protein called transferrin. Transferrin is produced by the liver, and its role is to pick up iron absorbed from the diet and deliver it to cells that need it, like those in the bone marrow that produce red blood cells. Iron saturation tells you how much of the available iron-carrying protein is actually carrying iron.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations):
   - Comments:

- Test Method Variance:
- Other:
  - Associated LOINC: 14801-5, 2502-3
  - Associated CPT:

4. FE_TIBC (Iron Binding Capacity)
   - Common Name: Iron Binding Capacity
   - Long Name: Iron Binding Capacity in Serum or Plasma
   - Test Type: FE_TIBC
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: There are different reasons that an iron saturation in blood test should be given:
     - Symptoms of Iron Deficiency: If the patient experiences symptoms like fatigue, weakness, pale skin, dizziness, shortness of breath, headache, or restless legs syndrome, your doctor might recommend an iron test to determine if iron deficiency is the cause.
     - Symptoms of Iron Overload: If the patient exhibits symptoms such as joint pain, abdominal pain, fatigue, weakness, loss of sex drive, or changes in skin color (bronze or gray), an iron blood test could be used to investigate the possibility of iron overload or hemochromatosis.
   - Description: The TIBC (Total Iron-Binding Capacity) blood test measures your blood's capacity to bind iron and transport it throughout your body. It's a crucial component of a comprehensive iron panel used to assess iron levels and metabolism, helping diagnose conditions like iron deficiency anemia and iron overload. The test primarily reflects the amount of transferrin, a protein produced by your liver, present in your blood. Transferrin is responsible for binding to and transporting iron to cells that need it. TIBC, therefore, indicates the total capacity of transferrin to bind iron.
   - Related Tests:

- Approximate Reference Range (Each site will have slight variations): The reference range for a Total Iron Binding Capacity (TIBC) blood test is typically 240 to 450 mcg/dL.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 14800-7,2500-7, 35215-3
  - Associated CPT:

5. FE_UIBC (Iron Binding Capacity unsaturated)

   - Common Name: Unsaturated Iron Binding Capacity
   - Long Name: Iron Binding Capacity in Serum or Plasma Unsaturated
   - Test Type: FE_UIBC
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: There are different reasons that an iron saturation in blood test should be given:
     - Symptoms of Iron Deficiency: If the patient experiences symptoms like fatigue, weakness, pale skin, dizziness, shortness of breath, headache, or restless legs syndrome, your doctor might recommend an iron test to determine if iron deficiency is the cause.
     - Symptoms of Iron Overload: If the patient exhibits symptoms such as joint pain, abdominal pain, fatigue, weakness, loss of sex drive, or changes in skin color (bronze or gray), an iron blood test could be used to investigate the possibility of iron overload or hemochromatosis.
   - Description: The Unsaturated Iron Binding Capacity (UIBC) blood test is a measurement of the reserve capacity of transferrin, the blood protein responsible for carrying iron throughout the body. Think of transferrin as a fleet of trucks transporting iron, and UIBC tells you how many of those trucks are currently empty and ready to carry more iron.

- Related Tests: What the UIBC blood test measures compared to other blood iron tests:
  - UIBC: The portion of transferrin not yet bound to iron.
  - Serum iron: Measures the amount of iron currently in the bloodstream.
  - TIBC: The total capacity of transferrin to bind iron (serum iron + UIBC).
  - Transferrin saturation: The percentage of transferrin that is currently carrying iron.
  - Ferritin: Measures the amount of stored iron in the body.
- Approximate Reference Range (Each site will have slight variations): The typical reference range for UIBC in adults is 150-375 mcg/dL.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 22753-8, 2501-5, 35216-1
  - Associated CPT:

6. FE_TRANSFERRIN_RATIO (Iron/Transferrin ratio)

- Common Name: Iron/Transferrin ratio
- Long Name: Iron/Transferrin ratio in Serum or Plasma
- Test Type: FE_TRANSFERRIN_RATIO
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: An iron transferrin ratio blood test is a key component of a comprehensive evaluation of iron status.
  - Symptoms of Iron Deficiency: If the patient experiences symptoms like fatigue, weakness, pale skin, dizziness, shortness of breath, headache, or restless legs syndrome, your doctor might recommend an iron test to determine if iron deficiency is the cause.

  - Symptoms of Iron Overload: If the patient exhibits symptoms such as joint pain, abdominal pain, fatigue, weakness, loss of sex drive, or changes in skin color (bronze or gray), an iron blood test could be used to investigate the possibility of iron overload or hemochromatosis.
- Description: A blood test measuring the iron-transferrin ratio assesses the body's iron levels and its ability to transport iron. It involves measuring serum iron and total iron-binding capacity (TIBC), which reflects transferrin levels, to calculate the percentage of transferrin that is saturated with iron.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The transferrin saturation, which reflects the ratio of serum iron to total iron-binding capacity (TIBC), typically falls within the range of 20% to 50%.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 13452-8, 60517-0
  - Associated CPT:

1.1.28 (FOB) Fecal Occult Blood Test Types

1. FOB_GUAI (Fecal Occult Blood Test qualitative, guaiac)

   - Common Name: FOBT; fecal occult blood test
   - Long Name: FOBT qualitative; fecal occult blood; occult blood in stool; guaiac
   - Test Type: FOB_GUAI (NOTE: This is likely to be the FOBT TEST_TYPE where the majority of your site's FOBT tests will be mapped as this is the "Home FOBT" [see below]).
   - Panels:
   - Equivalent tests: FOBT is a general name applied to stool tests that detect blood "hidden" (occult) in the stool. Two basic technologies are employed: the older guaiac tests and the newer immunochemical tests (see test type FOB_IMMUN). These two tests are sometimes distinguished as "guaiac test" or gFOBT versus fecal immunochemical test (FIT), or iFOBT. The information

on this sheet applies to the guaiac test (gFOBT) which can also be listed as Hemoccult® or similar (e.g., Hemoccult II®, Hemoccult II ® SENSA®, Hemoccult II ® SENSA®elite, ColoScreen®, Colocare®, EZ Detect®, Seracult®, Sure Vue®). Please refer to other sections of this Reference for information about other stool blood tests.

- Not equivalent tests: The test must be from stool (feces). Text searches for "hemoccult" may or may not be productive. Text searches on feces, "fecal", or "occult blood" may be useful, but many of these tests are unrelated to FOBT. Do not confuse FOBT with other tests related to blood. Be cautious when considering tests that contain the word "heme" because, although FOBT recognizes the "heme" portion of hemoglobin, hemoglobin is the oxygen-carrying molecule in blood and many tests containing the root "heme" are not FOBT. There were formerly two applications of the FOBT test. "Office FOBT" is no longer recommended. In that procedure, the provider collected a smear of fecal sample coincident with a digital rectal exam, and applied it to a single test card. "Home FOBT" is the recommended procedure, in which the provider gives the patient a test kit to complete at home (the kit often contains 3 cards, allowing for collection of 2 samples per day for 3 days). This TEST_TYPE is for the home FOBT.
- Indications: The FOBT detects blood in stool. The test can be used to screen for colorectal cancer (CRC), or to diagnose other conditions that cause bleeding into the gastrointestinal (GI) tract. Occult (not visible to the eye).blood in feces may be due to CRC, polyps, or other GI conditions (e.g., hemorrhoids, anal fissures, ulcers, inflammatory bowel disease).
- Description: The guaiac is qualitative, i.e., any color change indicates a positive test. A typical guaiac test card contains 3 or 6 "windows." The patient takes the test card home and collects fecal samples over 3 days. The patient applies a thin smear of feces to Window A of the card, and then applies a second smear from a different part of the stool to Window B, etc. A complete test consists of a card containing 6 smears. The patient returns the sample to a laboratory or medical office for development. The test is read by adding drops of developer to the guaiac paper and noting the color change (any trace of blue on any smear is a positive test). The test is qualitative in that the result is negative,

positive or not interpetable. At some sites, the results may be reported individually for each of the boxes (i.e., specimens). If all specimens are reported as negative the test result is negative. If even one specimen is reported as positive the test result is positive. Any color change for any of the specimens constitutes a positive test. The following LOINC codes correspond to the individual specimens when results are reported individually for each box:

  - 2nd specimen 14564-9
  - 3rd specimen 14565-6
  - 4th specimen 12503-9
  - 5th specimen 10504-7
  - 6th specimen 27401-9
  - 7th specimen 27925-7
  - 8th specimen 27926-5

- Related Tests: See the following TEST_TYPES: FOB1_GUAI, FOB_QN, and FOB_IMMUN
- Approximate Reference Range: N/A
- Comments: High does of vitamin C (or other strong anti-oxidant) may inhibit the color reaction and cause a false negative. Consumption of meat has been shown to cause false positives for guaiac testing by providing a source of hemoglobin other than blood.
- Test Method Variance:
- Other
  - Associated LOINC: 2335-8, (14564-9, 14565-6, 12503-9, 10504-7, 12504-7, 27401-9, 27925-7, 27926-5)
  - Associated CPT: 82270, 82272

2. FOB1_GUAI (Fecal Occult Blood Test One Specimen qualitative, guaiac)

   - Common Name: FOBT; fecal occult blood test, first specimen
   - Long Name: FOBT qualitative; fecal occult blood; occult blood in stool; guaiac, single specimen
   - Test Type: FOB1_GUAI (NOTE: This TEST_TYPE is intended to be used ONLY when the Office FOBT is performed [see below]. Some sites will not use this TEST_TYPE at all).

- Panels:
- Equivalent tests: FOBT is a general name applied to stool tests that detect blood "hidden" (occult) in the stool. Two basic technologies are employed: the guaiac test (older, routinely used), and newer immunochemical tests (see test type FOB_IMMUN). These two tests are sometimes distinguished as "guaiac test" or gFOBT, versus fecal immunochemical test (FIT), or iFOBT. This information applies to the older guaiac test (gFOBT).The FOBT test can be listed as Hemoccult® or similar (e.g., Hemoccult II®, Hemoccult II ® SENSA®, Hemoccult II ® SENSA®elite, ColoScreen®, Colocare®, EZ Detect®, Seracult®, Sure Vue®). Please refer to other section of the Reference Document for information about other stool blood tests. Not equivalent tests: The test must be from stool (feces). Text searches for "hemoccult" may or may not be productive. Text searches on "feces", "fecal", or "occult blood" may be useful, but do not over-generalize; many of these tests are unrelated to FOBT. Do not to confuse FOBT with other tests related to blood. Be cautious when considering tests that contain the word "heme" because, although FOBT recognizes the "heme" portion of hemoglobin, hemoglobin is the oxygen-carrying molecule in blood and many tests containing the root "heme" are not FOBT. There were formerly two applications of the FOBT test. "Office FOBT" is no longer recommended. In that procedure, the provider collects a smear of fecal sample coincident with a digital rectal exam, and applies it to a single test card. "Home FOBT" is the recommended procedure, in which the provider gives the patient a test kit to complete at home (the kit often contains 3 cards, allowing for collection of 2 samples per day for 3 days). This TEST_TYPE is for the office FOBT.
- Indications: The fecal occult blood test (FOBT) detects blood in stool. The test may be performed to screen for colorectal cancer (CRC), or to diagnose other conditions that cause bleeding into the gastrointestinal (GI) tract. Occult blood in feces may be due to CRC, polyps, or other GI conditions (e.g., hemorrhoids, anal fissures, ulcers, inflammatory bowel disease).
- Description: The guaiac is qualitative, i.e., any color change indicates a positive test. In the office FOBT, typically the smear is applied to a single card. The test is read by adding drops of developer to the guaiac paper and noting the color change. Any

trace of blue on any of the fecal smears is considered a positive test. The test is qualitative in that the result is negative, positive or not interpetable.

- Related Tests: See the following TEST_TYPES: FOB_GUAI, FOB_QN, and FOB_IMMUN
- Approximate Reference Range: N/A
- Comments: High does of vitamin C (or other strong anti-oxidant) may inhibit the color reaction and cause a false negative. Consumption of meat has been shown to cause false positives for guaiac testing by providing a source of hemoglobin other than blood.
- Test Method Variance:
- Other
  - Associated LOINC: 14563-1
  - Associated CPT: 82270, 82273

3. FOB_QN (Fecal Occult Blood Test, Quantitative)

   - Common Name: Quantitative FOBT test
   - Long Name: FOBT quantitative; fecal occult blood; occult blood in stool; guaiac
   - Test Type: FOB_QN (NOTE: We are not aware that any site is using a quantitative FOBT test with the exception of the newer immune test. We anticipate that this TEST_TYPE will rarely be populated in the VDW).
   - Panels:
   - Equivalent tests: FOBT is a general name applied to stool tests that detect blood "hidden" (occult) in the stool. Two basic technologies are employed: the guaiac test (older, routinely used), and newer immunochemical tests (see test type FOB_IMMUN). These two tests are sometimes distinguished as "guaiac test" or gFOBT, versus fecal immunochemical test (FIT), or iFOBT. Please refer to other section of the Reference Document for information about other stool blood tests.
   - Not equivalent tests: The test must be from stool (feces). Text searches for "hemoccult" may or may not be productive. Text searches on "feces", "fecal", or "occult blood" may be useful, but do not over-generalize; many of these tests are unrelated to FOBT.

Do not to confuse FOBT with other tests related to blood. Be cautious when considering tests that contain the word "heme" because, although FOBT recognizes the "heme" portion of hemoglobin, hemoglobin is the oxygen-carrying molecule in blood and many tests containing the root "heme" are not FOBT.

- Indications: The fecal occult blood test (FOBT) detects blood in stool. The test may be performed to screen for colorectal cancer (CRC), or to diagnose other conditions that cause bleeding into the gastrointestinal (GI) tract. Occult (blood not visible to the eye) blood in feces may be due to CRC, polyps, or other GI conditions (e.g., hemorrhoids, anal fissures, ulcers, inflammatory bowel disease).
- Description:
- Related Tests: See the following TEST_TYPES: FOB_GUAI, FOB1_GUAI, and FOB_IMMUN
- Approximate Reference Range:
- Comments:
- Test Method Variance:
- Other
  - Associated LOINC: 27396-1
  - Associated CPT: 82270, 82274

4. FOB_IMMUN (Fecal Immunologic [Immunochemical] qualitative)

   - Common Name: FIT (fecal immunochemical test0, iFOBT
   - Long Name: FOBT qualitative; quick test; HFH test; occult blood in stool; immunologic
   - Test Type: FOB_IMMUN
   - Panels:
   - Equivalent tests: FOBT is a general name applied to stool tests to detect blood "hidden" (occult) in the stool. Two basic technologies are employed: the guaiac test, which is older and used routinely, and the immunochemical tests, which are newer. The fecal immunochemical test is the "FIT" or iFOBT. The FIT can also be known as the Hemoccult ICT ® (formerly FlexSure OBT®), Immudia-Hem Sp® , HemSelect®, or Insure®. The information provided here applies to the iFOBT or FIT. Please see the

other FOBT sheets for information about other occult blood tests for stool.

- Not equivalent tests: The FOBT test must be from stool. Text searches for "hemoccult" may or may not be productive. Similarly, text searches on "feces", "fecal", or "occult blood" may be useful but do not over-generalize; many of these tests will be unrelated to FOBT. Do not to confuse FOBT with other tests related to blood. Be cautious when considering tests that contain the word "heme" because, although FOBT recognizes the "heme" portion of hemoglobin, hemoglobin is the oxygen-carrying molecule in blood and many tests containing the root "heme" are not FOBT.
- Indications: The FIT detects blood in stool. The test may be performed to screen for colorectal cancer (CRC), or to diagnose other conditions that cause bleeding into the gastrointestinal (GI) tract. Occult (blood not visible to the eye) blood in feces may be due to CRC, polyps, or other GI conditions (e.g., hemorrhoids, anal fissures, ulcers, inflammatory bowel disease).
- Description: Fecal immunochemical tests use antibodies to detect human hemoglobin in stool samples. They are specific for human blood (in contrast to the guaiac test which detects nonhuman blood and is influenced by other interferences). The immunochemical test causes a color change; i.e., with a change in color indicating a positive test. In routine clinical use the expected result would be qualitative; the number of windows tested, and the number that are negative/positive. FIT is a newer test than the guaiac. FIT sample collection procedures are designed to increase acceptability and adherence by patients.
- Related Tests: See the following TEST_TYPES: FOB_GUAI, FOB1_GUAI, and FOB_QN
- Approximate Reference Range: N/A
- Comments: Remember that this is a stool test; the material being tested is human feces. When conducting string searches of databases to locate iFOBT tests, it may be useful to use terms such as "fecal", "stool" and "occult", and to ascertain that stool is the medium being tested.
- Test Method Variance:
- Other
  - Associated LOINC: 29771-3, 56490-6, 56491-4, 57905-2

- Associated CPT: 82274

5. FOB_IMMUN_QN (Fecal Immunologic [Immunochemical] quantitative)

   - Common Name: FIT (fecal immunochemical test, iFOBT)
   - Long Name: FOBT quantitative; fecal occult blood; occult blood in stool; immunologic
   - Test Type: FOB_IMMUN_QN
   - Panels:
   - Equivalent tests: FOBT is a general name applied to stool tests to detect blood "hidden" (occult) in the stool. Two basic technologies are employed: the guaiac test, which is older and used routinely, and the immunochemical tests, which are newer. The fecal immunochemical test is the "FIT" or iFOBT. The FIT can also be known as the Hemoccult ICT ® (formerly FlexSure OBT®), Immudia-Hem Sp® , HemSelect®, or Insure®. The information provided here applies to the iFOBT or FIT. Please see the other FOBT sheets for information about other occult blood tests for stool.
   - Not equivalent tests: The FOBT test must be from stool. Text searches for "hemoccult" may or may not be productive. Similarly, text searches on "feces", "fecal", or "occult blood" may be useful but do not over-generalize; many of these tests will be unrelated to FOBT. Do not to confuse FOBT with other tests related to blood. Be cautious when considering tests that contain the word "heme" because, although FOBT recognizes the "heme" portion of hemoglobin, hemoglobin is the oxygen-carrying molecule in blood and many tests containing the root "heme" are not FOBT.
   - Indications: The FIT detects blood in stool. The test may be performed to screen for colorectal cancer (CRC), or to diagnose other conditions that cause bleeding into the gastrointestinal (GI) tract. Occult (blood not visible to the eye) blood in feces may be due to CRC, polyps, or other GI conditions (e.g., hemorrhoids, anal fissures, ulcers, inflammatory bowel disease).
   - Description: A fecal immunochemical test (FIT) (also called an immunochemical fecal occult blood test or iFOBT) uses antibodies specific to the human globulin part of the Hemoglobin. Fecal immunochemical tests use antibodies to detect human hemoglobin

in stool samples. They are specific for human blood; they do not react to animal blood as found in meat , nor plant analogues, and drugs like Vitamin C do not effect the testing, so dietary adjustment is not needed. More important, globulin does not survive the enzymes of the upper digestive tract, so it is specific to lower GI bleeding-that is important to colon cancer screening. Different kit manufacturers have different requirements for repeats and timing, but most experts say you need samples from 2 different days for good results.

- Related Tests: See the following TEST_TYPES: FOB_GUAI, FOB1_GUAI, and FOB_QN
- Approximate Reference Range: a common threshold for positivity is around 10 g Hb/g feces.
- Comments: Remember that this is a stool test; the material being tested is human feces. When conducting string searches of databases to locate iFOBT tests, it may be useful to use terms such as "fecal", "stool" and "occult", and to ascertain that stool is the medium being tested.
- Test Method Variance:
- Other
  - Associated LOINC: 58453-2
  - Associated CPT:

1.1.29 FOLATE (Folate in serum)

- Common Name Folate
- Long Name: Folate
- Test Type: FOLATE
- Panels:
- Equivalent Tests: Folic Acid in serum
- Not Equivalent Tests: RBC folate
- Indications: To help diagnose the cause of anemia or neuropathy (nerve damage); to evaluate nutritional status in some patients; to monitor effectiveness of treatment for B12 or folate deficiency

- Description: Fasting for 6-8 hours before sample collection may be required.
- Related Tests: B12; CBC; CMP; Methylmalonic acid (MMA); Homocysteine
- Approximate Reference Range (Each site will have slight variations: Lower limit of normal: Approximately 1.5 - 3 ng/mL (or mcg/L). Upper limit of normal approximately 17 – 100 ng/mL. (Note: If RBC folate is inadvertently included, the reference range is one useful way to check to see if incorrect folate tests have been pulled. The lower limit of the normal range of RBC folate levels is approximately 100 – 180 and the upper limit of the normal range is RBC folate is approximately 360 – 800.)
- Comments: B12 and folate are primarily ordered to help diagnose the cause of macrocytic anemia. They are ordered as follow-up tests when large RBCs and a decreased hemoglobin concentration are found during a CBC test. B12 may be ordered with folate, by itself, or with other screening laboratory tests to help diagnose the cause of neuropathy. Either a serum or RBC folate test may be used to help detect a deficiency. Some doctors feel that the RBC folate test is more clinically relevant than serum folate, but there is not widespread agreement on this. High levels of B12 and folate are not usually clinically monitored. Increased folate may be seen with pernicious anemia and with vegetarian diets.
- Test Method Variance:
- Other:
  - Associated LOINC: 35210-4, 2284-8, 14732-2, 2282-2, 25415-1
  - Associated CPT: 82746

1.1.30 FOLATE_RBC (Folate in red blood cells)

- Common Name Folate in red blood cells
- Long Name:
- Test Type: FOLATE_RBC
- Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Your healthcare provider might order an RBC folate test if you demonstrate fatigue, weakness, pale skin, irritability, sore tongue, diarrhea, weight loss, numbness/tingling, or memory problems. Megaloblastic anemia is characterized by abnormally large red blood cells and can be caused by folate deficiency. Conditions that affect folate absorption or metabolism can include celiac disease, Crohn's disease, chronic alcoholism, liver disease, kidney failure, or certain cancers. This test may also be ordered for patients who are pregnant or planning a pregnancy in order to ensure adequate folate levels.
- Description: The folate in red blood cells test is a blood test that measures the level of folate within the red blood cells. This test provides a more stable and accurate assessment of long-term folate status, reflecting the levels accumulated over the lifespan of the red blood cells, which is approximately 120 days. It is less affected by recent dietary intake compared to the serum folate test.
- Related Tests: B12; CBC; CMP; Methylmalonic acid (MMA); Homocysteine
- Approximate Reference Range (Each site will have slight variations: Lower limit of normal: a general range for adults is 140 to 628 ng/mL
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 2283-0, 14731-4
  - Associated CPT: 82747

1.1.31 (GFR) Glomerular Filtration Rate Test Types

1. GFR_AA_CKD (Glomerular Filtration Rate, CKD-EPI, African American)
   - Common Name: Glomerular Filtration Rate, CKD-EPI, African American

- Long Name: Glomerular Filtration Rate/1.73 sq M.predicted Black (CKD-EPI)
- Test Type: GFR_AA_CKD
- Panels: Comprehensive metabolic panel, basic metabolic panel
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status.
- Description: This test estimates how much blood passes through glomeruli, the kidneys' filters, each minute. A measure of creatinine is required to calculate GFR. The MDRD equation was created from data from the Modification of Diet and Renal Disease (MDRD) Study, is based on 4 variables (i.e. Scr, age, gender, race). The GFR test is a blood test used to estimate how well the kidneys are filtering waste from the blood. The race-based factor for estimated glomerular filtration rate (eGFR) calculations is no longer recommended. The 2021 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation, which removes the race variable, is now the standard used by most medical laboratories.
  - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.
- Related Tests:
- Approximate Reference Range (National Kidney Foundation): GFR Description Kidney Damage Stage

90 ml/min Normal GFR 1 60-89 ml/min Mild decrease in GFR 2 30-59 ml/min Moderate decrease in GFR 3 15-29 ml/min Severe decrease in GFR 4 <15 ml/min Kidney failure 5

- Comments: The GFR_AA_CKD test type should be used when your lab reports results include race. If GFR results are reported without regard to race, populate the GFR_MDRD test type. The GFR_AA_MD is only used for African Americans. Early on in GFR reporting there was no race adjustment; clinicians had to adjust the reported values themselves if their patient was African American. Still today many labs do not have race data associated with their samples, so they report two GFR values, one based on non-African Americans and the other for African Americans. This way clinicians just choose the appropriate result. This test is most

accurate for poor kidney function so it may only be reported for values less than 60 ml/min. Although the MDRD equation for African Americans is not intended for use in calculating eGFR for pediatric patients, some HMOs may report eGFR using the MDRD for children (i.e., one of the MDRD TEST_TYPEs would be populated in the VDW for children at your site if that is the case).

- Test Method Variance: Prior to 2005, the equation was as follows: GFR= 186 x (Scr)-1.154 x (Age)-0.203 x (0.742 if female) x (1.120 if African American) Therefore, if a 38 year old black man had a serum creatinine of 1.6 in 2003, his GFR would be 62.5, but in 2008 that man would have a GFR of 58.8. It may appear there was a decline in renal function over time, however part of the decline is an artifact of the equation change. Identify when your site lab changed equations.
- Other: –
  - Associated LOINC: 88293-6
  - Associated CPT:

2. GFR_AA_MD (Glomerular Filtration Rate, MDRD, African American)

   - Common Name: Estimated GFR, eGFR
   - Long Name: Glomerular Filtration Rate/1.73 sq M. predicted black (MDRD)
   - Test Type: GFR_AA_MD
   - Panels: Comprehensive metabolic panel, basic metabolic panel
   - Equivalent Tests: Calculated GFR
   - Not Equivalent Tests: GFR measured by inulin clearance; Creatinine [SCr]; Creatinine clearance [CrCl]; Microalbumin; Cystatin C; Urine protein; Beta-2 microglobulin; GFR predicted non-Black MDRD (GFR_NAA_MD)
   - Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status
   - Description: This test estimates how much blood passes through glomeruli, the kidneys' filters, each minute. A measure of creatinine is required to calculate GFR. The MDRD equation was

created from data from the Modification of Diet and Renal Disease (MDRD) Study, is based on 4 variables (i.e. Scr, age, gender, race), and was introduced in 2005. GFR= 175 x (Standardized Scr)-1.154 x (Age)-0.203 x (0.742 if female) x (1.120 if African American)

  - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.

- Related Tests: SCr; CrCl
- Approximate Reference Range (National Kidney Foundation):
  - GFR Description Kidney Damage Stage
  - 90 ml/min Normal GFR 1
  - 60-89 ml/min Mild decrease in GFR 2
  - 30-59 ml/min Moderate decrease in GFR 3
  - 15-29 ml/min Severe decrease in GFR 4
  - <15 ml/min Kidney failure 5
- Comments: The GFR_AA_MD test type should be used when your lab reports results include race. If GFR results are reported without regard to race, populate the GFR_MDRD test type. The GFR_AA_MD is only used for African Americans. Early on in GFR reporting there was no race adjustment; clinicians had to adjust the reported values themselves if their patient was African American. Still today many labs do not have race data associated with their samples, so they report two GFR values, one based on non-African Americans and the other for African Americans. This way clinicians just choose the appropriate result. This test is most accurate for poor kidney function so it may only be reported for values less than 60 ml/min. Although the MDRD equation for African Americans is not intended for use in calculating eGFR for pediatric patients, some HMOs may report eGFR using the MDRD for children (i.e., one of the MDRD TEST_TYPEs would be populated in the VDW for children at your site if that is the case).
- Test Method Variance: Prior to 2005, the equation was as follows:
  - GFR= 186 x (Scr)-1.154 x (Age)-0.203 x (0.742 if female) x (1.120 if African American)
  - Therefore, if a 38 year old black man had a serum creatinine of 1.6 in 2003, his GFR would be 62.5, but in 2008 that man would have a GFR of 58.8. It may appear there was a decline

in renal function over time, however part of the decline is an artifact of the equation change. Identify when your site lab changed equations.

- Other: –
  - Associated LOINC: 48643-1
  - Associated CPT: 82565

3. GFR_AA_NS (Glomerular Filtration Rate, method not-specified, African American)

   - Common Name: Glomerular Filtration Rate, method not-specified, African American
   - Long Name: Glomerular Filtration Rate/1.73 sq M. predicted black (method not specified)
   - Test Type: GFR_AA_NS
   - Panels:
   - Equivalent Tests: Calculated GFR
   - Not Equivalent Tests:
   - Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status
   - Description: This test estimates how much blood passes through glomeruli, the kidneys' filters, each minute. A measure of creatinine is required to calculate GFR. The MDRD equation was created from data from the Modification of Diet and Renal Disease (MDRD) Study, is based on 4 variables (i.e. Scr, age, gender, race).

   GFR= 175 x (Standardized Scr)-1.154 x (Age)-0.203 x (0.742 if female) x (1.120 if African American) See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.

   - Related Tests: SCr; CrCl
   - Approximate Reference Range (National Kidney Foundation):
     - GFR Description Kidney Damage Stage
     - 90 ml/min Normal GFR 1
     - 60-89 ml/min Mild decrease in GFR 2
     - 30-59 ml/min Moderate decrease in GFR 3

  - 15-29 ml/min Severe decrease in GFR 4
  - <15 ml/min Kidney failure 5
- Comments: The GFR_AA_MD test type should be used when your lab reports results include race. If GFR results are reported without regard to race, populate the GFR_MDRD test type. The GFR_AA_MD is only used for African Americans. Early on in GFR reporting there was no race adjustment; clinicians had to adjust the reported values themselves if their patient was African American. Still today many labs do not have race data associated with their samples, so they report two GFR values, one based on non-African Americans and the other for African Americans. This way clinicians just choose the appropriate result. This test is most accurate for poor kidney function so it may only be reported for values less than 60 ml/min. Although the MDRD equation for African Americans is not intended for use in calculating eGFR for pediatric patients, some HMOs may report eGFR using the MDRD for children (i.e., one of the MDRD TEST_TYPEs would be populated in the VDW for children at your site if that is the case).
- Test Method Variance: Prior to 2005, the equation was as follows:

GFR= 186 x (Scr)-1.154 x (Age)-0.203 x (0.742 if female) x (1.120 if African American) Therefore, if a 38 year old black man had a serum creatinine of 1.6 in 2003, his GFR would be 62.5, but in 2008 that man would have a GFR of 58.8. It may appear there was a decline in renal function over time, however part of the decline is an artifact of the equation change. Identify when your site lab changed equations.

- Other: –
  - Associated LOINC: 96591-3
  - Associated CPT:

4. GFR_ARV (Glomerular Filtration Rate, ArvRate)
   - Common Name: Glomerular Filtration Rate, ArvRate
   - Long Name: Glomerular Filtration Rate/1.73 sq M.predicted (Antiretroviral Rate)
   - Test Type: GFR_ARV
   - Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status
- Description: Glomerular Filtration Rate (GFR) is a key measure of kidney function. It estimates how much blood the kidneys' filtering units (glomeruli) process per minute. GFR is particularly important for individuals with HIV, as both the infection itself and some antiretroviral drugs can affect kidney function. The kidneys are crucial for filtering waste products, including some antiretroviral medications, from the body; therefore, GFR is used to assess if ARV dosages need adjustment to prevent potential drug buildup and minimize side effects.
  - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): An eGFR below 60 mL/min/1.73 m2 may indicate chronic kidney disease
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 69405-9
  - Associated CPT:

5. GFR_CKD (Glomerular Filtration Rate, CKD-EPI)
   - Common Name: Glomerular filtration rate
   - Long Name: Glomerular Filtration Rate/1.73 sq M.predicted (CKD-EPI)
   - Test Type: GFR_CKD
   - Panels: Comprehensive metabolic panel, basic metabolic panel
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status

- Description: Glomerular Filtration Rate (GFR) is a key measure of kidney function. It represents the rate at which blood is filtered by the kidneys each minute. This filtration process is essential for removing waste products and excess fluid from the blood, which are then excreted from the body as urine. The kidneys contain tiny filters called glomeruli, which are the main structures responsible for filtration. The glomeruli act like sieves, allowing small molecules like waste products (such as creatinine) and excess water to pass through, while preventing larger substances like blood cells and proteins from entering the filtrate. The GFR quantifies how efficiently this filtering process is occurring by measuring the volume of blood filtered per unit of time. This is typically expressed in milliliters per minute (mL/min). The MDRD equation was created from data from the Modification of Diet and Renal Disease (MDRD) Study, is based on 4 variables (i.e. Scr, age, gender, race), and was introduced in 2005. GFR= 175 x (Standardized Scr)-1.154 x (Age)-0.203 x (0.742 if female) x (1.120 if African American).
  - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.
- Related Tests: SCr; CrCl
- Approximate Reference Range (Each site will have slight variations): Normal GFR values typically fall within the range of 90 to 120 mL/min/1.73 m^2 (adjusted for body surface area). A GFR below 60 mL/min/1.73 m^2 for three months or more suggests potential kidney disease.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 62238-1, 97952-6
  - Associated CPT:

6. GFR_CKD_EPI (Glomerular Filtration Rate, CKD-EPI 2021, not based on race)

   - Common Name: Glomerular Filtration Rate, CKD-EPI 2021, not based on race
   - Long Name: Glomerular filtration rate/1.73 sq M.predicted (CKD-EPI 2021) creatinine formula not based on race

- Test Type: GFR_CKD_EPI
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Healthcare providers order this test to:
  - Diagnose Chronic Kidney Disease (CKD): Identifying individuals with an eGFR below 60 mL/min/1.73m^2 for more than three months.
  - Stage Kidney Disease: Classifying the severity of CKD from Stage 1 to Stage 5 to determine appropriate treatment plans.
  - Screen At-Risk Patients: Monitoring individuals with risk factors such as diabetes, high blood pressure (hypertension), heart disease, or a family history of kidney disease.
  - Determine Medication Dosing: Adjusting doses for drugs excreted by the kidneys to prevent toxicity.
  - Evaluate Transplant & Dialysis Eligibility: Assessing when a patient reaches the threshold for kidney transplant waitlists (eGFR 20 mL/min) or requires dialysis.
  - Monitor Kidney Health: Tracking function in patients with active kidney disease or those taking medications that may cause kidney damage.
- Description: Glomerular Filtration Rate (GFR) measures how well the kidneys filter waste from the blood. The CKD-EPI 2021 equation is the current clinical standard for estimating GFR (eGFR) in adults, specifically designed to exclude race as a variable to improve health equity and accuracy.
  - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Results are reported in mL/min/1.73m^2. A higher number indicates better kidney function.

GFR Value	Interpretation
90 or higher	Normal or high kidney function
60 to 89	Mildly decreased function; may be normal if no other signs of damage are prese
30 to 59	Moderate decrease; often indicates Chronic Kidney Disease (CKD) Stage 3
15 to 29	Severe decrease in function (Stage 4)
Below 15	Kidney failure (Stage 5)

- Comments: This test_type holds the 2021 non-race based calculation from the National Kidney Foundation and the American Society of Nephrology.
- Test Method Variance:
- Other:
  - Associated LOINC: 98979-8
  - Associated CPT:

7. GFR_CKD_EPI_CYS (Glomerular Filtration Rate, CKD-EPI 2021, creatine-cystatin not based on race)

   - Common Name: Glomerular Filtration Rate, CKD-EPI 2021, creatine-cystatin not based on race
   - Long Name: Glomerular filtration rate/1.73 sq M.predicted (CKD-EPI 2021) creatinine-cystatin not based on race
   - Test Type: GFR_CKD_EPI_CYS
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Healthcare providers order this test to:
     - Diagnose Chronic Kidney Disease (CKD): Identifying individuals with an eGFR below 60 mL/min/1.73m^2 for more than three months.
     - Stage Kidney Disease: Classifying the severity of CKD from Stage 1 to Stage 5 to determine appropriate treatment plans.
     - Screen At-Risk Patients: Monitoring individuals with risk factors such as diabetes, high blood pressure (hypertension), heart disease, or a family history of kidney disease.
     - Determine Medication Dosing: Adjusting doses for drugs excreted by the kidneys to prevent toxicity.

  - Evaluate Transplant & Dialysis Eligibility: Assessing when a patient reaches the threshold for kidney transplant waitlists (eGFR 20 mL/min) or requires dialysis.
  - Monitor Kidney Health: Tracking function in patients with active kidney disease or those taking medications that may cause kidney damage.
- Description: The CKD-EPI 2021 Creatinine-Cystatin C equation is a race-neutral method for calculating the Estimated Glomerular Filtration Rate (eGFR). Developed to address health inequities, it replaces older formulas that used race as a biological proxy. Combining both creatinine and cystatin C provides a more accurate estimate of GFR than using either marker alone.
  - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Results are reported in mL/min/1.73m^2. A higher number indicates better kidney function.

GFR Value	**Interpretation**
90 or higher	Normal or high kidney function
60 to 89	Mildly decreased function; may be normal if no other signs of damage are prese
30 to 59	Moderate decrease; often indicates Chronic Kidney Disease (CKD) Stage 3
15 to 29	Severe decrease in function (Stage 4)
Below 15	Kidney failure (Stage 5)

- Comments: This test_type holds the 2021 non-race based calculation from the National Kidney Foundation and the American Society of Nephrology.
- Test Method Variance:
- Other:
  - Associated LOINC: 98980-6
  - Associated CPT:

8. GFR_CYST (Glomerular Filtration Rate, Cystatin)
   - Common Name: Glomerular Filtration Rate, Cystatin
   - Long Name: Glomerular Filtration Rate/1.73 sq M.predicted (Cystatin)

- Test Type: GFR_CYST
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status
- Description: Cystatin C is an alternative and increasingly valuable marker for assessing kidney function. It is a protein produced at a constant rate by all nucleated cells and is primarily cleared from the blood by the kidneys through glomerular filtration. Because its production is less influenced by factors like muscle mass and diet, it can offer a more accurate assessment of GFR in certain situations.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal range for serum cystatin C is generally between 0.62-1.15 mg/L
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 102097-3, 50210-4, 87430-5, 94677-2
  - Associated CPT:

9. GFR_F_MD (Glomerular Filtration Rate, MDRD, Female)

- Common Name: Glomerular Filtration Rate, MDRD, Female
- Long Name: Glomerular Filtration Rate/1.73 sq M.predicted.female (MDRD)
- Test Type: GFR_F_MD
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The estimated glomerular filtration rate (eGFR) is a valuable tool used to assess kidney function and detect kidney disease. It's calculated from a blood test that measures creatinine, a waste product generated from muscle activity.

- Description: In women, glomerular filtration rate (GFR) is often estimated using a formula that includes serum creatinine levels, along with other factors like age, race, and body size. Creatinine is a waste product generated from muscle metabolism that is normally filtered out of the blood by the kidneys and excreted in urine. A creatinine blood test helps assess kidney function because higher levels in the blood might suggest that the kidneys aren't filtering waste products effectively.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal range for GFR in healthy, younger adult women is typically between 90 and 120 mL/min/1.73m^2
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 50044-7 (predict female)
  - Associated CPT:

10. GFR_M_MD (Glomerular Filtration Rate, MDRD, Male)

- Common Name: Glomerular Filtration Rate, MDRD, Male
- Long Name: Glomerular Filtration Rate/1.73 sq M.predicted.male (MDRD)
- Test Type: GFR_M_MD
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The estimated glomerular filtration rate (eGFR) is a valuable tool used to assess kidney function and detect kidney disease. It's calculated from a blood test that measures creatinine, a waste product generated from muscle activity.
- Description: The Glomerular Filtration Rate (GFR) is a crucial measure of kidney function, indicating how well the kidneys filter waste products from the blood. In men, GFR is often estimated using blood creatinine levels, leading to the term "estimated GFR" or eGFR. Creatinine is a waste product generated from muscle activity that is filtered out by the kidneys.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): a normal GFR for healthy adults, including men, falls between 90 and 120 mL/min/1.73 m^2
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 70969-1 (predict male)
  - Associated CPT:

11. GFR_MDRD (Glomerular Filtration Rate, MDRD)

    - Common Name: Estimated GFR, eGFR
    - Long Name: Glomerular Filtration Rate/1.73 sq M. predicted (MDRD)
    - Test Type: GFR_MDRD
    - Panels: Comprehensive metabolic panel, basic metabolic panel
    - Equivalent Tests: Calculated GFR
    - Not Equivalent Tests: GFR measured by inulin clearance; Creatinine [SCr]; Creatinine clearance [CrCl]; Microalbumin; Cystatin C; Urine protein; Beta-2 microglobulin
    - Indications: The GFR is used to measure kidney function. Clinicians may use GFR to screen for and detect kidney damage or monitor kidney status.
    - Description: This test estimates how much blood passes through glomeruli, the kidneys' filters, each minute. A measure of creatinine is required to calculate GFR. The MDRD equation was created from data from the Modification of Diet and Renal Disease (MDRD) Study and is based on 4 variables (i.e. Scr, age, gender, race). The equation is: GFR= 175 x (Standardized Scr)-1.154 x (Age)-0.203 x (0.742 if female) x (1.120 if African American).
      - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.
    - Related Tests: Serum creatinine; creatinine clearance
    - Approximate Reference Range (National Kidney Foundation):
      - GFR Description Kidney Damage Stage
      - 90 ml/min Normal GFR 1

  - 60-89 ml/min Mild decrease in GFR 2
  - 30-59 ml/min Moderate decrease in GFR 3
  - 15-29 ml/min Severe decrease in GFR 4
  - <15 ml/min Kidney failure 5
- Comments: This TEST_TYPE should only be used when 1) you cannot tell whether the lab results include race, or 2) eGFR results are reported without regard to race. If eGFR results are reported as African American or non-African American, populate those TEST_TYPEs instead. Although the MDRD equation is not intended for use in calculating eGFR for pediatric patients, some HMOs may report eGFR using the MDRD for children (i.e., one of the MDRD TEST_TYPEs would be populated in the VDW for children at your site if that is the case). The eGFR test is most accurate for poor kidney function so it may only be reported for values less than 60 ml/min. Because race is often not known, results under this test type name can be reported based on a non-African American (NAA) status and clinicians adjust the result if their patient was African American.
- Test Method Variance: Prior to 2005, the MDRD equation was:
  - GFR= 186 x (Scr)-1.154 x (Age)-0.203 x (0.742 if female) x (1.120 if African American)
  - Therefore if a 57 year old white man had a Scr of 1.1 in 2003 his GFR would be 73.3, but a man with the same characteristics in 2008 would have a GFR of 70.0. From a population level it may appear that there were declines in renal function over time, however part of the decline may be an artifact of the change in the equation. Identify when your site lab changed equations.
- Other: –
  - Associated LOINC: 33914-3, 76633-7, 77147-7
  - Associated CPT: 82565

12. GFR_NAA_CKD (Glomerular Filtration Rate, CKD-EPI, Non-African American)

- Common Name: Glomerular Filtration Rate, CKD-EPI, Non-African American
- Long Name: Glomerular Filtration Rate/1.73 sq M.predicted .non Black (CKD-EPI)

- Test Type: GFR_NAA_CKD
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status
- Description: The eGFR is a calculation based on the results of a blood test that measures the level of creatinine, a waste product from muscle metabolism, in your blood. For non-African American individuals, the calculation does not include a specific coefficient or adjustment for race. Your kidneys contain millions of tiny filters called glomeruli. The test estimates the volume of blood passing through these filters per minute. The eGFR is not a direct measurement. Instead, it is a calculation based on the results of a standard blood test that measures your serum creatinine level.
  - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
  - GFR Description Kidney Damage Stage
  - 90 ml/min Normal GFR 1
  - 60-89 ml/min Mild decrease in GFR 2
  - 30-59 ml/min Moderate decrease in GFR 3
  - 15-29 ml/min Severe decrease in GFR 4
  - <15 ml/min Kidney failure 5
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 88294-4
  - Associated CPT:

13. GFR_NAA_MD (Glomerular Filtration Rate, MDRD, Non-African American)

    - Common Name: Estimated GFR, eGFR, MDRD, Non-African American

- Long Name: Glomerular Filtration Rate/1.73 sq M. predicted non black (MDRD)
- Test Type: GFR_NAA_MD
- Panels: Comprehensive metabolic panel, basic metabolic panel
- Equivalent Tests: Calculated GFR; eGFR
- Not Equivalent Tests: GFR measured by inulin clearance; Creatinine [SCr]; Creatinine clearance [CrCl]; Microalbumin; Cystatin C; Urine protein; Beta-2 microglobulin; GFR predicted Black MDRD (GFR_AA_MD)
- Indications: The GFR is used to measure kidney function. Clinicians may use GFR to screen for and detect kidney damage or monitor kidney status.
- Description: This test estimates how much blood passes through glomeruli, the kidneys' filters, each minute. A measure of creatinine is required to calculate GFR. The MDRD equation was created from data from the Modification of Diet and Renal Disease (MDRD) Study, is based on 4 variables (i.e. Scr, age, gender, race), and was introduced in 2005. The equation is: GFR = 175 x (Standardized Scr)-1.154 x (Age)-0.203 x (0.742 if female)
  - See: `http://www.nkdep.nih.gov/professionals/gfr_calculators/`.
- Related Tests: SCr; CrCl
- Approximate Reference Range (National Kidney Foundation):
  - GFR Description Kidney Damage Stage
  - 90 ml/min Normal GFR 1
  - 60-89 ml/min Mild decrease in GFR 2
  - 30-59 ml/min Moderate decrease in GFR 3
  - 15-29 ml/min Severe decrease in GFR 4
  - <15 ml/min Kidney failure 5
- Comments: The "GFR_NAA_MD" TEST_TYPE should be used when your lab reports results that include race. If GFR results are reported without regard to race, populate the "GFR_MDRD" TEST_TYPE instead. Labs report race-specific GFR to allow clinicians to choose the correct result based on the race of their patients without having to do calculations. The GFR_NAA_MD is used for all races except African American. The eGFR test is most accurate for poor kidney function so it may only be reported for

values less than 60 ml/min. Although the MDRD equation is not intended for use in calculating eGFR for pediatric patients, some HMOs may report eGFR using the MDRD for children (i.e., one of the MDRD TEST_TYPEs would be populated in the VDW for children at your site if that is the case).

- Test Method Variance: Prior to 2005, the equation was as follows
  - GFR= 186 x (Scr)-1.154 x (Age)-0.203 x (0.742 if female)
  - Therefore, if a 35 year old Asian female had a serum creatinine of 2 in 2003 her GFR would be 30.1, but a female with the same characteristics in 2008 would have a GFR of 28.4. From a population level it appears there was a decline in renal function over time, however part of the decline is an artifact of the equation change. Identify when your site lab changed equations.
- Other: –
  - Associated LOINC: 48642-3
  - Associated CPT: 82565

14. GFR_NAA_NS (Glomerular Filtration Rate, Not Specified, Non-African American)

    - Common Name: Glomerular Filtration Rate, Not Specified, Non-African American
    - Long Name: Glomerular filtration rate/1.73 sq M.predicted.non Black
    - Test Type:
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status
    - Description: This test estimates how much blood passes through glomeruli, the kidneys' filters, each minute. A measure of creatinine is required to calculate GFR. The MDRD equation was created from data from the Modification of Diet and Renal Disease (MDRD) Study, is based on 4 variables (i.e. Scr, age, gender, race), and was introduced in 2005. The equation is: GFR = 175 x (Standardized Scr)-1.154 x (Age)-0.203 x (0.742 if female)

  - See: http://www.nkdep.nih.gov/professionals/gfr_calculators/.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
  - GFR Description Kidney Damage Stage
  - 90 ml/min Normal GFR 1
  - 60-89 ml/min Mild decrease in GFR 2
  - 30-59 ml/min Moderate decrease in GFR 3
  - 15-29 ml/min Severe decrease in GFR 4
  - <15 ml/min Kidney failure 5
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 96592-1
  - Associated CPT:

15. GFR_NS (Glomerular Filtration Rate, Not Specified)

- Common Name: Glomerular Filtration Rate, Not Specified
- Long Name: Glomerular filtration rate, method and Race not specified
- Test Type: GFR_NS
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Measure kidney function; screen for/detect kidney damage; monitor kidney status
- Description: This test estimates how much blood passes through glomeruli, the kidneys' filters, each minute. A measure of creatinine is required to calculate GFR.
  - See: http://www.nkdep.nih.gov/professionals/gfr_calculators/.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
  - GFR Description Kidney Damage Stage

  - 90 ml/min Normal GFR 1
  - 60-89 ml/min Mild decrease in GFR 2
  - 30-59 ml/min Moderate decrease in GFR 3
  - 15-29 ml/min Severe decrease in GFR 4
  - <15 ml/min Kidney failure 5
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 45064-3, 45065-0, 45066-8, 50261-7
  - Associated CPT:

16. GFR_SCH (Glomerular Filtration Rate, Schwartz)

- Common Name: Estimated GFR, eGFR
- Long Name: Glomerular Filtration Rate/1.73 sq M. predicted (Schwartz)
- Test Type: GFR_SCH
- Panels: Comprehensive metabolic panel, basic metabolic panel
- Equivalent Tests: Calculated GFR; eGFR
- Not Equivalent Tests: GFR measured by inulin clearance; Creatinine [SCr]; Creatinine clearance [CrCl]; Microalbumin; Cystatin C; Urine protein; Beta-2 microglobulin; estimated GFR MDRD
- Indications: The GFR is used to measure kidney function in children. Clinicians may use GFR to screen for and detect kidney damage or monitor kidney status.
- Description: This test estimates how much blood passes through glomeruli, the kidneys' filters, each minute. A measure of creatinine is required to calculate GFR. The Schwartz equation is for measuring GFR in the pediatric population (<18 years of age) and is based on 4 variables (i.e. Scr, height, age, gender).
  - GFR (mL/min/1.73 m2) = k x (height in cm)/ Scr in mg/dL Where k is a constant:
    * K=0.33 in premature infants
    * K=0.45 in term infants to 1 year of age
    * K=0.55 in children 1 to 13 years of age and adolescent females

        - ∗ K=0.70 in adolescent males (13 years and older)
    - Related Tests: SCr; CrCl
    - Approximate Reference Range:
        - GFR Description Kidney Damage Stage
        - 90 ml/min Normal GFR 1
        - 60-89 ml/min Mild decrease in GFR 2
        - 30-59 ml/min Moderate decrease in GFR 3
        - 15-29 ml/min Severe decrease in GFR 4
        - <15 ml/min Kidney failure 5
    - Comments: Many HMOs do not report GFR results using the Schwartz equation (even for pediatrics). Before populating this test type, please determine whether your health plan reports results using the Schwartz equation. This test is most accurate for poor kidney function so it may only be reported for values less than 75 ml/min/1.73 m2. Values any higher than 75 mL/min may be reported as "greater than or equal to 75 mL/min/1.73 m2".
    - Test Method Variance: –
    - Other: –
        - Associated LOINC: 50383-9, 50384-7, 102097-3
        - Associated CPT: 82565

1.1.32 GGT (GGTP, Gamma-glutamyl transpeptidase [or transferase] in blood/serum)

- Common Name: GGT or GGTP
- Long Name: GGTP, Gamma-glutamyl transpeptidase or transferase in blood/serum
- Test Type: GGT
- Panels: Liver Panel
- Equivalent Tests: N/A
- Not Equivalent Tests: GTT (glucose tolerance tests)
- Indications: To differentiate between liver and bone disease as a cause for elevated alkaline phosphatase (ALP). Both ALP and GGT are elevated in diseases of the bile ducts and some liver diseases, but only

ALP is elevated in bone disease, adolescence or pregnancy. If the GGT level is normal in a person with a high ALP, the cause is most likely bone disease. Also, GGT is used to help detect liver disease and/or alcohol abuse.

- Description: GGT is an enzyme found in many organs, (kidney, spleen, heart, brain and pancreas) but the main source of GGT in the blood is the liver. GGT is increased in most diseases that cause acute damage to the liver or bile ducts but is usually not helpful in distinguishing between different causes of liver damage. GGT levels are sometimes increased with consumption of even small amounts of alcohol. Higher levels are found in chronic heavy drinkers than in people who consume less than 2 to 3 drinks per day or who only drink heavily on occasion (binge drinkers).
- Related Tests: AST; ALT; ALP; Bilirubin; Liver panel
- Approximate Reference Range (Each site will have slight variations): 1 - 94 IU/L. GGT increases with age in women, and is always somewhat higher in men than in women.
- Comments: GGT levels fall after meals, so fasting for at least 8 hours is recommended. Even small amounts of alcohol within 24 hours of a GGT test may cause a temporary increase in the GGT. GGT levels may be elevated in heart failure. Drugs that may cause an elevated GGT level include phenytoin, carbamazepine, and barbiturates such as phenobarbital. Also, nonsteroidal anti-inflammatory drugs (NSAIDs), lipidlowering drugs, antibiotics, histamine receptor blockers (used to treat excess stomach acid production), antifungal agents, antidepressants, and hormones such as testosterone, can increase GGT. Clofibrate and oral contraceptives can decrease GGT. Smoking can increase GGT.
- Test Method Variance:
- Other:
  - Associated LOINC: 2324-2
  - Associated CPT: 82977

1.1.33 GLU_F (Glucose, Fasting Serum/Plasma)

- Common Name: Fasting serum glucose

- Long Name: Blood Sugar Fasting
- Test Type: GLU_F
- Panels:
- Equivalent Tests: Fasting plasma glucose, whole blood fasting glucose
- Not Equivalent Tests: Random glucose or fasting period not specified; glucose tolerance test (although may include a fasting glucose); 2-hour post parandial (after meal); glucose on other body fluid such as cerebrospinal fluid, urine, synovial, plueral or peritoneal fluids; glucose phosphate isomerase; glucose-6-phosphate dehydrogenase; glucocorticoids. Many tests using GLY prefix are not a blood glucose level.
- Indications: Fasting glucose is used for screening for diabetes or hypoglycemia and to monitor progression or control of diabetes. Also, fasting glucose is used to assist in determining medication adjustments for managing diabetes. Certain drugs used to treat conditions other than diabetes can also affect Fasting Blood Sugar.
- Description: In general, it is recommended that you fast 8 hours before having a fasting blood glucose test. In persons with diabetes, however, glucose levels are often checked both while fasting and after meals to provide the best control of diabetes.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Approximately 65- 115 mg/dL
- Comments: Delay in transport or delay in separating serum from cells can falsely decrease glucose values.
- Test Method Variance: Capillary specimens can vary from serum/plasma values by 10%. Some meters have different settings for different lot numbers of test strips. "Lack of circulation" may invalidate capillary glucose. Preferred specimen is a plasma from fluoride/oxalate anticoagulated specimen tube to prevent cells from metabolizing glucose during transport.
- Other:

  - Associated LOINC: 101476-0, 10450-5, 14768-6, 14769-4, 14770-2, 14771-0, 14996-3, 1547-9, 1549-5, 1550-3, 1552-9, 1554-5, 1556-0, 1557-8, 1558-6, 17865-7, 25680-0, 35184-1, 39997-2, 40148-9, 40193-5, 40858-3, 41604-0, 47622-6, 53049-3, 70208-4, 76629-5, 77145-1, 87421-4, 88365-2
  - Associated CPT: (CPT Code does not differentiate between random and fasting) 82947

1.1.34 GLU_RAN (Glucose, Random, not known if patient was fasting or known that patient was not fasting)

- Common Name: Glucose Random, Glucose-fasting period not indicated
- Long Name: Blood Sugar Random or Not Specified
- Test Type: GLU_RAN
- Panels: Basic metabolic panel, comprehensive metabolic profile
- Equivalent Tests: plasma glucose, whole blood glucose
- Not Equivalent Tests: Fasting glucose. Glucose tolerance (although may include a fasting glucose). 2-Hour post parandial (after meal). Glucose on other body fluid such as cerebral spinal fluid, urine, synovial, pleural or peritoneal fluids. Glucose phosphate isomerase, Glucose-6-Phosphate Dehydrogenase, Glucocorticoids. Many tests using GLY prefix are not a glucose level.
- Indications: Random glucose is used for screening for diabetes or hypoglycemia and to monitor progression or control of diabetes. Also used to adjust medication dosage for diabetes. Certain drugs used to treat conditions other than diabetes can also affect blood glucose.
- Description: In general, regardless of whether food or drink has been consumed, blood glucose test should be less than 200 mg/dL. Some will use 150 mg/dL if it has been $>$ 1 hour since consuming small meal or beverage. If random is greater than ~200, follow-up testing is indicated. Can also be used to validate glucose meter results.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Approximately $<$200 mg/dL

- Comments: Delay in transport or delay in separating serum from cells can falsely decrease glucose values.
- Test Method Variance: Capillary specimens can vary from serum/plasma values by 10%. Some meters have different settings for different lot numbers of test strips. "Lack of circulation" may invalidate capillary glucose. Preferred specimen is a plasma from fluoride/oxalate anticoagulated specimen tube to prevent cells from metabolizing glucose during transport.
- Other:
  - Associated LOINC: 100746-7, 14743-9, 14749-6, 15074-8, 2339-0, 2340-8, 2341-6, 2345-7, 32016-8, 35211-2, 39480-9, 39481-7, 41651-1, 41652-9, 41653-7, 51596-5, 5914-7, 72516-8, 74774-1, 77135-2, 96594-7
  - Associated CPT: (CPT Code does not differentiate between random and fasting) 82947

1.1.35 (HCO3) Bicarbonate Test Types

1. HCO3_A (Bicarbonate in Arterial blood)
   - Common Name: Bicarbonate in Arterial blood
   - Long Name: Bicarbonate [Moles/volume] in Arterial blood
   - Test Type: HCO3_A
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Low bicarbonate can indicate metabolic acidosis, where the body has too much acid. This can be caused by conditions like diabetic ketoacidosis, kidney disease, or severe diarrhea. High bicarbonate can indicate metabolic alkalosis, where the body has too much base (alkali). This can result from conditions like prolonged vomiting, dehydration, or certain kidney diseases.
   - Description: Bicarbonate (or HCO3), is often referred to as the "bicarbonate ion" or sometimes simply "bicarb", is a crucial element measured in an arterial blood gas (ABG) test. It plays a vital role in maintaining the acid-base balance (pH balance) in

the body, which is essential for various bodily functions to operate correctly.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal bicarbonate levels in adults typically range from 22 to 32 mmol/L or 23 to 29 mEq/L
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 19230-2, 1960-4
  - Associated CPT:

2. HCO3_C (Bicarbonate in Capillary blood)

   - Common Name: Bicarbonate in Capillary blood
   - Long Name: Bicarbonate [Moles/volume] in Capillary blood
   - Test Type: HCO3_C
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Metabolic Acidosis (Low Bicarbonate): Bicarbonate levels below the normal range may indicate metabolic acidosis, meaning the blood is too acidic. Causes include conditions like kidney disease, diabetic ketoacidosis, chronic diarrhea, and Addison's disease. Metabolic Alkalosis (High Bicarbonate): Bicarbonate levels above the normal range can signify metabolic alkalosis, where the blood is too alkaline. This can result from factors like severe vomiting, dehydration, chronic lung problems (e.g., emphysema), or certain hormonal disorders like Cushing's disease.
   - Description: Bicarbonate is a vital electrolyte in the blood, playing a crucial role in maintaining the body's acid-base (pH) balance. It's a calculated value typically obtained alongside other blood gas parameters like pH, pO2, and pCO2. Capillary Blood Gas' are a valuable alternative to arterial blood gases (ABGs), especially in situations where repeated arterial sampling is difficult, such as in pediatric patients.
   - Related Tests:

- Approximate Reference Range (Each site will have slight variations): In adults, normal bicarbonate levels typically range from 22 to 26 mEq/L.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 19231-0, 1961-2
  - Associated CPT:

3. HCO3_V (Bicarbonate in Venous blood)

   - Common Name: Bicarbonate in Venous blood
   - Long Name: Bicarbonate [Moles/volume] in Venous blood
   - Test Type: HCO3_V
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Metabolic Acidosis (Low Bicarbonate): Bicarbonate levels below the normal range may indicate metabolic acidosis, meaning the blood is too acidic. Causes include conditions like kidney disease, diabetic ketoacidosis, chronic diarrhea, and Addison's disease. Metabolic Alkalosis (High Bicarbonate): Bicarbonate levels above the normal range can signify metabolic alkalosis, where the blood is too alkaline. This can result from factors like severe vomiting, dehydration, chronic lung problems (e.g., emphysema), or certain hormonal disorders like Cushing's disease.
   - Description: Bicarbonate acts as a primary buffer, converting excess hydrogen ions into carbonic acid, then converts to carbon dioxide and water. While arterial blood gas analysis is often considered the gold standard for assessing acid-base status, venous bicarbonate, usually measured as part of a basic metabolic panel or comprehensive metabolic panel, can provide a reliable estimate of arterial bicarbonate.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Typically ranges from 22 to 32 mmol/L in adults.
   - Comments:

- Test Method Variance:
- Other:
  - Associated LOINC: 14627-4, 19229-4, 19232-8, 19233-6
  - Associated CPT:

4. HCO3_NS (Bicarbonate source not specified)
   - Common Name: Bicarbonate source not specified
   - Long Name: Bicarbonate [Moles/volume] in Serum, Plasma or Blood
   - Test Type: HCO3_NS
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Metabolic Acidosis (Low Bicarbonate): Bicarbonate levels below the normal range may indicate metabolic acidosis, meaning the blood is too acidic. Causes include conditions like kidney disease, diabetic ketoacidosis, chronic diarrhea, and Addison's disease. Metabolic Alkalosis (High Bicarbonate): Bicarbonate levels above the normal range can signify metabolic alkalosis, where the blood is too alkaline. This can result from factors like severe vomiting, dehydration, chronic lung problems (e.g., emphysema), or certain hormonal disorders like Cushing's disease.
   - Description: Bicarbonate is a vital electrolyte that plays a crucial role in maintaining the body's acid-base balance. It acts as a buffer, helping to neutralize acids and keep the blood's pH within a narrow, healthy range. It is a byproduct of metabolism, transported by the blood to the lungs to be exhaled as carbon dioxide and regulated by the kidneys.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Typically ranges from 22 to 32 mmol/L in adults.
   - Comments:
   - Test Method Variance:
   - Other:
     - Associated LOINC: 1959-6, 1962-0, 1963-8, 69964-5
     - Associated CPT:

1.1.36 HDL (High density lipoprotein cholesterol)

- Common Name: HDL
- Long Name: Cholesterol High Density Lipoprotein
- Test Type: HDL
- Panels: LIPID PROFILE
- Equivalent Tests: whole blood HDL
- Not Equivalent Tests: Total Cholesterol, VLDL - very low density lipoprotein cholesterol, LDL - low density lipoprotein cholesterol, any apolipoproteins
- Indications: High-density lipoproteins (HDL) is one of the 5 major groups of lipoproteins (chylomicrons, VLDL, IDL, LDL, HDL) which enable lipids like cholesterol and triglycerides to be transported within blood. In healthy individuals, about 30% of blood cholesterol is carried by HDL. It is hypothesized that HDL can remove cholesterol from atheroma within arteries and transport it back to the liver for excretion or re-utilization. HDL-bound cholesterol is sometimes called "good cholesterol", or HDL-C. A high level of HDL-C seems to protect against cardiovascular diseases, and low HDL cholesterol levels (less than 40 mg/dL) increase the risk for heart disease.
- Description: Indirect HDL is derived from values obtained by processing the serum/plasma (precipitation, magnetic separation, ultracentrifugation) to remove other cholesterol forms. The testing method for total cholesterol is used to quantify the remaining cholesterol. Direct HDL cholesterol is measured from unprocessed serum/plasma.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): greater than 40 mg/dl
- Comments: HDL is generally the inverse of LDL and considered the "good" cholesterol. HDL levels greater than 60 mg/dL may actually protect people from heart disease and remove atherosclerotic plaques.
- Test Method Variance:
- Other:

  - Associated LOINC: 14646-4, 2085-9, 35197-3 12772-0 (electrophoresis, prop is Acnc) 49130-8 (electrophoresis, prop is MCnc) 18263-4 (system is ser/plas. ultracentrifugate)
  - Associated CPT: 83718

1.1.37 HGBA1C (Hemoglobin A1C)

- Common Name: Hemoglobin A1c
- Long Name: Hemoglobin A1C (% of hemoglobin A1c to total hemoglobin in blood)
- Test Type: HGBA1C
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A1c levels can indicate if your blood sugar is higher than normal (prediabetes) or high enough to warrant a diagnosis of type 2 diabetes.
- Description: The hemoglobin A1c test is a blood test that provides insights into your average blood glucose levels over the preceding 2 to 3 months.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): an A1c below 5.7% is considered normal
- Comments: 43150-2 (system is ^patient)
- Test Method Variance:
- Other:
  - Associated LOINC: 17855-8, 17856-6, 4548-4, 4549-2, 59261-8, 62388-4 , 71875-9, 96595-4
  - Associated CPT: 83036, 83037

1.1.38 IL_6 (Interleukin 6)

- Common Name: Interleukin 6
- Long Name: Interleukin 6 in Serum or Plasma
- Test Type: IL_6
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Elevated IL-6 levels are often associated with the presence of inflammation or infection within the body and can also be linked to various health conditions.
- Description: Interleukin-6 (IL-6) is a cytokine, or cell-signaling protein, that plays a key role in the immune response and various physiological processes. It can be measured in both serum and plasma samples.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal range for IL-6 levels in healthy adults is generally considered to be less than 5 pg/mL
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 26881-3, 4655-7
  - Associated CPT:

1.1.39 IL_6_QL (Interleukin 6 presence)

- Common Name: Interleukin 6 presence
- Long Name: Interleukin 6 in Serum or Plasma - Presence
- Test Type: IL_6_QL
- Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The presence of IL-6 is often associated with the presence of inflammation or infection within the body and can also be linked to various health conditions.
- Description: Interleukin-6 (IL-6) is a cytokine, or cell-signaling protein, that plays a key role in the immune response and various physiological processes. It can be measured in both serum and plasma samples.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Not applicable for qualitative test
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 44322-6
  - Associated CPT:

1.1.40 K (Potassium, Serum)

- Common Name: Serum Potassium
- Long Name: Potassium Serum
- Test Type: K
- Panels: Basic metabolic profile, comprehensive metabolic profile, general health panel, renal function panel
- Equivalent Tests: Plasma potassium, whole blood potassium, electrolytes
- Not Equivalent Tests: Body fluids other than blood are commonly tested ie. Urine. KOH - fungal screen. Potassium salts of meds.
- Indications: Electrolyte balance, cardiac toxicity. Certain drugs can affect potassium homeostasis and potassium is monitored to assess the efficacy and safety of several drug therapies.

- Description: One of the few tests reported in SI units - mmol/L (also called mEq/dL. For K, mmol/L and mEq/dL have the same reference range.). Excreted by kidney at variable rates depending on hydration status, blood pressure and many other factors.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): ~ 3.5-5.0 mmol/L
- Comments: Low extracellular values and high intracellular values can be a source of specimen processing error that elevate K ie. order of blood collection tubes, capillary collection, leaving tourniquet or clenching fist too long, delay in separating from cells, hemolysis. Lipemia can cause faslely low values. Errors are common and test will often be repeated making it difficult to select among multiple test results..
- Test Method Variance:
- Other: Studies show fist pumping can increase a patient's potassium as much as 20%, in addition to ionized calcium.
  - Associated LOINC: 2823-3, 29349-8 (post dialysis), 32713-0, 39789-3, 39790-1, 41656-0, 51618-7 (pre_dialysis), 6298-4, 77142-8
  - Associated CPT: 80051, 84132

1.1.41 LDH (Lactate dehydrogenase, total in blood/serum)

- Common Name: LDH
- Long Name: Lactate dehydrogenase, Total in blood/serum (also known as: LD; Lactic dehydrogenase; Total LDH)
- Test Type: LDH
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications: As a screening tool to rule out various cellular or tissue damage. If total LDH is elevated, then LDH isoenzymes (i.e., subtypes) or other tests such as ALT, AST or ALP may be ordered to

help diagnose the condition and to help determine which organs are involved. LDH is used to monitor damage caused by muscle trauma or injury and to help identify hemolytic anemia. Occasionally LDH is ordered when a patient has symptoms of a heart attack.

- Description: LDH is widely distributed in body tissues. Elevated levels of LDH (and changes in the ratio of the LDH isoenzymes) are used to indicate acute or chronic tissue damage and to monitor progressive conditions. LDH levels rise as cellular destruction begins, peak, and then begin to fall. LDH exists in 5 subtypes or isoenzymes, which differ slightly in structure:
  - LDH-1 is found primarily in heart muscle and red blood cells.
  - LDH-2 is concentrated in white blood cells.
  - LDH-3 is highest in the lung.
  - LDH-4 is highest in the kidney, placenta, and pancreas.
  - LDH-5 is highest in the liver and skeletal muscle
  - Elevated levels of LDH may be seen with cerebrovascular accident (stroke), drugs (anesthetics, aspirin, narcotics, procainamide, alcohol), hemolytic anemia, pernicious anemia (megaloblastic anemia), infectious mononucleosis, intestinal and pulmonary infarction, kidney and liver disease, muscular dystrophy, pancreatitis and lymphoma or other cancers. Low and normal levels of LDH do not usually indicate a problem. Low levels are sometimes seen when a patient ingests large amounts of ascorbic acid (vitamin C).
- Related Tests: All CK tests; ALT AST; ALP; LDH isoenzymes
- Approximate Reference Range (Each site will have slight variations): 100-210 IU/L
- Comments: Various conditions can affect LDH results. For example, strenuous exercise can cause temporary elevations in LDH; hemolysis of the blood specimen can cause falsely elevated results; if platelet count is increased, serum LDH can be artificially high.
- Test Method Variance:
- Other:
  - Associated LOINC: 14804-9; 14805-6; 2532-0
  - Associated CPT: 83615

1.1.42 (LDL) Cholesterol Low Density Lipoprotein Test Types

1. LDL_CLC_F, Calculated, Fasting (Low density lipoprotein cholesterol, calculated, patient was fasting)

    - Common Name: LDL
    - Long Name: Cholesterol Low Density Lipoprotein Calculated Fasting
    - Test Type: LDL_CLC_F
    - Panels: LIPID PROFILE
    - Equivalent Tests: Whole blood LDL
    - Not Equivalent Tests: Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL
    - high density lipoprotein cholesterol, any apolipoproteins
    - Indications: LDL cholesterol is used to predict the risk of developing atherosclerosis heart disease. It is sometimes referred to as "bad cholesterol." Of all the forms of cholesterol in the blood, the LDL cholesterol is considered the most important in determining risk of heart disease. LDL cholesterol is the primary target of cholesterol-lowering treatment. Since treatment decisions are often based on LDL values, this test may be used to monitor levels after the start of diet or exercise programs or to determine whether or not prescribing one of the lipid-lowering drugs would be useful.
    - Description: LDL = total cholesterol level - HDL - [triglyceride level $\div$ 5]. Calculation is invalid when triglycerides are $>$400mg/dL
    - Related Tests: Lipid Profile, Triglycerides, Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL - high density lipoprotein cholesterol, any apolipoproteins
    - Approximate Reference Range (Each site will have slight variations): less than 130mg/dL
    - Comments: LOINC 13457-7 and 2089-1 are listed as multiple test_types because we see in source data that sometimes the LOINC is associated with results the site identified as fasting and sometimes it is not. Calculated LDL reflects an indirect method of testing. Chemicals are added to the serum/plasma and selected fractions are precipitated out of the sample and the test is performed on the remaining supernatant. The preferred specimen is

fasting. Calculated LDL is not accurate on specimens that have triglycerides levels exceeding 400.

- Test Method Variance:
- Other:
  - Associated LOINC: 13457-7, 2089-1
  - Associated CPT: 83721

2. LDL_CLC_NF, Calculated, Not Fasting (Cholesterol Low Density Lipoprotein, Calculated, patient was not fasting)

   - Common Name: LDL
   - Long Name: Cholesterol Low Density Lipoprotein Calculated Not Fasting
   - Test Type: LDL_CLC_NF
   - Panels: LIPID PROFILE
   - Equivalent Tests: Whole blood LDL
   - Not Equivalent Tests: Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL
   - high density lipoprotein cholesterol, any apolipoproteins
   - Indications: LDL cholesterol is used to predict the risk of developing atherosclerosis heart disease. It is sometimes referred to as "bad cholesterol." Of all the forms of cholesterol in the blood, the LDL cholesterol is considered the most important in determining risk of heart disease. LDL cholesterol is the primary target of cholesterol-lowering treatment. Since treatment decisions are often based on LDL values, this test may be used to monitor levels after the start of diet or exercise programs or to determine whether or not prescribing one of the lipid-lowering drugs would be useful.
   - Description: LDL = total cholesterol level - HDL - [triglyceride level $\div$ 5]. Calculation is invalid when triglycerides are $>$400mg/dL
   - Related Tests: Lipid Profile, Triglycerides, Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL - high density lipoprotein cholesterol, any apolipoproteins
   - Approximate Reference Range (Each site will have slight variations): less than 130mg/dL

- Comments: LOINC 13457-7 and 2089-1 are listed as multiple test_types because we see in source data that sometimes the LOINC is associated with results the site identified as fasting and sometimes it is not. Sites apply whatever is in their data with regard to the fasting status, because the LOINC is silent on fasting or not. Calculated LDL reflects an indirect method of testing. Chemicals are added to the serum/plasma and selected fractions are precipitated out of the sample and the test is performed on the remaining supernatant. The preferred specimen is fasting. Calculated LDL is not accurate on specimens that have triglycerides levels exceeding 400.
- Test Method Variance:
- Other:
  - Associated LOINC: 13457-7
  - Associated CPT: 83721

3. LDL_CLC_NS, Calculated, Random (Low density lipoprotein cholesterol, calculated measure, not known if patient was fasting)

   - Common Name: LDL
   - Long Name: Cholesterol Low Density Lipoprotein Calculated - fasting not specified
   - Test Type: LDL_CLC_NS
   - Panels: LIPID PROFILE
   - Equivalent Tests: Whole blood LDL
   - Not Equivalent Tests: Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL
   - high density lipoprotein cholesterol, any apolipoproteins
   - Indications: LDL cholesterol is used to predict the risk of developing atherosclerosis heart disease. It is sometimes referred to as "bad cholesterol." Of all the forms of cholesterol in the blood, the LDL cholesterol is considered the most important form in determining risk of heart disease. LDL cholesterol is the primary target of cholesterol-lowering treatment. Since treatment decisions are often based on LDL values, this test may be used to monitor levels after the start of diet or exercise programs or to determine whether or not prescribing one of the lipid-lowering drugs would be useful.

- Description: LDL = total cholesterol level - HDL - [triglyceride level ÷ 5]. Calculation is invalid when triglycerides are >400mg/dL
- Related Tests: Lipid Profile, Triglycerides, Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL - high density lipoprotein cholesterol, any apolipoproteins
- Approximate Reference Range (Each site will have slight variations): < 130mg/dL
- Comments: LOINC 13457-7 and 2089-1 are listed as multiple test_types because we see in source data that sometimes the LOINC is associated with results the site identified as fasting and sometimes it is not. Calculated LDL reflects an indirect method of testing. Chemicals are added to the serum/plasma and selected fractions are precipitated out of the sample and the test is performed on the remaining supernatant. The preferred specimen is fasting. Whether "fasting" is indicated with the lab result determines if it is categorized as fasting or fasting status unknown. Calculated LDL is not accurate on specimens that have triglycerides levels exceeding 400.
- Test Method Variance:
- Other:
  - Associated LOINC: 13457-7, 2089-1, 39469-2
  - Associated CPT: 83721

4. LDL_DIRECT (Low density lipoprotein cholesterol, direct measure, fasting status not a consideration)

   - Common Name: LDL, direct
   - Long Name: Cholesterol Low Density Lipoprotein Direct
   - Test Type: LDL_DIRECT
   - Panels: LIPID PROFILE
   - Equivalent Tests: whole blood LDL
   - Not Equivalent Tests: Total Cholesterol, VLDL - very low density liprotein cholesterol, HDL - high density lipoprotein cholesterol, any apolipoproteins (especially B)
   - Indications: LDL cholesterol is used to predict the risk of developing atherosclerosis heart disease. It is sometimes referred to as "bad cholesterol." Of all the forms of cholesterol in the blood, the

LDL cholesterol is considered the most important form in determining risk of heart disease. LDL cholesterol is the primary target of cholesterol-lowering treatment. Since treatment decisions are often based on LDL values, this test may be used to monitor levels after the start of diet or exercise programs or to determine whether or not prescribing one of the lipid-lowering drugs would be useful.

- Description: Direct measure LDL are run directly on neat serum or plasma. Direct measurements of LDL are not interfered with by elevated triglycerides.
- Related Tests: Lipid Profile, Triglycerides, Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL - high density lipoprotein cholesterol, any apolipoproteins
- Approximate Reference Range (Each site will have slight variations): $<$ 130mg/dL
- Comments: The direct method reflects the reagent system for testing. In contrast to calculated (indirectly measured) lipoproteins, direct methods can be reported for specimens that have triglyceride concentrations greater than 400.
- Test Method Variance:
- Other:
  - Associated LOINC: 18262-6, 69419-0
  - Associated CPT: 83721

5. LDL_NS Method not Indicated (Low density lipoprotein cholesterol, not known whether calculated or direct, fasting status not considered)

- Common Name: LDL
- Long Name: Cholesterol Low Density Lipoprotein Not Specified – can not determine if calculated or direct
- Test Type: LDL_NS
- Panels: LIPID PROFILE
- Equivalent Tests: whole blood LDL
- Not Equivalent Tests: Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL high density lipoprotein cholesterol

- Indications: LDL cholesterol is used to predict the risk of developing atherosclerosis heart disease. It is sometimes referred to as "bad cholesterol." Of all the forms of cholesterol in the blood, the LDL cholesterol is considered the most important form in determining risk of heart disease. LDL cholesterol is the primary target of cholesterol-lowering treatment. Since treatment decisions are often based on LDL values, this test may be used to monitor levels after the start of diet or exercise programs or to determine whether or not prescribing one of the lipid-lowering drugs would be useful.
- Description: LDL = total cholesterol level - HDL - [triglyceride level $\div$ 5]. Calculation is invalid when triglycerides are $>$400mg/dL
- Related Tests: Lipid Profile, Triglycerides, Total Cholesterol, VLDL - very low density lipoprotein cholesterol, HDL - high density lipoprotein cholesterol, any apolipoproteins
- Approximate Reference Range (Each site will have slight variations): $<$ 130mg/dL
- Comments: Use this test type only for those LDL test results where you cannot determine with the method was calculated or direct.
- Test Method Variance:
- Other:
  - Associated LOINC: 22748-8, 35198-1
  - Associated CPT: 83721

1.1.43 LIPASE (Lipase)

- Common Name: Lipase
- Long Name: Lipase
- Test Type: LIPASE
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: High lipase levels, particularly those three to 10 times higher than the normal range, strongly suggest acute pancreatitis. This test can help track the progression of existing pancreatic conditions, like acute or chronic pancreatitis, and determine if treatment is effective.
- Description: A lipase blood test, also known as a serum lipase or LPS test, measures the level of the enzyme lipase in your blood. Lipase, primarily produced in the pancreas, plays a vital role in digesting fats within the small intestine. While a small amount of lipase in the bloodstream is normal, elevated levels can indicate problems with the pancreas, such as inflammation (pancreatitis) or other pancreatic disorders.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Lipase blood test is typically 0 to 160 units per liter (U/L)
- Comments: 2572-6 is Lipoprotein lipase and not wanted in VDW
- Test Method Variance:
- Other:
  - Associated LOINC: 3040-3
  - Associated CPT: 83690

1.1.44 MAGNESIUM (Magnesium)

- Common Name: Serum Magnesium, Mg++
- Long Name: Magnesium, total in blood/serum
- Test Type: MAGNESIUM
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: Symptoms (such as weakness, irritability, cardiac arrhythmia, nausea, and/or diarrhea) that may be due to too much or too little magnesium or abnormal calcium or potassium levels. Renal function and certain drugs can affect magnesium levels. Abnormal levels of magnesium are most frequently seen in conditions or diseases that cause affect excretion of magnesium by the kidneys or that cause impaired absorption in the intestines.
- Description:
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): ~ 1.3-2.1 mEq/L
- Comments: Many antacids and laxatives contain magnesium and may affect blood levels. Magnesium blood levels may be low normally in the second and third trimesters of pregnancy.
- Test Method Variance:
- Other:
  - Associated LOINC: 11554-3, 19123-9, 21377-7, 2593-2, 2601-3, 35249-2, 55937-7
  - Associated CPT: 83735

1.1.45 (PG) Pregnancy Determination Test Types

1. PG_B_QL (Pregnancy Determination, Serum Beta Human Choriogonadotropin, qualitative)
   - Common Name: Pregnancy Determination, Serum Beta Human Choriogonadotropin, qualitative
   - Long Name: Pregnancy Determination - Beta Human Choriogonadotropin (HCG) in serum, qualitative
   - Test Type: PG_B_QL
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:

- Indications: A highly accurate way to confirm pregnancy, even in its very early stages. Assess Pregnancy Health: Tracking changes in hCG levels over time can help assess the health and viability of the pregnancy.
- Description: HCG (Human Chorionic Gonadotropin) is a hormone primarily produced by the placenta after fertilization and implantation. It plays a vital role in early pregnancy, supporting the corpus luteum (which produces progesterone to thicken the uterine lining) and maintaining the pregnancy.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Not applicable results are qualitative
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 2110-5
  - Associated CPT:

2. PG_B_QL_U (Pregnancy Determination, Urine Beta Human Choriogonadotropin, qualitative)

   - Common Name: Pregnancy Determination, Urine Beta Human Choriogonadotropin, qualitative
   - Long Name: Pregnancy Determination - Beta Human Choriogonadotropin (HCG) in urine, qualitative
   - Test Type: PG_B_QL_U
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A highly accurate way to confirm pregnancy, even in its very early stages. Assess Pregnancy Health: Tracking changes in hCG levels over time can help assess the health and viability of the pregnancy.
   - Description: Pregnancy tests work by detecting the presence of a hormone called human chorionic gonadotropin (hCG) in the body. This hormone is produced by the placenta shortly after a fertilized egg implants in the uterus. The presence of hCG in a woman's urine or blood is a strong indicator of pregnancy.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 2112-1
  - Associated CPT:

3. PG_B_QN (Pregnancy Determination, Beta Human Choriogonadotropin quantitative)

   - Common Name: Pregnancy Determination, Beta Human Choriogonadotropin quantitative
   - Long Name: Pregnancy Determination - Beta Human Choriogonadotropin (HCG) in serum or plasma, quantitative
   - Test Type: PG_B_QN
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A highly accurate way to confirm pregnancy, even in its very early stages. Assess Pregnancy Health: Tracking changes in hCG levels over time can help assess the health and viability of the pregnancy.
   - Description: HCG (Human Chorionic Gonadotropin) is a hormone primarily produced by the placenta after fertilization and implantation. It plays a vital role in early pregnancy, supporting the corpus luteum (which produces progesterone to thicken the uterine lining) and maintaining the pregnancy.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Generally, an hCG level above 25 mIU/mL is considered positive for pregnancy. An hCG level of less than 5 mIU/mL is typically considered negative.
   - Comments: 2115-4 is B-HCG Free, 45194-8 is sum of total not just beta

- Test Method Variance:
- Other:
  - Associated LOINC: 20415-6, 2111-3, 21198-7, 55869-2
  - Associated CPT:

4. PG_B_QN_U (Pregnancy Determination, Urine Beta Human Choriogonadotropin quantitative)
   - Common Name: Pregnancy Determination, Urine Beta Human Choriogonadotropin quantitative
   - Long Name: Pregnancy Determination - Beta Human Choriogonadotropin (HCG) in urine, quantitative
   - Test Type: PG_B_QN_U
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A highly accurate way to confirm pregnancy, even in its very early stages. Assess Pregnancy Health: Tracking changes in hCG levels over time can help assess the health and viability of the pregnancy.
   - Description: Pregnancy tests work by detecting the presence of a hormone called human chorionic gonadotropin (hCG) in the body. This hormone is produced by the placenta shortly after a fertilized egg implants in the uterus. The presence of hCG in a woman's urine or blood is a strong indicator of pregnancy.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations):
   - Comments:
   - Test Method Variance:
   - Other:
     - Associated LOINC: 2113-9, 2114-7, 56497-1
     - Associated CPT:

5. PG_QL (Pregnancy Determination - Human Choriogonadotropin qualitative)

- Common Name: Pregnancy Determination - Human Choriogonadotropin qualitative
- Long Name: Pregnancy Determination - Human Choriogonadotropin (HCG) in serum, qualitative
- Test Type: PG_QL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The main use of the hCG test is to confirm pregnancy, as hCG is produced after a fertilized egg implants in the uterine wall.
- Description: The human chorionic gonadotropin (hCG) test is a medical test used to detect and measure the levels of the hormone hCG in the body. HCG is primarily produced by the placenta during pregnancy, but it can also be indicative of other medical conditions.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): This is a qualitative test
- Comments: Use when data do not specify "beta" HCG
- Test Method Variance:
- Other:
  - Associated LOINC: 2116-2, 2118-8, 80385-8
  - Associated CPT:

6. PG_QL_U (Pregnancy Determination, Urine Beta Human Choriogonadotropin qualitative)

   - Common Name: Pregnancy Determination, Urine Beta Human Choriogonadotropin
   - Long Name: Pregnancy Determination - Human Choriogonadotropin (HCG) in urine, qualitative
   - Test Type: PG_QL_U
   - Panels:
   - Equivalent Tests:

- Not Equivalent Tests:
- Indications: The main use of the hCG test is to confirm pregnancy, as hCG is produced after a fertilized egg implants in the uterine wall.
- Description: The human chorionic gonadotropin (hCG) test is a medical test used to detect and measure the levels of the hormone hCG in the body. HCG is primarily produced by the placenta during pregnancy, but it can also be indicative of other medical conditions.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): This is a qualitative test
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 2106-3, 80384-1
  - Associated CPT:

7. PG_QN (Pregnancy Determination, Human Choriogonadotropin quantitative)

   - Common Name: Pregnancy Determination, Human Choriogonadotropin quantitative
   - Long Name: Pregnancy Determination - Human Choriogonadotropin (HCG) in serum, quantitative
   - Test Type: PG_QN
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The main use of the hCG test is to confirm pregnancy, as hCG is produced after a fertilized egg implants in the uterine wall.
   - Description: The human chorionic gonadotropin (hCG) test is a medical test used to detect and measure the levels of the hormone hCG in the body. HCG is primarily produced by the placenta during pregnancy, but it can also be indicative of other medical conditions.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Typically, hCG levels above 25 mIU/mL indicate pregnancy.
- Comments: Use when data do not specify "beta" HCG. (Choose to not add 30243-0 because 'intact' subunit 12/2013)
- Test Method Variance:
- Other:
  - Associated LOINC: 19080-1, 2117-0, 2119-6, 30243-0, 34670-0, 53957-7, 53958-5, 53959-3, 83086-9
  - Associated CPT:

8. PG_QN_U (Pregnancy Determination quantitative from Urine)

- Common Name: Pregnancy Determination quantitative from Urine
- Long Name: Pregnancy Determination - Beta Human Choriogonadotropin (hCG) in urine, quantitative
- Test Type: PG_QN_U
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The main use of the hCG test is to confirm pregnancy, as hCG is produced after a fertilized egg implants in the uterine wall.
- Description: This test measures the exact amount of hCG, a hormone produced during pregnancy, in a urine sample. If hCG levels are lower or rising more slowly than expected for the gestational age, it could indicate an ectopic pregnancy or a potential miscarriage. Abnormally high levels may indicate a multiple pregnancy (such as twins), or a molar pregnancy.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For a quantitative HCG urine test for pregnancy, a level above 25 mIU/mL is generally considered positive, while a level below 5 mIU/mL is negative.
- Comments:

- Test Method Variance:
- Other:
  - Associated LOINC: 2107-1, 25372-4, 99104-2
  - Associated CPT:

1.1.46 PHOSPHATE (Phosphate)

- Common Name: Phosphate
- Long Name: Phosphate in serum/plasma or blood
- Test Type: PHOSPHATE
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: This test is ordered to test for the following conditions:
  - Kidney Health: The kidneys filter excess phosphate; high levels often indicate kidney disease or failure.
  - Bone and Endocrine Disorders: It helps evaluate bone diseases (like rickets) and problems with the parathyroid glands, which regulate phosphate and calcium.
  - Nutritional Status: It is used to monitor people with severe malnutrition or malabsorption syndromes.
  - Diabetes Management: It can help assess diabetic ketoacidosis or uncontrolled diabetes.
- Description: A phosphate blood test (also called a phosphorus test) measures the level of inorganic phosphate in the blood. Phosphate is an essential electrolyte that helps build strong bones and teeth, enables nerve signaling, supports muscle contraction, and is vital for energy production.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal ranges vary by laboratory but generally fall between 2.5 and 4.5 mg/dL for adults. Children naturally have higher levels due to active bone growth.

  - High Levels (Hyperphosphatemia): May be caused by kidney failure, hypoparathyroidism, or excessive Vitamin D intake.
  - Low Levels (Hypophosphatemia): May indicate hyperparathyroidism, Vitamin D deficiency, alcoholism, or severe malnutrition.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 14879-1, 20941-1, 24519-1, 2774-8, 2777-1, 35221-1, 48641-5
  - Associated CPT:

1.1.47 PROCALC (Procalcitonin)

- Common Name: Procalcitonin
- Long Name: Procalcitonin in Serum or Plasma
- Test Type: PROCALC
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Procalcitonin testing is typically used in the following scenarios:
  - Diagnosis of sepsis and severe bacterial infections: PCT can help healthcare providers determine if a patient with symptoms suggestive of infection has a bacterial infection and assess their risk of developing sepsis.
  - Monitoring treatment response: Procalcitonin levels tend to decrease as the bacterial infection resolves. Monitoring Procalcitonin levels can help assess the effectiveness of antibiotic therapy.
  - Guiding antibiotic therapy: Procalcitonin levels can help clinicians make informed decisions about initiating, escalating, or discontinuing antibiotic treatment, especially in settings like the emergency department and intensive care unit.

  - Differentiating between bacterial and non-bacterial infections: Procalcitonin can help distinguish between bacterial infections and other inflammatory conditions, says Testing.com.
  - Assessing prognosis: Studies have shown a correlation between high Procalcitonin levels and increased mortality risk in patients with sepsis.

- Description: The procalcitonin test is a blood test that measures the levels of procalcitonin in the bloodstream. Normal levels in healthy individuals are very low, typically below 0.1 µg/L. However, in response to a serious bacterial infection, cells throughout the body release procalcitonin into the bloodstream, leading to significantly increased levels. This makes PCT a useful biomarker for diagnosing and assessing the risk of severe bacterial infections and sepsis.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): typically below 0.1 µg/L
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 33959-8, 75241-0
  - Associated CPT:

1.1.48 (PSA) Prostate specific antigen Test Types

1. PSA (Prostate specific antigen, total)
   - Common Name: PSA, Total (screening or diagnostic)
   - Long Name: Prostate Specific Antigen, Total PSA
   - Test Type: PSA
   - Panels: N/A
   - Equivalent Tests: N/A
   - Not Equivalent Tests: PAP (Prostatic Acid Phosphatase), Not equivalent to PSA from nonblood specimens. Free and protein bound forms are considered distinct tests not to be included with total.

- Indications:
- Description: Prostate-specific antigen (PSA) is a normal protein produced by the prostate gland and value does not necessarily indicate cancer. The PSA test measures the total level of PSA in the blood. PSA in other forms such as Free, "nicked"or pro-PSA are normally included in total PSA. PSA is a member of the kallikrein-related serine protease super family.
- Related Tests: Free PSA ratio
- Approximate Reference Range (Each site will have slight variations): 0 - 4.0 ng/mL is usual normal range.
- Comments: The U.S. Food and Drug Administration (FDA) has approved the use of the PSA test to help detect prostate cancer in men age 50 and older (40 if family history). The FDA has also approved the PSA test to monitor patients with a history of prostate cancer to see if the cancer has recurred. Most men with an elevated PSA test result turn out not to have cancer; only 25 to 35 percent of men who have a biopsy due to an elevated PSA level actually have prostate cancer. The test may indicate when biopsy is indicated.
- Test Method Variance: Different manufacturers have different molecular specificity. Comparing values over time need to consider different reagent systems if lab changes are made. Results between manufacturer/instrumentation may not be linear.
- Other:
  - Associated LOINC: 100716-0, 19195-7, 19197-3, 2857-1, 34611-4, 35741-8, 83112-3
  - Associated CPT: 84153, 84152

2. PSA_FREE(Prostate specific antigen free in blood/serum/plasma)

   - Common Name: Free PSA
   - Long Name: Prostate Specific Antigen Free
   - Test Type: PSA_FREE
   - Panels: N/A
   - Equivalent Tests: N/A
   - Not Equivalent Tests: PSA forms that are bound to blood proteins which includes total.

- Indications:
- Description: PSA circulates in the blood in two forms: Free or bound to a protein molecule.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Men with prostate cancer have a lower percentage of total PSA that is free than men without prostate cancer.
- Comments: The free PSA test may add value for elevated PSA values. With benign prostate conditions (such as BPH), there is usually more free PSA, while cancer usually produces more of the protein bound form.
- Test Method Variance:
- Other:
  - Associated LOINC: 10886-0, 19201-3, 19203-9, 83113-1
  - Associated CPT: 84154

3. PSA_P_FREE (Percent Free Prostate specific antigen [Ag], ratio of free to total)

   - Common Name: Percent Free PSA
   - Long Name: Prostate specific Ag free/Prostate specific Ag Total
   - Test Type: PSA_%_FREE
   - Panels: N/A
   - Equivalent Tests: N/A
   - Not Equivalent Tests: Individual PSA isoforms. Result is a ratio of two isoforms
   - Indications:
   - Description: PSA circulates in the blood in two forms: Free or bound to a protein molecule.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): 5% – 40 %. Expressed as a percent of unbound PSA over total PSA
   - Comments:
   - Test Method Variance:

- Other:
  - Associated LOINC: 12841-3, 14120-0, 72576-2
  - Associated CPT: 84154, 84153, 84152

1.1.49 SODIUM (Blood or Serum Sodium)

- Common Name: Serum Sodium; Na+
- Long Name: Sodium, serum/blood
- Test Type: Sodium
- Panels: Basic metabolic profile, comprehensive metabolic profile, electrolyte panel, general health panel, renal function panel
- Equivalent Tests:
- Not Equivalent Tests: body fluids other than blood (i.e. urine)
- Indications: Electrolyte balance; may be monitored to evaluate kidney function and in chronic conditions, like high or low blood pressure and congestive heart failure, or with dehydration or edema. Certain drugs can affect sodium levels.
- Description:
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): ~ 136-145 mEq/L or mmol/L
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 2947-0, 2951-2, 32717-1 , 39791-9, 39792-7, 41657-8, 42570-2, 77139-4
  - Associated CPT: 84295

1.1.50 (T4) Thyroxine Test Types

1. T4_FREE (Thyroxine Free)

    - Common Name: Free T4
    - Long Name: Thyroxine (T4) Free
    - Test Type: T4_FREE
    - Panels: Thyroid panel
    - Equivalent Tests: N/A
    - Not Equivalent Tests: Total Thyroxine (T4), Free Thyroxine Index, Thyroid stimulating hormone (TSH)
    - Indications: Free T4 is used in diagnosing hyper- or hypothyroidism in conjunction with TSH. It may be ordered at regular intervals to monitor the effectiveness of treatment when a patient is being treated for a known thyroid disorder.
    - Description: Thyroxine (T4) Free evaluates thyroid function by measuring the T4 produced by the thyroid gland. T4 Free measures the level of thyroxine circulating in the bloodstream that is not bound to protein (i.e., it is "free") as opposed to the Total T4 test which measures the total amount of thyroxine that is bound to thyroxine-binding globulin. Since free T4 is the active form of thyroxine, the free T4 test is considered a more accurate reflection of thyroid hormone function and, in most cases, it has replaced the total T4 test. Total T4 is being phased out due to the more sensitive and meaningful free T4 testing.
    - Related Tests: TSH; Free T4 Index
    - Approximate Reference Range: 0.8-1.5 ng/dL for adults (neonatal and pediatric reference ranges differ)
    - Comments: Only 0.02% of T4 circulates unbound to proteins. In general, high T4 free results may indicate an overactive thyroid gland (hyperthyroidism), and low T4 free results may indicate an underactive thyroid gland (hypothyroidism). Pregnancy can cause changes in the production and elimination of thyroid hormones resulting in altered reference ranges throughout the pregnancy.
    - Test Method Variance: There are 4 different assays to evaluate free T4. Results commonly reported as ng/dL but can also include pmol/L or pg/dL.

- Other:
  - Associated LOINC: 100978-6, 14920-3, 27987-7, 3024-7, 35228-6, 53349-7, 53350-5, 58838-4, 6892-4, 70217-5, 83121-4, 83122-2
  - Associated CPT: 84439

2. T4_FREE_I (Thyroxine Free Index)

   - Common Name: FTI
   - Long Name: Thyroxine (T4) Free Index; Serum Free Thyroxine Index
   - Test Type: T4_FREE_I
   - Panels: Thyroid Panel
   - Equivalent Tests: Also known as "T7"
   - Not Equivalent Tests: Thyroxine (T4) Total, Thyroxine (T4) Free
   - Indications: This test is used to monitor thyroid function in patients who have abnormal protein binding. It may be ordered at regular intervals to monitor the effectiveness of treatment when a patient is being treated for a known thyroid disorder.
   - Description: Thyroxine index is calculated by total T4 multiplied by T3 resin uptake (percentage) to estimate the amount of free thyroxine circulating in the blood. This calculation adjusts for the effects of alterations in thyroid-binding protein on the total serum T4 assay. This can be useful when evaluating patients who have factors that can increase or decrease thyroidbinding protein.
   - Related Tests: T4 Total; T3 Resin Uptake; T4 Free
   - Approximate Reference Range: 1.0-4.0 units; 0.8-2.7 ng/dL for adults (neonatal and pediatric reference ranges differ)
   - Comments: In general, high T4 index results may indicate an overactive thyroid gland (hyperthyroidism), and low T4 index results may indicate an underactive thyroid gland (hypothyroidism).
   - Test Method Variance:
   - Other: Thyroxine free index is rarely used today.
     - Associated LOINC: 3022-1, 32215-6
     - Associated CPT:

3. T4_TOTAL (Thyroxine Total)

- Common Name: Total T4
- Long Name: Thyroxine (T4) Total; Total Serum Thyroxine
- Test Type: T4_TOTAL
- Panels: Thyroid Panel
- Equivalent Tests: N/A
- Not Equivalent Tests: Thyroxine (T4) Free Thyroxine Index, Total T3, Thyroid stimulating hormone (TSH), newborn genetic screenings
- Indications: Total T4 is used to diagnose hyper- or hypothyroidism as well as for suspected subacute thyroiditis. It may be ordered at regular intervals to monitor the effectiveness of treatment when a patient is being treated for a known thyroid disorder.
- Description: Thyroxine (T4) Total evaluates thyroid function by measuring the total thyroxine in the bloodstream (both bound and unbound to thyroxine-binding globulin and other proteins).
- Related Tests: TSH; Serum T3 Resin Uptake; thyroxine-binding globulin (TBG); T4 Free; Free T4 Index
- Approximate Reference Range: 4.0-12.5 mcg/dL for adults (neonatal and pediatric reference ranges differ)
- Comments: In general, high T4 Total results indicate an overactive thyroid gland (hyperthyroidism), and low T4 Total results indicate an underactive thyroid gland (hypothyroidism). T4 Total results are affected by the amount of protein in the blood that is available to bind the T4. T3 resin uptake test is usually measured in conjunction with T4 total to factor out the interference of T3 binding to thyroid-binding protein. Total T4 is being phased out due to the more sensitive and meaningful free T4 testing – free T4 is not affected by the amount of protein in the blood.
- Test Method Variance:
- Other: Usually ordered along with or following a TSH test. Do not include T4 total from newborn genetic screening.
  - Associated LOINC: 14921-1, 20451-1, 3025-4, 3026-2, 31144-9, 31145-6, 35226-0, 38506-2, 83119-8, 83120-6
  - Associated CPT: 84436

1.1.51 TOT_CHOLES (Cholesterol, Total)

- Common Name: Cholesterol, Total
- Long Name: Cholesterol Total
- Test Type: TOT_CHOLES
- Panels: LIPID PROFILE
- Equivalent Tests:
- Not Equivalent Tests: LDL - very low density lipoprotein Cholesterol, LDL - Low Density lipoprotein Cholesterol, HDL - High Density lipoprotein Cholesterol, any apolipoproteins
- Indications:
- Description: Cholesterol is a molecule (chemically a steroid) that is essential for life. It forms the membranes for cells in all organs and tissues in your body. It is used to make hormones essential for development, growth, and reproduction. It forms bile acids that are needed to absorb nutrients from food. Levels of cholesterol are influenced by its production in the liver and the ingestion of dietary fats. Bile acids are released into the intestine, aid in digestion, and then are mostly reabsorbed. A small amount of your body's cholesterol circulates in the blood in complex particles called lipoproteins. These lipoproteins include some particles that carry excess cholesterol away for disposal (i.e., HDL cholesterol) and some particles that deposit cholesterol in tissues and organs (i.e., LDL cholesterol). The test for cholesterol measures total cholesterol (good [HDL] and bad [e.g., LDL]) that is carried in the blood by lipoproteins. Lowering LDL cholesterol reduces coronary heart disease and ischemic stroke.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
- Comments:
- Test Method Variance:
- Other:

  - Associated LOINC: 2093-3, 14647-2, 35200-5, 48620-9 (ultracentrig)
  - Associated CPT: 82465

1.1.52 TOT_PROT (Protein, total in blood/serum)

- Common Name: Total Protein
- Long Name: Total Protein in blood/serum
- Test Type: TOT_PROT
- Panels: Liver Panel or Liver Function Test (LFT); Comprehensive Metabolic Panel (CMP)
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications: To determine nutritional status as part of a general health checkup; to screen for certain liver and kidney disorders and other diseases; to investigate the cause of abnormal pooling of fluid in tissue (edema).
- Description: Proteins are important building blocks of cells and tissues; they are important for body growth, development, and health. They form the structural part of most organs and make up enzymes and hormones that regulate body functions. The total protein test measures the total amount of the two classes of proteins (albumin and globulin) in blood plasma. Low total protein levels can suggest a liver disorder, a kidney disorder, or a disorder in which protein is not digested or absorbed properly. Low levels may be seen in severe malnutrition and with conditions that cause malabsorption, such as Celiac disease or inflammatory bowel disease. High total protein levels may be seen with chronic inflammation, infections such as viral hepatitis or HIV, or bone marrow disorders such as multiple myeloma.
- Related Tests: Albumin; Liver panel; Protein electrophoresis
- Approximate Reference Range (Each site will have slight variations): 6.0 to 8.3 gm/dL (g/dL; grams per deciliter).

- Comments: Prolonged application of a tourniquet during blood collection can result in a blood sample that has a higher protein concentration than the rest of the circulation. This will mean that the test result for total protein will be falsely elevated. Drugs that may decrease protein levels include estrogens and oral contraceptives.
- Test Method Variance:
- Other:
  - Associated LOINC: 2885-2
  - Associated CPT: 84155

1.1.53 TRANSFERRIN (Transferrin)

- Common Name: Transferrin
- Long Name: Transferrin in Serum or Plasma
- Test Type: TRANSFERRIN
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Purpose of the Test:
  - Evaluate Iron Status: Helps determine if you have too much iron (overload) or too little (deficiency).
  - Diagnose Anemia: Used to distinguish between different types of anemia, particularly iron-deficiency anemia and anemia of chronic disease.
  - Assess Liver Function: Since transferrin is made in the liver, low levels can sometimes indicate liver disease.
  - Monitor Treatment: Tracks the body's response to iron supplements or treatments for iron overload.
- Description: The transferrin blood test measures the level of transferrin, a protein produced by the liver that binds to and transports iron throughout the body. It is typically ordered as part of an "iron panel" to help diagnose conditions like anemia or iron overload.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal ranges can vary slightly by laboratory, but common benchmarks include:
  - Transferrin: 215–380 mg/dL.
  - TIBC: 250–450 mcg/dL.
  - Transferrin Saturation: 15%–50%
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 22674-6, 3034-6, 35229-4
  - Associated CPT:

1.1.54 TRIGL_F (Triglycerides, Fasting)

- Common Name: Triglycerides, fasting
- Long Name: Triglycerides Fasting
- Test Type: TRIGL_F
- Panels: LIPID PROFILE
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications:
- Description: Triglycerides are the body's storage form for fat. Most triglycerides are found in adipose (fat) tissue. Some triglycerides circulate in the blood to provide fuel for muscles to work. Extra triglycerides are found in the blood after eating a meal - when fat is being sent from the gut to adipose tissue for storage.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): $<$150 mg/dL (1.70 mmol/L)

- Comments: Elevated triglycerides can be indicated by a "lipemic" comment in any other blood/serum/plasma laboratory test. Triglycerides are circulating forms of fat and are related both to heredity and to diet. You might think that a high fat diet will raise triglycerides and a low fat diet would lower triglycerides. However, carbohydrate appears to be the most important dietary predictor of triglycerides. Diets high in carbohydrates can lead to transient increases in triglycerides. This is why fasting specimens are usually preferred.
- Test Method Variance:
- Other:
  - Associated LOINC: 1644-4, 17081-1, 3048-6, 30524-3, 47210-0
  - Associated CPT: (CPT code doesnot differentiate between fasting or not) 84478

1.1.55 TRIGL_NS (Triglycerides, not known if patient was fasting

- Common Name: Triglycerides
- Long Name: Triglycerides - fasting not specified
- Test Type: TRIGL_NS
- Panels: LIPID PROFILE
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications:
- Description: Triglycerides are the body's storage form for fat. Most triglycerides are found in adipose (fat) tissue. Some triglycerides circulate in the blood to provide fuel for muscles to work. Extra triglycerides are found in the blood after eating a meal - when fat is being sent from the gut to adipose tissue for storage.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): $<$150 mg/dL (1.70 mmol/L)

- Comments: Elevated triglycerides can be indicated by a "lipemic" comment in any other blood/serum/plasma laboratory test. Triglycerides are circulating forms of fat and are related both to heredity and to diet. You might think that a high fat diet will raise triglycerides and a low fat diet would lower triglycerides. However, carbohydrate appears to be the most important dietary predictor of triglycerides. Diets high in carbohydrates can lead to transient increases in triglycerides. This is why fasting specimens are usually preferred. Occasionally, a non-fasting triglyceride sample can be useful. Non-fasting triglycerides are sometimes used to identify risk that was not predicted by a standard fasting lipid profile. The non-fasting test is most valuable when the sample is taken two to four hours after a meal.
- Test Method Variance:
- Other:
  - Associated LOINC: 14927-8, 2571-8, 28554-4, 3043-7, 3049-4, 35217-9, 70218-3, 96598-8
  - Associated CPT: (CPT code doesnot differentiate between fasting or not) 84478

1.1.56 (TROP) Troponin Test Types

1. TROP_I_QN (Troponin I Cardiac Quantitative)
   - Common Name: Cardiac-specific troponin (I)
   - Long Name: Cardiac-specific Troponin I Quantitative
   - Test Type: TROP_I_QN
   - Panels: N/A
   - Equivalent Tests: TnI; cTnI (see Description below)
   - Not Equivalent Tests: All CK tests; Myoglobin TROP_T_QN; TROP_T_QL; cTnT; TnT
   - Indications: The troponin test is used to help diagnose a heart attack, to detect and evaluate mild to severe heart injury, and to distinguish chest pain due to other causes. Either troponin I or troponin T test can be performed; usually a laboratory will offer one test or the other. Troponins are sometimes ordered along with other cardiac biomarkers such as CK_MB or myoglobin. Troponins are the preferred tests for a suspected heart attack

because they are more specific for heart injury than other tests and remain elevated for a longer period of time.

- Description: Troponins are a family of proteins found in skeletal and heart muscle fibers. The three different types of troponin are called troponin C (TnC), troponin T (TnT), and troponin I (TnI). Together, these three proteins regulate muscular contraction. Two of the proteins, TnI and TnT, occur in a form found only in the heart. These cardiac-specific troponins, called cTnI and cTnT, are normally present in very small quantities in the blood. When there is damage to heart muscle cells, cardiac troponins I and T are released into the circulation. The more damage there is, the greater the concentration of cardiac troponins I and T in the blood. The troponin test is usually ordered when a patient with a suspected heart attack comes into the emergency room and often repeated at 6 and 12 hours later. Typically, 2 or 3 troponin tests are done over a 12- to 16-hour period. In patients with stable angina, a troponin test may be ordered if the patient's symptoms get worse (e.g., increased risk of a heart attack in the near future).
- Related Tests: CK; CK_MB; Cardiac biomarkers; TROP_T_QL; TROP_T_QN; cTnT; TnT
- Approximate Reference Range: Cardiac troponin levels are normally so low they cannot be detected with most blood tests. (Troponin I: $<$ 10 microgram (mcg)/L).
- Comments: When a patient has significantly elevated troponin concentrations, some form of damage to the heart is likely. When a patient with chest pain/known stable angina has normal troponin values, it is likely that the heart has not been injured. The test is not affected by damage to other muscles.
- Test Method Variance:
- Other:
  - Associated LOINC: 10839-9, 16255-2, 42757-5, 49563-0, 76399-5, 89577-1, 89578-9, 89579-7
  - Associated CPT: 84484

2. TROP_T_QL (Troponin T Cardiac - Qualitative)

- Common Name: Cardiac-specific troponin (T qualitative)
- Long Name: Cardiac-specific Troponin T Qualitative

- Test Type: TROP_T_QL
- Panels: N/A
- Equivalent Tests:
- Not Equivalent Tests: CK; CK – MB; Myoglobin; Cardiac biomarkers; TROP_T_QN; TROP_I_QN; cTnI; TnI
- Indications: The troponin test is used to help diagnose a heart attack, to detect and evaluate mild to severe heart injury, and to distinguish chest pain due to other causes. Either troponin I or troponin T test can be performed; usually a laboratory will offer one test or the other. Troponins are sometimes ordered along with other cardiac biomarkers such as CK_MB or myoglobin. Troponins are the preferred tests for a suspected heart attack because they are more specific for heart injury than other tests and remain elevated for a longer period of time.
- Description: Troponins are a family of proteins found in skeletal and heart muscle fibers. The three different types of troponin are called troponin C (TnC), troponin T (TnT), and troponin I (TnI). Together, these three proteins regulate muscular contraction. Two of the proteins, TnI and TnT, occur in a form found only in the heart. These cardiac-specific troponins, called cTnI and cTnT, are normally present in very small quantities in the blood. When there is damage to heart muscle cells, cardiac troponins I and T are released into the circulation. The more damage there is, the greater the concentration of cardiac troponins I and T in the blood. The troponin test is usually ordered when a patient with a suspected heart attack comes into the emergency room and often repeated at 6 and 12 hours later. Typically, 2 or 3 troponin tests are done over a 12- to 16-hour period. In patients with stable angina, a troponin test may be ordered if the patient's symptoms get worse (e.g., increased risk of a heart attack in the near future).
- Related Tests: CK; CK_MB; Myoglobin; Cardiac biomarkers TROP_T_QN; TROP_I_QN
- Approximate Reference Range: Positive or negative for Troponin T elevated levels.
- Comments: When a patient has significantly elevated troponin concentrations, some form of damage to the heart is likely. When a patient with chest pain/known stable angina has normal tro-

ponin values, it is likely that the heart has not been injured. The test is not affected by damage to other muscles.

- Test Method Variance:
- Other:
  - Associated LOINC: 33204-9, 48426-1
  - Associated CPT: 84512

3. TROP_T_QN (Troponin T Cardiac - Quantitative)

   - Common Name: Cardiac-specific troponin (T)
   - Long Name: Cardiac-specific Troponin T Quantitative
   - Test Type: TROP_T_QN
   - Panels: N/A
   - Equivalent Tests: TnT; cTnT (see Description below)
   - Not Equivalent Tests: CK; CK – MB; Myoglobin; Cardiac biomarkers; TROP_T_QL; TROP_I_QN; cTnI; TnI
   - Indications: The troponin test is used to help diagnose a heart attack, to detect and evaluate mild to severe heart injury, and to distinguish chest pain due to other causes. Either troponin I or troponin T test can be performed; usually a laboratory will offer one test or the other. Troponins are sometimes ordered along with other cardiac biomarkers such as CK_MB or myoglobin. Troponins are the preferred tests for a suspected heart attack because they are more specific for heart injury than other tests and remain elevated for a longer period of time.
   - Description: Troponins are a family of proteins found in skeletal and heart muscle fibers. The three different types of troponin are called troponin C (TnC), troponin T (TnT), and troponin I (TnI). Two of the proteins, TnI and TnT, occur in a form found only in the heart. These cardiacspecific troponins (cTnI and cTnT) are normally present in very small quantities in the blood. When there is damage to heart cells, cardiac troponins I and T are released into the circulation. The troponin test is usually ordered when a patient with a suspected heart attack comes into the emergency room and often repeated at 6 and 12 hours later. Typically, 2 or 3 troponin tests are done over a 12- to 16-hour period. In patients with stable angina, a troponin test may be ordered if the patient's symptoms get worse (e.g., increased risk of a heart attack in the near future).

- Related Tests: CK; CK_MB; Myoglobin; Cardiac biomarkers TROP_T_QL; TROP_I_QN
- Approximate Reference Range: Cardiac troponin levels are normally so low they cannot be detected with most blood tests. (Troponin T: 0–0.1 g/L [micrograms per liter[).
- Comments: When a patient has significantly elevated troponins, heart damage is likely. When a patient with chest pain/known stable angina has normal troponin values, it is likely that the heart has not been injured. The test is not affected by damage to other muscles.
- Test Method Variance:
- Other:
  - Associated LOINC: 101911-6, 101912-4, 101915-7, 101916-5, 48425-3, 6597-9, 6598-7, 67151-1
  - Associated CPT: 84484

1.1.57 (TSH) Thyroid Stimulating Hormone Test Types

1. TSH (Thyroid Stimulating Hormone, Thyrotropin)

   - Common Name: TSH
   - Long Name: Thyroid Stimulating Hormone; Thyrotropin
   - Test Type: TSH
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The doctor may order a TSH test when a patient has symptoms of hyper- or hypothyroidism and/or when a patient has an enlarged thyroid gland. It may be ordered at regular intervals to monitor the effectiveness of treatment when a patient is being treated for a known thyroid disorder.
   - Description: The TSH test is often the test of choice for evaluating thyroid function and/or symptoms of hyper- or hypothyroidism. It is frequently ordered along with or preceding a T4 test. Other thyroid tests that may be ordered include a T3 test and thyroid antibodies (if autoimmunerelated thyroid disease is suspected). TSH testing is used to:

  - diagnose a thyroid disorder in a person with symptoms,
  - screen newborns for an underactive thyroid,
  - monitor thyroid replacement therapy in people with hypothyroidism
  - diagnose and monitor female infertility problems,
  - help evaluate the function of the pituitary gland (occasionally), and
  - screen adults for thyroid disorders, although expert opinions vary on who can benefit from screening and at what age to begin.
- Related Tests: T4, T3, Thyroid Antibodies
- Approximate Reference Range (Each site will have slight variations): 0.32-5.50 uIU/mL.
- Comments: No test preparation is needed. Certain medications — including aspirin and thyroid-hormone replacement therapy — can interfere with the TSH test, so the doctor should be informed about any drugs that are being taken.
- Test Method Variance:
- Other: This test measures the amount of thyroid-stimulating hormone (TSH) in your blood. TSH is produced by the pituitary gland, a tiny organ located below the brain and behind the sinus cavities. It is part of the body's feedback system to maintain stable amounts of the thyroid hormones thyroxine (T4) and triiodothyronine (T3) in the blood. Thyroid hormones help control the rate at which the body uses energy. When concentrations decrease in the blood, the hypothalamus (an organ in the brain) releases thyrotropin releasing hormone (TRH). This stimulates the release of TSH by the pituitary gland. The TSH in turn stimulates the production and release of T4 and T3 by the thyroid gland, a small butterfly-shaped gland that lies in the neck flat against the windpipe. When all three organs are functioning normally, thyroid production turns on and off to maintain constant blood thyroid hormone levels. If there is pituitary dysfunction, then increased or decreased amounts of TSH may result. When TSH concentrations are decreased, the thyroid will make and release inappropriate amounts of T4 and T3 and the patient may experience symptoms associated with hyperthyroidism, such as rapid heart rate, weight loss, nervousness, hand tremors, irritated eyes,

and difficulty sleeping. If there is elevated TSH and subsequent decreased production of thyroid hormones (hypothyroidism), the patient may experience symptoms such as weight gain, dry skin, constipation, cold intolerance, and fatigue. In addition to pituitary dysfunction, hyper- or hypothyroidism can occur if there is a problem with the hypothalamus (insufficient or excessive TRH). Thyroid hormone levels may also be altered by a variety of thyroid diseases regardless of the amount of TSH present in the blood.

  - Associated LOINC: 11579-0, 11580-8, 14297-6, 14999-7, 1642-8, 27975-2, 29575-8, 3015-5, 3016-3
  - Associated CPT: 84443

2. (TSH_BSU-DEPRECATED) Beta Subunit (Thyroid Stimulating Hormone Beta Subunit), Quantitative

   - Common Name: TSH Beta Subunit, Quantitative
   - Long Name: Thyroid Stimulating Hormone Beta Subunit Quantitative
   - Test Type: TSH_BSU
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The doctor may order a TSH test when a patient has symptoms of hyper- or hypothyroidism and/or when a patient has an enlarged thyroid gland. It may be ordered at regular intervals to monitor the effectiveness of treatment when a patient is being treated for a known thyroid disorder.
   - Description: The TSH test is often the test of choice for evaluating thyroid function and/or symptoms of hyper- or hypothyroidism. It is frequently ordered along with or preceding a T4 test. Other thyroid tests that may be ordered include a T3 test and thyroid antibodies (if autoimmunerelated thyroid disease is suspected). TSH testing is used to:
     - diagnose a thyroid disorder in a person with symptoms,
     - screen newborns for an underactive thyroid,
     - monitor thyroid replacement therapy in people with hypothyroidism
     - diagnose and monitor female infertility problems,

  - help evaluate the function of the pituitary gland (occasionally), and
  - screen adults for thyroid disorders, although expert opinions vary on who can benefit from screening and at what age to begin.
- Related Tests: TSH, T4, T3, Thyroid Antibodies
- Approximate Reference Range (Each site will have slight variations):
- Comments: No test preparation is needed. Certain medications — including aspirin and thyroid-hormone replacement therapy — can interfere with the TSH test, so the doctor should be informed about any drugs that are being taken.
- Test Method Variance:
- Other: This test measures the amount of thyroid-stimulating hormone (TSH) in your blood. TSH is produced by the pituitary gland, a tiny organ located below the brain and behind the sinus cavities. It is part of the body's feedback system to maintain stable amounts of the thyroid hormones thyroxine (T4) and triiodothyronine (T3) in the blood. Thyroid hormones help control the rate at which the body uses energy. When concentrations decrease in the blood, the hypothalamus (an organ in the brain) releases thyrotropin releasing hormone (TRH). This stimulates the release of TSH by the pituitary gland. The TSH in turn stimulates the production and release of T4 and T3 by the thyroid gland, a small butterfly-shaped gland that lies in the neck flat against the windpipe. When all three organs are functioning normally, thyroid production turns on and off to maintain constant blood thyroid hormone levels. If there is pituitary dysfunction, then increased or decreased amounts of TSH may result. When TSH concentrations are increased, the thyroid will make and release inappropriate amounts of T4 and T3 and the patient may experience symptoms associated with hyperthyroidism, such as rapid heart rate, weight loss, nervousness, hand tremors, irritated eyes, and difficulty sleeping. However, because of the feedback loop, usually TSH production has been "turned off" (i.e., TSH concentrations are low) when there is hyperthyroidism. If there is increased production of TSH and subsequent decreased production of thyroid hormones (hypothyroidism), the patient may

experience symptoms such as weight gain, dry skin, constipation, cold intolerance, and fatigue. In addition to pituitary dysfunction, hyper- or hypothyroidism can occur if there is a problem with the hypothalamus (insufficient or excessive TRH). Usually, by the time TSH is measured in a patient with hypothyroidism, the TSH is very high (i.e., the pituitary is trying to get the thyroid to make more thyroid hormone). Thyroid hormone levels may also be altered by a variety of thyroid diseases regardless of the amount of TSH present in the blood.

  - Associated LOINC:
  - Associated CPT:

3. (TSH_LA-DEPRECATED) Long Acting (Thyroid Stimulator Long Acting)

   - Common Name: Long Acting Thyroid Stimulator
   - Long Name: Thyroid Stimulator Long Acting
   - Test Type: TSH_LA
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The doctor may order a TSH test when a patient has symptoms of hyper- or hypothyroidism and/or when a patient has an enlarged thyroid gland. It may be ordered at regular intervals to monitor the effectiveness of treatment when a patient is being treated for a known thyroid disorder.
   - Description:
   - Related Tests: TSH, T4, T3, Thyroid Antibodies
   - Approximate Reference Range (Each site will have slight variations):
   - Comments: No test preparation is needed. Certain medications — including aspirin and thyroid-hormone replacement therapy — can interfere with the TSH test, so the doctor should be informed about any drugs that are being taken.
   - Test Method Variance:
   - Other:

4. (TSH_SCRN-DEPRECATED) Screen (Thyroid Stimulating Hormone, Screen), Blood

    - Common Name: Thyroid stimulating hormone screen in blood
    - Long Name: Thyroid Stimulating Hormone, Screen, Blood
    - Test Type: TSH_SCRN
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The doctor may order a TSH test when a patient has symptoms of hyper- or hypothyroidism and/or when a patient has an enlarged thyroid gland. It may be ordered at regular intervals to monitor the effectiveness of treatment when a patient is being treated for a known thyroid disorder. TSH screening is routinely performed in the United States on newborns as part of each state's newborn screening program. The American Thyroid Association recommends that adults older than age 35 be screened for thyroid disease with a TSH test every five years, although other organizations, such as the U.S. Preventive Services Task Force, challenge this recommendation. Several organizations recommend instead screening women over 50 or those at high risk for thyroid disorders, such as pregnant and postpartum women.
    - Description: The TSH test is often the test of choice for evaluating thyroid function and/or symptoms of hyper- or hypothyroidism. It is frequently ordered along with or preceding a T4 test. Other thyroid tests that may be ordered include a T3 test and thyroid antibodies (if autoimmunerelated thyroid disease is suspected).

    TSH testing is used to:

    - diagnose a thyroid disorder in a person with symptoms,
    - screen newborns for an underactive thyroid,
    - monitor thyroid replacement therapy in people with hypothyroidism
    - diagnose and monitor female infertility problems,
    - help evaluate the function of the pituitary gland (occasionally), and

- screen adults for thyroid disorders, although expert opinions vary on who can benefit from screening and at what age to begin.
- Related Tests: TSH, T4, T3, Thyroid Antibodies
- Approximate Reference Range (Each site will have slight variations): Low, normal or high.
- Comments: No test preparation is needed. Certain medications — including aspirin and thyroid-hormone replacement therapy — can interfere with the TSH test, so the doctor should be informed about any drugs that are being taken.
- Test Method Variance:
- Other: This test measures the amount of thyroid-stimulating hormone (TSH) in your blood. TSH is produced by the pituitary gland, a tiny organ located below the brain and behind the sinus cavities. It is part of the body's feedback system to maintain stable amounts of the thyroid hormones thyroxine (T4) and triiodothyronine (T3) in the blood. Thyroid hormones help control the rate at which the body uses energy. When concentrations decrease in the blood, the hypothalamus (an organ in the brain) releases thyrotropin releasing hormone (TRH). This stimulates the release of TSH by the pituitary gland. The TSH in turn stimulates the production and release of T4 and T3 by the thyroid gland, a small butterfly-shaped gland that lies in the neck flat against the windpipe. When all three organs are functioning normally, thyroid production turns on and off to maintain constant blood thyroid hormone levels. If there is pituitary dysfunction, then increased or decreased amounts of TSH may result. If there is thyroid dysfunction, then the amount of TSH will be inversely related to the thyroid dysfunction (i.e., TSH will be low with overactive thyroid).

1.1.58 (U_ACR) Microalbumin to Creatine Ratio Test Types

1. U_ACR (Microalbumin to creatinine ratio)
    - Common Name: ACR
    - Long Name: Microalbumin or albumin to creatinine ratio in urine
    - Test Type: U_ACR
    - Panels: N/A

- Equivalent Tests: N/A
- Not Equivalent Tests: Serum or urine Albumin, serum prealbumin, serum or urine microalbumin, Urinalysis
- Indications: To screen for possible kidney disorder or for early kidney damage in diabetics
- Description: The microalbumin test is an early indicator of kidney failure. It measures the tiny amounts of albumin that the body begins to release into the urine several years before significant kidney damage becomes apparent. Albumin is a protein that is produced in the liver and is present in high concentrations in the blood. When the kidneys are functioning properly, virtually no albumin is present in the urine. Damaged or diseased kidneys lose their ability to filter proteins out of the urine. Patients who have consistently detectible amounts of albumin in their urine (known as microalbuminuria) have an increased risk of developing progressive kidney failure and cardiovascular disease. Microalbumin measurements can be obtained using urine collected over a 24-hour period, for a specified amount of time (e.g., 4 hours or overnight), or randomly (spot). When a creatinine measurement is performed along with a random microalbumin, the result is the ACR. Creatinine, a byproduct of muscle metabolism, is normally excreted into the urine on a consistent basis. Its level in the urine is relatively stable. Since the concentration (or dilution) of urine varies throughout the day, this property of creatinine allows its measurement to be used as a corrective factor in random/spot urine samples. When a creatinine measurement is performed along with a random microalbumin, the result is the ACR (which the American Diabetes Association states is the preferred test for screening for microalbuminuria).
- Related Tests: Albumin; Creatinine; Microalbumin
- Approximate Reference Range: Normal is $<$ 30 mcg albumin/mg creatinine; microalbuminuria = 30 - 299 mcg albumin/mg creatinine (spot or random urine sample)
- Comments: Elevated results may also be caused by vigorous exercise, blood in the urine, urinary tract infection, dehydration, and some drugs.
- Test Method Variance:
- Other:

  - Associated LOINC: 14585-4, 14959-1, 30000-4, 32294-1, 77253-3, 89998-9, 9318-7
  - Associated CPT:

2. U_ACR12 (12-hour Microalbumin to creatinine ratio)

   - Common Name: 12 hour Microalbumin to creatinine ratio
   - Long Name: 12-hour Microalbumin to creatinine ratio in urine
   - Test Type: U_ACR12
   - Panels: N/A
   - Equivalent Tests: N/A
   - Not Equivalent Tests: Serum or urine Albumin, serum prealbumin, serum or urine microalbumin, Urinalysis
   - Indications: To screen for possible kidney disorder or for early kidney damage in diabetics
   - Description: This test involves collecting all urine samples over a 12-hour period. The collected urine is stored in a special container in a refrigerator during the collection period and then returned to the lab or healthcare provider for analysis. The microalbumin test is an early indicator of kidney failure. It measures the tiny amounts of albumin that the body begins to release into the urine several years before significant kidney damage becomes apparent. Albumin is a protein that is produced in the liver and is present in high concentrations in the blood. When the kidneys are functioning properly, virtually no albumin is present in the urine. Damaged or diseased kidneys lose their ability to filter proteins out of the urine. Patients who have consistently detectible amounts of albumin in their urine (known as microalbuminuria) have an increased risk of developing progressive kidney failure and cardiovascular disease. Microalbumin measurements can be obtained using urine collected over a 24-hour period, for a specified amount of time (e.g., 4 hours or overnight), or randomly (spot). When a creatinine measurement is performed along with a random microalbumin, the result is the ACR. Creatinine, a byproduct of muscle metabolism, is normally excreted into the urine on a consistent basis. Its level in the urine is relatively stable. Since the concentration (or dilution) of urine varies throughout the day, this property of creatinine allows its measurement to be used as a corrective factor in random/spot urine samples. When a creatinine

measurement is performed along with a random microalbumin, the result is the ACR (which the American Diabetes Association states is the preferred test for screening for microalbuminuria).

- Related Tests: Albumin; Creatinine; Microalbumin
- Approximate Reference Range: Normal is < 30 mcg albumin/mg creatinine; microalbuminuria = 30 - 299 mcg albumin/mg creatinine (spot or random urine sample)
- Comments: Elevated results may also be caused by vigorous exercise, blood in the urine, urinary tract infection, dehydration, and some drugs.
- Test Method Variance:
- Other:
  - Associated LOINC: 44292-1
  - Associated CPT:

3. U_ACR24 (24-hour Microalbumin to creatinine ratio in urine)

   - Common Name: 24-hour Microalbumin to creatinine ratio in urine24-hour Microalbumin to creatinine ratio in urine
   - Long Name: 24-hour Microalbumin to creatinine ratio in urine; albumin to creatinine ratio in urine
   - Test Type: U_ACR24
   - Panels: 58447-4 (panel)
   - Equivalent Tests: N/A
   - Not Equivalent Tests: Serum or urine Albumin, serum prealbumin, serum or urine microalbumin, Urinalysis
   - Indications: To screen for possible kidney disorder or for early kidney damage in diabetics
   - Description: This test involves collecting all urine samples over a 24-hour period. The microalbumin test is an early indicator of kidney failure. It measures the tiny amounts of albumin that the body begins to release into the urine several years before significant kidney damage becomes apparent. Albumin is a protein that is produced in the liver and is present in high concentrations in the blood. When the kidneys are functioning properly, virtually no albumin is present in the urine. Damaged or diseased kidneys lose their ability to filter proteins out of the urine.

Patients who have consistently detectible amounts of albumin in their urine (known as microalbuminuria) have an increased risk of developing progressive kidney failure and cardiovascular disease. Microalbumin measurements can be obtained using urine collected over a 24-hour period, for a specified amount of time (e.g., 4 hours or overnight), or randomly (spot). When a creatinine measurement is performed along with a random microalbumin, the result is the ACR. Creatinine, a byproduct of muscle metabolism, is normally excreted into the urine on a consistent basis. Its level in the urine is relatively stable. Since the concentration (or dilution) of urine varies throughout the day, this property of creatinine allows its measurement to be used as a corrective factor in random/spot urine samples. When a creatinine measurement is performed along with a random microalbumin, the result is the ACR (which the American Diabetes Association states is the preferred test for screening for microalbuminuria).

- Related Tests: Albumin; Creatinine; Microalbumin
- Approximate Reference Range: Normal is $<$ 30 mcg albumin/mg creatinine; microalbuminuria = 30 - 299 mcg albumin/mg creatinine (spot or random urine sample)
- Comments: Elevated results may also be caused by vigorous exercise, blood in the urine, urinary tract infection, dehydration, and some drugs.
- Test Method Variance:
- Other:
  - Associated LOINC: 13705-9, 14958-3, 59159-4, 76401-9, 77254-1
  - Associated CPT:

1.1.59 U_CREAT (Creatinine in Urine)

- Common Name: Urine Creatinine (spot or random)
- Long Name: Creatinine in Urine
- Test Type: U_CREAT
- Panels: N/A

- Equivalent Tests: This TEST_TYPE should only be used for spot or random (single point in time) urine creatinine measurements.
- Not Equivalent Tests: 24 hour urine creatinine; BUN; serum creatinine; BUN/creatinine ratio; eGFR; Creatinine clearance; Urinalysis; Urine protein to creatinine ratio; microalbumin; albumin/creatinine ratio
- Indications: To determine kidney functioning and to monitor treatment for kidney disease
- Description: Creatinine, a byproduct of muscle metabolism, is normally excreted into the urine on a consistent basis. It is a good measure of how well the kidneys are working. Urine creatinine may be used with other urine tests as a correction factor. Since it is produced and removed at a relatively constant rate, the amount of urine creatinine can be compared to the amount of another substance being measured. Examples include the urine protein/creatinine ratio (UP/CR) and the microalbumin/creatinine ratio (also known as albumin/creatinine ratio, ACR).
- Related Tests: 24 hour urine creatinine; BUN; BUN/creatinine ratio; eGFR; Creatinine clearance; Urinalysis; Urine protein to creatinine ratio; microalbumin; albumin/creatinine ratio
- Approximate Reference Range: Although spot or random urine creatinine values are usually reported in mg, random urine creatinine levels have no standard reference ranges. They are usually used with other tests to reference levels of other substances measured in the urine.
- Comments: The quantity of urine creatinine produced depends on the size of the person and their muscle mass. For this reason, creatinine concentrations will be slightly higher in men than in women and children. Drugs such as aminoglycosides (vancomycin, gentamicin) can cause kidney damage and so creatinine is monitored. Other drugs, such as cephalosprins (cefoxitin), may increase creatinine concentration without reflecting kidney damage.
- Test Method Variance:
- Other:
  - Associated LOINC: 14683-7, 2161-8, 35204-7, 35674-1, 39977-4, 39982-4, 40129-9, 40133-1, 55593-8
  - Associated CPT: 82570

1.1.60 U_CREAT24 (24-Hour Creatinine in Urine)

- Common Name: 24 Hour Urine Creatinine
- Long Name: 24-Hour Creatinine in Urine
- Test Type: U_CREAT24
- Panels: N/A
- Equivalent Tests: This TEST_TYPE should only be used for 24 hour urine creatinine collections.
- Not Equivalent Tests: U_CREAT: BUN; serum creatinine; BUN/creatinine ratio; eGFR; Creatinine clearance; Urinalysis; Urine protein to creatinine ratio; microalbumin; albumin/creatinine ratio
- Indications: To determine kidney functioning and to monitor treatment for kidney disease
- Description: Creatinine, a byproduct of muscle metabolism, is normally excreted into the urine on a consistent basis. It is used to measure kidney functioning. A combination of blood and urine creatinine levels may be used to calculate a creatinine clearance (how effectively the kidney is filtering small molecules such as creatinine out of the blood). Although shorter collection periods (3-8 hours) appear to be adequate and more reliable, creatinine clearance is routinely calculated using a 24 hour urine collection.
- Related Tests: U_CREAT: BUN; BUN/creatinine ratio; eGFR; Creatinine clearance; Urinalysis; Urine protein to creatinine ratio; microalbumin; albumin/creatinine ratio
- Approximate Reference Range: 20 – 28 mg/kg24 hours (men); 15 – 21 mg/kg/24 hours (women)
- Comments: The quantity of creatinine produced depends on a person's size and muscle mass. Therefore, creatinine concentrations will be slightly higher in men than in women and children. Drugs such as aminoglycosides (vancomycin, gentamicin) can cause kidney damage and so creatinine is monitored. Other drugs, such as cephalosprins (cefoxitin), may increase creatinine concentration without reflecting kidney damage.

- Test Method Variance:
- Other:
  - Associated LOINC: 14684-5, 20624-3, 2162-6, 25886-3, 35251-8, 35262-5
  - Associated CPT: 82575

1.1.61 U_HEME_DIP (Hemoglobin in urine by dipstick)

- Common Name: Hemoglobin in urine by dipstick
- Long Name: Hemoglobin in urine by dipstick
- Test Type: U_HEME_DIP
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A positive dipstick result might indicate:
  - Hematuria: Presence of intact red blood cells. Usually confirmed by seeing >3 RBCs per high-power field under a microscope.
  - Hemoglobinuria: Presence of free hemoglobin without red blood cells. Often caused by intravascular hemolysis (breakdown of RBCs in the blood vessels).
  - Myoglobinuria: Presence of myoglobin from muscle breakdown (rhabdomyolysis).
- Description: A urine dipstick lab test detects hemoglobin by measuring its pseudoperoxidase activity, which catalyzes a color change on a reagent pad. While often referred to as a "blood" test, the dipstick does not distinguish between intact red blood cells, free hemoglobin, and myoglobin.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Possible outcomes of the test are:

  - False-Positive Alkaline urine (pH $>$ 9), semen, or contamination with oxidizing cleaning agents (e.g., bleach/peroxide).
  - False-Negative High concentrations of Vitamin C (ascorbic acid), which inhibits the oxidation reaction.
  - Normal Result Hemoglobin is normally negative in urine.

- Comments: Is LOINC 752-2 a dipstick even though the words dipstick or test strip do not appear in the definition? We are assuming Yes.
- Test Method Variance:
- Other:
  - Associated LOINC: 49137-3, 50559-4, 57751-0, 5794-3, 725-2
  - Associated CPT: 81000

1.1.62 U_MIC_ALB (Microalbumin [or albumin] in urine)

- Common Name: Urine Microalbumin
- Long Name: Microalbumin or albumin in urine
- Test Type: U_MIC_ALB
- Panels: N/A
- Equivalent Tests: This TEST_TYPE should only be used for spot or random urine microalbumin determinations.
- Not Equivalent Tests: Albumin, Prealbumin, Urinalysis, U_ACR; U_MICALB24
- Indications: To screen for possible kidney disorder or for early kidney damage in diabetics
- Description: The microalbumin test is an early indicator of kidney failure. It measures the tiny amounts of albumin that the body begins to release into the urine several years before significant kidney damage becomes apparent. Albumin is a protein that is produced in the liver. It is present in high concentrations in the blood, but when the kidneys are functioning properly, virtually no albumin is allowed to leak through into the urine. If a person's kidneys become damaged or diseased, however, they begin to lose their ability to filter proteins

out of the urine. This is frequently seen in chronic diseases, such as diabetes and hypertension, with increasing amounts of protein in the urine reflecting increasing kidney failure. Since the albumin molecule is small, it is one of the first proteins to be detected in the urine with kidney damage. Patients who have consistently detectible amounts of albumin in their urine (microalbuminuria) have an increased risk of developing progressive kidney failure and cardiovascular disease in the future. Microalbumin measurements can be obtained using urine collected over a 24-hour period, for a specified amount of time (e.g., 4 hours or overnight), or randomly (spot).

- Related Tests: Albumin; Creatinine; U_ACR; U_MICALB24
- Approximate Reference Range: Normal is < 30 mcg albumin; microalbuminuria = 30-299 mcg albumin (spot or random urine)
- Comments: The National Kidney Foundation recommends that diabetics between 12 and 70 years of age have a urine microalbumin test annually. If microalbuminuria is detected, it should be confirmed by retesting and, if positive on 2 of 3 determinations over a 3-6 month period, it is considered to be present and appropriate treatment is advised.
- Test Method Variance:
- Other:
  - Associated LOINC: 100158-5, 14957-5, 1754-1, 53531-0, 77158-4, 89999-7
  - Associated CPT: 82043, 82042

1.1.63 U_MICALB24 (24-Hour Microalbumin in urine)

- Common Name: 24-hour Urine Microalbumin
- Long Name: 24-hour Microalbumin in urine
- Test Type: U_MICALB24
- Panels: N/A
- Equivalent Tests: This TEST_TYPE should only be used for 24-hour urine collections for microalbumin determinations.

- Not Equivalent Tests: Albumin, Prealbumin, Urinalysis; U_ACR; U_MIC_ALB
- Indications: To screen for possible kidney disorder or for early kidney damage in diabetics
- Description: The microalbumin test is an early indicator of kidney failure. It measures the tiny amounts of albumin that the body begins to release into the urine several years before significant kidney damage becomes apparent. Albumin is a protein that is produced in the liver. It is present in high concentrations in the blood, but when the kidneys are functioning properly, virtually no albumin is allowed to leak through into the urine. If a person's kidneys become damaged or diseased, however, they begin to lose their ability to filter proteins out of the urine. This is frequently seen in chronic diseases, such as diabetes, hypertension, heart failure, cirrhosis, or systemic lupus erythematosus (SLE) with increasing amounts of protein in the urine reflecting increasing kidney failure. Since the albumin molecule is small, it is one of the first proteins to be detected in the urine with kidney damage. Patients who have consistently detectible amounts of albumin in their urine (microalbuminuria) have an increased risk of developing progressive kidney failure and cardiovascular disease in the future. Microalbumin measurements can be obtained using urine collected over a 24-hour period, for a specified amount of time (e.g., 4 hours or overnight), or randomly (spot).
- Related Tests: tests: Albumin; Creatinine; U_ACR; U_MIC_ALB
- Approximate Reference Range: Normal $<=$ 30 mg/24 hours; or 20 – 199 mcg/min (timed urine collection)
- Comments: The National Kidney Foundation recommends that diabetics between 12 and 70 years of age have a urine microalbumin test annually. If microalbuminuria is detected, it should be confirmed by retesting and, if positive on 2 of 3 determinations over a 3-6 month period, it is considered to be present and appropriate treatment is advised.
- Test Method Variance:
- Other:
  - Associated LOINC: 21059-1, 30003-8 , 53530-2

- Associated CPT: 82043

1.1.64 U_PCR (Protein to creatinine ratio in urine)

- Common Name: Urine Protein to Creatinine Ratio (UPCR)
- Long Name: Protein to creatinine ratio in urine
- Test Type: U_PCR
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: 24-Hour Urine Protein; Total Urine Protein
- Indications: To evaluate and monitor kidney function, and to detect kidney damage
- Description: Protein is not normally found in the urine, and urine protein tests detect and/or measure protein being excreted in the urine. U_PCR is one of several different urine protein tests, and involves the amount of protein in a random urine sample measured along with urine creatinine and reported as the ratio of urine protein to creatinine. Creatinine, a byproduct of muscle metabolism, is normally excreted into the urine at a constant rate. When both a urine creatinine and a random urine protein test are performed, the resulting U_PCR approaches the accuracy of the 24-hour urine protein test. Since saving all of the urine for a 24-hour period can be cumbersome for adults and difficult for infants and children, U_PCR may be substituted for a 24-hour urine protein sample.

U_PCR is used when a child shows evidence of significant and persistent protein in their urine with the dipstick urine test. It is also utilized to monitor kidney function over time on a patient with known kidney disease or damage. A U_CPR may used as a screen for kidney involvement when a patient is taking a medication that may potentially affect kidney function.

- Related Tests: Urinalysis; Albumin; Microalbumin; Protein Electrophoresis; Total Protein; ACR
- Approximate Reference Range: zero to trace

- Comments: The protein to creatinine ratio is more of a snapshot of how much protein is in the urine at the time the sample is collected. Children, and sometimes adults, occasionally have some degree of transient proteinuria without apparent kidney dysfunction and may have a higher excretion of protein into their urine during the day than at night. Monitoring by U_PCR at intervals is conducted to determine if the amount of proteinuria changes over time.
- Test Method Variance:
- Other:
  - Associated LOINC: 2890-2, 34366-5
  - Associated CPT:

1.1.65 U_PCR24 (24-hour Protein to creatinine ratio in urine)

- Common Name: 24 hour Urine Protein to Creatinine Ratio
- Long Name: 24 hour Protein to creatinine ratio in urine
- Test Type: U_PCR24
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests:
- Indications: To evaluate and monitor kidney function, and to detect kidney damage
- Description: Protein is not normally found in the urine, and urine protein tests detect and/or measure protein being excreted in the urine. U_PCR is one of several different urine protein tests, and involves the amount of protein in a random urine sample measured along with urine creatinine and reported as the ratio of urine protein to creatinine. Creatinine, a byproduct of muscle metabolism, is normally excreted into the urine at a constant rate. When both a urine creatinine and a random urine protein test are performed, the resulting U_PCR approaches the accuracy of the 24-hour urine protein test. Since saving all of the urine for a 24-hour period can be cumbersome for adults and difficult for infants and children, U_PCR may be substituted for

a 24-hour urine protein sample. U_PCR is used when a child shows evidence of significant and persistent protein in their urine with the dipstick urine test. It is also utilized to monitor kidney function over time on a patient with known kidney disease or damage. A U_CPR may used as a screen for kidney involvement when a patient is taking a medication that may potentially affect kidney function.

- Related Tests: Urinalysis; Albumin; Microalbumin; Protein Electrophoresis; Total Protein; ACR
- Approximate Reference Range: zero to trace
- Comments: The protein to creatinine ratio is more of a snapshot of how much protein is in the urine at the time the sample is collected. Children, and sometimes adults, occasionally have some degree of transient proteinuria without apparent kidney dysfunction and may have a higher excretion of protein into their urine during the day than at night. Monitoring by U_PCR at intervals is conducted to determine if the amount of proteinuria changes over time.
- Test Method Variance:
- Other:
  - Associated LOINC: 13801-6, 40486-3
  - Associated CPT:

1.1.66 (U_PROT) Protein to Creatinine Ratio in Urine Test Types

1. U_PROT (Protein in urine; total protein in urine)
   - Common Name: Urine Protein
   - Long Name: Protein in urine; total protein in urine
   - Test Type: U_PROT
   - Panels: N/A
   - Equivalent Tests: This TEST_TYPE should only be used for spot or random urine protein determinations.
   - Not Equivalent Tests: 24-Hour Urine Protein; Urine Protein to Creatinine Ratio; U_PCR

- Indications: To detect excessive protein escaping into the urine (proteinuria), to assist in evaluating and monitoring kidney function, and to detect kidney damage. In diagnosing the cause of proteinuria, a serum protein electrophoresis may also be ordered to analyze the proteins in the blood.
- Description: Protein is not normally found in the urine, and urine protein tests detect and/or measure protein being excreted in the urine. When the kidneys are functioning normally, they retain or reabsorb filtered proteins and return them to the blood. However, if the kidneys are damaged, they become less effective at filtering, and detectible amounts of protein begin to find their way into the urine. Often, it is the smaller albumin molecules that are detected first. If the damage continues, the amount of protein in the urine increases, and globulins may also begin to be lost. Proteinuria is frequently seen in chronic diseases, such as diabetes and hypertension, with increasing amounts of protein in the urine reflecting increasing kidney damage. With early kidney damage, the patient is often asymptomatic. As damage progresses, or if protein loss is severe, the patient may have symptoms such as edema, shortness of breath, nausea, and fatigue. Excess protein production, such as may be seen with multiple myeloma, can also lead to proteinuria. The presence of albumin in the urine is a sensitive indicator of kidney disease in patients with diabetes and with hypertension. Therefore, in some situations the doctor may test specifically for albumin in the urine, as opposed to total urine protein.
- Related Tests: Urinalysis; Albumin; Microalbumin; Protein Electrophoresis; Total Protein; U_PCR
- Approximate Reference Range: <8mg/dL
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 21482-5, 2888-6, 32551-4, 35663-4
  - Associated CPT: 84156, 84160

2. U_PROT_24 (24-hour urine protein; total protein in urine)
   - Common Name: 24-Hour Urine Protein

- Long Name: 24-hour urine protein; total protein in urine
- Test Type: U_PROT_24
- Panels: N/A
- Equivalent Tests: This TEST_TYPE should only be used for 24-hour urine collections for protein determinations.
- Not Equivalent Tests: Urine Total Protein; U_PCR; U_PROT
- Indications: To detect excessive protein escaping into the urine, to help evaluate and monitor kidney function, and to detect kidney damage. Since the dipstick protein test primarily measures albumin, a follow-up 24-hour urine protein test may be used to determine if proteins other than albumin are being released.
- Description: Protein is not normally found in the urine, and urine protein tests detect and/or measure protein being excreted in the urine. The quantity of protein in a 24-hour urine sample will be measured and reported as the amount of protein excreted per 24 hours. Proteinuria is frequently seen in chronic diseases, such as diabetes and hypertension, with increasing amounts of protein in the urine reflecting increasing kidney damage. With early kidney damage, the patient is often asymptomatic. As damage progresses, or if protein loss is severe, the patient may have symptoms such as edema, shortness of breath, nausea, and fatigue. Excess protein production, such as may be seen with multiple myeloma, can also lead to proteinuria. Either a 24-hour urine protein may be used to monitor a patient with known kidney disease or damage.
- Related Tests: Urinalysis; Albumin; Microalbumin; Protein Electrophoresis; Total Protein; U_PCR; U_PROT
- Approximate Reference Range: $<$ 150mg/day (in healthy individuals)
- Comments: A 24-hour urine sample gives the protein excretion rate over 24 hours and is accurate only if all of the urine is collected.
- Test Method Variance:
- Other:
  - Associated LOINC: 2889-4
  - Associated CPT: 81060

3. U_PROT_DIP (Protein in Urine by Dipstick, qualitative)

   - Common Name: Urine Protein Dipstick
   - Long Name: Protein in Urine by Dipstick, qualitative
   - Test Type: U_PROT_DIP
   - Panels: N/A
   - Equivalent Tests: N/A
   - Not Equivalent Tests: 24-Hour Urine Protein; U_PCR; ACR
   - Indications: To detect excessive protein escaping into the urine, to help evaluate and monitor kidney function, and to detect kidney damage.
   - Description: Protein is not normally found in the urine, and urine protein dipstick test detects protein being excreted in the urine. A protein "dipstick" may be performed as part of a urinalysis, generally on a random urine sample. A dipstick urine protein is measured frequently as a screening test, when a urinalysis is performed. This may be done as part of a routine physical, a pregnancy workup, when a urinary tract infection is suspected, as part of a hospital admission, or when evaluating kidney function. It may also be done when a previous dipstick has been positive for protein to see if the protein excretion persists. Since the dipstick primarily measures albumin, a follow-up 24-hour urine protein test may be ordered to determine if proteins other than albumin are being released. A dipstick urine protein on a random urine sample may be used as a screen for kidney involvement when a patient is taking a medication that may potentially affect kidney function. Proteinuria is frequently seen in chronic diseases, such as diabetes and hypertension, with increasing amounts of protein in the urine reflecting increasing kidney damage. With early kidney damage, the patient is often asymptomatic. As damage progresses, or if protein loss is severe, the patient may have symptoms such as edema, shortness of breath, nausea, and fatigue. Excess protein production, such as may be seen with multiple myeloma, can also lead to proteinuria.
   - Related Tests: Urinalysis; Albumin; Microalbumin; Protein Electrophoresis; Total Protein
   - Approximate Reference Range: zero to trace

- Comments: A positive dipstick protein may be elevated due to other sources of protein, such as blood, semen, or vaginal secretions in the urine. Since it measures primarily albumin, the dipstick may occasionally be normal when significant quantities of other proteins are present in the urine. Children, and sometimes adults, occasionally have some degree of transient proteinuria without apparent kidney dysfunction and may have a higher excretion of protein into their urine during the day than at night.
- Test Method Variance:
- Other:
  - Associated LOINC: 20454-5, 27298-9, 2887-8, 32209-9, 50561-0, 50949-7, 53525-2 ,57735-3, 5804-0
  - Associated CPT: 81000

1.1.67 U_RBC_DIP (Erythrocytes in urine by dipstick)

- Common Name: Erythrocytes in urine by dipstick
- Long Name: Erythrocytes (red blood cells) in urine by dipstick
- Test Type: U_RBC_DIP
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Some clinical findings that may be found are:
  - Hemoglobinuria: Free hemoglobin from red blood cells that have already broken down in the bloodstream.
  - Myoglobinuria: Muscle protein released into the urine during muscle injury (e.g., from rhabdomyolysis).
  - Contamination: Menstrual blood or dehydration can also cause positive readings.
- Description: In a urine dipstick test, the "blood" or "erythrocytes" pad detects the presence of red blood cells (RBCs) through a chemical reaction. A positive result is called dipstick hematuria.
- Related Tests:

- Approximate Reference Range (Each site will have slight variations): Results from dipstick are:
  - Negative: No color change; normal.
  - Trace: May represent a small number of RBCs (roughly 5–10 cells/µL).
  - 1+ to 4+ (Small to Large): Increasing intensities of green indicate higher concentrations of blood.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 20409-9, 33051-4, 53292-9, 57747-8
  - Associated CPT:

1.1.68 URIC_ACID (Urate or Uric Acid in serum or plasma)

- Common Name: Uric acid
- Long Name: Urate; Uric acid in serum or plasma
- Test Type: URIC_ACID
- Panels: Arthritis panel, Bone and Joint panel
- Equivalent Tests: N/A
- Not Equivalent Tests: Urine uric acid concentration
- Indications: Physicians can use a serum uric acid test to determine if a patient has gout and to assess the effectiveness of its treatment. It is also used with other electrolyte tests to determine if oncology patients have tumor lysis syndrome during their chemotherapy regimen.
- Description: Uric acid is the metabolic end-product of purines. High serum uric acid levels can be the result of decreased kidney function, acidic urine, a diet high in purine-containing foods, or use of chemotherapy drugs.
- Related Tests: Urine uric acid concentration, serum potassium, serum calcium, serum phosphate

- Approximate Reference Range: Males: 2.5 - 8 mg/dL or 150 - 480 micromol/L; females: 1.5 - 6 mg/dL or 90 - 360 micromol/L
- Comments: In gouty arthritis, uric acid may not be properly cleared from the body and can deposit crystals into the joints and kidneys. High uric acid levels can also be due to increased consumption of purine rich foods, such as meats and yeast, as well as some fishes, beans, and vegetables. When oncology patients are using certain chemotherapy drugs, the increased amount of cell death can increase purine degradation from DNA therefore increasing uric acid levels. In the diagnosis of tumor lysis syndrome, serum uric acid levels are increased in addition to serum potassium and phosphate levels while serum calcium levels are lowered.
- Test Method Variance:
- Other:
  - Associated LOINC: 14933-6 , 3084-1, 35232-8, 98981-4
  - Associated CPT: 84550

1.1.69 VIT_B12 (Vitamin B12; Cobalamin)

- Common Name: Vitamin B12
- Long Name: Cyanocobalamin; Vitamin B12; cobalamin
- Test Type: VIT_B12
- Panels: N/A
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications: Physicians will order this test if they suspect vitamin B12 deficiency, megaloblastic anemia, pernicious anemia, or neuropathy (nerve damage). It can also be used to evaluate nutritional status and to monitor the effectiveness of treatment for B12 or folate deficiency.
- Description: Serum vitamin B12 evaluation is used to determine whether a patient has a deficiency of vitamin B12, usually as a result of inadequate dietary intake or decreased absorption. Vitamin B12 is highly

protein bound in the bloodstream, with 70% of vitamin B12 bound to transcobalamin I (not physiologically active), and 30% to transcobalamin II (physiologically active). Vitamin B12 is necessary for generating DNA, neurotransmitters, and for metabolizing homocysteine.

- Related Tests: methylmalonic acid (MMA), homocysteine, folate (folic acid)
- Approximate Reference Range: 140–820 pg/mL; 100–600 pmol/L
- Comments: All vitamin B12 comes from animal-based foods such as meats, eggs, and dairy products. Intrinsic factor is necessary for the absorption of oral vitamin B12 and may also lead to vitamin B12 deficiency if it is not properly secreted from the gastric mucosa. B12 and folate are primarily ordered to help diagnose the cause of macrocytic anemia. They are ordered as followup tests when large RBCs and a decreased hemoglobin concentration are found during a complete blood count (CBC) test. B12 may be ordered with folate, by itself, or with other screening laboratory tests – such as a comprehensive metabolic panel – to help diagnose the cause of neuropathy. High levels of B12 and folate are not usually clinically monitored. Increased B12 may be seen in conditions such as leukemia or liver dysfunction.
- Test Method Variance:
- Other:
  - Associated LOINC: 14685-2, 16695-9, 2132-9, 26916-7, 35201-3
  - Associated CPT: 82607

1.1.70 VIT_D_250H (Vitamin D; 25-hydroxy-vitamin D)

- Common Name: Vitamin D
- Long Name: Vitamin D (25 Hydroxy); 25-OH vitamin D; Calcidiol; 25-hydroxycholecalciferol
- Test Type: VIT_D_25OH
- Panels: Osteoporosis panel
- Equivalent Tests: N/A
- Not Equivalent Tests: 1,25 – dihydroxy-vitamin D

- Indications: 25-hydroxy-vitamin D testing is most commonly used to determine vitamin D deficiency. It is also indicated in hypocalcemic disorders associated with increased parathyroid hormone (PTH) levels, in children and adults with rickets, in adults with osteomalacia, and in vitamin D toxicity.
- Description: 25-hydroxy-cholecalciferol is generated in the liver by hydroxylation of cholecalciferol. Cholecalciferol is synthesized in the epidermis after exposure to UV radiation (sunlight). This value is used to determine a patient's vitamin D status.
- Related Tests: Serum calcium; PTH; Serum phosphorus
- Approximate Reference Range: 10 – 100 ng/mL; preferred range is $>$ 30 ng/mL since most laboratories consider 10-20 ng/mL to be insufficient
- Comments: The function of vitamin D is to help maintain appropriate serum calcium levels via absorption and distribution. Vitamin D toxicity is usually presented with hypercalcemia, hyperphosphatemia, soft tissue calcification, and renal failure.
- Test Method Variance:
- Other:
  - Associated LOINC: 14635-7, 1989-3, 35196-5, 95594-8
  - Associated CPT: 82306

1.2 Hematology and Coagulation

1.2.1 ANC (Absolute neutrophil count)

- Common Name: Absolute neutrophil count
- Long Name: Absolute neutrophil count
- Test Type: ANC
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: A low ANC, known as neutropenia, signals a weakened immune system and a higher risk of infection. Severe neutropenia (ANC <500) can leave a patient highly vulnerable to serious infections. A high ANC, known as neutrophilia, suggests your body is under stress from an infection or inflammation.
- Description: An absolute neutrophil count (ANC) test measures the total number of neutrophils in a blood sample. As the most abundant type of white blood cell, neutrophils are a primary component of the immune system and help fight infection.
- Related Tests: An ANC is almost always performed as part of a complete blood count (CBC) with differential.
- Approximate Reference Range (Each site will have slight variations): A doctor will interpret your ANC result based on your overall health, age, medical history, and other factors. A normal range for a healthy adult is generally 2,500 to 7,000 cells per microliter, but values vary by lab.
- Comments: Take CAUTION when using 753-4 (manual)! / 752-6 is depricated. / LOINC does not state the LOINCs listed in this row are specific to ANC, but some sites may find indication in their lab data "local name" that these LOINCS are used for ANC.
- Test Method Variance:
- Other:
  - Associated LOINC: 26499-4, 751-8, 752-6, 753-4
  - Associated CPT: 85048

1.2.2 (BASO) Basophil Test Types

1. BASO_C_A (Basophil automated count)
   - Common Name: Basophil automated count
   - Long Name: Basophil number, automated count
   - Test Type: BASO_C_A
   - Panels:
   - Equivalent Tests:

- Not Equivalent Tests:
- Indications:
  - An elevated basophil count can be a sign of the following conditions:
    * Myeloproliferative disorders: These are blood cancers that cause the bone marrow to produce too many blood cells. Basophilia is a common finding in:
    * Allergic diseases: Basophils release histamine and other chemicals during allergic reactions, so their numbers may increase with:
    * Chronic inflammation and autoimmune diseases: These conditions involve a long-term inflammatory response.
    * Endocrine conditions: Some hormonal disorders can affect basophil levels.
    * Infections: Persistent or certain parasitic infections can trigger an increase in basophils.
  - A decreased basophil count, though less common as a significant finding, may occur during certain immune responses. Basopenia can be a sign of:
    * Acute allergic reactions: During a sudden, severe reaction, basophils release their granules and migrate to tissues, causing their numbers in the blood to drop.
    * Severe acute infections: As part of the immune response, basophils can be used up or mobilized to the site of infection.
    * Hyperthyroidism: An overactive thyroid gland is associated with lower basophil levels.
    * Stress reactions: Emotional or physical stress can cause a temporary drop in basophils.
    * Medication side effects: The use of corticosteroids, for example, can cause a decrease in basophils.
- Description: Basophils are granular white blood cells characterized by a pale-staining nucleus with 2 to 3 lobes and a cytoplasm containing coarse, dark-staining granules of variable size. Automated basophil counting is performed by modern hematology analyzers, which are instruments used to conduct a complete blood count (CBC) with a differential. These machines use a combination of technologies to quickly classify and count basophils along

with other white blood cells, but the results can be prone to inconsistencies.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For adults, the typical reference ranges for an absolute basophil count (ABC) is 0 to 200 cells per microliter.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 704-7
  - Associated CPT:

2. BASO_C_M (Basophil manual count)

   - Common Name: Basophil manual count
   - Long Name: Basophil number, manual count
   - Test Type: BASO_C_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications:
     - An elevated basophil count can be a sign of the following conditions:
       * Myeloproliferative disorders: These are blood cancers that cause the bone marrow to produce too many blood cells. Basophilia is a common finding in:
       * Allergic diseases: Basophils release histamine and other chemicals during allergic reactions, so their numbers may increase with:
       * Chronic inflammation and autoimmune diseases: These conditions involve a long-term inflammatory response.
       * Endocrine conditions: Some hormonal disorders can affect basophil levels.
       * Infections: Persistent or certain parasitic infections can trigger an increase in basophils.

  - A decreased basophil count, though less common as a significant finding, may occur during certain immune responses. Basopenia can be a sign of:
    * Acute allergic reactions: During a sudden, severe reaction, basophils release their granules and migrate to tissues, causing their numbers in the blood to drop.
    * Severe acute infections: As part of the immune response, basophils can be used up or mobilized to the site of infection.
    * Hyperthyroidism: An overactive thyroid gland is associated with lower basophil levels.
    * Stress reactions: Emotional or physical stress can cause a temporary drop in basophils.
    * Medication side effects: The use of corticosteroids, for example, can cause a decrease in basophils.

- Description: Basophils are granular white blood cells characterized by a pale-staining nucleus with 2 to 3 lobes and a cytoplasm containing coarse, dark-staining granules of variable size. A basophil manual count is a lab procedure for a medical professional to identify and count basophils, a type of white blood cell, by examining a blood smear under a microscope, often as a secondary check when automated analyzers indicate an issue. This manual method is used to provide an accurate absolute basophil count (ABC).
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For adults, the typical reference ranges for an absolute basophil count (ABC) is 0 to 200 cells per microliter.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 705-4
  - Associated CPT:

3. BASO_C (Basophil count)

- Common Name: Basophil count
- Long Name: Basophil number , count method not stated

- Test Type: BASO_C
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications:
  - An elevated basophil count can be a sign of the following conditions:
    * Myeloproliferative disorders: These are blood cancers that cause the bone marrow to produce too many blood cells. Basophilia is a common finding in:
    * Allergic diseases: Basophils release histamine and other chemicals during allergic reactions, so their numbers may increase with:
    * Chronic inflammation and autoimmune diseases: These conditions involve a long-term inflammatory response.
    * Endocrine conditions: Some hormonal disorders can affect basophil levels.
    * Infections: Persistent or certain parasitic infections can trigger an increase in basophils.
  - A decreased basophil count, though less common as a significant finding, may occur during certain immune responses. Basopenia can be a sign of:
    * Acute allergic reactions: During a sudden, severe reaction, basophils release their granules and migrate to tissues, causing their numbers in the blood to drop.
    * Severe acute infections: As part of the immune response, basophils can be used up or mobilized to the site of infection.
    * Hyperthyroidism: An overactive thyroid gland is associated with lower basophil levels.
    * Stress reactions: Emotional or physical stress can cause a temporary drop in basophils.
    * Medication side effects: The use of corticosteroids, for example, can cause a decrease in basophils.
- Description: Basophils are granular white blood cells characterized by a pale-staining nucleus with 2 to 3 lobes and a cytoplasm containing coarse, dark-staining granules of variable size.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For adults, the typical reference ranges for an absolute basophil count (ABC) is 0 to 200 cells per microliter.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 26444-0
  - Associated CPT:

4. BASO_P_A (Basophil percent, automated)
   - Common Name: Basophil percent, automated
   - Long Name: Basophil percent, automated
   - Test Type: BASO_P_A
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications:
     - An elevated basophil count can be a sign of the following conditions:
       * Myeloproliferative disorders: These are blood cancers that cause the bone marrow to produce too many blood cells. Basophilia is a common finding in:
       * Allergic diseases: Basophils release histamine and other chemicals during allergic reactions, so their numbers may increase with:
       * Chronic inflammation and autoimmune diseases: These conditions involve a long-term inflammatory response.
       * Endocrine conditions: Some hormonal disorders can affect basophil levels.
       * Infections: Persistent or certain parasitic infections can trigger an increase in basophils.
     - A decreased basophil count, though less common as a significant finding, may occur during certain immune responses. Basopenia can be a sign of:

        * Acute allergic reactions: During a sudden, severe reaction, basophils release their granules and migrate to tissues, causing their numbers in the blood to drop.
        * Severe acute infections: As part of the immune response, basophils can be used up or mobilized to the site of infection.
        * Hyperthyroidism: An overactive thyroid gland is associated with lower basophil levels.
        * Stress reactions: Emotional or physical stress can cause a temporary drop in basophils.
        * Medication side effects: The use of corticosteroids, for example, can cause a decrease in basophils.
    - Description: Basophils are granular white blood cells characterized by a pale-staining nucleus with 2 to 3 lobes and a cytoplasm containing coarse, dark-staining granules of variable size. A manual basophil percentage is the result of a white blood cell (WBC) differential performed by a trained laboratory technician using a microscope. The technician examines a stained blood smear and counts and classifies at least 100, and sometimes 200 to 400, white blood cells to determine the percentage of each type.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): For adults, the typical reference ranges for a relative basophil count (differential): 0.5% to 1% of the total white blood cell count. Some labs may report a slightly wider range, such as 0% to 2%.
    - Comments:
    - Test Method Variance:
    - Other:
        - Associated LOINC: 706-2
        - Associated CPT:

5. BASO_P_M (Basophil percent, manual)
    - Common Name: Basophil percent, manual
    - Long Name: Basophil percent, manual
    - Test Type: BASO_P_M

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications:
  - An elevated basophil percent can be a sign of the following conditions:
    * Myeloproliferative disorders: These are blood cancers that cause the bone marrow to produce too many blood cells. Basophilia is a common finding in:
    * Allergic diseases: Basophils release histamine and other chemicals during allergic reactions, so their numbers may increase with:
    * Chronic inflammation and autoimmune diseases: These conditions involve a long-term inflammatory response.
    * Endocrine conditions: Some hormonal disorders can affect basophil levels.
    * Infections: Persistent or certain parasitic infections can trigger an increase in basophils.
  - A low basophil percent, though less common as a significant finding, may occur during certain immune responses. Basopenia can be a sign of:
    * Acute allergic reactions: During a sudden, severe reaction, basophils release their granules and migrate to tissues, causing their numbers in the blood to drop.
    * Severe acute infections: As part of the immune response, basophils can be used up or mobilized to the site of infection.
    * Hyperthyroidism: An overactive thyroid gland is associated with lower basophil levels.
    * Stress reactions: Emotional or physical stress can cause a temporary drop in basophils.
    * Medication side effects: The use of corticosteroids, for example, can cause a decrease in basophils.
- Description: Basophils are granular white blood cells characterized by a pale-staining nucleus with 2 to 3 lobes and a cytoplasm containing coarse, dark-staining granules of variable size. A manual basophil percentage is the result of a white blood cell (WBC)

differential performed by a trained laboratory technician using a microscope. The technician examines a stained blood smear and counts and classifies at least 100, and sometimes 200 to 400, white blood cells to determine the percentage of each type.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For adults, the typical reference ranges for a relative basophil count (differential): 0.5% to 1% of the total white blood cell count. Some labs may report a slightly wider range, such as 0% to 2%.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 707-0
  - Associated CPT:

6. BASO_P (Basophil percent)

   - Common Name: Basophil percent
   - Long Name: Basophil percent, method not stated
   - Test Type: BASO_P
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications:
     - An elevated basophil percent can be a sign of the following conditions:
       * Myeloproliferative disorders: These are blood cancers that cause the bone marrow to produce too many blood cells. Basophilia is a common finding in:
       * Allergic diseases: Basophils release histamine and other chemicals during allergic reactions, so their numbers may increase with:
       * Chronic inflammation and autoimmune diseases: These conditions involve a long-term inflammatory response.
       * Endocrine conditions: Some hormonal disorders can affect basophil levels.

        * Infections: Persistent or certain parasitic infections can trigger an increase in basophils.
    - A low basophil percent, though less common as a significant finding, may occur during certain immune responses. Basopenia can be a sign of:
        * Acute allergic reactions: During a sudden, severe reaction, basophils release their granules and migrate to tissues, causing their numbers in the blood to drop.
        * Severe acute infections: As part of the immune response, basophils can be used up or mobilized to the site of infection.
        * Hyperthyroidism: An overactive thyroid gland is associated with lower basophil levels.
        * Stress reactions: Emotional or physical stress can cause a temporary drop in basophils.
        * Medication side effects: The use of corticosteroids, for example, can cause a decrease in basophils.
- Description: Basophils are granular white blood cells characterized by a pale-staining nucleus with 2 to 3 lobes and a cytoplasm containing coarse, dark-staining granules of variable size.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For adults, the typical reference ranges for an absolute basophil count (ABC) is 0 to 200 cells per microliter.
- Comments:
- Test Method Variance:
- Other:
    - Associated LOINC: 30180-4
    - Associated CPT:

1.2.3 (EOS) Eosinophil Test Types

1. EOS_C_A (Eosinophil number, automated count)

    - Common Name: Eosinophil number, automated count
    - Long Name: Eosinophil number, automated count
    - Test Type: EOS_C_A

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Eosinophil testing is indicated when a complete blood count (CBC) reveals abnormal levels or when a patient's symptoms suggest an eosinophil-related disorder.
  - Indications for high eosinophil counts (eosinophilia):
    * Allergic rhinitis and asthma: Eosinophils are a key driver of inflammation in allergic diseases. An elevated count can indicate the presence or severity of these conditions, especially eosinophilic asthma.
    * Eosinophilic gastrointestinal disorders: This includes conditions like eosinophilic esophagitis, gastritis, and colitis, which cause inflammation in parts of the digestive tract.
    * Skin conditions: High eosinophil levels are associated with various dermatological issues, including eczema and atopic dermatitis.
    * Drug reactions: An allergic reaction to certain medications, such as DRESS syndrome (Drug Rash with Eosinophilia and Systemic Symptoms), can cause eosinophilia.
    * Parasitic infestations: Eosinophils play a critical role in combating parasitic infections like roundworms, hookworms, and schistosomes. This is a very common cause of high eosinophil levels, particularly in tropical regions.
    * Fungal infections: Certain fungal infections, such as aspergillosis, can lead to elevated eosinophil counts.
    * Blood cancers: Some malignancies, including Hodgkin lymphoma, certain types of leukemia, and other myeloproliferative disorders, are associated with high eosinophil levels.
    * Hypereosinophilic syndrome (HES): A rare disorder characterized by extremely high eosinophil counts that can lead to organ damage.
    * Primary eosinophilia: A rare disorder involving the clonal proliferation of eosinophils, often linked to specific genetic mutations.
    * Systemic autoimmune diseases: Conditions like vasculitis (e.g., Churg-Strauss syndrome) and inflammatory bowel

diseases (Crohn's disease, ulcerative colitis) are known to cause eosinophilia.

  - Indications for low eosinophil counts (eosinopenia):
    * Excess cortisol: The overproduction of cortisol, as seen in Cushing's syndrome, can lead to lower-than-normal eosinophil counts.
    * Steroid use: The use of corticosteroids can suppress eosinophil levels.
    * Sepsis: A severe, life-threatening infection of the bloodstream can cause a sharp drop in eosinophil levels.
    * Acute inflammation or stress: The body's acute inflammatory response, often from severe infection, can result in low eosinophil counts.

- Description: A type of white blood cell characterized by the presence of granules in its cytoplasm with a nucleus that usually has two lobes connected by a slender thread of chromatin. The cytoplasm contains coarse, round and uniform granules stainable by eosin.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):

Normal Counts by Age:

Age	**Absolute Eosinophil Count**
Newborn to 6 months	0 to 900 cells/µL
6 months to 1 year	150 to 585 cells/µL
1 to 2 years	180 to 510 cells/µL
3 to 11 years	0 to 500 cells/µL
12 to 17 years	100 to 200 cells/µL
18+ years (Adults)	0 to 500 cells/µL (less than 0.48 x 10/L)

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 711-2
  - Associated CPT:

2. EOS_C_M (Eosinophil number, manual count)

   - Common Name: Eosinophil number, manual count
   - Long Name: Eosinophil number, manual count
   - Test Type: EOS_C_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Eosinophil testing is indicated when a complete blood count (CBC) reveals abnormal levels or when a patient's symptoms suggest an eosinophil-related disorder.
     - Indications for high eosinophil counts (eosinophilia):
       * Allergic rhinitis and asthma: Eosinophils are a key driver of inflammation in allergic diseases. An elevated count can indicate the presence or severity of these conditions, especially eosinophilic asthma.
       * Eosinophilic gastrointestinal disorders: This includes conditions like eosinophilic esophagitis, gastritis, and colitis, which cause inflammation in parts of the digestive tract.
       * Skin conditions: High eosinophil levels are associated with various dermatological issues, including eczema and atopic dermatitis.
       * Drug reactions: An allergic reaction to certain medications, such as DRESS syndrome (Drug Rash with Eosinophilia and Systemic Symptoms), can cause eosinophilia.
       * Parasitic infestations: Eosinophils play a critical role in combating parasitic infections like roundworms, hookworms, and schistosomes. This is a very common cause of high eosinophil levels, particularly in tropical regions.
       * Fungal infections: Certain fungal infections, such as aspergillosis, can lead to elevated eosinophil counts.
       * Blood cancers: Some malignancies, including Hodgkin lymphoma, certain types of leukemia, and other myeloproliferative disorders, are associated with high eosinophil levels.
       * Hypereosinophilic syndrome (HES): A rare disorder characterized by extremely high eosinophil counts that can lead to organ damage.

        * Primary eosinophilia: A rare disorder involving the clonal proliferation of eosinophils, often linked to specific genetic mutations.
        * Systemic autoimmune diseases: Conditions like vasculitis (e.g., Churg-Strauss syndrome) and inflammatory bowel diseases (Crohn's disease, ulcerative colitis) are known to cause eosinophilia.
    - Indications for low eosinophil counts (eosinopenia):
        * Excess cortisol: The overproduction of cortisol, as seen in Cushing's syndrome, can lead to lower-than-normal eosinophil counts.
        * Steroid use: The use of corticosteroids can suppress eosinophil levels.
        * Sepsis: A severe, life-threatening infection of the bloodstream can cause a sharp drop in eosinophil levels.
        * Acute inflammation or stress: The body's acute inflammatory response, often from severe infection, can result in low eosinophil counts.
- Description: A type of white blood cell characterized by the presence of granules in its cytoplasm with a nucleus that usually has two lobes connected by a slender thread of chromatin. The cytoplasm contains coarse, round and uniform granules stainable by eosin.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):

Normal Counts by Age:

Age	**Absolute Eosinophil Count**
Newborn to 6 months	0 to 900 cells/µL
6 months to 1 year	150 to 585 cells/µL
1 to 2 years	180 to 510 cells/µL
3 to 11 years	0 to 500 cells/µL
12 to 17 years	100 to 200 cells/µL
18+ years (Adults)	0 to 500 cells/µL (less than 0.48 x 10/L)

- Comments:

- Test Method Variance:
- Other:
  - Associated LOINC: 712-0
  - Associated CPT:

3. EOS_C (Eosinophil number)

   - Common Name: Eosinophil number
   - Long Name: Eosinophil number, count method not stated
   - Test Type: EOS_C
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Eosinophil testing is indicated when a complete blood count (CBC) reveals abnormal levels or when a patient's symptoms suggest an eosinophil-related disorder.
     - Indications for high eosinophil counts (eosinophilia):
       * Allergic rhinitis and asthma: Eosinophils are a key driver of inflammation in allergic diseases. An elevated count can indicate the presence or severity of these conditions, especially eosinophilic asthma.
       * Eosinophilic gastrointestinal disorders: This includes conditions like eosinophilic esophagitis, gastritis, and colitis, which cause inflammation in parts of the digestive tract.
       * Skin conditions: High eosinophil levels are associated with various dermatological issues, including eczema and atopic dermatitis.
       * Drug reactions: An allergic reaction to certain medications, such as DRESS syndrome (Drug Rash with Eosinophilia and Systemic Symptoms), can cause eosinophilia.
       * Parasitic infestations: Eosinophils play a critical role in combating parasitic infections like roundworms, hookworms, and schistosomes. This is a very common cause of high eosinophil levels, particularly in tropical regions.
       * Fungal infections: Certain fungal infections, such as aspergillosis, can lead to elevated eosinophil counts.

        * Blood cancers: Some malignancies, including Hodgkin lymphoma, certain types of leukemia, and other myeloproliferative disorders, are associated with high eosinophil levels.
        * Hypereosinophilic syndrome (HES): A rare disorder characterized by extremely high eosinophil counts that can lead to organ damage.
        * Primary eosinophilia: A rare disorder involving the clonal proliferation of eosinophils, often linked to specific genetic mutations.
        * Systemic autoimmune diseases: Conditions like vasculitis (e.g., Churg-Strauss syndrome) and inflammatory bowel diseases (Crohn's disease, ulcerative colitis) are known to cause eosinophilia.
    - Indications for low eosinophil counts (eosinopenia):
        * Excess cortisol: The overproduction of cortisol, as seen in Cushing's syndrome, can lead to lower-than-normal eosinophil counts.
        * Steroid use: The use of corticosteroids can suppress eosinophil levels.
        * Sepsis: A severe, life-threatening infection of the bloodstream can cause a sharp drop in eosinophil levels.
        * Acute inflammation or stress: The body's acute inflammatory response, often from severe infection, can result in low eosinophil counts.
- Description: A type of white blood cell characterized by the presence of granules in its cytoplasm with a nucleus that usually has two lobes connected by a slender thread of chromatin. The cytoplasm contains coarse, round and uniform granules stainable by eosin.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):

Normal Counts by Age:

Age	Absolute Eosinophil Count
Newborn to 6 months	0 to 900 cells/µL
6 months to 1 year	150 to 585 cells/µL
1 to 2 years	180 to 510 cells/µL
3 to 11 years	0 to 500 cells/µL
12 to 17 years	100 to 200 cells/µL
18+ years (Adults)	0 to 500 cells/µL (less than 0.48 x 10/L)

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 26449-9
  - Associated CPT:

4. EOS_P_A (Eosinophil percent, automated)
   - Common Name: Eosinophil percent, automated
   - Long Name: Eosinophil percent, automated
   - Test Type: EOS_P_A
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Eosinophil testing is indicated when a complete blood count (CBC) reveals abnormal levels or when a patient's symptoms suggest an eosinophil-related disorder.
     - Indications for high eosinophil counts (eosinophilia):
       * Allergic rhinitis and asthma: Eosinophils are a key driver of inflammation in allergic diseases. An elevated count can indicate the presence or severity of these conditions, especially eosinophilic asthma.
       * Eosinophilic gastrointestinal disorders: This includes conditions like eosinophilic esophagitis, gastritis, and colitis, which cause inflammation in parts of the digestive tract.
       * Skin conditions: High eosinophil levels are associated with various dermatological issues, including eczema and atopic dermatitis.

    * Drug reactions: An allergic reaction to certain medications, such as DRESS syndrome (Drug Rash with Eosinophilia and Systemic Symptoms), can cause eosinophilia.
    * Parasitic infestations: Eosinophils play a critical role in combating parasitic infections like roundworms, hookworms, and schistosomes. This is a very common cause of high eosinophil levels, particularly in tropical regions.
    * Fungal infections: Certain fungal infections, such as aspergillosis, can lead to elevated eosinophil counts.
    * Blood cancers: Some malignancies, including Hodgkin lymphoma, certain types of leukemia, and other myeloproliferative disorders, are associated with high eosinophil levels.
    * Hypereosinophilic syndrome (HES): A rare disorder characterized by extremely high eosinophil counts that can lead to organ damage.
    * Primary eosinophilia: A rare disorder involving the clonal proliferation of eosinophils, often linked to specific genetic mutations.
    * Systemic autoimmune diseases: Conditions like vasculitis (e.g., Churg-Strauss syndrome) and inflammatory bowel diseases (Crohn's disease, ulcerative colitis) are known to cause eosinophilia.
  - Indications for low eosinophil counts (eosinopenia):
    * Excess cortisol: The overproduction of cortisol, as seen in Cushing's syndrome, can lead to lower-than-normal eosinophil counts.
    * Steroid use: The use of corticosteroids can suppress eosinophil levels.
    * Sepsis: A severe, life-threatening infection of the bloodstream can cause a sharp drop in eosinophil levels.
    * Acute inflammation or stress: The body's acute inflammatory response, often from severe infection, can result in low eosinophil counts.
- Description: A type of white blood cell characterized by the presence of granules in its cytoplasm with a nucleus that usually has two lobes connected by a slender thread of chromatin. The cytoplasm contains coarse, round and uniform granules stainable by eosin.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for eosinophil percentage is typically 1% to 4% of the total white blood cell (WBC) count.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 713-8
  - Associated CPT:

5. EOS_P_M (Eosinophil percent, manual)

   - Common Name: Eosinophil percent, manual
   - Long Name: Eosinophil percent, manual
   - Test Type: EOS_P_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Eosinophil testing is indicated when a complete blood count (CBC) reveals abnormal levels or when a patient's symptoms suggest an eosinophil-related disorder.
     - Indications for high eosinophil counts (eosinophilia):
       * Allergic rhinitis and asthma: Eosinophils are a key driver of inflammation in allergic diseases. An elevated count can indicate the presence or severity of these conditions, especially eosinophilic asthma.
       * Eosinophilic gastrointestinal disorders: This includes conditions like eosinophilic esophagitis, gastritis, and colitis, which cause inflammation in parts of the digestive tract.
       * Skin conditions: High eosinophil levels are associated with various dermatological issues, including eczema and atopic dermatitis.
       * Drug reactions: An allergic reaction to certain medications, such as DRESS syndrome (Drug Rash with Eosinophilia and Systemic Symptoms), can cause eosinophilia.

        * Parasitic infestations: Eosinophils play a critical role in combating parasitic infections like roundworms, hookworms, and schistosomes. This is a very common cause of high eosinophil levels, particularly in tropical regions.
        * Fungal infections: Certain fungal infections, such as aspergillosis, can lead to elevated eosinophil counts.
        * Blood cancers: Some malignancies, including Hodgkin lymphoma, certain types of leukemia, and other myeloproliferative disorders, are associated with high eosinophil levels.
        * Hypereosinophilic syndrome (HES): A rare disorder characterized by extremely high eosinophil counts that can lead to organ damage.
        * Primary eosinophilia: A rare disorder involving the clonal proliferation of eosinophils, often linked to specific genetic mutations.
        * Systemic autoimmune diseases: Conditions like vasculitis (e.g., Churg-Strauss syndrome) and inflammatory bowel diseases (Crohn's disease, ulcerative colitis) are known to cause eosinophilia.
    - Indications for low eosinophil counts (eosinopenia):
        * Excess cortisol: The overproduction of cortisol, as seen in Cushing's syndrome, can lead to lower-than-normal eosinophil counts.
        * Steroid use: The use of corticosteroids can suppress eosinophil levels.
        * Sepsis: A severe, life-threatening infection of the bloodstream can cause a sharp drop in eosinophil levels.
        * Acute inflammation or stress: The body's acute inflammatory response, often from severe infection, can result in low eosinophil counts.
- Description: A type of white blood cell characterized by the presence of granules in its cytoplasm with a nucleus that usually has two lobes connected by a slender thread of chromatin. The cytoplasm contains coarse, round and uniform granules stainable by eosin.
- Related Tests:

- Approximate Reference Range (Each site will have slight variations): The normal reference range for eosinophil percentage is typically 1% to 4% of the total white blood cell (WBC) count.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 714-6
  - Associated CPT:

6. EOS_P (Eosinophil percent)

   - Common Name: Eosinophil percent
   - Long Name: Eosinophil percent, method not stated
   - Test Type: EOS_P
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Eosinophil testing is indicated when a complete blood count (CBC) reveals abnormal levels or when a patient's symptoms suggest an eosinophil-related disorder.
     - Indications for high eosinophil counts (eosinophilia):
       * Allergic rhinitis and asthma: Eosinophils are a key driver of inflammation in allergic diseases. An elevated count can indicate the presence or severity of these conditions, especially eosinophilic asthma.
       * Eosinophilic gastrointestinal disorders: This includes conditions like eosinophilic esophagitis, gastritis, and colitis, which cause inflammation in parts of the digestive tract.
       * Skin conditions: High eosinophil levels are associated with various dermatological issues, including eczema and atopic dermatitis.
       * Drug reactions: An allergic reaction to certain medications, such as DRESS syndrome (Drug Rash with Eosinophilia and Systemic Symptoms), can cause eosinophilia.
       * Parasitic infestations: Eosinophils play a critical role in combating parasitic infections like roundworms, hookworms, and schistosomes. This is a very common cause of high eosinophil levels, particularly in tropical regions.

        * Fungal infections: Certain fungal infections, such as aspergillosis, can lead to elevated eosinophil counts.
        * Blood cancers: Some malignancies, including Hodgkin lymphoma, certain types of leukemia, and other myeloproliferative disorders, are associated with high eosinophil levels.
        * Hypereosinophilic syndrome (HES): A rare disorder characterized by extremely high eosinophil counts that can lead to organ damage.
        * Primary eosinophilia: A rare disorder involving the clonal proliferation of eosinophils, often linked to specific genetic mutations.
        * Systemic autoimmune diseases: Conditions like vasculitis (e.g., Churg-Strauss syndrome) and inflammatory bowel diseases (Crohn's disease, ulcerative colitis) are known to cause eosinophilia.
    - Indications for low eosinophil counts (eosinopenia):
        * Excess cortisol: The overproduction of cortisol, as seen in Cushing's syndrome, can lead to lower-than-normal eosinophil counts.
        * Steroid use: The use of corticosteroids can suppress eosinophil levels.
        * Sepsis: A severe, life-threatening infection of the bloodstream can cause a sharp drop in eosinophil levels.
        * Acute inflammation or stress: The body's acute inflammatory response, often from severe infection, can result in low eosinophil counts.
- Description: A type of white blood cell characterized by the presence of granules in its cytoplasm with a nucleus that usually has two lobes connected by a slender thread of chromatin. The cytoplasm contains coarse, round and uniform granules stainable by eosin.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for eosinophil percentage is typically 1% to 4% of the total white blood cell (WBC) count.
- Comments:

- Test Method Variance:
- Other:
  - Associated LOINC: 26450-7
  - Associated CPT:

1.2.4 FERRITIN (Ferritin in blood)

- Common Name: Ferritin
- Long Name: Ferritin in blood, serum, or plasma
- Test Type: FERRITIN
- Panels: Iron Panel
- Equivalent Tests: None
- Not Equivalent Tests: Serum iron
- Indications: Ferritin testing is often used in detecting iron deficiency anemia or iron overload.
- Description: Ferritin is a globular protein complex that plays a significant role in the absorption, storage, and release of iron for the body. The amount of ferritin reflects the amount of iron stored as a soluble and non-toxic form to buffer against iron deficiency and iron overload.
- Related Tests: Serum iron, transferrin, total iron-binding capacity
- Approximate Reference Range: Males: 30-300 mcg/L; Females: 10-200 mcg/L
- Comments: A low serum ferritin value is thought to be the best laboratory indicator of iron depletion (iron deficiency). Ferritin levels are elevated in those with hemochromatosis and other excess iron storage disorders and in those who have had multiple blood transfusions. Ferritin is an acute phase reactant, therefore serum ferritin concentrations can be increased by chronic infections, fever, and inflammatory reactions. The ferritin test may be ordered, along with other iron tests, when a complete blood count (CBC) shows that a person's hemoglobin and hematocrit are low and their red blood cells are smaller and paler than normal (i.e., potentially iron deficiency anemia).

- Test Method Variance: Results can also be reported as ng/mL
- Other:
  - Associated LOINC: 2276-4, 14723-1, 20567-4, 35209-6, 14724-9, 24373-3
  - Associated CPT: 82728

1.2.5 FIBRINOGEN (Fibrinogen in platelet poor plasma)

- Common Name: Fibrinogen in platelet poor plasma
- Long Name: Fibrinogen activity in platelet poor plasma (PPP)
- Test Type: FIBRINOGEN
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Testing fibrinogen levels in a platelet-poor blood sample is indicated to assess blood clotting function and evaluate bleeding or clotting disorders. This test measures the amount and function of fibrinogen, a crucial protein produced by the liver that is converted into fibrin to form blood clots. The use of platelet-poor plasma (PPP), which has most platelets removed, is critical for accuracy because residual platelets can interfere with the test results.
- Description: A fibrinogen test on platelet-poor plasma (PPP) measures the amount or activity of the clotting protein fibrinogen (Factor I). Using PPP is essential for the accuracy of coagulation tests because it removes platelets, which are also involved in clotting and can interfere with results. This test helps diagnose bleeding disorders, evaluate abnormal clotting, and monitor conditions that affect the liver, where fibrinogen is produced. The blood sample is centrifuged (spun at high speed) to separate the cellular components, including red blood cells and platelets, from the liquid plasma. The plasma is then removed for testing.
- Related Tests:

- Approximate Reference Range (Each site will have slight variations): Normal fibrinogen reference ranges can vary between labs, but are typically between 200 and 400 mg/dL.
  - Low fibrinogen levels: These can indicate a congenital deficiency (rare), liver disease (which impairs production), malnutrition, or rapid consumption of fibrinogen due to an underlying condition like DIC.
  - High fibrinogen levels: As an acute phase reactant, fibrinogen can increase during inflammation or tissue damage. Elevated levels may be seen in cases of infection, cancer, rheumatoid arthritis, pregnancy, or in patients with coronary heart disease or stroke.
  - Discrepancy between activity and antigen tests: If the fibrinogen activity test is low but the antigen test shows a normal quantity, it may indicate a functional problem with the fibrinogen protein, known as dysfibrinogenemia.
- Comments: Does not include fibrinogen fragments codes
- Test Method Variance:
- Other:
  - Associated LOINC: 3255-7, 16859-1, 30902-1, 42568-6, 48664-7, 49045-8
  - Associated CPT: 85384

1.2.6 HGB (Hemoglobin)

- Common Name: Hemoglobin; Hgb; Hb
- Long Name: Hemoglobin
- Test Type: HGB
- Panels: Complete blood count
- Equivalent Tests: –
- Not Equivalent Tests: Red blood cell count (RBC); hematocrit (HCT); hemoglobin A1C (HGBA1C)

- Indications: To assess whether either anemia or polycythemia is present, and if so, to assess the severity of the condition. Also, hemoglobin is used to monitor response to treatment and to make decisions about the need for, and effectiveness of, transfusions.
- Description: Hemoglobin is a protein found in red blood cells. The amount of hemoglobin in the blood is an indication of the ability of the blood to deliver oxygen to organs and tissues and to transport carbon dioxide to the lungs.
- Related Tests: One component of the complete blood count (CBC) is hemoglobin. (Other components of the CBC include hematocrit, red blood cells, platelets, and white blood cells).
- Approximate Reference Range (Each site will have slight variations): 12 to 16 or 18 g/dL (12 to 18 g/100 milliliters or 12 to 18 grams/dL) among normal adults. There are separate age related reference ranges for males and female. The ranges are higher in newborns than in adults and children have slightly lower normal ranges.
- Comments: Hemoglobin increases when the number of red blood cells increases. Hemoglobin falls when production of red blood cells decreases, when there is increased destruction of red blood cells, or when there is significant bleeding. Elevated hemoglobin concentrations can be seen with dehydration, excess production of red blood cells by the bone marrow, severe lung disease, and other conditions. Low hemoglobin (anemia) can be the result of conditions such as iron, B12, or folate deficiency, inherited hemoglobin defects (e.g., sickle cell anemia or thalassemia), other inherited conditions, cirrhosis of the liver, bleeding, red cell destruction, kidney disease, aplastic anemia, other chronic illnesses, and certain cancers. Heavy smokers have higher hemoglobin levels than nonsmokers. Hemoglobin values are higher among those who live in high altitudes because the body produces more red blood cells in response to the decreased oxygen available at high altitude.
- Test Method Variance:
- Other:
  - Associated LOINC: 14775-1 (arterial; oximetry), 20509-6, 24360-0, 30313-1 (arterial), 30350-3, 30351-1, 30352-9 (capillary), 55782-7, 59260-0, 718-7

– Associated CPT: 85018, 83026

1.2.7 HCT (Hematocrit)

- Common Name: Hematocrit, Hct, Crit;
- Long Name: Hematocrit
- Test Type: HCT
- Panels: Complete blood count
- Equivalent Tests: Packed cell volume (PCV)
- Not Equivalent Tests: Hemoglobin (HBG); red blood cell count (RBC); mean corpuscular

volume (MCV)

- Indications: To assess whether anemia, polycythemia, or dehydration is present. To assess response to treatment of anemia or polycythemia. To make decisions about need for transfusion and assess effectiveness of transfusions.
- Description: The hematocrit is a measurement of the proportion of blood that is made up of red blood cells. The value is expressed as a percentage (fraction, proportion) of cells in blood. For example, a hematocrit of 36% means that there are 36 milliliters of red blood cells in 100 milliliters of blood.
- Related Tests: HGB; red blood cells; MCV; One component of the complete blood count (CBC) is hematocrit. (Other components of the CBC include hemoglobin, red blood cells, platelets, and white blood cells).
- Approximate Reference Range (Each site will have slight variations): 36% to 48% (usually a higher reference range for higher altitudes)
- Comments: The hematocrit rises when the number of red blood cells increases or when the plasma volume is reduced such as with dehydration. Elevated hematocrit can also indicate polycythemia vera. The hematocrit falls when the body decreases its production of red blood cells, increases destruction of red blood cells, or when there is blood loss (e.g., with bleeding). Decreased hematocrit indicates anemia. The

hematocrit reflects both the number and the volume of red blood cells. Hematocrit values are higher among those who live in high altitudes because of an increase in the number of red blood cells (i.e., the body produces more red blood cells in response to the decreased oxygen available at high altitude). See also the comments for the related test hemoglobin.

- Test Method Variance: –
- Other: –

1.2.8 INR (International Normalization Ratio)

- Common Name: INR
- Long Name: International Normalization Ratio
- Test Type: INR
- Panels: –
- Equivalent Tests: –
- Not Equivalent Tests: Prothrombin time [PT]; activated clotting time [ACT]; partial thromboplastin time [PTT]
- Indications: To monitor the effectiveness of anti-coagulants such as warfarin which are prescribed to prevent the formation of blood clots.
- Description: The INR was created to standardize PT readings which vary from lab to lab. It is calculated by taking the PT ratio and exponeniating it by the international sensitivity index of the thromboplastin used to measure PT.
- Related Tests: PT; ACT; PTT; platelet count, platelet function test
- Approximate Reference Range: The goal of therapy for most patients is an INR of 2.0 to 3.0. The goal INR for some patients with a high risk of clot formation is 2.5 to 3.5.
- Comments: Patients' warfarin therapy must be titrated to reach these therapeutic ranges. Therefore, INR values may vary from higher or lower than the recommended ranges as regimens are being titrated or adjusted. The risk of bleeding increases as INR increases.

- Test Method Variance:
- Other:
  - Associated LOINC: 6301-6, 34714-6, 46418-0
  - Associated CPT: 85610

1.2.9 (LYMPH) Lymphocyte Test Types

1. LYMPH_C_A (Lymphocyte number, automated count)
   - Common Name: Lymphocyte number, automated count
   - Long Name: Lymphocyte number, automated count
   - Test Type: LYMPH_C_A
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A lymphocyte count is indicated to assess the immune system and diagnose or monitor various conditions, including infections, autoimmune disorders, and blood cancers. It is commonly ordered as part of a complete blood count (CBC) with a differential.
   - Description: A lymphocyte count test is a measurement of the number of lymphocytes in a blood sample. These white blood cells are a vital part of your immune system and help protect the body from infection and disease.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): The normal absolute lymphocyte count (ALC) for adults is generally between 1,000 and 4,800 lymphocytes per microliter. Children's lymphocyte counts are naturally higher than adults', especially in younger children, as their immune systems are actively developing. A normal count for children can be between 3,000 and 9,500 lymphocytes per microliter.
   - Comments:
   - Test Method Variance:
   - Other:

  - Associated LOINC: 731-0
  - Associated CPT:

2. LYMPH_C_M (Lymphocyte number, manual count)

   - Common Name: Lymphocyte number, manual count
   - Long Name: Lymphocyte number, manual count
   - Test Type: LYMPH_C_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A lymphocyte count is indicated to assess the immune system and diagnose or monitor various conditions, including infections, autoimmune disorders, and blood cancers. It is commonly ordered as part of a complete blood count (CBC) with a differential.
   - Description: A lymphocyte count test is a measurement of the number of lymphocytes in a blood sample. These white blood cells are a vital part of your immune system and help protect the body from infection and disease.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): The normal absolute lymphocyte count (ALC) for adults is generally between 1,000 and 4,800 lymphocytes per microliter. Children's lymphocyte counts are naturally higher than adults', especially in younger children, as their immune systems are actively developing. A normal count for children can be between 3,000 and 9,500 lymphocytes per microliter.
   - Comments:
   - Test Method Variance:
   - Other:
     - Associated LOINC: 732-8
     - Associated CPT:

3. LYMPH_C (Lymphocyte number)

   - Common Name: Lymphocyte number

- Long Name: Lymphocyte number, count method not stated
- Test Type: LYMPH_C
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A lymphocyte count is indicated to assess the immune system and diagnose or monitor various conditions, including infections, autoimmune disorders, and blood cancers. It is commonly ordered as part of a complete blood count (CBC) with a differential.
- Description: A lymphocyte count test is a measurement of the number of lymphocytes in a blood sample. These white blood cells are a vital part of your immune system and help protect the body from infection and disease.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal absolute lymphocyte count (ALC) for adults is generally between 1,000 and 4,800 lymphocytes per microliter. Children's lymphocyte counts are naturally higher than adults', especially in younger children, as their immune systems are actively developing. A normal count for children can be between 3,000 and 9,500 lymphocytes per microliter.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 26474-7
  - Associated CPT:

4. LYMPH_P_A (Lymphocyte percent, automated)

- Common Name: Lymphocyte percent, automated
- Long Name: Lymphocyte percent, automated
- Test Type: LYMPH_P_A
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: A lymphocyte percentage test, or relative lymphocyte count, is a component of a complete blood count (CBC) with differential. Changes in the percentage of lymphocytes can reveal an underlying problem with the immune system.
  - An increased percentage of lymphocytes is a common sign of a viral infection, such as infectious mononucleosis (Epstein-Barr virus), measles, mumps, or hepatitis. It can also occur with some bacterial infections, like whooping cough or tuberculosis.
  - A decreased percentage can be caused by certain viral infections, including HIV and influenza, especially in severe or later stages.
- Description: A lymphocyte percentage test measures the proportion of lymphocytes relative to the total number of white blood cells (WBCs) in a blood sample. The test is typically part of a complete blood count (CBC) with a differential, which provides a detailed breakdown of all the different types of WBCs.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): In adults, the typical reference range for a lymphocyte percentage is 20% to 40% of the total white blood cell (WBC) count. This measurement is also known as a lymphocyte differential.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 736-9
  - Associated CPT:

5. LYMPH_P_M (Lymphocyte percent, manual)

- Common Name: Lymphocyte percent, manual
- Long Name: Lymphocyte percent, manual
- Test Type: LYMPH_P_M
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: A lymphocyte percentage test, or relative lymphocyte count, is a component of a complete blood count (CBC) with differential. Changes in the percentage of lymphocytes can reveal an underlying problem with the immune system.
  - An increased percentage of lymphocytes is a common sign of a viral infection, such as infectious mononucleosis (Epstein-Barr virus), measles, mumps, or hepatitis. It can also occur with some bacterial infections, like whooping cough or tuberculosis.
  - A decreased percentage can be caused by certain viral infections, including HIV and influenza, especially in severe or later stages.
- Description: A lymphocyte percentage test measures the proportion of lymphocytes relative to the total number of white blood cells (WBCs) in a blood sample. The test is typically part of a complete blood count (CBC) with a differential, which provides a detailed breakdown of all the different types of WBCs.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): In adults, the typical reference range for a lymphocyte percentage is 20% to 40% of the total white blood cell (WBC) count. This measurement is also known as a lymphocyte differential.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 737-7
  - Associated CPT:

6. LYMPH_P (Lymphocyte percent)

- Common Name: Lymphocyte percent
- Long Name: Lymphocyte percent, method not stated
- Test Type: LYMPH_P
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: A lymphocyte percentage test, or relative lymphocyte count, is a component of a complete blood count (CBC) with differential. Changes in the percentage of lymphocytes can reveal an underlying problem with the immune system.
  - An increased percentage of lymphocytes is a common sign of a viral infection, such as infectious mononucleosis (Epstein-Barr virus), measles, mumps, or hepatitis. It can also occur with some bacterial infections, like whooping cough or tuberculosis.
  - A decreased percentage can be caused by certain viral infections, including HIV and influenza, especially in severe or later stages.
- Description: A lymphocyte percentage test measures the proportion of lymphocytes relative to the total number of white blood cells (WBCs) in a blood sample. The test is typically part of a complete blood count (CBC) with a differential, which provides a detailed breakdown of all the different types of WBCs.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): In adults, the typical reference range for a lymphocyte percentage is 20% to 40% of the total white blood cell (WBC) count. This measurement is also known as a lymphocyte differential.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 26478-8
  - Associated CPT:

1.2.10 (MONO) Monocyte Test Types

1. MONO_C_A (Monocyte number, automated count)
   - Common Name: Monocyte number, automated count
   - Long Name: Monocyte number, automated count
   - Test Type: MONO_C_A
   - Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A high monocyte count can signal that the body is fighting off an infection or inflammation. A low monocyte count may indicate issues with bone marrow production or a suppressed immune system.
- Description: A monocyte count is a blood test that measures the number of monocytes, a type of white blood cell, in a blood sample. Monocytes are identified by their distinct size and the expression of specific surface markers, particularly CD14, and are classified into classical, intermediate, and non-classical subsets based on the levels of CD14 and CD16 expression detected through flow cytometry. While historically identified by their morphology (large size, indented or kidney-shaped nucleus, and blue-gray cytoplasm), modern identification relies on flow cytometry to analyze the levels of these cell surface markers.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For adults, the normal range of monocyte absolute count of approximately 200 to 800 monocytes per microliter of blood.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 742-7
  - Associated CPT:

2. MONO_C_M (Monocyte number, manual count)
   - Common Name: Monocyte number, manual count
   - Long Name: Monocyte number, manual count
   - Test Type: MONO_C_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A high monocyte count can signal that the body is fighting off an infection or inflammation. A low monocyte count may indicate issues with bone marrow production or a suppressed immune system.

- Description: A monocyte count is a blood test that measures the number of monocytes, a type of white blood cell, in a blood sample. Monocytes are identified by their distinct size and the expression of specific surface markers, particularly CD14, and are classified into classical, intermediate, and non-classical subsets based on the levels of CD14 and CD16 expression detected through flow cytometry. While historically identified by their morphology (large size, indented or kidney-shaped nucleus, and blue-gray cytoplasm), modern identification relies on flow cytometry to analyze the levels of these cell surface markers.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For adults, the normal range of monocyte absolute count of approximately 200 to 800 monocytes per microliter of blood.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 743-5
  - Associated CPT:

3. MONO_C (Monocyte number)
   - Common Name: Monocyte number
   - Long Name: Monocyte number, count method not stated
   - Test Type: MONO_C
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A high monocyte count can signal that the body is fighting off an infection or inflammation. A low monocyte count may indicate issues with bone marrow production or a suppressed immune system.
   - Description: A monocyte count is a blood test that measures the number of monocytes, a type of white blood cell, in a blood sample. Monocytes are identified by their distinct size and the expression of specific surface markers, particularly CD14, and are classified into classical, intermediate, and non-classical subsets based

on the levels of CD14 and CD16 expression detected through flow cytometry. While historically identified by their morphology (large size, indented or kidney-shaped nucleus, and blue-gray cytoplasm), modern identification relies on flow cytometry to analyze the levels of these cell surface markers.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For adults, the normal range of monocyte absolute count of approximately 200 to 800 monocytes per microliter of blood.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 26484-6
  - Associated CPT:

4. MONO_P_A (Monocyte percent, automated)

   - Common Name: Monocyte percent, automated
   - Long Name: Monocyte percent, automated
   - Test Type: MONO_P_A
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A monocyte percentage test is typically ordered as part of a complete blood count (CBC) with differential, which measures the number of different white blood cells. High monocytes can signal that the body is fighting an acute or chronic infection, responding to inflammation, or dealing with other medical conditions. Low monocyte levels can be caused by conditions or treatments that suppress the immune system or affect the bone marrow.
   - Description: A monocyte percentage test is a component of a standard blood differential test, which is often included in a complete blood count (CBC). This test measures the proportion of monocytes, a type of white blood cell (WBC), relative to the total number of WBCs in a blood sample.
   - Related Tests:

- Approximate Reference Range (Each site will have slight variations): For adults, the normal range for a monocyte count is 2% to 8% of the total white blood cell (WBC) count.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 5905-5
  - Associated CPT:

5. MONO_P_M (Monocyte percent, manual)
   - Common Name: Monocyte percent, manual
   - Long Name: Monocyte percent, manual
   - Test Type: MONO_P_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A monocyte percentage test is typically ordered as part of a complete blood count (CBC) with differential, which measures the number of different white blood cells. High monocytes can signal that the body is fighting an acute or chronic infection, responding to inflammation, or dealing with other medical conditions. Low monocyte levels can be caused by conditions or treatments that suppress the immune system or affect the bone marrow.
   - Description: A monocyte percentage test is a component of a standard blood differential test, which is often included in a complete blood count (CBC). This test measures the proportion of monocytes, a type of white blood cell (WBC), relative to the total number of WBCs in a blood sample.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): For adults, the normal range for a monocyte count is 2% to 8% of the total white blood cell (WBC) count.
   - Comments:
   - Test Method Variance:

- Other:
  - Associated LOINC: 744-3
  - Associated CPT:

6. MONO_P (Monocyte percent)

   - Common Name: Monocyte percent
   - Long Name: Monocyte percent, method not stated
   - Test Type: MONO_P
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A monocyte percentage test is typically ordered as part of a complete blood count (CBC) with differential, which measures the number of different white blood cells. High monocytes can signal that the body is fighting an acute or chronic infection, responding to inflammation, or dealing with other medical conditions. Low monocyte levels can be caused by conditions or treatments that suppress the immune system or affect the bone marrow.
   - Description: A monocyte percentage test is a component of a standard blood differential test, which is often included in a complete blood count (CBC). This test measures the proportion of monocytes, a type of white blood cell (WBC), relative to the total number of WBCs in a blood sample.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): For adults, the normal range for a monocyte count is 2% to 8% of the total white blood cell (WBC) count.
   - Comments:
   - Test Method Variance:
   - Other:
     - Associated LOINC: 26485-3
     - Associated CPT:

1.2.11 (NTBAND) Band Neutrophil Test Types

1. NTBAND_C_A (Band neutrophil number, automated count)
    - Common Name: Band neutrophil number, automated count
    - Long Name: Band neutrophil number, automated count
    - Test Type: NTBAND_C_A
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: This test is ordered in the following situations:
        - A suspicion of bacterial infection
        - An assessment of sepsis
        - Monitoring patients with bone marrow issues
        - Evaluating febrile infants and children
        - Monitoring recovery from chemotherapy
    - Description: An automated band neutrophil number test is a lab measurement of the concentration of immature neutrophils in the blood, calculated as part of an automated complete blood count (CBC). Automated analyzers report the percentage and absolute number of immature granulocytes (IG), which often includes band neutrophils.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): In a healthy person there should be no band neutrophils.
    - Comments: Ordering an automated band neutrophil count has become increasingly limited in clinical practice due to its inaccuracy and imprecision compared to manual counting and other inflammatory markers. Can only use for ANC if combined with Ntseg and Ntimmature. Although this summing of segs, bands and immature might be possible it is NOT recommended. (34524-9 is qualitative)
    - Test Method Variance:
    - Other:
        - Associated LOINC: 30229-9, 34524-9
        - Associated CPT:

2. NTBAND_C_M (Band neutrophil number, manual count)

   - Common Name: Band neutrophil number, manual count
   - Long Name: Band neutrophil number, manual count
   - Test Type: NTBAND_C_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: This test is ordered in the following situations:
     - A suspicion of bacterial infection
     - An assessment of sepsis
     - Monitoring patients with bone marrow issues
     - Evaluating febrile infants and children
     - Monitoring recovery from chemotherapy
   - Description: A manual band neutrophil count is a lab test that involves the microscopic examination of a blood sample to determine the percentage of immature neutrophils, known as band neutrophils. An elevated number of band neutrophils, or "bandemia," is a key indicator of infection or inflammation. The body's bone marrow produces neutrophils, a type of white blood cell (WBC), to fight infection. In response to an infection, the bone marrow ramps up production and releases immature neutrophils (bands) into the bloodstream prematurely. This phenomenon is known as a "left shift" and suggests the body is fighting a bacterial infection or another inflammatory process.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): In a healthy person there should be no band neutrophils.
   - Comments: Can only use for ANC if combined with Ntseg and Ntimmature. Although this summing of segs, bands and immature might be possible it is NOT recommended.
   - Test Method Variance:
   - Other:
     - Associated LOINC: 763-3
     - Associated CPT:

3. NTBAND_C (Band neutrophil number)

    - Common Name: Band neutrophil number
    - Long Name: Band neutrophil number, count method unknown
    - Test Type: NTBAND_C
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: This test is ordered in the following situations:
        - A suspicion of bacterial infection
        - An assessment of sepsis
        - Monitoring patients with bone marrow issues
        - Evaluating febrile infants and children
        - Monitoring recovery from chemotherapy
    - Description: A band neutrophil test, part of a blood differential count, measures immature neutrophils, which are a type of white blood cell that fights infection. An elevated number of band neutrophils can indicate the body is fighting an infection or dealing with inflammation, as the bone marrow is releasing these young cells into the bloodstream. This is often called a "left shift".
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): In a healthy person there should be no band neutrophils.
    - Comments: Can only use for ANC if combined with Ntseg and Ntimmature. Although this summing of segs, bands and immature might be possible it is NOT recommended.
    - Test Method Variance:
    - Other:
        - Associated LOINC: 26507-4
        - Associated CPT:

4. NTBAND_P_A (Band neutrophil percent, automated)

    - Common Name: Band neutrophil percent, automated
    - Long Name: Band neutrophil percent, automated
    - Test Type: NTBAND_P_A

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: This test is ordered in the following situations:
  - A suspicion of bacterial infection
  - An assessment of sepsis
  - Monitoring patients with bone marrow issues
  - Evaluating febrile infants and children
  - Monitoring recovery from chemotherapy
- Description: A band neutrophil test measures the percentage of immature neutrophils in a blood sample, indicating the body's immune response to infection or inflammation. An automated percent result shows this percentage calculated by a machine, with normal values typically between 0% and 6%. A result higher than normal, called a left shift or bandemia, suggests the body is releasing immature neutrophils before they are fully mature, which can be triggered by bacterial infections, inflammation, or physical stress.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): In a healthy person there should be 0% band neutrophils. Up to 3% can still be considered within a typical range.
- Comments: Can only use to calculate ANC if Ntseg and Ntimmature percentages are also used. Although this calculation using segs, bands and immature might be possible it is NOT recommended.
- Test Method Variance:
- Other:
  - Associated LOINC: 35332-6
  - Associated CPT:

5. NTBAND_P_M (Band neutrophil percent, manual)
   - Common Name: Band neutrophil percent, manual
   - Long Name: Band neutrophil percent, manual
   - Test Type: NTBAND_P_M

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: This test is ordered in the following situations:
  - A suspicion of bacterial infection
  - An assessment of sepsis
  - Monitoring patients with bone marrow issues
  - Evaluating febrile infants and children
  - Monitoring recovery from chemotherapy
- Description: A manual band neutrophil percentage test is performed through a microscopic examination of a stained blood smear, which is part of a complete manual white blood cell (WBC) differential count. Band neutrophils are immature WBCs with a horseshoe-shaped nucleus, unlike mature segmented neutrophils, which have a segmented, lobular nucleus. An increased percentage of band neutrophils is known as a "left shift" and often indicates a bacterial infection or significant inflammation.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): In a healthy person there should be 0% band neutrophils. Up to 3% can still be considered within a typical range.
- Comments: Can only use to calculate ANC if Ntseg and Ntimmature percentages are also used. Although this calculation using segs, bands and immature might be possible it is NOT recommended.
- Test Method Variance:
- Other:
  - Associated LOINC: 764-1, 71670-4
  - Associated CPT:

6. NTBAND_P (Band neutrophil percent)

- Common Name: Band neutrophil percent
- Long Name: Band neutrophil percent, method unknown
- Test Type: NTBAND_P
- Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: This test is ordered in the following situations:
  - A suspicion of bacterial infection
  - An assessment of sepsis
  - Monitoring patients with bone marrow issues
  - Evaluating febrile infants and children
  - Monitoring recovery from chemotherapy
- Description: The band neutrophil percentage is a lab test that measures the proportion of immature neutrophils, a type of white blood cell, in your bloodstream. This test is a component of a complete blood count (CBC) with differential, which provides a breakdown of all the different white blood cell types.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): In a healthy person there should be 0% band neutrophils. Up to 3% can still be considered within a typical range.
- Comments: Can only use to calculate ANC if Ntseg and Ntimmature percentages are also used. Although this calculation using segs, bands and immature might be possible it is NOT recommended.
- Test Method Variance:
- Other:
  - Associated LOINC: 26508-2
  - Associated CPT:

1.2.12 (NTHYP) Hypersegmented Neutrophil Test Types

1. NTHYP_C_A (Hypersegmented neutrophil number, automated count)

- Common Name: Hypersegmented neutrophil number, automated count
- Long Name: Hypersegmented neutrophil number, automated count
- Test Type: NTHYP_C_A
- Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The presence of hypersegmented neutrophils is a hallmark diagnostic feature for several conditions: Megaloblastic Anemia, Medication Side Effects, Myelodysplatic Syndromes, and Chronic Inflammation.
- Description: Hypersegmented neutrophils are mature white blood cells with an abnormally high number of nuclear segments (lobes). While automated hematology analyzers perform the initial count, definitive identification of these cells typically requires a manual microscopic review. An automated Complete Blood Count (CBC) with differential uses flow cytometry or computerized analyzers to count and categorize white blood cells.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): A neutrophil is classified as hypersegmented if it meets either of the following criteria:
  - A single neutrophil with 6 or more nuclear lobes
  - More than 3% to 5% of neutrophils in a sample having 5 or more lobes
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 34199-0
  - Associated CPT:

2. NTHYP_C_M (Hypersegmented neutrophil number, manual count)
   - Common Name: Hypersegmented neutrophil number, manual count
   - Long Name: Hypersegmented neutrophil number, manual count
   - Test Type: NTHYP_C_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:

- Indications: A manual differential blood smear is typically ordered to look for hypersegmented neutrophils when a patient exhibits:
  - Suspected Megaloblastic Anemia: Often caused by Vitamin B12 or Folate deficiency.
  - Macrocytosis: An elevated Mean Corpuscular Volume indicating unusually large red blood cells.
  - Pancytopenia: Unexplained low counts across all blood cell types (red cells, white cells, and platelets).
  - Drug Toxicity: Monitoring patients on medications like Methotrexate or Hydroxyurea, which interfere with DNA synthesis.
  - Myelodysplastic Syndromes: Assessing for bone marrow maturation defects or clonal disorders.
- Description: A microscopic lab test where a pathologist examines a stained blood smear, counting neutrophils with six or more nuclear lobes, indicating abnormal maturation, most strongly suggesting megaloblastic anemia.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): If more than 3% to 5% of neutrophils have five lobes.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 766-6
  - Associated CPT: 85007

3. NTHYP_C (Hypersegmented neutrophil number)

- Common Name: Hypersegmented neutrophil number
- Long Name: Hypersegmented neutrophil number, method not specified or other method
- Test Type: NTHYP_C
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: Hypersegmented neutrophils contain more than six nuclear lobes are a key sign of megaloblastic anemia (B12/folate deficiency), but also point to myelodysplastic syndromes (MDS), certain medications (affecting DNA), chronic liver issues, and rare genetic disorders, often accompanied by large red blood cells (macrocytes).
- Description: Identifies white blood cells with an abnormally high number of nuclear lobes.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): While normal, mature neutrophils contain 3 to 5 lobes, hypersegmented cells are characterized by having 6 or more lobes, or a significant percentage (over 3%–5%) having 5 or more lobes.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 30449-3, 765-8
  - Associated CPT:

4. NTHYP_P_A (Hypersegmented neutrophil percent, automated) ***NEEDS TO BE UPDATED*** There are no LOINCs that have been identified for this Test Type.

5. NTHYP_P_M (Hypersegmented neutrophil percent, manual)

   - Common Name: Hypersegmented neutrophil percent, manual
   - Long Name: Hypersegmented neutrophil percent, manual
   - Test Type: NTHYP_P_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Hypersegmented neutrophils contain more than six nuclear lobes are a key sign of megaloblastic anemia (B12/folate deficiency), but also point to myelodysplastic syndromes (MDS), certain medications (affecting DNA), chronic liver issues, and rare genetic disorders, often accompanied by large red blood cells (macrocytes).

- Description: Measures the proportion of mature white blood cells (neutrophils) that have an abnormally high number of segments (lobes) in their nucleus.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The total count of neutrophils in a standard test typically falls within these ranges:
  - Absolute Neutrophil Count (ANC): 1,500 to 7,700 cells/µL.
  - Relative Percentage: 40% to 70% of total white blood cells.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 767-4
  - Associated CPT: 85007

6. NTHYP_P (Hypersegmented neutrophil percent, method unknown)

- Common Name: Hypersegmented neutrophil percent
- Long Name: Hypersegmented neutrophil percent, method unknown
- Test Type: NTHYP_P
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Hypersegmented neutrophils contain more than six nuclear lobes are a key sign of megaloblastic anemia (B12/folate deficiency), but also point to myelodysplastic syndromes (MDS), certain medications (affecting DNA), chronic liver issues, and rare genetic disorders, often accompanied by large red blood cells (macrocytes).
- Description: Measures the proportion of mature white blood cells (neutrophils) that have an abnormally high number of segments (lobes) in their nucleus.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The total count of neutrophils in a standard test typically falls within these ranges:

  - Absolute Neutrophil Count (ANC): 1,500 to 7,700 cells/µL.
  - Relative Percentage: 40% to 70% of total white blood cells.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 30450-1
  - Associated CPT:

1.2.13 (NTIMMATURE) Immature Neutrophil Test Types

1. NTIMMATURE_C_A (Immature neutrophil number, automated count) ***UNDER REVIEW*** There are no LOINCs associated with this test type.

2. NTIMMATURE_C_M (Immature neutrophil number, manual count) ***UNDER REVIEW*** There are no LOINCs associated with this test type.

3. NTIMMATURE_C (Immature neutrophil number, count method unknown)

   - Common Name: Immature neutrophil number
   - Long Name: Immature neutrophil number, count method unknown
   - Test Type: NTIMMATURE_C
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Mature neutrophils are the body's primary defense against infection, their immature counterparts are normally confined to the bone marrow.
   - Description: Measures young neutrophils (like bands, metamyelocytes, myelocytes) in your blood, indicating the bone marrow's stress or release of reserves, typically due to infection or inflammation, revealing a "left shift" that helps diagnose issues like sepsis or bone marrow disorders.
   - Related Tests:

- Approximate Reference Range (Each site will have slight variations): Immature neutrophils are usually measured in two ways:
  - Band Neutrophils (Bands):
    * Percentage: Typically 0% to 3% of total white blood cells.
    * Absolute Count: Generally less than 0.7 x 10/L (or <700 cells/µL).
  - Immature Granulocytes (IG):
    * This category includes earlier stages than bands (metamyelocytes, myelocytes, and promyelocytes).
    * Percentage: Usually 0% to 1.0%.
    * Absolute Count: Often reported as 0.00 to 0.04 x 10^3/µL.

Clinical Significance

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 26523-1, 30433-7
  - Associated CPT:

4. NTIMMATURE_P_A (Immature neutrophil percent, automated) ***UNDER REVIEW*** There are no LOINCs associated with this test type.

5. NTIMMATURE_P_M (Immature neutrophil percent, manual) ***UNDER REVIEW***

   - Common Name: Immature neutrophil percent, manual
   - Long Name: Immature neutrophil percent, manual
   - Test Type: NTIMMATURE_P_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Mature neutrophils are the body's primary defense against infection, their immature counterparts are normally confined to the bone marrow.

- Description: Measures young neutrophils (like bands, metamyelocytes, myelocytes) in your blood, indicating the bone marrow's stress or release of reserves, typically due to infection or inflammation, revealing a "left shift" that helps diagnose issues like sepsis or bone marrow disorders.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Immature neutrophils are usually measured in two ways:
  - Band Neutrophils (Bands):
    * Percentage: Typically 0% to 3% of total white blood cells.
    * Absolute Count: Generally less than 0.7 x 10/L (or <700 cells/µL).
  - Immature Granulocytes (IG):
    * This category includes earlier stages than bands (metamyelocytes, myelocytes, and promyelocytes).
    * Percentage: Usually 0% to 1.0%.
    * Absolute Count: Often reported as 0.00 to 0.04 x 10^3/µL.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 74425-0, 74428-4
  - Associated CPT:

6. NTIMMATURE_P (Immature neutrophil percent, method unknown)

- Common Name: Immature neutrophil percent
- Long Name: Immature neutrophil percent, method unknown
- Test Type: NTIMMATURE_P
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Mature neutrophils are the body's primary defense against infection, their immature counterparts are normally confined to the bone marrow.

- Description: Measures young neutrophils (like bands, metamyelocytes, myelocytes) in your blood, indicating the bone marrow's stress or release of reserves, typically due to infection or inflammation, revealing a "left shift" that helps diagnose issues like sepsis or bone marrow disorders.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Immature neutrophils are usually measured in two ways:
  - Band Neutrophils (Bands):
    - Percentage: Typically 0% to 3% of total white blood cells.
    - Absolute Count: Generally less than 0.7 x 10/L (or <700 cells/µL).
  - Immature Granulocytes (IG):
    - This category includes earlier stages than bands (metamyelocytes, myelocytes, and promyelocytes).
    - Percentage: Usually 0% to 1.0%.
    - Absolute Count: Often reported as 0.00 to 0.04 x 10^3/µL.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 53797-7, 26498-6, 26524-9, 28541-1
  - Associated CPT:

1.2.14 (NTSEG) Segmented Neutrophil Test Types

1. NTSEG_C_A (Segmented neutrophil number, automated count) ***UNDER REVIEW*** There are no LOINCs associated with this test type.
2. NTSEG_C_M (Segmented neutrophil number, manual count)
   - Common Name: Segmented neutrophil number, manual count
   - Long Name: Segmented neutrophil number, manual count
   - Test Type: NTSEG_C_M
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:

- Indications: Healthcare providers order this test when a patient exhibits signs of an active immune response:
  - Systemic symptoms: Persistent fever, shaking chills, and night sweats.
  - Respiratory signs: Rapid pulse or rapid, shallow breathing.
  - Gastrointestinal distress: Nausea, vomiting, or chronic diarrhea.
  - Localized issues: Wounds that do not heal or the presence of an abscess.
- Description: A Segmented Neutrophil lab test measures the number of mature neutrophils in your blood. Segmented neutrophils, often called "segs" or "polys," are the primary white blood cells (WBCs) responsible for fighting bacterial and fungal infections.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The reference range for segmented neutrophils (mature neutrophils) in adults is typically an absolute count of 1,500 to 8,000 cells/µL (or x 10^9/L)
- Comments: Can only use for ANC if combined with Ntband and Ntimmature counts. Although this summing of segs, bands and immature might be possible it is NOT recommended.
- Test Method Variance:
- Other:
  - Associated LOINC: 768-2
  - Associated CPT: 85007, 85009, 85032

3. NTSEG_C (Segmented neutrophil number, count method unknown)
   - Common Name: Segmented neutrophil number
   - Long Name: Segmented neutrophil number, count method unknown
   - Test Type: NTSEG_C
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Healthcare providers order this test when a patient exhibits signs of an active immune response:

  - Systemic symptoms: Persistent fever, shaking chills, and night sweats.
  - Respiratory signs: Rapid pulse or rapid, shallow breathing.
  - Gastrointestinal distress: Nausea, vomiting, or chronic diarrhea.
  - Localized issues: Wounds that do not heal or the presence of an abscess.
- Description: A Segmented Neutrophil lab test measures the number of mature neutrophils in your blood. Segmented neutrophils, often called "segs" or "polys," are the primary white blood cells (WBCs) responsible for fighting bacterial and fungal infections.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The reference range for segmented neutrophils (mature neutrophils) in adults is typically an absolute count of 1,500 to 8,000 cells/µL (or x 10^9/L)
- Comments: Can only use for ANC if combined with Ntband and Ntimmature counts. Although this summing of segs, bands and immature might be possible it is NOT recommended.
- Test Method Variance:
- Other:
  - Associated LOINC: 30451-9
  - Associated CPT: 85048

4. NTSEG_P_A (Segmented neutrophil percent, automated)

- Common Name: Segmented neutrophil percent, automated
- Long Name: Segmented neutrophil percent, automated
- Test Type: NTSEG_P_A
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Healthcare providers order this test when a patient exhibits signs of an active immune response:
  - Systemic symptoms: Persistent fever, shaking chills, and night sweats.

  - Respiratory signs: Rapid pulse or rapid, shallow breathing.
  - Gastrointestinal distress: Nausea, vomiting, or chronic diarrhea.
  - Localized issues: Wounds that do not heal or the presence of an abscess.
- Description: A Segmented Neutrophil lab test measures the percentage of mature neutrophils in your blood. Segmented neutrophils, often called "segs" or "polys," are the primary white blood cells (WBCs) responsible for fighting bacterial and fungal infections.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The reference range for segmented neutrophils (mature neutrophils) in adults is typically 40-60% of total white blood cells.
- Comments: Can only use for ANC if combined with Ntband and Ntimmature counts. Although this summing of segs, bands and immature might be possible it is NOT recommended.
- Test Method Variance:
- Other:
  - Associated LOINC: 32200-8, 770-8, 19023-1
  - Associated CPT: 85048, 85004, 85025

5. NTSEG_P_M (Segmented neutrophil percent, manual)

- Common Name: Segmented neutrophil percent, manual
- Long Name: Segmented neutrophil percent, manual
- Test Type: NTSEG_P_M
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Healthcare providers order this test when a patient exhibits signs of an active immune response:
  - Systemic symptoms: Persistent fever, shaking chills, and night sweats.
  - Respiratory signs: Rapid pulse or rapid, shallow breathing.

  - Gastrointestinal distress: Nausea, vomiting, or chronic diarrhea.
  - Localized issues: Wounds that do not heal or the presence of an abscess.
- Description: A Segmented Neutrophil lab test measures the percentage of mature neutrophils in your blood. Segmented neutrophils, often called "segs" or "polys," are the primary white blood cells (WBCs) responsible for fighting bacterial and fungal infections.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The reference range for segmented neutrophils (mature neutrophils) in adults is typically 40-60% of total white blood cells.
- Comments: Can only use for ANC if combined with Ntband and Ntimmature counts. Although this summing of segs, bands and immature might be possible it is NOT recommended.
- Test Method Variance:
- Other:
  - Associated LOINC: 769-0, 23761-0, 31160-5
  - Associated CPT: 85007

6. NTSEG_P (Segmented neutrophil percent, method unknown)

- Common Name: Segmented neutrophil percent
- Long Name: Segmented neutrophil percent, method unknown
- Test Type: NTSEG_P
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Healthcare providers order this test when a patient exhibits signs of an active immune response:
  - Systemic symptoms: Persistent fever, shaking chills, and night sweats.
  - Respiratory signs: Rapid pulse or rapid, shallow breathing.
  - Gastrointestinal distress: Nausea, vomiting, or chronic diarrhea.

  - Localized issues: Wounds that do not heal or the presence of an abscess.
- Description: A Segmented Neutrophil lab test measures the percentage of mature neutrophils in your blood. Segmented neutrophils, often called "segs" or "polys," are the primary white blood cells (WBCs) responsible for fighting bacterial and fungal infections.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The reference range for segmented neutrophils (mature neutrophils) in adults is typically 40-60% of total white blood cells.
- Comments: Can only use for ANC if combined with Ntband and Ntimmature counts. Although this summing of segs, bands and immature might be possible it is NOT recommended.
- Test Method Variance:
- Other:
  - Associated LOINC: 26505-8, 34999-3, 30395-8, 26511-6
  - Associated CPT: 85048

1.2.15 PLATELETS (Platelet count in blood)

- Common Name: Platelet count; thrombocyte count; PLT
- Long Name: Platelets
- Test Type: PLATELETS
- Panels: Complete blood count
- Equivalent Tests:
- Not Equivalent Tests: Platelet aggregation; other tests of platelet function
- Indications: Platelet counts are used to diagnose bleeding disorders and diseases of the bone marrow. Platelet counts are ordered when an individual has unexplained bruising. Platelet counts are also used when an individual seems to take an excessive amount of time to stop bleeding from cuts and wounds that are not large.

- Description: The platelet count is used to determine the number of platelets in your blood. Platelets, also known as thrombocytes, are small pieces of larger cells known as megakarycytes. Megakarycytes are made in the bone marrow. Platelets are released from the bone marrow and circulate in the blood. Platelets are the "first responders" when there is injury to a tissue or blood vessel. Platelets begin the formation of a blood clot in response to injury.

- Related Tests: Platelet aggregation; One component of the complete blood count (CBC) is platelets. (Other components of the CBC include hemoglobin, red blood cells, hematocrit, and white blood cells).

- Approximate Reference Range (Each site will have slight variations): 150,000 to 450,000 per microliter (mcl or L or 10-6/Liter) of blood.

- Comments: Platelet counts are used to help diagnose bleeding disorders and bone marrow diseases such as leukemia. Other causes of low platelets include autoimmune diseases, drug reactions, chemotherapy, diseases such as thrombocytopenic purpura (TTP), heparin induced thrombocytopenia, hemolytic uremia syndrome, and pooling of platelets in the spleen. When platelet counts fall below 10,000 – 20,000 per microliter, bleeding can occur even in the absence of apparent injury. Very low platelet counts are called thrombocytopenia. Platelet counts can also be too high (aka thrombocytosis). Thrombocytosis sometimes occurs without any other medical condition or with myeloproliferative disorders. The platelets in individuals with myeloproliferative disorders often do not function normally.

- Test Method Variance:

- Other:

  - Associated LOINC: 777-3, 24361-8, 26515-7, 778-1, 13056-7, 97995-5

    * Exclusion Criteria: Plasma is Excluded (26516-5), no qualitative results (9317-9)

  - Associated CPT:

1.2.16 PT (Prothrombin time)

- Common Name: PT; Prothrombin Time; Pro Time; Protime

- Long Name: Prothrombin time
- Test Type: PT
- Panels: Coagulation
- Equivalent Tests: –
- Not Equivalent Tests: International normalization ratio [INR]; activated clotting time [ACT]; partial thromboplastin time [PTT]; thrombin time, prothrombin factor assay (factor 2), prothrombin gene polymorphism, i.e., prothrombin 20210A
- Indications: There are several indications for measuring PT. It can be used to: monitor coagulation for patients on blood thinners; assess bleeding disorders/screening for coagulopathies in patients not on blood thinners; assure normal clotting ability among patients undergoing surgery; and assess liver function.
- Description: The prothrombin time test measures how long it takes for blood to clot. More specifically it measures the time it takes for the clotting factor prothrombin to be converted to thrombin. Clotting factors in the intrinsic and extrinsic coagulation pathways impact prothrombin time. If either factors I (fibrinogen), II (prothrombin), V, VII, or X are missing, deficient, or defective, clotting time may be delayed.
- Related Tests: INR; ACT; PTT; platelet count, platelet function test
- Approximate Reference Range: The unit of measurement is "seconds." Each site will have slight variations due to the reagent and instrumentation used.
- Comments: A PT of 20 seconds in one lab may not represent the same degree of anticoagulation as a PT of 20 seconds in another lab. Because of the variation in PT reporting, the INR was created to adjust for the sensitivity of the thromboplastin reagent used to measure PT.
- Test Method Variance: The test results depend on the method used. .
- Other: –
  - Associated LOINC:
  - Associated CPT:

1.2.17 PTT (activated Partial throomboplastin time)

- Common Name: PT; Prothrombin Time; Pro Time; Protime
- Long Name: Prothrombin time
- Test Type: PT
- Panels: Coagulation
- Equivalent Tests: –
- Not Equivalent Tests: International normalization ratio [INR]; activated clotting time [ACT]; partial thromboplastin time [PTT]; thrombin time, prothrombin factor assay (factor 2), prothrombin gene polymorphism, i.e., prothrombin 20210A
- Indications: There are several indications for measuring PT. It can be used to: monitor coagulation for patients on blood thinners; assess bleeding disorders/screening for coagulopathies in patients not on blood thinners; assure normal clotting ability among patients undergoing surgery; and assess liver function.
- Description: The prothrombin time test measures how long it takes for blood to clot. More specifically it measures the time it takes for the clotting factor prothrombin to be converted to thrombin. Clotting factors in the intrinsic and extrinsic coagulation pathways impact prothrombin time. If either factors I (fibrinogen), II (prothrombin), V, VII, or X are missing, deficient, or defective, clotting time may be delayed.
- Related Tests: INR; ACT; PTT; platelet count, platelet function test
- Approximate Reference Range: The unit of measurement is "seconds." Each site will have slight variations due to the reagent and instrumentation used.
- Comments: A PT of 20 seconds in one lab may not represent the same degree of anticoagulation as a PT of 20 seconds in another lab. Because of the variation in PT reporting, the INR was created to adjust for the sensitivity of the thromboplastin reagent used to measure PT.
- Test Method Variance: The test results depend on the method used. .
- Other: –

- Associated LOINC:
- Associated CPT:

1.2.18 RBC (Red Blood Cell Count [in blood])

- Common Name: Red blood cell count; RBC count; erythrocyte count; red count
- Long Name: Red blood cell count
- Test Type: RBC
- Panels: Complete blood count
- Equivalent Tests: –
- Not Equivalent Tests: Hemoglobin (HGB); hematocrit (HCT); mean corpuscular volume (MCV)
- Indications: To assess whether the number of RBCs in whole blood is within the normal range, and to aid in diagnosing hematologic disorders, bleeding problems, anemias, and polycythemia.
- Description: The RBC is used to evaluate the number of red blood cells in blood.
- Related Tests: Blood smear; complete blood count (CBC) (Other components of the CBC include hemoglobin, hematocrit, platelets, and white blood cells).
- Approximate Reference Range (Each site will have slight variations): Male: 4.7 – 6.1 million cells per microliter (mcl or L or 10-6/Liter) of blood; Female: 4.2 – 5.4 million cells per microliter (mcl or L or 10-6/Liter) of blood
- Comments: RBC counts must be interpreted along with HGB, HCT, and/or red blood cell indices (MCV, mean corpuscular hemoglobin concentration [MCHC], and mean corpuscular hemoglobin [MCH]). RBCs are made in the bone marrow and carry oxygen from the lungs to body cells and transport carbon dioxide from cells back to the lungs. Changes in RBC count are usually associated with changes in HGB levels. When the RBC falls below normal, anemia occurs. A decreased number of RBCs can result from blood loss. A slight decrease in RBC

is seen in pregnancy as a result of expansion of body fluids. Disorders such as RBC destruction (e.g., hemolytic anemia) or decreased RBC production (e.g., iron deficiency anemia) can result in low RBC counts. When RBC rises about normal, polycythemia occurs. Dehydration, congenital heart diseases, some pulmonary diseases, and tissue hypoxia over an extended period of time can result in increases in RBCs. RBC counts are higher among those who live in high altitudes because the body produces more red RBCs in response to decreased oxygen available at high altitude.

- Test Method Variance: –
- Other: –
  - Associated LOINC:
  - Associated CPT:

1.2.19 (WBC) White Blood Cell Test Types

1. WBC (Total White Blood Cell Count)
   - Common Name: WBC count; Leukocyte count; White count
   - Long Name: Total white blood cell count
   - Test Type: WBC
   - Panels: Complete blood count
   - Equivalent Tests:
   - Not Equivalent Tests: Tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils).
   - Indications: To assist in determining the presence of an infection or a disease (e.g., leukemia) that affects the production of white blood cells; to monitor treatment and bone marrow function.
   - Description: The white blood cell (WBC) count is used to evaluate the number of white blood cells in blood. White blood cells are made in the bone marrow and protect the body against infection as well as aiding in the immune response. If there is an infection, white blood cells will attack and destroy the pathogen causing the infection.
   - Related Tests: Blood smear; tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils).

One component of the complete blood count (CBC); (Other components of the CBC include hemoglobin, red blood cells, hematocrit, and platelets).

- Approximate Reference Range (Each site will have slight variations): 4,500 – 10,000 WBCs per microliter (mcl or L or 10-6/Liter) of blood
- Comments: White blood cells are also called leukocytes. Conditions that inhibit WBC proliferation and/or weaken the immune system (e.g., HIV infection, chemotherapy) cause a decrease in the number of WBCs. An elevated number of WBCs is called leukocytosis. A decreased WBC count is called leukopenia. On average, infants have higher WBC counts than do adults. Some drugs and smoking can affect the WBC count.
- Test Method Variance:
- Other:
  - Associated LOINC: 26464-8, 49498-9
  - Associated CPT: 85048

2. WBC_A (White Blood Cell, automated count)

   - Common Name: White Blood Cell, automated count
   - Long Name: White Blood Cell, automated count
   - Test Type: WBC_A
   - Panels: Complete blood count
   - Equivalent Tests: –
   - Not Equivalent Tests: Tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils).
   - Indications: To assist in determining the presence of an infection or a disease (e.g., leukemia) that affects the production of white blood cells; to monitor treatment and bone marrow function.
   - Description: The white blood cell (WBC) count is used to evaluate the number of white blood cells in blood. White blood cells are made in the bone marrow and protect the body against infection as well as aiding in the immune response. If there is an infection, white blood cells will attack and destroy the pathogen causing the infection.

- Related Tests: Blood smear; tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils). One component of the complete blood count (CBC); (Other components of the CBC include hemoglobin, red blood cells, hematocrit, and platelets).
- Approximate Reference Range (Each site will have slight variations): 4,500 – 10,000 WBCs per microliter (mcl or L or 10-6/Liter) of blood
- Comments: White blood cells are also called leukocytes. Conditions that inhibit WBC proliferation and/or weaken the immune system (e.g., HIV infection, chemotherapy) cause a decrease in the number of WBCs. An elevated number of WBCs is called leukocytosis. A decreased WBC count is called leukopenia. On average, infants have higher WBC counts than do adults. Some drugs and smoking can affect the WBC count.
- Test Method Variance: –
- Other: –
  - Associated LOINC: 6690-2
  - Associated CPT: 85025, 85027, 85004, 85048

3. WBC_COR (White blood cell total, corrected for nucleated red blood cells)

   - Common Name: White blood cell total, corrected for nucleated red blood cells
   - Long Name: White blood cell total, corrected for nucleated red blood cells
   - Test Type: WBC_COR
   - Panels: Complete blood count
   - Equivalent Tests:
   - Not Equivalent Tests: Tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils).
   - Indications: To assist in determining the presence of an infection or a disease (e.g., leukemia) that affects the production of white blood cells; to monitor treatment and bone marrow function.
   - Description: The white blood cell (WBC) count is used to evaluate the number of white blood cells in blood. White blood cells

are made in the bone marrow and protect the body against infection as well as aiding in the immune response. If there is an infection, white blood cells will attack and destroy the pathogen causing the infection.

- Related Tests: Blood smear; tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils). One component of the complete blood count (CBC); (Other components of the CBC include hemoglobin, red blood cells, hematocrit, and platelets).
- Approximate Reference Range (Each site will have slight variations): 4,500 – 10,000 WBCs per microliter (mcl or L or 10-6/Liter) of blood
- Comments: White blood cells are also called leukocytes. Conditions that inhibit WBC proliferation and/or weaken the immune system (e.g., HIV infection, chemotherapy) cause a decrease in the number of WBCs. An elevated number of WBCs is called leukocytosis. A decreased WBC count is called leukopenia. On average, infants have higher WBC counts than do adults. Some drugs and smoking can affect the WBC count.
- Test Method Variance:
- Other:
  - Associated LOINC: 12227-5
  - Associated CPT:

4. WBC_COR_A (White blood cell total, corrected for nucleated red blood cells)

   - Common Name: White blood cell total number, corrected for nucleated red blood cells, automated count
   - Long Name: White blood cell total number, corrected for nucleated red blood cells, automated count
   - Test Type: WBC_COR_A
   - Panels: Complete blood count
   - Equivalent Tests:
   - Not Equivalent Tests: Tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils).

- Indications: To assist in determining the presence of an infection or a disease (e.g., leukemia) that affects the production of white blood cells; to monitor treatment and bone marrow function.
- Description: The white blood cell (WBC) count is used to evaluate the number of white blood cells in blood. White blood cells are made in the bone marrow and protect the body against infection as well as aiding in the immune response. If there is an infection, white blood cells will attack and destroy the pathogen causing the infection.
- Related Tests: Blood smear; tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils). One component of the complete blood count (CBC); (Other components of the CBC include hemoglobin, red blood cells, hematocrit, and platelets).
- Approximate Reference Range (Each site will have slight variations): 4,500 – 10,000 WBCs per microliter (mcl or L or 10-6/Liter) of blood
- Comments: White blood cells are also called leukocytes. Conditions that inhibit WBC proliferation and/or weaken the immune system (e.g., HIV infection, chemotherapy) cause a decrease in the number of WBCs. An elevated number of WBCs is called leukocytosis. A decreased WBC count is called leukopenia. On average, infants have higher WBC counts than do adults. Some drugs and smoking can affect the WBC count.
- Test Method Variance:
- Other:
  - Associated LOINC: 33256-9
  - Associated CPT:

5. WBC_M (White Blood Cell Count, manual count)

   - Common Name: White Blood Cell Count, manual count
   - Long Name: White Blood Cell Count, manual count
   - Test Type: WBC_M
   - Panels: Complete blood count
   - Equivalent Tests:
   - Not Equivalent Tests: Tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils).

- Indications: To assist in determining the presence of an infection or a disease (e.g., leukemia) that affects the production of white blood cells; to monitor treatment and bone marrow function.
- Description: The white blood cell (WBC) count is used to evaluate the number of white blood cells in blood. White blood cells are made in the bone marrow and protect the body against infection as well as aiding in the immune response. If there is an infection, white blood cells will attack and destroy the pathogen causing the infection.
- Related Tests: Blood smear; tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils). One component of the complete blood count (CBC); (Other components of the CBC include hemoglobin, red blood cells, hematocrit, and platelets).
- Approximate Reference Range (Each site will have slight variations): 4,500 – 10,000 WBCs per microliter (mcl or L or 10-6/Liter) of blood
- Comments: White blood cells are also called leukocytes. Conditions that inhibit WBC proliferation and/or weaken the immune system (e.g., HIV infection, chemotherapy) cause a decrease in the number of WBCs. An elevated number of WBCs is called leukocytosis. A decreased WBC count is called leukopenia. On average, infants have higher WBC counts than do adults. Some drugs and smoking can affect the WBC count. Test

Method Variance:

- Other:
  - Associated LOINC: 804-5
  - Associated CPT:

6. WBC_OTH_C (White blood cell other leukocytes, count)

- Common Name: White blood cell other leukocytes, count
- Long Name: White blood cell other leukocytes, count
- Test Type: WBC_OTH_C
- Panels: Complete blood count
- Equivalent Tests:

- Not Equivalent Tests: Tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils).
- Indications: To assist in determining the presence of an infection or a disease (e.g., leukemia) that affects the production of white blood cells; to monitor treatment and bone marrow function.
- Description: The white blood cell (WBC) count is used to evaluate the number of white blood cells in blood. White blood cells are made in the bone marrow and protect the body against infection as well as aiding in the immune response. If there is an infection, white blood cells will attack and destroy the pathogen causing the infection.
- Related Tests: Blood smear; tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils). One component of the complete blood count (CBC); (Other components of the CBC include hemoglobin, red blood cells, hematocrit, and platelets).
- Approximate Reference Range (Each site will have slight variations): 4,500 – 10,000 WBCs per microliter (mcl or L or 10-6/Liter) of blood
- Comments: White blood cells are also called leukocytes. Conditions that inhibit WBC proliferation and/or weaken the immune system (e.g., HIV infection, chemotherapy) cause a decrease in the number of WBCs. An elevated number of WBCs is called leukocytosis. A decreased WBC count is called leukopenia. On average, infants have higher WBC counts than do adults. Some drugs and smoking can affect the WBC count.
- Test Method Variance: –
- Other: –
  - Associated LOINC: 30406-3, 51383-8, 729-4, 15190-2
  - Associated CPT:

7. WBC_OTH_P (White blood cell other leukocytes, percent)

- Common Name: WBC count; Leukocyte count; White count
- Long Name: Total white blood cell count
- Test Type: WBC
- Panels: Complete blood count

- Equivalent Tests:
- Not Equivalent Tests: Tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils).
- Indications: To assist in determining the presence of an infection or a disease (e.g., leukemia) that affects the production of white blood cells; to monitor treatment and bone marrow function.
- Description: The white blood cell (WBC) count is used to evaluate the number of white blood cells in blood. White blood cells are made in the bone marrow and protect the body against infection as well as aiding in the immune response. If there is an infection, white blood cells will attack and destroy the pathogen causing the infection.
- Related Tests: Blood smear; tests of individual types of WBCs (neutrophils, lymphocytes, monocytes, eosinophils, or basophils). One component of the complete blood count (CBC); (Other components of the CBC include hemoglobin, red blood cells, hematocrit, and platelets).
- Approximate Reference Range (Each site will have slight variations): 4,500 – 10,000 WBCs per microliter (mcl or L or 10-6/Liter) of blood
- Comments: White blood cells are also called leukocytes. Conditions that inhibit WBC proliferation and/or weaken the immune system (e.g., HIV infection, chemotherapy) cause a decrease in the number of WBCs. An elevated number of WBCs is called leukocytosis. A decreased WBC count is called leukopenia. On average, infants have higher WBC counts than do adults. Some drugs and smoking can affect the WBC count.
- Test Method Variance: –
- Other: –
  - Associated LOINC: 26471-3, 730-2, 40646-2
  - Associated CPT:

1.3 Influenza

1.3.1 INF_A_VTC (Influenza virus A organism-specific culture)

- Common Name: Influenza virus A organism-specific culture
- Long Name: Influenza virus A organism-specific culture

- Test Type: INF_A_VTC
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Primarily indicated for public health surveillance, outbreak investigation, and diagnostic confirmation when other tests are inconclusive.
- Description: An Influenza A virus organism-specific culture test isolates the actual virus from a patient's respiratory sample (like a nasopharyngeal swab) by growing it in live cell lines or embryonated eggs, then identifies it using techniques like fluorescent antibody staining or hemadsorption, providing a definitive diagnosis but taking longer (days) than rapid tests, and is considered a gold standard for confirmation.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): There is no presence of the Influenza virus in a healthy person.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 48310-7
  - Associated CPT:

1.3.2 INF_A_NS (Influenza virus A)

- Common Name: Influenza virus A
- Long Name: Influenza virus A method not specified
- Test Type: INF_A_NS
- Panels:
- Equivalent Tests:

- Not Equivalent Tests:
- Indications: Laboratory testing for Influenza Virus A is primarily indicated when results will directly influence clinical management, such as the decision to initiate antiviral therapy or implement infection control measures.
- Description: A laboratory test for Influenza Virus A is a diagnostic procedure used to detect the presence of the virus in a patient's respiratory system.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): There is no presence of the Influenza virus in a healthy person.
- Comments: LOINCs indicates this is an antigen test
- Test Method Variance:
- Other:
  - Associated LOINC: 31859-2 (NS), 43874-7 (NPH), 44562-7 (BAL), 31858-4 (THRT), 44563-5 (NSWAB)
  - Associated CPT:

1.3.3 INF_A_EIA (Influenza virus A immunoassay)

- Common Name: Influenza virus A immunoassay
- Long Name: Influenza virus A immunoassay
- Test Type: INF_A_EIA
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Indications for an influenza virus A immunoassay lab test include:
  - Symptomatic Patients: Testing is indicated for individuals presenting with "influenza-like illness," which includes fever (100.4°F), cough, sore throat, muscle aches, fatigue, and headaches.

  - Influencing Treatment Decisions: The test is indicated if the result will determine whether to initiate antiviral therapy, especially for outpatients who are not at high risk but seek care within 48 hours of symptom onset.
  - High-Risk Populations: Testing is strongly recommended for symptomatic patients at higher risk of complications to guide prompt treatment, including:
    * Adults 65 years and children <2 years.
    * Pregnant women or those up to two weeks postpartum.
    * Individuals with chronic conditions (e.g., asthma, COPD, heart failure, or diabetes).
    * Immunocompromised persons.

- Description: An Influenza A immunoassay lab test, often a rapid antigen test (RIDT) or a more sensitive molecular test (like RT-PCR), detects viral proteins (antigens) or genetic material from Influenza A in respiratory samples (swabs, washes) to diagnose infection quickly, differentiating it from Influenza B but not subtypes, offering results in minutes to hours, though RIDTs have lower sensitivity and may miss early infections compared to PCR.

- Related Tests:

- Approximate Reference Range (Each site will have slight variations): The index values are:

  - Negative 0.89 IV No significant level of influenza A antibody detected.
  - Equivocal 0.90 – 1.10 IV Questionable presence; repeat testing in 10–14 days is recommended.
  - Positive 1.11 IV Antibodies detected, suggesting current, recent, or past infection.

- Comments: LOINCs indicates this is an antigen test

- Test Method Variance:

- Other:

  - Associated LOINC: 46082-4 (NPH), 5860-2 (THRT), 5862-8 (NS), 44564-3 (NSWAB)
  - Associated CPT:

1.3.4 INF_A_IF (Influenza virus A immunofluorescence)

- Common Name: Influenza virus A immunofluorescence
- Long Name: Influenza virus A immunofluorescence
- Test Type: INF_A_IF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Immunofluorescence (IF) lab tests—including Direct (DFA) and Indirect (IFA) fluorescent antibody staining—are antigen detection assays used to identify Influenza virus A and B in respiratory specimens.
- Description: An Influenza A immunofluorescence (IF) lab test detects viral antigens in respiratory samples (like nasal swabs) using fluorescently labeled antibodies, allowing rapid identification (1-4 hours) under a microscope, providing quick diagnosis for flu, especially Influenza A, with moderate sensitivity but high specificity, helping differentiate it from other respiratory viruses.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): If the virus is detectected then the test is positive.
- Comments: LOINCs indicates this is an antigen test
- Test Method Variance:
- Other:
  - Associated LOINC: 44558-5 (NPH), 5863-6 (NS), 44559-3 (BAL), 5861-0 (THRT), 44560-1 (NSWAB)
  - Associated CPT:

1.3.5 INF_A_PCR (Influenza virus A probe and target amplification)

- Common Name: Influenza virus A probe and target amplification
- Long Name: Influenza virus A probe and target amplification
- Test Type: INF_A_PCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A lab test is indicated for hospitalized patients, high-risk outpatients, clinical management decisions, surveillance and public health, and preferred timing and method.
- Description: Influenza A probe and target amplification in diagnostics involves amplifying the virus's genetic material (RNA) using primers, often isothermally (constant temp), then detecting it with specific probes (like molecular beacons or barcoded oligos) that light up or change signal when they bind to the viral target (e.g., M gene), allowing rapid, sensitive detection and sometimes subtyping.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): No reference range presence indicates a positive result.
- Comments: LOINC indicates this is H1 2009 pandemic RNA
- Test Method Variance:
- Other:
  - Associated LOINC: 55465-9 (NS), 61101-2 (NS), 59423-4 (NS), 55464-2 (NS), 55463-4 (NS), 44263-2 (NS), 40981-3 (NS), 39103-7 (NS), 38381-0 (NS), 39102-9 (NS), 39025-2 (NS), 34487-9 (NS), 53250-7 (NS), 49521-8 (NS), 49524-2 (NS) (updated 12/2014), 92142-9 (NS)
  - Associated CPT:

1.3.6 INF_B_VTC (Influenza virus B organism-specific culture)

- Common Name: Influenza virus B organism-specific culture
- Long Name: Influenza virus B organism-specific culture
- Test Type: INF_B_VTC
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The organism-specific culture remains clinically indicated in the following high-priority scenarios: outbreak management & surveillance, confirming ambiguous results, monitoring treatment resistance, and severe or atypical illness.
- Description: An organism-specific culture lab test for Influenza B involves growing the virus from a patient's respiratory sample (like a swab) in living cells (cell culture), identifying its presence, and then confirming it's Influenza B. While molecular tests like PCR are now more common for rapid diagnosis, viral culture remains the "gold standard" for confirmed identification and detailed characterization of the virus.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Not applicable since presence results in a positive diagnosis.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 38382-8 (NS)
  - Associated CPT:

1.3.7 INF_B_NS (Influenza virus B)

- Common Name: Influenza virus B
- Long Name: Influenza virus B method not specified
- Test Type: INF_B_NS
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Testing is indicated for individuals presenting with acute respiratory illness symptoms, which typically include:
  - Abrupt onset of fever, chills, cough, sore throat, or body aches.
  - Atypical symptoms, such as confusion in older adults or sepsis-like syndromes in infants.
  - Respiratory distress or shortness of breath.
  - Gastrointestinal symptoms, such as vomiting and diarrhea, which are more common in children.
- Description: An Influenza B lab test is a diagnostic procedure used to detect the presence of the Influenza type B virus in respiratory secretions. These tests are primarily used to confirm a diagnosis, distinguish between flu types A and B, and guide treatment decisions, such as the use of antiviral medications.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments: LOINCs indicates this is an antigen test
- Test Method Variance:
- Other:
  - Associated LOINC: 31864-2 (NS), 43895-2 (NPH), 44576-7 (BAL), 31863-4 (THRT), 44577-5 (NSWAB)
  - Associated CPT:

1.3.8 INF_B_EIA (Influenza virus B immunoassay)

- Common Name: Influenza virus B immunoassay
- Long Name: Influenza virus B immunoassay
- Test Type: INF_B_EIA
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Testing is indicated for individuals presenting with classic "flu-like" symptoms, including:
  - Acute respiratory illness: Sudden onset of cough, sore throat, or runny/stuffy nose, typically with or without fever.
  - Systemic symptoms: Fever, chills, muscle/body aches (myalgias), headache, and significant fatigue.
  - Atypical presentations: In infants (e.g., sepsis-like syndrome) or older adults (e.g., new-onset confusion), even if typical respiratory symptoms are absent.
- Description: An Influenza B immunoassay is a rapid laboratory test used to detect specific viral proteins (antigens) of the influenza B virus in respiratory specimens.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments: LOINCs indicates this is an antigen test
- Test Method Variance:
- Other:
  - Associated LOINC: 46083-2 (NPH), 5864-4 (THRT), 5866-9 (NS), 44575-9 (NSWAB)
  - Associated CPT:

1.3.9 INF_B_IF (Influenza virus B immunofluorescence)

- Common Name: Influenza virus B immunofluorescence
- Long Name: Influenza virus B immunofluorescence
- Test Type: INF_B_IF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Clinicians typically indicate this test in the following scenarios:
  - Suspected Influenza Infection: Presentation of symptoms such as fever, cough, sore throat, muscle aches, and fatigue, especially during documented community influenza activity.
  - Hospital Admission: Recommended for all patients with suspected influenza being admitted to the hospital to inform infection control and treatment.
  - High-Risk Patients: Testing is indicated for individuals at high risk of complications, including:
- Description: An Influenza B immunofluorescence lab test (DFA/IFA) quickly detects viral antigens in respiratory swabs by using fluorescently-labeled antibodies that bind to specific influenza B proteins (like nucleoprotein) in infected cells, appearing as yellow-gold glowing cells under a fluorescence microscope for rapid diagnosis, though it's less sensitive than PCR but faster than viral culture.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments: LOINCs indicates this is an antigen test
- Test Method Variance:
- Other:

  - Associated LOINC: 5865-1(THRT), 5867-7 (NS), 44572-6 (BAL), 44573-4 (NSWAB), 44571-8 (NPH)
  - Associated CPT:

1.3.10 INF_B_PCR (Influenza virus B probe and target amplification)

- Common Name: Influenza virus B probe and target amplification
- Long Name: Influenza virus B probe and target amplification
- Test Type: INF_B_PCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Hospitalized Patients: Testing is recommended for all patients with suspected influenza who are being admitted to the hospital.
  - High-Risk Outpatients: For patients at higher risk of complications who present with influenza-like illness, pneumonia, or non-specific respiratory symptoms.
  - Outbreak Management: To identify the cause of a respiratory illness outbreak in closed settings like hospitals, long-term care facilities, or schools.
  - Confirmation of Other Tests: To confirm a negative rapid antigen test result when community influenza activity is high, or to confirm a positive result when activity is low.
- Description: The Influenza virus B probe and target amplification test is a molecular diagnostic tool used for the qualitative detection and differentiation of influenza B viral RNA in respiratory specimens. These tests, often referred to as Nucleic Acid Amplification Tests (NAATs), are considered the gold standard for influenza diagnosis due to their high sensitivity and specificity.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".

- Comments: Viral RNA test
- Test Method Variance:
- Other:
  - Associated LOINC: 40982-1 (NS), 53251-5 (NS), 92141-1 (NS)
  - Associated CPT:

1.3.11 INF_AB_NS (Influenza virus A+B)

- Common Name: Influenza virus A+B
- Long Name: Influenza virus A+B method not specified
- Test Type: INF_AB_NS
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Common symptoms are:
  - Respiratory Symptoms: Abrupt onset of cough, sore throat, runny or stuffy nose, and difficulty breathing.
  - Systemic Symptoms: Fever (though not always present), chills, extreme fatigue, muscle or body aches, and headaches.
  - Atypical Symptoms: Confusion in older adults, or vomiting and diarrhea, which are more common in children.
- Description: Influenza A+B lab tests are diagnostic procedures used to detect and differentiate between the two main types of seasonal influenza viruses. These tests are critical for guiding clinical treatment, such as the administration of antivirals like Tamiflu within the effective 48-hour window.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".

- Comments: Only use INF_AB_NS if one of the LOINCs listed here is in the data stream. LOINCs indicates this is an antigen test
- Test Method Variance:
- Other:
  - Associated LOINC: 33535-6 (NPH) 24015-0 (NS), 44566-8 (BAL), 31860-0 (THRT), 31861-8 (THRT), 31862-6 (NS), 44567-6 (NSWAB)
  - Associated CPT:

1.3.12 INF_AB_EIA (Influenza virus A+B immunoassay)

- Common Name: Influenza virus A+B immunoassay
- Long Name: Influenza virus A+B immunoassay
- Test Type: INF_AB_EIA
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Influenza A+B immunoassay (rapid antigen) tests are indicated for patients with acute respiratory illness (fever, cough, sore throat, headache, myalgia) to guide treatment, especially for high-risk individuals, during active influenza season.
- Description: An influenza virus A+B immunoassay is a rapid, qualitative lateral flow test that detects viral nucleoprotein antigens from respiratory samples (nasal/throat swabs, washes) within 15 minutes. It differentiates between Influenza A and B, aiding in immediate diagnosis, but cannot distinguish subtypes. The test uses specific antibodies on a cassette, producing colored lines for positive results.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments: Only use INF_AB_EIA if the LOINC listed here is in the data stream. LOINC indicates this is an antigen test.

- Test Method Variance:
- Other:
  - Associated LOINC: 6437-8 (NS), 6441-0 (NS), 6435-2 (THRT), 6439-4 (THRT)
  - Associated CPT:

1.3.13 INF_AB_IF (Influenza virus A+B immunofluorescence)

- Common Name: Influenza virus A+B immunofluorescence
- Long Name: Influenza virus A+B immunofluorescence
- Test Type: INF_AB_IF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Influenza A+B immunofluorescence tests are indicated for patients with acute respiratory illness (fever, cough, sore throat, headache, myalgia) to guide treatment, especially for high-risk individuals, during active influenza season.
- Description: An Influenza A+B immunofluorescence (IF) test is a rapid diagnostic assay that detects influenza antigens in respiratory specimens (nasal/throat swabs) using fluorescently labeled antibodies, providing results in 1–4 hours. It uses a fluorescent microscope to distinguish influenza A (apple-green fluorescence) from influenza B (yellow-gold) with moderate-to-high sensitivity.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments: Only use INF_AB_IF if the LOINC listed here is in the data stream. LOINC indicates this is an antigen test.
- Test Method Variance:

- Other:
  - Associated LOINC: 6438-6 (NS), 6442-8 (NS), 6440-2 (THRT), 6436-0 (THRT), 61102-0 (NS)
  - Associated CPT:

1.3.14 INF_AB_PCR (Influenza virus A+B probe and target amplification)

- Common Name: Influenza virus A+B probe and target amplification
- Long Name: Influenza virus A+B probe and target amplification
- Test Type: INF_AB_PCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Influenza A+B probe and target amplification tests are indicated for patients with acute respiratory illness (fever, cough, sore throat, headache, myalgia) to guide treatment, especially for high-risk individuals, during active influenza season.
- Description: An Influenza A+B probe and target amplification PCR test (commonly Reverse Transcription-PCR or RT-PCR) is a highly sensitive and specific molecular assay designed to detect and differentiate influenza A and B viral RNA in respiratory specimens. It uses fluorescent-labeled probes to detect amplified genetic material, providing rapid, qualitative results.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments: Only use INF_AB_PCR if one of the LOINCs listed here is in the data stream. 62462-7 is Viral RNA test.
- Test Method Variance:
- Other:

  - Associated LOINC: 62462-7 (NS) 48509-4 (NS)
  - Associated CPT:

1.3.15 INF_NS_VTC (Influenza virus not specified organism-specific culture)

- Common Name: Influenza virus not specified organism-specific culture
- Long Name: Influenza virus not specified organism-specific culture
- Test Type: INF_NS_VTC
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Influenza virus testing is indicated for patients with acute respiratory illness (fever, cough, sore throat, fatigue) when results will guide clinical management, such as initiating antiviral therapy, reducing unnecessary diagnostic testing, or implementing infection control. Testing is highly recommended for hospitalized patients, those at high risk for complications, or during outbreaks in closed settings.
- Description: An influenza virus organism-specific test is a diagnostic method designed to identify the presence of the influenza virus and often determine its specific type (A or B) or subtype (e.g., H1N1, H3N2) in a clinical specimen.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 6604-3 (NS) 6601-9 (SPUTUM), 6603-5 (THRT), 6602-7 (SPUTUM)
  - Associated CPT:

1.3.16 INF_NS_NS (Influenza virus not specified method not specified)

- Common Name:
- Long Name: Influenza virus not specified method not specified
- Test Type: INF_NS_NS
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Influenza virus testing is indicated for patients with acute respiratory illness (fever, cough, sore throat, fatigue) when results will guide clinical management, such as initiating antiviral therapy, reducing unnecessary diagnostic testing, or implementing infection control. Testing is highly recommended for hospitalized patients, those at high risk for complications, or during outbreaks in closed settings.
- Description: An influenza (flu) test is a medical procedure used to determine if a person's respiratory symptoms—such as fever, cough, and body aches—are caused by an influenza virus. The process typically involves collecting a fluid sample from the respiratory tract using a sterile swab.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 54244-9 (NS); 54240-7 (NS)
  - Associated CPT:

1.3.17 INF_NS_PCR (Influenza virus not specified probe and target amplification)

- Common Name:
- Long Name: Influenza virus not specified probe and target amplification
- Test Type: INF_NS_PCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Influenza virus testing is indicated for patients with acute respiratory illness (fever, cough, sore throat, fatigue) when results will guide clinical management, such as initiating antiviral therapy, reducing unnecessary diagnostic testing, or implementing infection control. Testing is highly recommended for hospitalized patients, those at high risk for complications, or during outbreaks in closed settings.
- Description: A PCR (polymerase chain reaction) influenza test is a highly sensitive and specific molecular diagnostic tool that detects the genetic material (RNA) of influenza A and B viruses in respiratory samples. It is more accurate than rapid antigen tests, often providing results in 15-30 minutes for rapid molecular assays or within a few hours for laboratory-based tests.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal reference range for these tests is "Not detected" or "Negative".
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 54243-1 (NS)
  - Associated CPT:

1.4 COVID-19

1.4.1 SARS_COV_2_AB_IGA (SARS-CoV-2 (COVID19) IgA Ab [Presence] in Serum, Plasma or Blood by Immunoassay)

- Common Name: SARS-CoV-2 (COVID19) IgA Ab [Presence] in Serum, Plasma or Blood by Immunoassay
- Long Name: SARS-CoV-2 (COVID19) IgA Ab [Presence] in Serum, Plasma or Blood by Immunoassay
- Test Type: SARS_COV_2_AB_IGA
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: IgA antibody tests for COVID-19 (SARS-CoV-2) are primarily indicated for identifying early infection, assessing immune response to vaccination or infection (particularly in mucosal areas like saliva), and investigating potential, less common, post-acute complications. IgA is often used as a complement to IgG and IgM testing because it appears rapidly, sometimes within 2-5 days of symptom onset.
- Description: An IgA (Immunoglobulin A) antibody test for COVID-19 detects specific antibodies in the blood, saliva, or mucus that appear early in the immune response to SARS-CoV-2. It helps identify recent or acute infection, particularly when PCR tests are negative, and is valuable for evaluating mucosal immunity, though it is not used to confirm active infection.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): IgA antibody tests for COVID-19 are generally reported as a ratio of the sample to a calibrator, not a simple numerical range. Results are typically defined as negative (ratio < 0.8), borderline (0.8 to 1.0), or positive (ratio ≥ 1.1), appearing 5–7 days post-infection.
- Comments:
- Test Method Variance:

- Other:
  - Associated LOINC: 94562-6, 94720-0, 94768-9, 95427-1
  - Associated CPT:

1.4.2 SARS_COV_2_AB_IGG (SARS-CoV-2 (COVID19) IgG Ab [Presence] in Serum, Plasma or Blood by Immunoassay)

- Common Name: SARS-CoV-2 (COVID19) IgG Ab [Presence] in Serum, Plasma or Blood by Immunoassay
- Long Name: SARS-CoV-2 (COVID19) IgG Ab [Presence] in Serum, Plasma or Blood by Immunoassay
- Test Type: SARS_COV_2_AB_IGG
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: An IgG antibody COVID-19 test is primarily indicated for identifying individuals with an adaptive immune response to the SARS-CoV-2 virus, which suggests a previous or recent infection. Because antibodies take time to develop, it is generally indicated at least 14 days after symptom onset.
- Description: A COVID-19 IgG antibody test is a blood test that detects specific IgG antibodies, indicating a past infection with the SARS-CoV-2 virus or response to vaccination, rather than an active, current infection. These antibodies typically appear 1–3 weeks after infection.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): SARS-CoV-2 IgG antibody test reference ranges vary by manufacturer, but typically a result of greater than or equal to 50 AU/mL.
- Comments: These tests measure response to the virus's Spike protein, but do not determine immunity.
- Test Method Variance:
- Other:

  - Associated LOINC: 94505-5, 94507-1, 94563-4, 94761-4, 95429-7,96742-2, 99596-9, 99597-7
  - Associated CPT:

1.4.3 SARS_COV_2_AB_IGM (SARS-CoV-2 (COVID19) IgM Ab [Presence] in Serum, Plasma or Blood by Immunoassay)

- Common Name: SARS-CoV-2 (COVID19) IgM Ab [Presence] in Serum, Plasma or Blood by Immunoassay
- Long Name: SARS-CoV-2 (COVID19) IgM Ab [Presence] in Serum, Plasma or Blood by Immunoassay
- Test Type: SARS_COV_2_AB_IGM
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: An IgM antibody test for COVID-19 is primarily used to identify a recent or prior infection by detecting the body's early immune response to the SARS-CoV-2 virus.
- Description: An IgM antibody COVID-19 test is a blood-based immunoassay designed to detect the first, short-term antibodies produced by the immune system in response to a SARS-CoV-2 infection. Often part of a rapid lateral flow test, it identifies recent exposure, typically appearing within 1-2 weeks of infection.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Often set at 10 AU/mL
- Comments: Do not confuse COVID-specific tests with a "Total IgM" blood test, which has a normal reference range of 40 to 250 mg/dL and measures overall immune health.
- Test Method Variance:
- Other:
  - Associated LOINC: 94506-3, 94508-9, 94564-2, 95416-4, 95428-9
  - Associated CPT:

1.4.4 SARS_COV_2_AB_IGMA (SARS-CoV-2 (COVID19) IgA+IgM [Presence] in Serum or Plasma by Immunoassay)

- Common Name: SARS-CoV-2 (COVID19) IgA+IgM [Presence] in Serum or Plasma by Immunoassay
- Long Name: SARS-CoV-2 (COVID19) IgA+IgM [Presence] in Serum or Plasma by Immunoassay
- Test Type: SARS_COV_2_AB_IGMA
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: IgA and IgM COVID-19 antibody tests are indicated for the early detection of acute SARS-CoV-2 infection, typically 5–7 days post-symptom onset. They are useful when RT-PCR results are uncertain or unavailable, particularly because IgA may detect infections earlier than IgM, with studies showing high sensitivity in the first week.
- Description: An IgA/IgM COVID-19 test is a serological antibody test that detects specific antibodies produced by the immune system in response to SARS-CoV-2 infection.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): No standard reference range, results are Positive or Negative.
- Comments: IgA and IgM COVID-19 tests do not have a standard "reference range" like blood tests; they are typically reported as Positive (Reactive) or Negative (Non-reactive) based on a cut-off index (COI) determined by the specific laboratory's assay.
- Test Method Variance:
- Other:
  - Associated LOINC: 95125-1
  - Associated CPT:

1.4.5 SARS_COV_2_AB_IGMG (SARS-CoV-2 (COVID19) IgG+IgM Ab [Presence] in Serum or Plasma by Immunoassay)

- Common Name: SARS-CoV-2 (COVID19) IgG+IgM Ab [Presence] in Serum or Plasma by Immunoassay
- Long Name: SARS-CoV-2 (COVID19) IgG+IgM Ab [Presence] in Serum or Plasma by Immunoassay
- Test Type: SARS_COV_2_AB_IGMG
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: IgA/IgM COVID-19 antibody tests are used to detect recent infection or immune response, typically appearing within days to weeks after exposure or vaccination. Primary indications include aiding in the diagnosis of MIS-C/MIS-A (Multisystem Inflammatory Syndrome in Children/Adults). They are not used to diagnose acute infection.
- Description: An IgA/IgM antibody test for COVID-19 is a blood-based serology test designed to detect early-to-mid-stage immune responses to SARS-CoV-2. IgM acts as the initial antibody response, while IgA plays a key role in mucosal immunity.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The results are not reported as a numeric result. Results are positive if the test is above a manufacturer's index.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 94503-0, 94547-7, 95542-7
  - Associated CPT:

1.4.6 SARS_COV_2_AB_NEUT (SARS-CoV-2 (COVID19) neutralizing antibody in Serum or Plasma)

- Common Name: SARS-CoV-2 (COVID19) neutralizing antibody in Serum or Plasma
- Long Name: SARS-CoV-2 (COVID19) neutralizing antibody in Serum or Plasma
- Test Type: SARS_COV_2_AB_NEUT
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Neutralizing antibody COVID-19 tests, which measure the ability of antibodies to inhibit SARS-CoV-2 in cell cultures, are primarily used for research and clinical studies rather than routine diagnosis. They are generally indicated for assessing immune response following vaccination or infection, or evaluating immune protection in immunocompromised individuals.
- Description: A neutralizing antibody COVID-19 test is a specialized blood test that measures the presence of antibodies capable of inhibiting the SARS-CoV-2 virus from infecting cells, unlike standard antibody tests that only detect binding antibodies. These tests assess the functional ability of immune system proteins to block the virus.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): the test threshold is >22 IU/mL, >34.47 IU/mL
- Comments: Test_Type value originally contained NUET. Corrected to NEUT on 10/15/2021.
- Test Method Variance:
- Other:
  - Associated LOINC: 95410-7, 95411-5, 96603-6, 98734-7
  - Associated CPT:

1.4.7 SARS_COV_2_AB_NEUT_P (Percent neutralization by SARS-CoV-2 spike protein RBD neutralizing antibody-Serum/Plasma)

- Common Name: Percent neutralization by SARS-CoV-2 spike protein RBD neutralizing antibody-Serum/Plasma
- Long Name: Percent neutralization by SARS-CoV-2 spike protein RBD neutralizing antibody-Serum/Plasma
- Test Type: SARS_COV_2_AB_NEUT_P
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indication for a SARS-CoV-2 spike protein RBD neutralizing antibody test is to identify an adaptive immune response to the virus, which indicates a recent or prior infection. Unlike standard antibody tests that only detect if antibodies are present (binding antibodies), this specific test measures "functional" antibodies that can actively block the virus from entering cells.
- Description: The SARS-CoV-2 spike protein RBD neutralizing antibody-serum/plasma test measures the ability of antibodies to block the interaction between the viral receptor-binding domain (RBD) and the human ACE2 receptor. Percentage neutralization (% inhibition) indicates the proportion of this binding interaction blocked, with higher percentages reflecting stronger neutralizing potency.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): a negative result as less than 20% to 30% inhibition over 30% is a positive result
- Comments: Test_Type value originally contained NUET. Corrected to NEUT on 10/15/2021.
- Test Method Variance:
- Other:
  - Associated LOINC: 98733-9
  - Associated CPT:

1.4.8 SARS_COV_2_AB_NEUT_T (SARS-CoV-2 (COVID-19) S protein RBD neutralizing antibody [Titer] in Serum or Plasma by Immunoassay)

- Common Name: SARS-CoV-2 (COVID-19) S protein RBD neutralizing antibody [Titer] in Serum or Plasma by Immunoassay
- Long Name: SARS-CoV-2 (COVID-19) S protein RBD neutralizing antibody [Titer] in Serum or Plasma by Immunoassay
- Test Type: SARS_COV_2_AB_NEUT_T
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: SARS-CoV-2 neutralizing antibody (NAb) titer tests are primarily used to assess the functional immune response rather than for routine diagnosis of a current infection.
- Description: A SARS-CoV-2 spike protein RBD neutralizing antibody serum/plasma test measures the concentration (titer) of antibodies that specifically block the receptor-binding domain (RBD) from attaching to host cell ACE2 receptors. High titers indicate stronger protection, often determined by plaque reduction neutralization tests (PRNT) or surrogate ELISA-based assays that measure the inhibition of virus-host cell interaction.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Positive: 0.80 U/mL is a common threshold
- Comments: Test_Type value originally contained NUET. Corrected to NEUT on 10/15/2021.
- Test Method Variance:
- Other:
  - Associated LOINC: 98732-1
  - Associated CPT:

1.4.9 SARS_COV_2_AB_TOT (SARS-CoV-2 (COVID19) Ab in Serum, Plasma or Saliva)

- Common Name: SARS-CoV-2 (COVID19) Ab in Serum, Plasma or Saliva
- Long Name: SARS-CoV-2 (COVID19) Ab in Serum, Plasma or Saliva
- Test Type: SARS_COV_2_AB_TOT
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indication for a COVID-19 antibody test is to identify individuals with an adaptive immune response to SARS-CoV-2, which suggests a prior infection or vaccination.
- Description: A COVID-19 antibody test (also known as a serology test) is a blood test used to determine if your body has developed an immune response to the SARS-CoV-2 virus from a previous infection or vaccination. Unlike diagnostic "viral tests" (PCR or antigen tests), it cannot detect a current, active infection.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Positive: 0.80 U/mL is a common threshold
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 94504-8, 94661-6, 94762-2, 94769-7, 95825-6, 98069-8
  - Associated CPT:

1.4.10 SARS_COV_2_IA (SARS-CoV-2 (COVID19) Ag [Presence] in Respiratory specimen by Rapid immunoassay)

- Common Name: SARS-CoV-2 (COVID19) Ag [Presence] in Respiratory specimen by Rapid immunoassay
- Long Name: SARS-CoV-2 (COVID19) Ag [Presence] in Respiratory specimen by Rapid immunoassay
- Test Type: SARS_COV_2_IA
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A rapid COVID-19 antigen immunoassay test is indicated for several diagnostic and screening scenarios, primarily to identify current active infection when viral load is highest and symptoms are present.
- Description: A COVID-19 rapid antigen (Ag) test is a lateral flow immunoassay used to qualitatively detect specific viral proteins (antigens) from the SARS-CoV-2 virus, most commonly the nucleocapsid (N) protein. It provides results quickly, typically within 15 to 30 minutes, and is commonly used for both at-home self-testing and point-of-care diagnosis.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): positive/reative is $>$ .80 U/mL
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 94558-4, 96119-3, 97097-0
  - Associated CPT:

1.4.11 SARS_COV_2_NS (SARS-CoV-2 (COVID19) unspecified or other specimen or method)

- Common Name: SARS-CoV-2 (COVID19)
- Long Name: SARS-CoV-2 (COVID19) unspecified or other specimen or method
- Test Type: SARS_COV_2_NS
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Common Symptoms: Fever or chills, cough, shortness of breath, fatigue, muscle aches, new loss of taste or smell, sore throat, congestion, nausea, or diarrhea.
- Description: COVID-19 tests, including rapid antigen and molecular PCR tests, detect the SARS-CoV-2 virus to diagnose infection.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Test reference ranges vary by the type of test.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 94763-0, 94764-8, 94822-4
  - Associated CPT:

1.4.12 SARS_COV_2_NAAT (SARS-CoV-2 (COVID19) (Nucleic Acid Amp Testing or gene presence in Resp, ser/plas or unspec specimen)

- Common Name: SARS-CoV-2 (COVID19) (Nucleic Acid Amp Testing or gene presence in Resp, ser/plas or unspec specimen
- Long Name: SARS-CoV-2 (COVID19) (Nucleic Acid Amp Testing or gene presence in Resp, ser/plas or unspec specimen

- Test Type: SARS_COV_2_NAAT
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Indications for a COVID-19 Nucleic Acid Amplification Test (NAAT) primarily focus on diagnosing current infection in both symptomatic and asymptomatic individuals.
- Description: A COVID-19 Nucleic Acid Amplification Test (NAAT) is a highly sensitive diagnostic test that detects the genetic material (RNA) of the SARS-CoV-2 virus from respiratory samples, such as nasal swabs. It works by amplifying, or creating many copies of, the virus's genetic material if present, making it a reliable, high-accuracy method for detecting current infections.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Doesn't provide a quantitative result
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 94306-8, 94307-6, 94308-4, 94309-2, 94314-2, 94315-9, 94316-7, 94500-6, 94531-1, 94533-7, 94534-5, 94559-2, 94565-9, 94639-2, 94640-0, 94641-8, 94660-8, 94756-4, 94757-2, 94758-0, 94759-8, 94760-6, 94765-5, 94766-3, 94767-1, 94845-5, 95406-5, 95409-9, 95424-8, 95425-5, 95608-6, 95609-4, 96091-4, 95823-1, 95824-9, 95826-4, 96448-6, 96120-1, 96123-5, 96797-6, 96763-8, 96765-3, 96897-4, 96957-6, 96958-4, 96986-5, 97098-8, 101289-7
  - Associated CPT:

1.4.13 SARS_COV_2_T_STIM (SARS_CoV-2 stimulated gamma interferon release by Helper CD4+ T-cells [Units/volume] by Immunoassay)

- Common Name: SARS_CoV-2 stimulated gamma interferon release by Helper CD4+ T-cells [Units/volume] by Immunoassay
- Long Name: SARS_CoV-2 stimulated gamma interferon release by Helper CD4+ T-cells [Units/volume] by Immunoassay
- Test Type: SARS_COV_2_T_STIM
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: While not typically used for diagnosing an acute infection, the SARS-CoV-2 IGRA is indicated for evaluating a patient's long-term cellular immune status. It is particularly useful in clinical scenarios where standard antibody tests are insufficient or unreliable.
- Description: A COVID-19 SARS-CoV-2 IGRA (Interferon-Gamma Release Assay) is a specialized blood test that measures the cell-mediated immune response of T-cells, particularly Helper CD4+ T-cells, to the virus. Unlike standard antibody tests that look for proteins in the blood, this test evaluates the ability of "memory" T-cells to recognize and react to specific SARS-CoV-2 viral components. The test uses synthetic peptide pools (often from the Spike (S) and Nucleocapsid (N) proteins) to stimulate T-cells in a whole blood sample.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
  - Negative: <100 mIU/mL (suggests no detectable T-cell response to SARS-CoV-2).
  - Borderline: 100–200 mIU/mL (requires interpretation or repeat testing).
  - Positive: >200 mIU/mL (indicates T-cell reactivity).
- Comments:

- Test Method Variance:
- Other:
  - Associated LOINC: 98846-9, 98847-7, 99772-6, 99773-4, 99774-2
  - Associated CPT:

1.4.14 SARS_COV_2_VAR (SARS-CoV-2 (COVID-19) variant [Type] in Specimen)

- Common Name: SARS-CoV-2 (COVID-19) variant [Type] in Specimen
- Long Name: SARS-CoV-2 (COVID-19) variant [Type] in Specimen
- Test Type: SARS_COV_2_VAR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Testing for a specific SARS-CoV-2 variant (genomic sequencing) is distinct from a standard diagnostic test. While anyone can take a standard test to see if they have COVID-19, testing to identify the type of variant is typically reserved for broader public health and clinical research purposes.
- Description: SARS-CoV-2 variants are identified by analyzing the virus's genetic sequence within a specimen using genomic surveillance. CDC scientists use this genetic data alongside observed characteristics (phenotypic data) to classify variants based on their relationships, tracking how they evolve and ensuring tests, treatments, and vaccines remain effective.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Variant testing (genomic sequencing) is a qualitative analysis used to identify a specific viral lineage; therefore, it does not have a traditional "normal reference range" like a blood sugar or cholesterol test.
- Comments:
- Test Method Variance:

- Other:
  - Associated LOINC: 96741-4, 96756-2, 96755-4, 96757-0, 96751-3, 96752-1, 96894-1, 96895-8, 96896-6, 99314-7
  - Associated CPT:

1.4.15 SARS_COV_2_VL (SARS-CoV-2 (COVID-19) N gene [#/volume] (viral load) in Resp specimen by NAA with probe detection)

- Common Name: SARS-CoV-2 (COVID-19) N gene [#/volume] (viral load) in Resp specimen by NAA with probe detection
- Long Name: SARS-CoV-2 (COVID-19) N gene [#/volume] (viral load) in Resp specimen by NAA with probe detection
- Test Type: SARS_COV_2_VL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: This quantitative test is primarily indicated for diagnosing and managing active SARS-CoV-2 infections. Because it measures the specific amount of viral RNA (the "viral load") rather than just confirming its presence, it is used in clinical, research, and public health settings.
- Description: This laboratory test measures the viral load of the SARS-CoV-2 (COVID-19) nucleocapsid (N) gene in a respiratory specimen using Nucleic Acid Amplification (NAA) with probe-based detection. It provides a quantitative count of viral RNA copies per unit volume of the sample.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): There is no universal "normal" range for an infected person, clinicians often categorize viral loads based on Cycle Threshold (Ct) values to assess infectiousness.

  - High Viral Load: Often defined as a Ct value below 25 (roughly >1,000,000 copies/mL).
  - Moderate Viral Load: Typically a Ct value between 25 and 30.
  - Low Viral Load: Generally a Ct value above 30 (fewer than 1,000 copies/mL).

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 95521-1, 97104-4
  - Associated CPT:

1.4.16 SARS_COV_2_VL_LOG (SARS-CoV-2 (COVID19) RNA [Log #/volume] (viral load))

- Common Name: SARS-CoV-2 (COVID19) RNA [Log #/volume] (viral load)
- Long Name: SARS-CoV-2 (COVID19) RNA [Log #/volume] (viral load) Unspecified specimen by NAA - probe detection
- Test Type: SARS_COV_2_VL_LOG
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: While most COVID-19 PCR tests provide a simple "positive" or "negative" result, a quantitative RNA viral load test is typically indicated in specialized clinical or research scenarios.
- Description: A SARS-CoV-2 RNA viral load test is a highly sensitive diagnostic tool—most commonly a PCR test—that detects and measures the amount of genetic material (RNA) of the virus in a clinical sample, typically collected via nasal or throat swab. These tests are essential for confirming active infection, often by identifying minute amounts of the virus, making them more precise than antigen tests.
- Related Tests:

- Approximate Reference Range (Each site will have slight variations): Because viral loads can span nine orders of magnitude (from under 100 to over 1 billion copies/mL), results are typically binned into broad categories:
  - High Viral Load: Usually defined as >1,000,000 (10) copies/mL or a Ct value <25. This level is strongly associated with active shedding of infectious virus.
  - Intermediate/Moderate: Often between 10,000 and 1,000,000 copies/mL or Ct values 26–30.
  - Low Viral Load: Often <10,000 (10) copies/mL or Ct values >31. At this level, the risk of transmission is significantly lower, and the sample may contain only "dead" viral fragments.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 94819-0, 95522-9
  - Associated CPT:

1.5 Opioid Urine Toxicity

1.5.1 Opiates from natural sources

1. U_CODEINE_CNF (Codeine in Urine by Confirmation)
   - Common Name: Codeine in Urine by Confirmation
   - Long Name: Codeine in Urine by Confirmatory method
   - Test Type: U_CODEINE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The clinical indications for a urine codeine confirmation lab test are:
     - Pain Management Monitoring: Ensuring compliance for patients prescribed codeine for chronic or post-surgical pain MedlinePlus.

  - Substance Use Disorder Treatment: Monitoring abstinence or detecting unauthorized use during recovery programs NIDA.
  - Emergency Assessment: Identifying potential causes for altered mental status, unexplained seizures, or suspected opioid overdose PMC.
  - Signs of Misuse: Behavioral changes, such as seeking early refills, social withdrawal, or physiological signs like constricted pupils and flushed skin MedlinePlus.
- Description: In forensic and clinical settings, a urine drug test for codeine is performed in two steps: an initial immunoassay (IA) screening, followed by a highly sensitive and specific confirmation method, typically mass spectrometry. This two-tiered approach ensures accuracy and helps rule out false positives. The immunoassay uses antibodies that bind to a specific drug or its metabolites, triggering a color change or other signal if the drug is present above a set cutoff level.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Common Cutoff Levels: 20 ng/mL, 25 ng/mL, or 50 ng/mL.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 13641-6 (Codeine),16197-6 (Codeine),16211-5 (Dihydrocodeine),16250-3 (Codeine),19448-0 (Dihydrocodeine),19449-8 (Dihydrocodeine),3508-9 (Codeine),49829-5 (Norcodeine),51448-9 (Dihydrocodeine+Hydrocodol),51739-1 (Codeine.free),60514-7 (6_Acetylcodeine),70206-8 (Codeine)
  - Associated CPT:

2. U_CODEINE_SCR (Codeine in Urine by Screen)
   - Common Name: Codeine in Urine by Screen method
   - Long Name: Codeine in Urine by Confirmatory method

- Test Type: U_CODEINE_SCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The clinical indications for a urine codeine confirmation lab test are:
  - Pain Management Monitoring: Ensuring compliance for patients prescribed codeine for chronic or post-surgical pain MedlinePlus.
  - Substance Use Disorder Treatment: Monitoring abstinence or detecting unauthorized use during recovery programs NIDA.
  - Emergency Assessment: Identifying potential causes for altered mental status, unexplained seizures, or suspected opioid overdose PMC.
  - Signs of Misuse: Behavioral changes, such as seeking early refills, social withdrawal, or physiological signs like constricted pupils and flushed skin MedlinePlus.
- Description: A urine codeine screen is a qualitative lab test used to detect the presence of codeine and its metabolites in a urine sample. It is typically performed as part of a broader "opiate" or "opioid" panel to monitor prescription compliance, screen for substance misuse, or fulfill workplace safety requirements.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Returns a "positive" result when concentration is 300ng/mL.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 14066-5 (Dihydrocodeine),19411-8 (Codeine),19446-4 (Dihydrocodeine),3507-1 (Codeine),89310-7 (Codeine_6_glucuronide)
  - Associated CPT:

3. U_MORPHINE_CNF (Morphine in Urine by Confirmation)

- Common Name: Morphine in Urine by Confirmation
- Long Name: Morphine in Urine by Confirmatory method
- Test Type: U_MORPHINE_CNF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A morphine confirmation test is typically ordered to follow up on a preliminary screening or to provide definitive clinical or legal evidence.
- Description: A morphine confirmation test is a highly specific, quantitative laboratory procedure, usually Gas Chromatography-Mass Spectrometry (GC/MS) or Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS). It definitively identifies the exact drug—morphine—in urine, distinguishing it from other opiates like codeine, and confirms positive results from preliminary, less precise immunoassay screens.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Common Levels:
  - None Detected (Negative): No morphine was found, or the amount was too low to reach the laboratory's cutoff.
  - Quantitative Positive: If the test is positive, the lab reports a specific number (e.g., 4,500 ng/mL). Higher levels typically suggest recent direct use of morphine or heroin rather than poppy seed consumption.
  - Poppy Seed Exception: Ingesting poppy seeds can result in morphine levels up to 2,000 ng/mL. This is why federal guidelines use the higher 2,000 ng/mL cutoff to ensure workplace positives are likely due to drug misuse.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:

- Associated LOINC: 10975-1 (6_Monoacetylmorphine),13648-1 (Morphine),14845-2 (6_Monoacetylmorphine),16196-8 (Morphine),16198-4 (Diamorphine),16251-1 (Morphine),16755-1 (Diamorphine),19322-7 (6_Monoacetylmorphine),19463-9 (Ethylmorphine),19592-5 (6_Monoacetylmorphine),19593-3 (6_Monoacetylmorphine),19594-1 (6_Monoacetylmorphine),20514-6 (6_Monoacetylmorphine),20527-8 (Ethylmorphine),21050-0 (6_Monoacetylmorphine),27073-6 (6_Monoacetylmorphine),3547-7 (Diamorphine),3619-4 (Ethylmorphine),3830-7 (Morphine),3831-5 (Morphine),3851-3 (Nicomorphine),49831-1 (Normorphine),55520-1 (6_Monoacetylmorphine),70197-9 (6_Monoacetylmorphine),70210-0 (Morphine),74817-8 (Desomorphine)
- Associated CPT:

4. U_MORPHINE_SCR (Morphine in Urine by Screen)

   - Common Name: Morphine in Urine by Screen
   - Long Name: Morphine in Urine by Screen method
   - Test Type: U_MORPHINE_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A morphine screen test is indicated for several clinical, legal, and administrative reasons, primarily to monitor the use of opioid medications or to detect the presence of illicit substances.
   - Description: A urine morphine screen is a laboratory test used to detect the presence of morphine and other natural opiates in a urine sample. It is typically the first step in a two-stage process for identifying opioid use or monitoring prescription compliance.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Returns a "positive" result when concentration is 300ng/mL.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.

- Test Method Variance:
- Other:
  - Associated LOINC: 10976-9 (6_Monoacetylmorphine),16851-8 (Ethylmorphine),19321-9 (6_Monoacetylmorphine),19439-9 (Diamorphine),19462-1 (Ethylmorphine),19591-7 (6_Monoacetylmorphine),19597-4 (Morphine),3546-9 (Diamorphine),3618-6 (Ethylmorphine),74818-6 (Desomorphine),89308-1 (Morphine_6_glucuronide)
  - Associated CPT:

5. U_OPIATE_CNF (Opiate in Urine by Confirmation)

   - Common Name: Opiates in Urine by Confirmatory method
   - Long Name: Opiates in Urine by Confirmatory method
   - Test Type: U_OPIATE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Indications for an opiate confirmation test generally involve the need for definitive identification and quantification of specific substances, especially when an initial screening result is inconclusive or inconsistent with clinical expectations.
   - Description: An opiate confirmation test is a highly specific, quantitative laboratory procedure used to verify the results of an initial drug screen. While screening tests are fast and inexpensive, they are prone to false positives from cross-reactivity with other substances.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): In an opiate confirmation test, the "reference range" is typically expressed as a cutoff concentration. Because these tests are used to detect the presence of substances that should not normally be in your system (unless prescribed), the expected reference result is generally "None Detected" or "Negative".
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.

- Test Method Variance:
- Other:
  - Associated LOINC: 17384-9 (Opiates),18390-5 (Opiates),19298-9 (Opiates positive),52952-9 (Opiates),8220-6 (Opiates),8221-4 (Opiates)
  - Associated CPT:

6. U_OPIATE_SCR (Opiate in Urine by Screen)
   - Common Name: Opiates in Urine by Screen method
   - Long Name: Opiates in Urine by Screen method
   - Test Type: U_OPIATE_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Opiate screening is used in diverse medical, legal, and professional contexts to identify the use or misuse of specific opioid substances.
   - Description: An opiate screen test is a rapid, qualitative immunoassay, usually performed on urine, that detects natural and semi-synthetic opiates (morphine, codeine, heroin metabolite 6-AM) by measuring if drug levels exceed a specific cutoff (typically 25-300 ng/mL). It often misses semi-synthetic/synthetic opioids like oxycodone or fentanyl, which require specialized testing.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Cutoff levels can vary depending on whether the test is for clinical medical use or workplace/employment purposes.
     - Standard Clinical 300 ng/mL
     - Federal/Workplace 2,000 ng/mL
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
   - Test Method Variance:

- Other:
  - Associated LOINC: 19295-5 (Opiates),19296-3 (Opiates tested for),19297-1 (Opiates tested for),21431-2 (Opiates),3879-4 (Opiates),5706-7 (Opiates),5707-5 (Opiates),70150-8 (Opiates),70151-6 (Opiates),8222-2 (Opiates)
  - Associated CPT:

1.5.2 Semi-synthetic opioids

1. U_HYDROCODONE_CNF (Hydrocodone in Urine by Confirmation)

   - Common Name: Hydrocodone in Urine by Confirmatory method
   - Long Name: Hydrocodone in Urine by Confirmatory method
   - Test Type: U_HYDROCODONE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A hydrocodone confirmation test is generally ordered when a high level of precision is needed to verify the presence, absence, or exact amount of the drug.
   - Description: A hydrocodone confirmation test is a highly specific laboratory procedure used to definitively identify and quantify hydrocodone and its metabolites in a biological sample—most commonly urine. It is typically performed as a follow-up to a presumptive positive result from a faster, less specific initial screening to ensure accuracy and rule out false positives.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): The "reference range" for a hydrocodone confirmation test is generally listed as "Negative" or "None Detected".
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
   - Test Method Variance:

- Other:
  - Associated LOINC: 16252-9 (HYDROcodone),19483-7 (HYDROcodone),3681-4 (HYDROcodone),61421-4 (Norhydrocodone),61422-2 (Norhydrocodone),74760-0 (HYDROcodone & metabolites panel)
  - Associated CPT:

2. U_HYDROCODONE_SCR (Hydrocodone in Urine by Screen)
   - Common Name: Hydrocodone in Urine by Screen method
   - Long Name: Hydrocodone in Urine by Screen method
   - Test Type: U_HYDROCODONE_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A hydrocodone screen test is typically indicated for monitoring, safety, or diagnostic purposes. Healthcare providers, employers, and legal entities use these tests to ensure medication adherence or to identify substance misuse.
   - Description: A hydrocodone screen test is a medical or forensic procedure used to detect the presence of hydrocodone, a semi-synthetic opioid, in a person's system. It is often part of a broader "expanded" opiate panel since standard 5-panel drug tests typically only look for natural opiates like morphine and codeine.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Most labs use the following thresholds to determine if hydrocodone is present:
     - Initial Screen (Immunoassay) 300 ng/mL Higher threshold to reduce "false positives" from other substances.
     - Confirmatory Test (LC-MS/MS) 100 ng/mL More precise; identifies hydrocodone specifically.
     - Clinical/Pain Management 50 ng/mL Sometimes set lower to ensure patients are taking their prescribed dose.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.

- Test Method Variance:
- Other:
  - Associated LOINC: 12308-3 (HYDROcodone),19482-9 (HYDROcodone),89304-0 (Norhydrocodone)
  - Associated CPT:

3. U_HYDROMORPHONE_CNF (Hydromorphone in Urine by Confirmation)

   - Common Name: Hydromorphone in Urine by Confirmatory method
   - Long Name: Hydromorphone in Urine by Confirmatory method
   - Test Type: U_HYDROMORPHONE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A hydromorphone confirmation test is typically ordered to provide high-precision data that basic screenings cannot.
   - Description: A hydromorphone confirmation test is a highly sensitive and specific laboratory analysis used to verify the presence and concentration of hydromorphone (Dilaudid) in a biological sample, typically urine. It is performed as a second step after an initial "screening" test (like an immunoassay) returns a positive result, or when high precision is needed for clinical compliance monitoring.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): In drug testing, the "reference range" is typically reported as Negative or None Detected. Because hydromorphone is a potent opioid, any amount found in the system of a person not prescribed the medication is considered a "positive" finding.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
   - Test Method Variance:
   - Other:

  - Associated LOINC: 16998-7 (HYDROmorphone),18473-9 (HYDROmorphone),9835-0 (HYDROmorphone)
  - Associated CPT:

4. U_HYDROMORPHONE_SCR (Hydromorphone in Urine by Screen)

- Common Name: Hydromorphone in Urine by Screen method
- Long Name: Hydromorphone in Urine by Screen method
- Test Type: U_HYDROMORPHONE_SCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Clinical and forensic indications for a hydromorphone screening test generally fall into four categories: therapeutic monitoring, emergency assessment, employment/legal requirements, and addiction management.
- Description: A hydromorphone screening test is a rapid, urine-based, immunoassay that detects the presence of hydromorphone (Dilaudid) or its metabolites, typically within 6–24 hours of use. It identifies drug usage, such as heroin, morphine, or hydrocodone, which metabolize into hydromorphone.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The initial screening is a qualitative "Positive" or "Negative" result.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 19486-0 (HYDROmorphone),89309-9 (HYDROmorphone_3_glucur
  3 (HYDROmorphone)
  - Associated CPT:

5. U_OXYCODONE_CNF (Oxycodone in Urine by Confirmation)

   - Common Name: Oxycodone in Urine by Confirmation
   - Long Name: Oxycodone in Urine by Confirmatory method
   - Test Type: U_OXYCODONE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: An oxycodone confirmation test is generally indicated when a provider needs a definitive answer rather than the presumptive result provided by a standard drug screen.
   - Description: An oxycodone confirmation test is a highly sensitive and specific laboratory analysis used to definitively identify and quantify oxycodone and its metabolites in a biological sample, typically urine. Unlike initial screening tests, which are often "presumptive" and prone to false positives from cross-reactivity with other substances, a confirmation test is considered "definitive".
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Laboratories use specific numerical thresholds to determine if a result is positive. While these can vary by lab, common confirmation cutoffs include:
     - Standard Clinical Lab: Often 25 ng/mL to 50 ng/mL.
     - Federal/Workplace (DOT): Typically 100 ng/mL.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
   - Test Method Variance:
   - Other:
     - Associated LOINC: 11246-6 (oxyCODONE),16249-5 (oxyCODONE),19643-6 (oxyCODONE),61197-0 (oxyCODONE+Oxymorphone),61424-8 (Noroxycodone),61425-5 (Noroxycodone),70215-9 (oxyCODONE), 95135-0 (Noroxycodone)

- Associated CPT:

6. U_OXYCODONE_SCR (Oxycodone in Urine by Screen)
   - Common Name: Oxycodone in Urine by Screen
   - Long Name: Oxycodone in Urine by Screen method
   - Test Type: U_OXYCODONE_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Oxycodone screening tests are indicated for several medical, legal, and professional reasons, primarily centered on monitoring safety and compliance. Common categories are: therapeutic monitoring, identifying misuse or addiction, emergency & safety settings, and legal or institutional requirements.
   - Description: An oxycodone screening test is a preliminary, qualitative immunoassay (often urine) designed to detect oxycodone and its metabolite, oxymorphone, within a specific timeframe, usually within 6–24 hours to a few days after use.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Most labs use 100 ng/mL as the threshold for a positive result.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
   - Test Method Variance:
   - Other:
     - Associated LOINC: 10998-3 (oxyCODONE),19642-8 (oxyCODONE),58430-0 (oxyCODONE+Oxymorphone), 89303-2 (Noroxycodone)
     - Associated CPT:

7. U_OXYMORPHONE_CNF (Oxymorphone in Urine by Confirmation)
   - Common Name: Oxymorphone in Urine by Confirmation

- Long Name: Oxymorphone in Urine by Confirmatory method
- Test Type: U_OXYMORPHONE_CNF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: An oxymorphone confirmation test is generally indicated when a provider needs a definitive answer rather than the presumptive result provided by a standard drug screen.
- Description: An oxymorphone confirmation test is a specialized laboratory procedure used to definitively identify and quantify the presence of oxymorphone and its metabolites in a biological specimen, most commonly urine. Unlike initial screening tests, which are often subject to cross-reactivity and false positives, confirmation tests provide a "gold standard" level of accuracy for clinical and forensic monitoring.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The minimum amount of the drug that must be present for the lab to report a "positive" result is 25 ng/mL to 100 ng/mL.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 17395-5 (oxyMORphone),18325-1 (oxyMORphone),19648-5 (oxyMORphone),90894-7 (Noroxymorphone)
  - Associated CPT:

8. U_OXYMORPHONE_SCR (Oxymorphone in Urine by Screen)
   - Common Name: Oxymorphone in Urine by Screen
   - Long Name: Oxymorphone in Urine by Screen method
   - Test Type: U_OXYMORPHONE_SCR
   - Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: This test is commonly ordered to monitor medication compliance in pain management or to screen for potential drug misuse.
- Description: An oxymorphone screening test is a medical procedure used to detect the presence of oxymorphone, a powerful semi-synthetic opioid, in a person's system.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Can be as low as 25 ng/mL to 50 ng/mL depending on the lab, though some federal guidelines also use 100 ng/mL for confirmation.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 11247-4 (oxyMORphone),19646-9 (oxyMORphone),89301-6 (oxyMORphone_3_glucuronide), 89302-4 (Noroxymorphone)
  - Associated CPT:

1.5.3 Synthetic opioids

1. U_DIPIPANONE_CNF (Dipipanone in Urine by Confirmation)
   - Common Name: Dipipanone in Urine by Confirmation
   - Long Name: Dipipanone in Urine by Confirmatory method
   - Test Type: U_DIPIPANONE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The primary indications for ordering this specific test include:

  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: A confirmation test for dipipanone is typically performed using Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS) or Gas Chromatography-Mass Spectrometry (GC-MS) to definitively identify the drug.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): A positive ranges from 5 to 50 ng/mL depending on the lab.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 3574-1 (Dipipanone)
  - Associated CPT:

2. U_DIPIPANONE_SCR (Dipipanone in Urine by Screen)
   - Common Name: Dipipanone in Urine by Screen
   - Long Name: Dipipanone in Urine by Screen method
   - Test Type: U_DIPIPANONE_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Indications for a dipipanone screening test generally include:
     - Monitoring Prescription Compliance
     - Clinical Assessment of Opioid Use Disorder (OUD)
     - Emergency and Forensic Toxicology
     - Legal and Regulatory Requirements

  - Pre-operative and Specialist Review
- Description: Dipipanone is a potent synthetic opioid, often found in the combination drug Diconal.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): often around 25–100 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 3573-3 (Dipipanone)
  - Associated CPT:

3. U_FENTANYL_CNF (Fentanyl in Urine by Confirmation)

- Common Name: Fentanyl in Urine by Confirmation
- Long Name: Fentanyl in Urine by Confirmatory method
- Test Type: U_FENTANYL_CNF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indications for ordering this specific test include:
  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: A fentanyl confirmation test is a highly sensitive laboratory analysis used to definitively verify the presence of fentanyl and its metabolites in a biological sample.
- Related Tests:

- Approximate Reference Range (Each site will have slight variations): Typical confirmation cutoffs for urine testing often fall within 0.2 ng/mL to 1.0 ng/mL.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 11073-4 (Despropionylfentanyl),11075-9 (Norfentanyl),3637-6 (fentaNYL),40839-3 (fentaNYL),43200-5 (Norfentanyl),58381-5 (fentaNYL),58383-1 (Norfentanyl),66129-8 (Fentanyl+Norfentanyl),67822-7 (Fentanyl & Norfentanyl panel),74372-4 (Acetyl fentaNYL),74810-3 (Acetyl fentaNYL)
  - Associated CPT:

4. U_FENTANYL_SCR (Fentanyl in Urine by Screen)
   - Common Name: Fentanyl in Urine by Screen
   - Long Name: Fentanyl in Urine by Screen method
   - Test Type: U_FENTANYL_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Indications for a fentanyl screen test vary depending on whether the test is being used as a harm reduction tool or as a clinical/forensic diagnostic.
   - Description: A fentanyl screen test—most commonly in the form of fentanyl test strips (FTS)—is a low-cost, rapid harm reduction tool used to detect the presence of fentanyl and its analogs in drug samples or urine.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Laboratory cutoff levels can vary by facility, but typical range can be set at 1 to 2 ng/mL. Some labs may use higher cutoffs, such as 5 ng/mL.

- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 11235-9 (fentaNYL), 43199-9 (Norfentanyl), 58379-9 (Fentanyl+Norfentanyl), 59673-4 (fentaNYL)
  - Associated CPT:

5. U_LEVORPHANOL_CNF (Levorphanol in Urine by Confirmation)
   - Common Name: Levorphanol in Urine by Confirmation
   - Long Name: Levorphanol in Urine by Confirmatory method
   - Test Type: U_LEVORPHANOL_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The primary indications for ordering this specific test include:
     - Verification of Screening Results
     - Clinical Monitoring
     - Forensic and Emergency Situations
     - Workplace and Regulatory Requirements
   - Description: A levorphanol confirmation test is a definitive, laboratory-based analytical procedure used to confirm the presence of levorphanol in urine after an initial presumptive screening test indicates a positive result. Because levorphanol can cross-react with immunoassays for other opiates or be confused with the cough suppressant dextromethorphan, confirmation is required to definitively distinguish it.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Confirmation cutoff is typically 50 ng/mL to 100 ng/mL

- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 16213-1 (Levorphanol), 20540-1 (Levorphanol), 3712-7 (Levorphanol)
  - Associated CPT:

6. U_LEVORPHANOL_SCR (Levorphanol in Urine by Screen)

   - Common Name: Levorphanol in Urine by Screen
   - Long Name: Levorphanol in Urine by Screen method
   - Test Type: U_LEVORPHANOL_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A levorphanol screen test is typically indicated for patients in specific clinical scenarios involving chronic pain management or suspected substance use. Because levorphanol is a potent, long-acting synthetic opioid, it requires specific monitoring.
   - Description: A levorphanol screen test detects the presence of levorphanol, a potent prescription opioid, or its metabolites in serum, plasma, or urine, typically for medication compliance. Due to similarities in molecular structure, many assays cannot distinguish between levorphanol and the over-the-counter cough suppressant dextromethorphan, often requiring further confirmatory testing.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): generally described as "Negative", cutoff levels for these tests are often around 20 ng/mL to 50 ng/mL.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT

table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.

- Test Method Variance:
- Other:
  - Associated LOINC: 19516-4 (Levorphanol),27076-9 (Dextromethorphan+Levorphanol),39 (Levorphanol)
  - Associated CPT:

7. U_MEPERIDINE_CNF (Meperidine in Urine by Confirmation)
   - Common Name: Meperidine in Urine by Confirmation
   - Long Name: Meperidine in Urine by Confirmatory method
   - Test Type: U_MEPERIDINE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The primary indications for ordering this specific test include:
     - Verification of Screening Results
     - Clinical Monitoring
     - Forensic and Emergency Situations
     - Workplace and Regulatory Requirements
   - Description: A meperidine confirmation test is a definitive laboratory analysis used to verify the presence of meperidine (brand name Demerol) and its primary metabolite, normeperidine, in a biological sample (typically urine). It is performed after an initial screening—such as an immunoassay—returns a presumptive positive result to rule out false positives and provide a quantitative measurement.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): cutoffs can be as low as 50 ng/mL to detect even trace amounts. Some high-sensitivity labs use a "decision level" as low as 5.0 ng/mL.

- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 16207-3 (Meperidine),16253-7 (Meperidine),3747-3 (Meperidine),3869-5 (Normeperidine),4076-6 (Trimeperidine),58389-8 (Normeperidine),72384-1 (Meperidine+Normeperidine)
  - Associated CPT:

8. U_MEPERIDINE_SCR (Meperidine in Urine by Screen)

   - Common Name: Meperidine in Urine by Screen
   - Long Name: Meperidine in Urine by Screen method
   - Test Type: U_MEPERIDINE_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Meperidine (Demerol) screening is typically indicated for clinical monitoring, safety investigations, and emergency medical evaluations.
   - Description: A meperidine screening test is a preliminary medical or forensic assay used to detect the presence of meperidine (commonly known by the brand name Demerol) and its primary metabolite, normeperidine, in biological samples.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Typical cutoff is 100 ng/mL or 200 ng/mL.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
   - Test Method Variance:

- Other:
  - Associated LOINC: 19532-1 (Meperidine),27920-8 (Normeperidine),3746-5 (Meperidine),4075-8 (Trimeperidine),58385-6 (Meperidine+Normeperidine),58388-0 (Normeperidine),58953-1 (Meperidine & Normeperidine panel)
  - Associated CPT:

9. U_METHADONE_CNF (Methadone in Urine by Confirmation)
   - Common Name: Methadone in Urine by Confirmation
   - Long Name: Methadone in Urine by Confirmatory method
   - Test Type: U_METHADONE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The primary indications for ordering this specific test include:
     - Verification of Screening Results
     - Clinical Monitoring
     - Forensic and Emergency Situations
     - Workplace and Regulatory Requirements
   - Description: A methadone confirmation test is a high-complexity laboratory analysis used to definitively verify the presence of methadone and its primary metabolite, EDDP, in a specimen. Unlike standard "screening" tests that provide a quick, preliminary result, a confirmation test uses advanced technology to provide a specific, quantitative measurement of the drug and its metabolite.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): results are usually reported as follows:
     - Negative: Concentrations below the laboratory's defined cutoff (e.g., <100 ng/mL).
     - Positive: Concentrations at or above the cutoff (e.g., 100 ng/mL), followed by a specific quantitative value.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.

- Test Method Variance:
- Other:
  - Associated LOINC: 16199-2 (Methadone),16246-1 (Methadone),33527-3 (Methadone.long acting metabolite),3774-7 (Methadone)
  - Associated CPT:

10. U_METHADONE_SCR (Methadone in Urine by Screen)
    - Common Name: Methadone in Urine by Screen
    - Long Name: Methadone in Urine by Screen method
    - Test Type: U_METHADONE_SCR
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: A methadone screening test is primarily indicated for monitoring medical treatment, ensuring safety in workplace or legal settings, and diagnosing potential substance misuse or overdose.
    - Description: A methadone screening test is a preliminary medical or forensic assessment used to detect the presence of methadone or its primary metabolite, EDDP in a biological sample. Because methadone is a synthetic opioid, it is typically not detected by standard "opiate" screens, which primarily look for natural opiates like morphine or codeine.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): The "reference range" is typically reported as Negative.
    - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
    - Test Method Variance:
    - Other:
      - Associated LOINC: 19550-3 (Methadone),3773-9 (Methadone),42251-9 (Methadone+Metabolite),52958-6 (Methadone),70147-4 (Methadone),70148-2 (Methadone),70149-0 (Methadone)

- Associated CPT:

11. U_NALTREXOL_CNF (Naltrexol in Urine by Confirmation)
    - Common Name: Naltrexol in Urine by Confirmation
    - Long Name: Naltrexol in Urine by Confirmatory method
    - Test Type: U_NALTREXOL_CNF
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The primary indications for ordering this specific test include:
        - Verification of Screening Results
        - Clinical Monitoring
        - Forensic and Emergency Situations
        - Workplace and Regulatory Requirements
    - Description: A naltrexol confirmation test is a highly sensitive laboratory analysis used to verify the presence of naltrexone and its primary active metabolite in a patient's system.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Negative/Non-Adherent: $<$ 5 ng/mL or $<$ 50 ng/mL (depending on the lab's cutoff).
    - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
    - Test Method Variance:
    - Other:
        - Associated LOINC: 79377-8 (Naltrexol),82524-0 (6_Beta naltrexol)
        - Associated CPT:

12. U_NALTREXOL_SCR (Naltrexol in Urine by Screen)
    - Common Name: Naltrexol in Urine by Screen

- Long Name:
- Test Type:
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indication for a naltrexol screen test is to objectively monitor treatment adherence in patients prescribed naltrexone for alcohol or opioid use disorders.
- Description: A naltrexol screen test is a specialized laboratory analysis, usually via urine or blood, used to measure adherence to naltrexone treatment for opioid or alcohol dependence. It detects the primary metabolite of naltrexone, which remains in the body longer than the medication itself, often using chromatography or mass spectrometry to provide quantitative results.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): (Negative): < 5 ng/mL.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC:
  - Associated CPT:

13. U_PENTAZOCINE_CNF (Pentazocine in Urine by Confirmation)

- Common Name: Pentazocine in Urine by Confirmation
- Long Name: Pentazocine in Urine by Confirmatory method
- Test Type: U_PENTAZOCINE_CNF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: The primary indications for ordering this specific test include:
  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: A pentazocine confirmation test is a highly specific laboratory analysis used to definitively identify and quantify pentazocine (often known by the brand name Talwin) in a biological sample.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): cutoff is around around 50 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 16219-8 (Pentazocine),3918-0 (Pentazocine)
  - Associated CPT:

14. U_PENTAZOCINE_SCR (Pentazocine in Urine by Screen)

- Common Name: Pentazocine in Urine by Screen
- Long Name: Pentazocine in Urine by Screen method
- Test Type: U_PENTAZOCINE_SCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A pentazocine screening test is indicated when there is a clinical or legal need to confirm the presence of the drug in a person's system.

- Description: A pentazocine screening test is a preliminary, qualitative analysis used to detect the presence of pentazocine (often known by the brand name Talwin) or its metabolites in a biological sample or unknown substance.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The most common cutoff is 100 ng/mL.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 3917-2 (Pentazocine),59951-4 (Pentazocine)
  - Associated CPT:

15. U_PROPOXYPHENE_CNF (Propoxyphene in Urine by Confirmation)

    - Common Name: Propoxyphene in Urine by Confirmation
    - Long Name: Propoxyphene in Urine by Confirmatory method
    - Test Type: U_PROPOXYPHENE_CNF
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The primary indications for ordering this specific test include:
      - Verification of Screening Results
      - Clinical Monitoring
      - Forensic and Emergency Situations
      - Workplace and Regulatory Requirements
    - Description: A propoxyphene confirmation test is a highly specific laboratory procedure used to definitively identify and quantify propoxyphene and its primary metabolite, norpropoxyphene.
    - Related Tests:

- Approximate Reference Range (Each site will have slight variations): cutoff is often 200 ng/mL to 300 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 16200-8 (Propoxyphene),16242-0 (Propoxyphene),17376-5 (Norpropoxyphene),17377-3 (Norpropoxyphene),19434-0 (Propoxyphene+Norpropoxy
    7 (Propoxyphene+Norpropoxyphene),19436-5 (Propoxyphene+Norpropoxyphene),1963
    2 (Norpropoxyphene),22065-7 (Propoxyphene),3545-1 (Propoxyphene),59960-5 (Propoxyphene)
  - Associated CPT:

16. U_PROPOXYPHENE_SCR (Propoxyphene in Urine by Screen)

- Common Name: Propoxyphene in Urine by Screen
- Long Name: Propoxyphene in Urine by Screen method
- Test Type: U_PROPOXYPHENE_SCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A propoxyphene screen test is indicated when there is a need to detect the presence of propoxyphene or its metabolite, norpropoxyphene, for medical, legal, or occupational reasons.
- Description: A propoxyphene screen test is a preliminary diagnostic procedure used to detect the presence of propoxyphene and its primary metabolite, norpropoxyphene, in a biological sample—most commonly urine. Propoxyphene is a synthetic opioid pain reliever that was withdrawn from the U.S. market in 2010 due to serious heart toxicity risks.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Positive result when concentration is 300 ng/mL

- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 16749-4 (Propoxyphene+Norpropoxyphene),19141-1 (Propoxyphene),19429-0 (Propoxyphene),19433-2 (Propoxyphene+Norpropoxyphene),9 (Norpropoxyphene),3544-4 (Propoxyphene),3871-1 (Norpropoxyphene),53781-1 (Acetaminophen & Propoxyphene panel)
  - Associated CPT:

17. U_SUFENTANIL_CNF (Sufentanil in Urine by Confirmation)

- Common Name: Sufentanil in Urine by Confirmation
- Long Name: Sufentanil in Urine by Confirmatory method
- Test Type: U_SUFENTANIL_CNF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indications for ordering this specific test include:
  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: A sufentanil confirmation test is a high-specificity laboratory analysis used to definitively identify and quantify the presence of sufentanil and its metabolites in a biological specimen, typically following an initial positive screen. This second-tier test is critical because standard opioid screens often do not detect synthetic opioids like sufentanil, or they may produce false positives from other substances.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Confirmation Cutoff (Urine) 5 ng/mL

- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 9396-3 (SUFentanil)
  - Associated CPT:

18. U_SUFENTANIL_SCR (Sufentanil in Urine by Screen)

- Common Name: Sufentanil in Urine by Screen
- Long Name: Sufentanil in Urine by Screen method
- Test Type: U_SUFENTANIL_SCR
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A sufentanil screen test is typically indicated when standard opioid panels are insufficient to detect this specific, high-potency drug. It is used across various clinical, forensic, and occupational settings to verify use, monitor safety, or ensure compliance.
- Description: A sufentanil screen test is a specialized laboratory procedure designed to detect the presence of sufentanil, an extremely potent synthetic opioid, typically in urine or blood samples. Because sufentanil is 5 to 10 times more powerful than fentanyl, it is often excluded from standard "opiate" or even "fentanyl" screening panels and must be ordered as a specific test or as part of a high-level comprehensive panel.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Urine: 5 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.

- Test Method Variance:
- Other:
  - Associated LOINC: 18435-8 (SUFentanil)
  - Associated CPT:

19. U_TAPENTADOL_CNF (Tarpentadol in Urine by Confirmation)

    - Common Name: Tarpentadol in Urine by Confirmation
    - Long Name: Tapentadol in Urine by Confirmatory method
    - Test Type: U_TAPENTADOL_CNF
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The primary indications for ordering this specific test include:
      - Verification of Screening Results
      - Clinical Monitoring
      - Forensic and Emergency Situations
      - Workplace and Regulatory Requirements
    - Description: A tapentadol confirmation test is a definitive laboratory analysis used to identify and quantify the drug and its metabolites in biological samples, typically following a presumptive positive result from a screening test.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Positive (Cutoff): Quantifiable levels (e.g., 50–2,500 ng/mL)
    - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
    - Test Method Variance:
    - Other:

  - Associated LOINC: 58401-1 (Tapentadol),58402-9 (Tapentadol),65807-0 (Tapentadol),65808-8 (Nortapentadol),81754-4 (Nortapentadol)
  - Associated CPT:

20. U_TAPENTADOL_SCR (Tarpentadol in Urine by Screen)

    - Common Name: Tarpentadol in Urine by Screen
    - Long Name: Tapentadol in Urine by Screen method
    - Test Type: U_TAPENTADOL_SCR
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: It is primarily used by healthcare providers to monitor treatment compliance for chronic pain or to identify potential misuse or acute intoxication.
    - Description: A tapentadol screening test is a diagnostic tool used to detect the presence of tapentadol (commonly known by brand names like Nucynta, Palexia, or Tapal) and its metabolites in a person's system.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Typical Cutoff: 200 ng/mL.
    - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
    - Test Method Variance:
    - Other:
      - Associated LOINC: 72485-6 (Tapentadol),75362-4 (Tapentadol glucuronide),89300-8 (Tapentadol glucuronide),89306-5 (Nortapentadol)
      - Associated CPT:

21. U_TRAMADOL_CNF (Tramadol in Urine by Confirmation)

    - Common Name: Tramadol in Urine by Confirmation

- Long Name: Tramadol in Urine by Confirmatory method
- Test Type: U_TRAMADOL_CNF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indications for ordering this specific test include:
  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: A tramadol confirmation test is a definitive laboratory analysis used to verify a preliminary positive result from a screening test or to accurately measure the specific amount of tramadol and its metabolites in a sample.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Common Cutoff: 25 ng/mL for both tramadol and its metabolite, O-desmethyltramadol.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 17718-8 (traMADol),17719-6 (traMADol),18338-4 (Nortramadol),20561-7 (traMADol)
  - Associated CPT:

22. U_TRAMADOL_SCR (Tramadol in Urine by Screen)

- Common Name: Tramadol in Urine by Screen
- Long Name: Tramadol in Urine by Screen method
- Test Type: U_TRAMADOL_SCR
- Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indications for ordering this specific test include:
  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: A tramadol screen test is a specialized drug test used to detect the presence of tramadol—a synthetic opioid analgesic—and its main metabolite, O-desmethyltramadol, in a person's system. Unlike standard 5-panel or 10-panel drug tests (SAMHSA-5), which typically look for opiates like codeine or morphine, tramadol requires a specific test because its chemical structure is different.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Negative when below 100 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 19710-3 (traMADol),43219-5 (traMADol),86453-8 (O-nortramadol),86454-6 (N-nortramadol)
  - Associated CPT:

23. U_NARCOTIC_SCR (Narcotics and opioids in Urine by Screen)
    - Common Name: Narcotics and opioids in Urine by Screen
    - Long Name: Narcotics and opioids in Urine by Screen method
    - Test Type: U_NARCOTIC_SCR
    - Panels:
    - Equivalent Tests:

- Not Equivalent Tests:
- Indications: The primary indications for ordering this specific test include:
  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: An active laboratory test used to detect the presence or identity of narcotic substances—specifically derivatives of opium or morphine, or synthetic substances with similar effects—within a urine sample.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): If the test is positive, it means the drug was detected above one of the following typical cutoff levels:
  - Opiates (Morphine, Codeine) 2,000 ng/mL (standard) or 300 ng/mL (clinical)
  - Hydrocodone / Hydromorphone 300 ng/mL
  - Oxycodone / Oxymorphone 100 ng/mL
  - Heroin Metabolite (6-AM) 10 ng/mL
  - Methadone 300 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 12554-2 (Narcotics)
  - Associated CPT:

1.5.4 Loosely referred to as opioids

1. U_BUPRENORPHINE_CNF (Buprenorphine in Urine by Confirmation)
   - Common Name: Buprenorphine in Urine by Confirmation

- Long Name: Buprenorphine in Urine by Confirmatory method
- Test Type: U_BUPRENORPHINE_CNF
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indications for ordering this specific test include:
  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: A buprenorphine confirmation test is a definitive laboratory analysis used to verify the presence and concentration of buprenorphine and its primary metabolite, norbuprenorphine.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Typically < 5.0 ng/mL is considered negative.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 16208-1 (Buprenorphine),16496-2 (Buprenorphine),3415-7 (Buprenorphine),49751-1 (Norbuprenorphine),49752-9 (Buprenorphine),49753-7 (Norbuprenorphine),58362-5 (Norbuprenorphine),69033-9 (Buprenorphine & Norbuprenorphine panel)
  - Associated CPT:

2. U_BUPRENORPHINE_SCR (Buprenorphine in Urine by Screen)

- Common Name: Buprenorphine in Urine by Screen
- Long Name: Buprenorphine in Urine by Screen method
- Test Type: U_BUPRENORPHINE_SCR
- Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indications for ordering this specific test include:
  - Verification of Screening Results
  - Clinical Monitoring
  - Forensic and Emergency Situations
  - Workplace and Regulatory Requirements
- Description: A buprenorphine (BUP) screen test is a specialized medical assay used to detect the presence of buprenorphine and its primary metabolite, norbuprenorphine, in a person's system. Buprenorphine is a partial opioid agonist commonly prescribed for opioid use disorder (e.g., Suboxone or Subutex) and chronic pain.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Negative is less than 5 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 3414-0 (Buprenorphine),38373-7 (Buprenorphine+Norbuprenorphine
    1 (Buprenorphine+Norbuprenorphine),82371-6 (Norbuprenorphine),89305-
    7 (Norbuprenorphine_3_glucuronide),91027-3 (Buprenorphine_3_glucuronide),
    93494-3 (all metabolites)
  - Associated CPT:

3. U_BUTORPHANOL_CNF (Butorphanol in Urine by Confirmation)

- Common Name: Butorphanol in Urine by Confirmation
- Long Name: Butorphanol in Urine by Confirmatory method
- Test Type: U_BUTORPHANOL_CNF

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: A butorphanol confirmation test is indicated in clinical, legal, or workplace settings to definitively verify the drug's presence and concentration after an initial screening has occurred or is deemed insufficient.
- Description: A butorphanol confirmation test is a specialized laboratory procedure used to definitively identify and quantify the presence of butorphanol (brand name Stadol) in a biological sample, typically following a positive or non-negative screening result. Unlike screening tests that only detect a class of drugs, a confirmation test provides a "fingerprint" of the specific molecule.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Urine Confirmation Cut-off is frequently set at 300 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 12395-0 (Butorphanol),16499-6 (Butorphanol),19378-9 (Butorphanol)
  - Associated CPT:

4. U_BUTORPHANOL_SCR (Butorphanol in Urine by Screen)
   - Common Name: Butorphanol in Urine by Screen
   - Long Name: Butorphanol in Urine by Screen method
   - Test Type: U_BUTORPHANOL_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:

- Indications: Since butorphanol is a Schedule IV controlled substance, these tests are used to monitor for misuse, abuse, or compliance with medical treatment.
- Description: A butorphanol screen test is a toxicology screening designed to detect the presence of butorphanol (brand name Stadol) or its metabolites in biological samples.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Urine Screen cutoff is between 2.5 – 5.0 ng/mL
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 12333-1 (Butorphanol),19377-1 (Butorphanol)
  - Associated CPT:

5. U_NALBUPHINE_CNF (Nalbuphine in Urine by Confirmation)
   - Common Name: Nalbuphine in Urine by Confirmation
   - Long Name: Nalbuphine in Urine by Confirmatory method
   - Test Type: U_NALBUPHINE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A nalbuphine confirmation test is typically indicated when specific, high-certainty evidence of the drug's presence or concentration is required for clinical, legal, or occupational reasons.
   - Description: A nalbuphine confirmation test is a high-sensitivity laboratory analysis used to definitively identify and quantify the presence of nalbuphine (an opioid agonist-antagonist) in a urine specimen, typically following an initial positive screening. Unlike standard "opiates" screenings that look for morphine or codeine, nalbuphine often requires a specific test panel.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): typical analytic cutoff is 50 ng/mL.
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 16218-0 (Nalbuphine),20551-8 (Nalbuphine),3840-6 (Nalbuphine)
  - Associated CPT:

6. U_NALBUPHINE_SCR (Nalbuphine in Urine by Screen)

   - Common Name: Nalbuphine in Urine by Screen
   - Long Name: Nalbuphine in Urine by Screen method
   - Test Type: U_NALBUPHINE_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: A Nalbuphine screen test is clinically indicated primarily to monitor for misuse, ensure treatment compliance, or evaluate for toxicity.
   - Description: A Nalbuphine screen test is a specialized laboratory procedure designed to detect the presence of Nalbuphine (brand name Nubain) and its metabolites in biological specimens. Because Nalbuphine is a synthetic "mixed agonist-antagonist" opioid, it is often "invisible" to standard, routine drug screens that typically target common opiates like morphine or codeine.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): specialized Nalbuphine screens may have specific laboratory cutoffs, such as 1,250 ng/mL.

- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 19605-5 (Nalbuphine),3839-8 (Nalbuphine)
  - Associated CPT:

7. U_NALOXONE_CNF (Naloxone in Urine by Confirmation)

   - Common Name: Naloxone in Urine by Confirmation
   - Long Name: Naloxone in Urine by Confirmatory method
   - Test Type: U_NALOXONE_CNF
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: It is typically performed as a "reflex" test to follow up on a presumptive positive result from a simpler screening test (like an immunoassay) or to monitor medication adherence in treatment programs.
   - Description: A Naloxone confirmation test is a highly specific laboratory analysis—usually using Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS)—to verify and quantify the presence of naloxone in a biological sample like urine or oral fluid.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): A common reference range is <2 ng/mL (meaning any result below this is considered "not detected")
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
   - Test Method Variance:

- Other:
  - Associated LOINC: 77207-9 (Naloxone), 92649-3 (Naloxone)
  - Associated CPT:

8. U_NALOXONE_SCR (Naloxone in Urine by Screen)

   - Common Name: Naloxone in Urine by Screen
   - Long Name: Naloxone in Urine by Screen method
   - Test Type: U_NALOXONE_SCR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Naloxone screen tests are primarily used in clinical settings to monitor patients receiving Medication-Assisted Treatment (MAT) for opioid use disorder. Because naloxone is not a drug of abuse, it is rarely included in standard workplace or forensic drug panels.
   - Description: A naloxone screen test is a specialized laboratory analysis, typically of urine, used to detect the presence of naloxone or its metabolites. Unlike standard drug tests that look for substances of abuse (like cocaine or marijuana), naloxone is often tested to monitor medical treatment or verify the integrity of a sample.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): the most common "not detected" reference range is <2 ng/mL.
   - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
   - Test Method Variance:
   - Other:
     - Associated LOINC: 42618-9 (Naloxone),89307-3 (Naloxone_3_glucuronide)
     - Associated CPT:

9. U_NALTREXONE_CNF (Naltrexone in Urine by Confirmation)

    - Common Name: Naltrexone in Urine by Confirmation
    - Long Name: Naltrexone in Urine by Confirmatory method
    - Test Type: U_NALTREXONE_CNF
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indications for a naltrexone confirmation test (typically via GC-MS or LC-MS/MS) revolve around managing treatment for opioid or alcohol use disorders.
    - Description: A naltrexone confirmation test is a specialized laboratory procedure used to definitively identify and quantify naltrexone and its metabolites in a biological sample, typically following an initial "non-negative" screening. Unlike standard drug screens, this test is designed to verify medication adherence or resolve false-positive results on other drug panels.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Results below 5 ng/mL are reported as "None Detected" or "Negative".
    - Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
    - Test Method Variance:
    - Other:
        - Associated LOINC: 18334-3 (Naltrexone),19610-5 (Naltrexone),19611-3 (Naltrexone),72787-5 (6_Beta naltrexone)
        - Associated CPT:

10. U_NALTREXONE_SCR (Naltrexone in Urine by Screen)

    - Common Name: Naltrexone in Urine by Screen
    - Long Name: Naltrexone in Urine by Screen method
    - Test Type: U_NALTREXONE_SCR

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The most common indications for a naltrexone urine screen—or the broader urine toxicology screens used alongside it—revolve around safe initiation and maintenance of treatment for Alcohol Use Disorder (AUD) and Opioid Use Disorder (OUD).
- Description: A Naltrexone urine screen test is a specialized laboratory analysis designed to detect the presence of naltrexone and its primary metabolite, 6-beta-naltrexol, in a person's urine. While not part of standard drug panels, it is frequently used to monitor treatment adherence for individuals recovering from alcohol or opioid use disorders.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Generally <5 ng/mL is considered "negative"
- Comments: For opioid results place the component value from the EverLOINC.SAS table into the local_cd field of the LAB_RESULT table. The component value is seen in parentheses in this spreadsheet after each LOINC and names the form of the drug (metabolite) tested for.
- Test Method Variance:
- Other:
  - Associated LOINC: 12309-1 (Naltrexone),19609-7 (Naltrexone)
  - Associated CPT:

1.6 Glucose Tolerance Tests (GTT)

1.6.1 GTT Test_Types

(NOTE: Information about all the numerous GTT TEST_TYPES is included in this document)

- Common Name: OGTT; GTT; Oral glucose tolerance test; glucose tolerance test

Test Types and Long Names: See Table 1. Table 1. Glucose Tolerance Test TEST_TYPE and Long Name TEST_TYPE LONGNAME

1. GTT_0_5 (Glucose Tolerance 1/2 hour post XXX challenge)
    - Common Name: Glucose Tolerance 1/2 hour post XXX challenge
    - Long Name: Glucose Tolerance 1/2 hour post XXX challenge (glucose dose not specified)
    - Test Type: GTT_0_5
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
    - Description: In a glucose tolerance test, the 30-minute post-dose mark is the first measurement taken after you consume the glucose solution. This specific interval is used to observe how rapidly your body begins to process sugar and is a key indicator of early insulin response.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Research suggests that a high 30-minute glucose level (e.g., above 182 mg/dL) can be a strong predictor of future type 2 diabetes.
    - Comments:
    - Test Method Variance:
    - Other:
        - Associated LOINC: 14763-7, 20439-6, 25671-9, 40263-6
        - Associated CPT:
2. GTT_1 (Glucose Tolerance 1 hour post XXX challenge)
    - Common Name: Glucose Tolerance 1 hour post XXX challenge
    - Long Name: Glucose Tolerance 1 hour post XXX challenge (glucose dose not specified)
    - Test Type: GTT_1

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
- Description: A one-hour glucose tolerance test (also called a glucose challenge test) is a standard screening used during pregnancy, typically between weeks 24 and 28, to check for gestational diabetes. Unlike longer versions of the test, it usually does not require fasting beforehand.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): A result below 140 mg/dL (7.8 mmol/L) is typically considered within the standard range.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 12646-6, 14756-1, 20438-8, 25665-1
  - Associated CPT: 82947, 82950, 82951

3. (GTT_1_5 Deprecated) (Glucose Tolerance 1 1/2 hour post XXX challenge (glucose dose not specified) ***NEEDS TO BE UPDATED***

4. GTT_2 (Glucose Tolerance 2 hour post XXX challenge)

   - Common Name: Glucose Tolerance 2 hour post XXX challenge
   - Long Name: Glucose Tolerance 2 hour post XXX challenge (glucose dose not specified)
   - Test Type: GTT_2
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:

- Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
- Description: A 2-hour glucose tolerance test (OGTT) measures how your body handles sugar by checking your blood glucose levels exactly two hours after you consume a standardized sugary drink. This test is primarily used to diagnose prediabetes, type 2 diabetes, and gestational diabetes.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): For non-pregnant adults the 2-hour post-dose value is critical for diagnosis:
  - Normal: Less than 140 mg/dL (7.8 mmol/L).
  - Prediabetes (Impaired Glucose Tolerance): Between 140 and 199 mg/dL (7.8–11 mmol/L).
  - Diabetes: 200 mg/dL (11.1 mmol/L) or higher.
  - For pregnant individuals, a 2-hour result above 153 mg/dL may indicate gestational diabetes, though thresholds can vary by lab.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 12610-2, 14759-5, 20436-2, 25668-5, 49134-0
  - Associated CPT: 82951, 82950

5. GTT_3 (Glucose Tolerance 3 hour post XXX challenge (glucose dose not specified)

   - Common Name: Glucose Tolerance 3 hour post XXX challenge
   - Long Name: Glucose Tolerance 3 hour post XXX challenge (glucose dose not specified)
   - Test Type: GTT_3
   - Panels:

- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
- Description: A 3-hour glucose tolerance test (GTT) is a diagnostic procedure used primarily to confirm gestational diabetes if an initial screening test was abnormal. It measures how your body clears sugar from the blood over several hours.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The most common expected expected healthy range is less than 140 mg/dL (7.8 mmol/L) 3 hours after glucose dose.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 14765-2, 18342-6, 20437-0, 40162-0
  - Associated CPT: 82951

6. (GTT_4 Deprecated) (Glucose Tolerance 4 hour post XXX challenge (glucose dose not specified)

7. (GTT_5 Deprecated) (Glucose Tolerance 5 hour post XXX challenge (glucose dose not specified)

8. (GTT_6 Deprecated) (Glucose Tolerance 6 hour post XXX challenge (glucose dose not specified)

9. (GTT_7 Deprecated) (Glucose Tolerance 7 hour post XXX challenge (glucose dose not specified)

10. GTT50_1 (Glucose Tolerance 1 hour post 50 g oral challenge)

    - Common Name: Glucose Tolerance 1 hour post 50 g oral challenge
    - Long Name: Glucose Tolerance 1 hour post 50 g oral challenge
    - Test Type: GTT50_1

- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
- Description: The 1-hour, 50-gram glucose tolerance test (also known as a glucose challenge test) is a routine screening performed between 24 and 28 weeks of pregnancy to identify the risk of gestational diabetes. It measures how efficiently your body processes sugar after a concentrated dose.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): A blood sugar level below 140 mg/dL (7.8 mmol/L) is generally considered standard.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 14754-6, 1504-0
  - Associated CPT: 82950

11. GTT50_2 (Glucose Tolerance 2 hour post 50 g oral challenge)

    - Common Name: Glucose Tolerance 2 hour post 50 g oral challenge
    - Long Name: Glucose Tolerance 2 hour post 50 g oral challenge
    - Test Type: GTT50_2
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.

- Description: A 2-hour glucose tolerance test with a 50-gram dose is an uncommon variation of standard glucose testing, typically used as a screening tool during pregnancy rather than a definitive diagnostic test. While the standard diagnostic 2-hour test uses 75 grams of glucose, some specific protocols or "glucose challenges" may use 50 grams with a 2-hour draw to assess how the body processes sugar over a longer interval than the standard 1-hour screen.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): There is no universal standard for this test:
  - Normal: Less than 140 mg/dL (7.8 mmol/L)
  - Prediabetes (Impaired Glucose Tolerance): 140–199 mg/dL (7.8–11.0 mmol/L)
  - Diabetes: 200 mg/dL (11.1 mmol/L) or higher
- Comments: If your provider has ordered a 2-hour test specifically with 50 grams, they may be looking for reactive hypoglycemia (low blood sugar after eating) or using a specialized protocol for your specific health history.
- Test Method Variance:
- Other:
  - Associated LOINC: 41024-1
  - Associated CPT:

12. GTT75_0_5 (Glucose Tolerance 1/2 hour post 75 g oral challenge)

    - Common Name: Glucose Tolerance 1/2 hour post 75 g oral challenge
    - Long Name: Glucose Tolerance 1/2 hour post 75 g oral challenge
    - Test Type: GTT75_0_5
    - Panels: 81324-6 OB panel
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose

type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.

- Description: In a 75-gram oral glucose tolerance test (OGTT), the 30-minute blood draw is an intermediate measurement used primarily to assess how quickly your body begins secretes insulin in response to sugar.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range is generally less than 200 mg/dL (11.1 mmol/L).
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 1527-1, 32319-6
  - Associated CPT: 82951

13. GTT75_1 (Glucose Tolerance 1 hour post 75 g oral challenge)

    - Common Name: Glucose Tolerance 1 hour post 75 g oral challenge
    - Long Name: Glucose Tolerance 1 hour post 75 g oral challenge
    - Test Type: GTT75_1
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
    - Description: In a 75-gram Oral Glucose Tolerance Test (OGTT), the 1-hour blood draw is a critical measurement used to evaluate how your body processes a concentrated sugar load. While the 2-hour mark is the traditional diagnostic standard for diabetes, the 1-hour level is increasingly recognized as a sensitive early indicator of metabolic health.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range is less than 155 mg/dL (8.6 mmol/L)
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 1507-3, 1508-1, 51597-3
  - Associated CPT:

14. (GTT75_1_5 Deprecated) (Glucose Tolerance 1 1/2 hour post 75 g oral challenge)

15. GTT75_2 (Glucose Tolerance 2 hour post 75 g oral challenge)

    - Common Name: Glucose Tolerance 2 hour post 75 g oral challenge
    - Long Name: Glucose Tolerance 2 hour post 75 g oral challenge
    - Test Type: GTT75_2
    - Panels: 81324-6 OB panel
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
    - Description: A 75-gram, 2-hour oral glucose tolerance test (OGTT) is a diagnostic procedure used to evaluate how your body processes sugar. You must fast (no food or drink except water) for at least 8 to 14 hours before the test.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Normal is $<$ 140 mg/dL
    - Comments:
    - Test Method Variance:
    - Other:

  - Associated LOINC: 14995-5, 1518-0, 1519-8
  - Associated CPT: 82951, 82950

16. GTT75_3 (Glucose Tolerance 3 hour post 75 g oral challenge)

    - Common Name: Glucose Tolerance 3 hour post 75 g oral challenge
    - Long Name: Glucose Tolerance 3 hour post 75 g oral challenge
    - Test Type: GTT75_3
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
    - Description: A 3-hour glucose tolerance test (OGTT) using a 75-gram dose is a diagnostic procedure that measures how your body processes sugar over time. While a 75-gram dose is more commonly associated with a 2-hour test for prediabetes or type 2 diabetes, certain protocols use it for a 3-hour duration to screen for gestational diabetes or other metabolic issues.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Normal range is < 140 mg/dL
    - Comments:
    - Test Method Variance:
    - Other:
      - Associated LOINC: 1533-9, 32320-4
      - Associated CPT:

17. GTT100_0_5 (Glucose Tolerance 1/2 hour post 100 g oral challenge)

    - Common Name: Glucose Tolerance 1/2 hour post 100 g oral challenge

- Long Name: Glucose Tolerance 1/2 hour post 100 g oral challenge
- Test Type: GTT100_0_5
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
- Description: During a 100-gram oral glucose tolerance test (OGTT), the 30-minute mark represents a critical point where your body is actively absorbing the sugar load, causing blood glucose levels to begin their sharpest climb toward a peak.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range is around 140–160 mg/dL
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 1525-5
  - Associated CPT: 82952

18. GTT100_1 (Glucose Tolerance 1 hour post 100 g oral challenge)

    - Common Name: Glucose Tolerance 1 hour post 100 g oral challenge
    - Long Name: Glucose Tolerance 1 hour post 100 g oral challenge
    - Test Type: GTT100_1
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:

- Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
- Description: In the context of prenatal care, the 1-hour blood draw after a 100-gram glucose dose is the first post-drink measurement of a diagnostic 3-hour Oral Glucose Tolerance Test (OGTT). This measurement evaluates how efficiently your body processes a concentrated sugar load during a period of peak insulin resistance in pregnancy.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range is 180 mg/dL (10.0 mmol/L) or lower.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 14753-8, 1501-6, 1502-4
  - Associated CPT: 82951

19. (GTT100_1_5 Deprecated) (Glucose Tolerance 1 1/2 hour post 100 g oral challenge)

20. GTT100_2 (Glucose Tolerance 2 hour post 100 g oral challenge)

    - Common Name: Glucose Tolerance 2 hour post 100 g oral challenge
    - Long Name: Glucose Tolerance 2 hour post 100 g oral challenge
    - Test Type: GTT100_2
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.

- Description: A 2-hour 100-gram Glucose Tolerance Test (GTT) is a diagnostic procedure, often used for gestational diabetes, measuring how the body handles glucose.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Normal range is $<$ 155 mg/dL (8.6 mmol/L)
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 14757-9, 1514-9, 1515-6
  - Associated CPT: 82951, 82950

21. GTT100_3 (Glucose Tolerance 3 hour post 100 g oral challenge)

    - Common Name:Glucose Tolerance 3 hour post 100 g oral challenge
    - Long Name: Glucose Tolerance 3 hour post 100 g oral challenge
    - Test Type: GTT100_3
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The most common indication for an oral glucose tolerance test (OGTT) is to screen for and diagnose gestational diabetes in pregnant women. It is also frequently used to diagnose type 2 diabetes and prediabetes in non-pregnant adults, particularly when other tests (like fasting glucose or A1c) provide borderline or inconclusive results.
    - Description: The 3-hour 100-gram Oral Glucose Tolerance Test (OGTT) is a diagnostic procedure used primarily to determine if a pregnant woman has gestational diabetes. It is typically performed as the second step of a "two-step" screening process if the initial 1-hour glucose challenge result was elevated.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Normal range is $<$ 140 mg/dL
    - Comments:

- Test Method Variance:
- Other:
  - Associated LOINC: 14764-5, 1530-5, 1531-3
  - Associated CPT: 82951

22. (GTT100_3_5 Deprecated) (Glucose Tolerance 3 1/2 hour post 100 g oral challenge)

23. (GTT100_4 Deprecated) (Glucose Tolerance 4 hour post 100 g oral challenge)

24. (GTT100_5 Deprecated) (Glucose Tolerance 5 hour post 100 g oral challenge)

25. (GTT100_6 Deprecated) (Glucose Tolerance 6 hour post 100 g oral challenge)

26. (GTT100_PRE Deprecated) (Glucose pre 100 g oral challenge, fasting)

27. (GTT50_PRE Deprecated) (Glucose pre 50 g oral challenge, fasting)

28. (GTT75_PRE Deprecated) (Glucose pre 75 g oral challenge, fasting)

29. (GTT_PRE Deprecated) (Glucose pre oral challenge, fasting; glucose dose not specified) It is anticipated that each site will populate some subset of the TEST_TYPEs listed above. Sites are not expected to find that all are in use at their locations. It is also anticipated that the 1 hour and 2 hour TEST_TYPEs will be the more common time points (see below).

    - Panels: NA
    - Equivalent Tests: The test types in Table 1 that contain "pre" in the test type name are fasting glucose values (i.e., values obtained prior to the patient drinking the glucose load). For research projects where fasting glucose tests are assessed, it may be appropriate to ask the investigator whether he/she also wants to include these fasting glucose tests that were obtained as part of a GTT challenge test. The GTT "pre" tests are not included in the GLU_F (fasting glucose test type) in the VDW since they are included in the GTT test types.

- Not Equivalent Tests: Do not include any tests where the name or title of the test includes "post prandial" or "lactose tolerance." GGT (or GGTP, gamma-glutamyl transpeptidase) is an unrelated test. Random glucose tests are related but not equivalent tests.
- Indications: The oral glucose tolerance test (GTT) is considered a "challenge test" in that it is used to assess the presence of gestational diabetes in pregnant women based on a "challenge" or loading dose of glucose. It is also used (not common) to assess patients who have signs and symptoms of diabetes mellitus but whose fasting plasma glucose is normal or suggests prediabetes. The GTT provides information about both the ability of the pancreas to secrete insulin after a glucose "load" and the body's response to insulin. Interpretation of the results of the GTT is usually based on both the blood glucose concentrations measured before the glucose load is ingested and at set time points after drinking the glucose load. In women who had gestational diabetes, a postpartum glucose evaluation is also often done. In this situation, either a 2 hour post 75 g glucose load test (see below) or a standard fasting glucose is obtained within 6 weeks after delivery.
- Description: A standard 50-, 75-, or 100-gram dose of an oral glucose solution is ingested (over 5 minutes for the 75-g dose) after an overnight fast. Blood samples are usually drawn before the test, between 0 and 2 hours, and at 2 hours. In pregnant women, a "screening" challenge GTT test is often done that measures plasma glucose 1 hour after a 50-g glucose load. If the glucose level is $>$ 130 mg/dL, then a 100-g glucose load is done (see Table 2). Glucose concentrations are drawn before drinking the glucose load for the test, and at 1 hour, 2 hours, and 3 hours after drinking the glucose load. Time periods beyond 4 hours are not routinely used for gestational diabetes assessment. We anticipate that the 75 g GTT test will be used more often as we go forward in time. This is because the 2011 American Diabetes Association (ADA) criteria for gestational diabetes mellitus recommend a 75g GTT in all pregnant women at 24-28 weeks gestation. The times and values used in this newly-recommended standard include a fasting result, and a 1 hr and 2 hr result as part of the GTT. A 75g 2hr value $>$153 mg/dl during pregnancy is diagnostic for gestational diabetes mellitus.
- Related Tests: Fasting glucose, random glucose

- Approximate Reference Range: Values above which the test results are considered abnormal or high or indicative of diabetes at 1, 2, and/or 3 hours are shown in Table 2.

Table 2. Threshold Values for Diagnosis Diabetes Mellitus Using the Glucose Tolerance Test Glucose Load for GTT Fasting Plasma Glucose (mg/dL) GTT Value at 1 hour (or 30 or 90 minutes) GTT Value at 2 hours GTT Value at 3 hours Gestational diabetes 50 g $<$ 95 $>$ 130 75 g $<$ 95 $>$ 153 100 g $<$ 95 $>$180 * $>$155 * $>$140 * Assessing glucose tolerance in individual who is not pregnant (75 g load) Normal $<$ 100 $<$ 200 $<$ 140 Prediabetes 100 – 125 $>$ 200 140-199 Diabetes $>$ 126 $>$ 200 $>$ 200 If 2 or more glucose levels exceed the threshold, then a diagnosis of gestational diabetes is made.

- Comments: Component test names for the GTT vary widely from site to site. Sometimes the words "gestational," "gestational diabetes," or "OB screen" are in the name. Sometimes the word "tolerance" is in the name, but not always.
- Test Method Variance: See statements above about the varying glucose loads that are used.
- Other: Some sites do not use unique CPT or LOINC codes for different glucose loads. There may be unique codes for the different time increments (1hr, 2 hr, etc), but not always. The text string field usually specifies the hour (for example, a GTT 3 hr may have three rows of data in which the text string specifies the specific hour [1 hr, 2 hr, 3 hr]). Loading dose of glucose is not included in electronic data at all sites. Because the loading dose of glucose is not always indicated in the electronic lab data stream, it is crucial to ensure the abnormal flags are populated in the VDW. As shown in Table 2, the criteria for determining whether or not the lab results are abnormal vary, not only based on the time the sample was obtained, but also based on the glucose load the individual ingested. The abnormal flag, along with the result value, may be the only way to conclusively know whether or not the test result was abnormal.

30. (GTTPM) (Glucose Tolerance post meal)

- Common Name: Glucose Tolerance post meal
- Long Name: Glucose Tolerance post meal

- Test Type: GTTPM
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Healthcare providers order a 2-hour postprandial glucose (PPG) test to assess how effectively the body regulates blood sugar after a meal or a standardized sugar load. The conditions that they are trying to are: Diabetes or Prediabetes, Gestational Diabetes (GDM), Identifying Impaired Glucose Tolerance (IGT).
- Description: A post-meal (postprandial) glucose tolerance test measures how your body handles sugar after eating. Typically, it involves a baseline blood draw, consumption of a high-carbohydrate meal or glucose drink, and a follow-up blood test 2 hours later to check how effectively insulin lowers blood sugar levels.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
  - Normal (No Diabetes): Less than 140 mg/dL.
  - Prediabetes: Between 140 and 199 mg/dL.
  - Diabetes: 200 mg/dL or higher.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 16915-1, 53094-9
  - Associated CPT:

31. (GTTPM_0_5) (Glucose Tolerance 1/2 hour post meal)

    - Common Name: Glucose Tolerance 1/2 hour post meal
    - Long Name: Glucose Tolerance 1/2 hour post meal
    - Test Type: GTTPM_0_5
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:

- Indications: Healthcare providers order a 2-hour postprandial glucose (PPG) test to assess how effectively the body regulates blood sugar after a meal or a standardized sugar load. The conditions that they are trying to are: Diabetes or Prediabetes, Gestational Diabetes (GDM), Identifying Impaired Glucose Tolerance (IGT).
- Description: Thirty minutes post-meal, postprandial glucose (PPG) is actively rising as the body breaks down carbohydrates, typically reaching its peak between 30 to 90 minutes later.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): healthy levels usually remain under 140 mg/dL.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 95105-3
  - Associated CPT:

32. (GTTPM_1) (Glucose Tolerance 1 hour post meal)

    - Common Name: Glucose Tolerance 1 hour post meal
    - Long Name: Glucose Tolerance 1 hour post meal
    - Test Type: GTTPM_1
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: Healthcare providers order a 2-hour postprandial glucose (PPG) test to assess how effectively the body regulates blood sugar after a meal or a standardized sugar load. The conditions that they are trying to are: Diabetes or Prediabetes, Gestational Diabetes (GDM), Identifying Impaired Glucose Tolerance (IGT).
    - Description: A postprandial glucose (PPG) level at 1 hour represents the measurement of blood sugar roughly at its peak after eating, as glucose concentrations typically reach their highest point between 60 and 90 minutes after a meal starts. Measuring at this specific interval provides a sensitive look at how effectively the body handles a glucose load, often revealing metabolic issues that fasting tests might miss.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Healthy adults typically below 140 mg/dL.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 10449-7, 40287-5
  - Associated CPT:

33. (GTTPM_2) (Glucose Tolerance 2 hour post meal)

- Common Name: Glucose Tolerance 2 hour post meal
- Long Name: Glucose Tolerance 2 hour post meal
- Test Type: GTTPM_2
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Healthcare providers order a 2-hour postprandial glucose (PPG) test to assess how effectively the body regulates blood sugar after a meal or a standardized sugar load. The conditions that they are trying to are: Diabetes or Prediabetes, Gestational Diabetes (GDM), Identifying Impaired Glucose Tolerance (IGT).
- Description: A postprandial glucose (PPG) test measures the amount of sugar (glucose) in your blood exactly two hours after you begin a meal. This specific timeframe is used because blood sugar levels typically peak between one and two hours after eating as carbohydrates are broken down and enter the bloodstream.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The following values are standard for non-pregnant adults:
  - Normal: Less than 140 mg/dL (7.8 mmol/L).
  - Prediabetes: Between 140 and 199 mg/dL (7.8 to 11.0 mmol/L).
  - Diabetes Diagnosis: 200 mg/dL (11.1 mmol/L) or higher.
- Comments:
- Test Method Variance:

- Other:
  - Associated LOINC: 14760-3, 14761-1, 1521-4, 6689-4, 77677-3
  - Associated CPT:

34. (GTTPM_3) (Glucose Tolerance 3 hour post meal)

    - Common Name: Glucose Tolerance 3 hour post meal
    - Long Name: Glucose Tolerance 3 hour post meal
    - Test Type: GTTPM_3
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: Healthcare providers order a 2-hour postprandial glucose (PPG) test to assess how effectively the body regulates blood sugar after a meal or a standardized sugar load. The conditions that they are trying to are: Diabetes or Prediabetes, Gestational Diabetes (GDM), Identifying Impaired Glucose Tolerance (IGT).
    - Description: A 3-hour postprandial glucose (PPG) measurement refers to the level of sugar in your blood exactly three hours after the start of a meal. While 1-hour and 2-hour tests are more common for routine diabetes monitoring, the 3-hour mark is a critical diagnostic point in specialized testing.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): the 3-hour mark is a return to fasting levels, typically between 70–100 mg/dL.
    - Comments:
    - Test Method Variance:
    - Other:
      - Associated LOINC: 95106-1
      - Associated CPT:

1.7 Cell Markers

1.7.1 CD4 (CD4 Count)

- Common Name: CD4 Count

- Long Name: CD4 Count, T-cell helper subset
- Test Type: CD4
- Panels: 65758-5, 65759-3
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: CD4 is used for HIV/AIDS Management, Immunodeficiency Disorders, Transplant Monitoring, and Screening for Immune Issues.
- Description: A CD4 count is a blood test measuring helper T-cells (CD4 cells) to assess immune system health, crucial for managing HIV by showing how the virus weakens immunity and if treatment works, with results guiding infection prevention and treatment.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Reported as cells per cubic millimeter (cells/mm^3), with a typical range for healthy adults around 500 to 1500 cells/mm^3
- Comments: Panels (65758-5, 65759-3) are included, but not expected to be seen associated with results. If a site does see results associated with these LOINCs please advise the lab workgroup. LOINC 5474-2 is a ratio defined as CD3+CD4+ (T4 helper) cells/CD3+CD8+ (T8 suppressor cells) cells [# Ratio] in Blood. This ratio is acceptable to use as a proxy to identify HIV patients.
- Test Method Variance:
- Other:
  - Associated LOINC: 24467-3, 26570-2, 54218-3, 5472-6, 5474-2, 92736-8
  - Associated CPT:

1.8 Microbiology

1.8.1 (HCV) Hepatitis C Virus Test Types

1. HCV_AB (Hepatitis C virus antibody)

    - Common Name: Hepatitis C virus antibody
    - Long Name: Hepatitis C virus antibody [Units/volume] or Signal/Cutoff in Serum or Plasma
    - Test Type: HCV_AB
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: Indications for a Hepatitis C virus (HCV) antibody lab test are categorized into universal screening for all adults, targeted screening for high-risk individuals, and diagnostic testing for those with clinical symptoms.
    - Description: A Hepatitis C virus (HCV) antibody test is a blood screen used to determine if a person has ever been infected with the virus. It identifies proteins called antibodies that the immune system produces to fight the virus; these antibodies typically remain in the bloodstream for life, even after successful treatment or natural clearance.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Interpretive Values:
        - Non-Reactive (<0.9 COI): No HCV antibodies detected.
        - Borderline (0.9 to <1.0 COI): Inconclusive result; further testing or retesting may be required.
        - Reactive (1.0 COI): Indicates past or current infection. A reactive result should be followed by an HCV RNA test for confirmation.
        - S/CO Ratio: Specimens with a signal-to-cutoff (S/CO) value 1.00 are considered reactive.
    - Comments: Link to excellent site for Hep C test explanation in lay terms: `https://labtestsonline.org/understanding/analytes/hepatitis-c/tab/glance/`

  - Expected values are integers from two digits up to almost 1,000,000 IU/mL. Beaware that units can differ. Decimal values have been seen and are suspected to be LOG values. Inspect the unit value!
- Test Method Variance:
- Other:
  - Associated LOINC: 16936-7,22327-1,48159-8,5198-7,57006-9
  - Associated CPT:

2. HCV_AB_QL (Hepatitis C virus antibody presence found)

- Common Name: Hepatitis C virus antibody presence found
- Long Name: Hepatitis C virus [Presence] or band pattern [Interpretation] in Serum, Plasma or Blood
- Test Type: HCV_AB_QL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Indications for a Hepatitis C virus (HCV) antibody lab test are categorized into universal screening for all adults, targeted screening for high-risk individuals, and diagnostic testing for those with clinical symptoms.
- Description: The Hepatitis C Virus (HCV) Antibody Presence test (also called the anti-HCV test) is a blood test used to determine if an individual has ever been infected with the Hepatitis C virus.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Interpretive Values:
  - Non-Reactive (<0.9 COI): No HCV antibodies detected.
  - Borderline (0.9 to <1.0 COI): Inconclusive result; further testing or retesting may be required.
  - Reactive (1.0 COI): Indicates past or current infection. A reactive result should be followed by an HCV RNA test for confirmation.
  - S/CO Ratio: Specimens with a signal-to-cutoff (S/CO) value 1.00 are considered reactive.

- Comments: Only KPNC to sometimes use 48767-8. DO NOT use 48767-8 if the data is a note about the RNA test. Use 48767-8 only when the result represents HEPATITIS C ANTIBODY CONFIRMATORY TEST INTERPETATION with result such as POSITIVE or NEGATIVE.
- Test Method Variance:
- Other:
  - Associated LOINC: 13955-0, 16128-1, 16129-9, 40726-2, 72376-7, 13125-1, 5199-5
  - Associated CPT:

3. HCV_AB_QL_PR (Hepatitis C virus antibody presence found using probe method)

   - Common Name: Hepatitis C virus antibody presence found using probe method
   - Long Name: Hepatitis C virus [Presence] or band pattern [Interpretation] in Serum, Plasma or Blood found using probe method
   - Test Type: HCV_AB_QL_PR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Indications for a Hepatitis C virus (HCV) antibody lab test are categorized into universal screening for all adults, targeted screening for high-risk individuals, and diagnostic testing for those with clinical symptoms.
   - Description: Using a probe method, such as real-time Reverse Transcription-PCR (RT-PCR) or Transcription-Mediated Amplification (TMA), the presence of HCV is determined by the binding of a fluorescently labeled probe to specific viral sequences. Modern probe methods often use digital or fluorescent signals, older confirmatory methods like the Recombinant Immunoblot Assay (RIBA) or specialized electrophoretic NAT assays use band patterns to interpret results.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Interpretive Values:

  - Non-Reactive (<0.9 COI): No HCV antibodies detected.
  - Borderline (0.9 to <1.0 COI): Inconclusive result; further testing or retesting may be required.
  - Reactive (1.0 COI): Indicates past or current infection. A reactive result should be followed by an HCV RNA test for confirmation.
  - S/CO Ratio: Specimens with a signal-to-cutoff (S/CO) value 1.00 are considered reactive.
- Comments: The LOINCs in this test type are provisional. The workgroup needs data from sites to review before final determination of inclusion can be made.
- Test Method Variance:
- Other:
  - Associated LOINC: 22324-8, 22326-3, 22329-7, 34162-8, 11076-7, 33462-3, 51649-2, 56926-9, 56927-7, 56928-5, 56929-3, 56930-1, 9609-9
  - Associated CPT:

4. HCV_GENO (Hepatitis C virus genotype)

- Common Name: Hepatitis C virus genotype
- Long Name: Hepatitis C virus genotype [Identifier] in Ser, Plas or Bld by Probe and target amplification method
- Test Type: HCV_GENO
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Indications for a Hepatitis C virus (HCV) antibody lab test are categorized into universal screening for all adults, targeted screening for high-risk individuals, and diagnostic testing for those with clinical symptoms.
- Description: Hepatitis C Virus (HCV) genotyping, identifying specific viral strains (1-6) in serum/plasma/blood (Ser, Plas, Bld) via Nucleic Acid Amplification (NAA) (like RT-PCR), uses genotype-specific probes (fluorescent or colorimetric) targeting variable regions (5'UTR, NS5B) to detect sequence differences, crucial for guiding targeted antiviral treatments, differentiating subtypes (like

1a/1b), and guiding therapy duration, with results reported as genotype/subtype identifiers (e.g., 1a, 3a).

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Genotyping tests usually require a minimum HCV RNA level (e.g., 500 IU/mL or higher) for reliable results.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 32286-7,48574-8
  - Associated CPT:

5. HCV_GENO_PR (Hepatitis C virus genotype probe method)

   - Common Name: Hepatitis C virus genotype probe method
   - Long Name: Hepatitis C virus genotype [Identifier] in Ser, Plas or Bld by Probe and target amplification method
   - Test Type: HCV_GENO_PR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Indications for a Hepatitis C virus (HCV) antibody lab test are categorized into universal screening for all adults, targeted screening for high-risk individuals, and diagnostic testing for those with clinical symptoms.
   - Description: A Hepatitis C virus (HCV) genotype test identifies the specific viral strain (e.g., Genotype 1, 2, 3, 4, 5, 6) from a patient's serum, plasma, or blood using Reverse Transcriptase Polymerase Chain Reaction (RT-PCR) to amplify viral RNA, followed by detection with genotype-specific probes that bind to unique sequences, often in the 5' UTR or NS5B regions, to determine treatment guidance. Common methods include real-time PCR with fluorescent probes or hybridization assays (like Line Probe Assays) that create visible bands for each genotype/subtype.
   - Related Tests:

- Approximate Reference Range (Each site will have slight variations): Genotyping tests usually require a minimum HCV RNA level (e.g., 500 IU/mL or higher) for reliable results.
- Comments: The LOINCs in this test type are provisional. The workgroup needs data from sites to review before final determination of inclusion can be made.
- Test Method Variance:
- Other:
  - Associated LOINC: 82513-3
  - Associated CPT:

6. HCV_RNA (Hepatitis C virus RNA viral load)

   - Common Name: Hepatitis C virus RNA viral load
   - Long Name: Hepatitis C virus RNA [Units or #/volume] (viral load) in Ser or Plas by Probe with amplification
   - Test Type: HCV_RNA
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Indications for a Hepatitis C virus (HCV) antibody lab test are categorized into universal screening for all adults, targeted screening for high-risk individuals, and diagnostic testing for those with clinical symptoms.
   - Description: A Hepatitis C virus (HCV) RNA viral load test is a diagnostic tool that measures the amount of genetic material (RNA) of the hepatitis C virus in your blood. Unlike antibody tests, which only show if you have ever been exposed, this test confirms if the virus is currently active and replicating.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): Viral Load Classification:
     - Low Viral Load: Typically less than 800,000 IU/mL.
     - High Viral Load: Typically more than 800,000 IU/mL.
     - Undetectable: Often defined as <12 or <15 IU/mL, depending on the specific laboratory's sensitivity.

- Comments: Expected values are integers from two digits up to almost 1,000,000 IU/mL. Beaware that units can differ. Decimal values have been seen and are suspected to be LOG values. Inspect the unit value!
- Test Method Variance:
- Other:
  - Associated LOINC: 10676-5,11011-4,20416-4,20571-6,29609-5,34703-9,34704-7,49758-6, 48576-3
  - Associated CPT:

7. HCV_RNA_LOG (Hepatitis C virus RNA Log Units of viral load)
   - Common Name: Hepatitis C virus RNA Log Units of viral load
   - Long Name: Hep C virus RNA [Log units of #/vol] (viral load) in Ser or Plas by Probe with amplification
   - Test Type: HCV_RNA_LOG
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: This test is primarily indicated to confirm active infection and to guide and monitor antiviral treatment.
   - Description: A Hepatitis C Virus (HCV) RNA log units viral load test is a quantitative blood test that measures the amount of HCV genetic material (RNA) in the bloodstream. While results are often reported in International Units per milliliter (IU/mL), they are also frequently expressed in log units (logarithmic scale) to make large, complex numbers easier for clinicians to interpret and compare over time.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): 1.0 log IU/mL = 10 IU/mL, a result over 800,000 IU/mL (roughly 5.9 log IU/mL) is generally considered a "high" viral load:
     - Low Viral Load: Typically less than 800,000 IU/mL.
     - High Viral Load: Typically more than 800,000 IU/mL.
     - Undetectable: Often defined as <12 or <15 IU/mL, depending on the specific laboratory's sensitivity.

- Comments: Number generally between 1.00 log IU/mL and 8.00 log IU/ML-possibly a little lower or higher
- Test Method Variance:
- Other:
  - Associated LOINC: 38180-6,42003-4,42617-1,47252-2, 49605-9
  - Associated CPT:

8. HCV_RNA_QL_PR (Hepatitis C virus RNA Presence by probe) ***LOINCS in this test type have been moved to HCV_RNA_QL***

   - Common Name: Hepatitis C virus RNA Presence by probe
   - Long Name: Hepatitis C virus RNA [Presence] in Ser, Plas or Bld by Probe and target amplification method
   - Test Type: HCV_RNA_QL_PR
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications:
   - Description:
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations):
   - Comments: The LOINCs in this test type are provisional. The workgroup needs data from sites to review before final determination of inclusion can be made.
   - Test Method Variance:
   - Other:
     - Associated LOINC: deprecated: (11259-9, 5010-4)
     - Associated CPT:

9. HCV_RNA_QL (Hepatitis C virus RNA Presence)

   - Common Name: Hepatitis C virus RNA Presence
   - Long Name: Hepatitis C virus RNA [Presence] in Ser, Plas or Bld by Probe and target amplification method

- Test Type: HCV_RNA_QL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indication for an HCV RNA test is to determine if a person has a current, active infection rather than just a past exposure. While an antibody test acts as the initial screen, the RNA test is the definitive tool for clinical diagnosis and management.
- Description: A Hepatitis C virus (HCV) RNA presence by probe test is a sensitive molecular diagnostic tool used to directly detect the genetic material of the virus in the blood. This is test is capable of identifying that the virus is actively replicating. Specialized nucleic acid probes (small pieces of DNA or RNA) are added to the sample. These probes are designed to "stick" (hybridize) only to HCV genetic material.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
  - Low Viral Load: Typically defined as less than 800,000 IU/mL.
  - High Viral Load: Typically defined as greater than 800,000 IU/mL.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 11259-9, 5010-4
  - Associated CPT:

1.8.2 (HIV) Human Immunodeficiency Virus Test Types

1. HIV1_AB_CNF_QL (HIV 1 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative)

   - Common Name: HIV 1 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative
   - Long Name: HIV 1 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative

- Test Type: HIV1_AB_CNF_QL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indication for an HIV-1 antibody confirmatory test (specifically the differentiation assay) is a reactive (positive) result from an initial 4th-generation HIV-1/2 antigen-antibody screening test.
- Description: A qualitative confirmatory test for HIV-1 antibodies is a laboratory procedure used to verify a reactive (preliminary positive) result from an initial screening assay. Its primary purpose is to differentiate between HIV-1 and HIV-2 antibodies and to rule out false positives from the first stage of testing. The assay uses synthetic or recombinant peptides representing specific HIV-1 proteins (such as gp160, gp41, p31, and p24).
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The test does not measure the amount of virus (viral load); instead, it provides a "Reactive" (Positive), "Non-reactive" (Negative), or "Indeterminate" result based on the presence of specific antibodies.
- Comments: The confirmatory antibody test may be negative or indeterminate during the "window period" (the time before antibodies develop). If an acute (early) infection is suspected—typically within 10–33 days of exposure—a Nucleic Acid Test (NAT) may be indicated instead.
- Test Method Variance:
- Other:
  - Associated LOINC: 13499-9, 21009-6, 40732-0, 5221-7, 86233-4
  - Associated CPT:

2. HIV1_AB_SCR_QL (HIV 1 Antibody in Serum, Plasma or Blood by Screen method - qualitative)

   - Common Name: HIV 1 Antibody in Serum, Plasma or Blood by Screen method - qualitative

- Long Name: HIV 1 Antibody in Serum, Plasma or Blood by Screen method - qualitative
- Test Type: HIV1_AB_SCR_QL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The qualitative HIV-1 antibody screen is indicated for routine health maintenance, specific clinical symptoms, or following potential exposure to the virus.
- Description: The HIV-1 Antibody screen is a qualitative test used to detect the presence of antibodies to Human Immunodeficiency Virus Type 1 (HIV-1) in human blood, serum, or plasma.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): A qualitative result does not measure the amount of antibody but instead provides a simple "Reactive" or "Non-Reactive" status.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 14092-1, 16975-5, 29893-5, 33866-5, 41145-4, 68961-2, 7917-8, 85686-4
  - Associated CPT:

3. HIV1_AB_SCR_QN (HIV 1 Antibody in Serum, Plasma or Blood by Screen method - quantitative)

    - Common Name: HIV 1 Antibody in Serum, Plasma or Blood by Screen method - quantitative
    - Long Name: HIV 1 Antibody in Serum, Plasma or Blood by Screen method - quantitative
    - Test Type: HIV1_AB_SCR_QN
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:

- Indications: The HIV-1 Antibody/Antigen Screening Test (commonly referred to as a 4th-generation assay) is the primary method for detecting HIV infection in clinical settings. Healthcare providers indicate this test for several distinct reasons, ranging from routine preventive care to specific diagnostic needs following potential exposure.
- Description: A quantitative HIV-1 antibody screening test is a laboratory procedure used to detect the presence and relative amount of HIV-1 antibodies and/or the p24 antigen in human blood, serum, or plasma. While most screening tests are reported as "reactive" or "non-reactive" (qualitative), they often use a quantitative measurement—known as a Signal-to-Cutoff (S/CO) ratio—to determine that result.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Non-reactive: S/CO ratio < 1.00
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 22356-0, 43599-0, 5220-9
  - Associated CPT:

4. HIV12_AB_CNF_QL (HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative)

   - Common Name: HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative
   - Long Name: HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative
   - Test Type: HIV12_AB_CNF_QL
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The primary indication for this qualitative confirmatory test is a reactive (positive) result from a preliminary HIV screening assay. It is a critical "second step" in the standard diagnostic algorithm used to verify infection and distinguish between the two virus types.

- Description: A confirmatory HIV-1/HIV-2 antibody test is a qualitative assay used to verify a reactive result from an initial screening. Unlike screening tests that only detect the presence of HIV markers, confirmatory tests are designed to differentiate between HIV-1 and HIV-2 antibodies and confirm their specific presence.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): A "Positive" or "Confirmed" result requires a specific pattern of reactivity across the various viral proteins:
  - HIV-1 Positive: Reactivity to multiple HIV-1 specific bands (e.g., two envelope proteins or one envelope plus a core/polymerase protein).
  - HIV-2 Positive: Reactivity to specific HIV-2 envelope proteins (gp36 and gp140).
  - Indeterminate: Some bands are present, but they do not meet the full criteria for a positive result. This may occur during early infection (seroconversion) or due to cross-reactivity.
  - Negative: No specific HIV antibodies are detected, suggesting the initial screening may have been a false positive.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 43185-8, 44873-8, 69668-2, 77685-6, 80203-3, 95524-5
  - Associated CPT:

5. HIV12_AB_SCR_QL (HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Screen method - qualitative)
   - Common Name: HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Screen method - qualitative
   - Long Name: HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Screen method - qualitative
   - Test Type: HIV12_AB_SCR_QL
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:

- Indications: The primary indications for an HIV-1/HIV-2 qualitative screening test involve both routine preventative care and targeted diagnostic evaluation based on risk or symptoms. Clinical symptoms include presenting with signs of Acute Retroviral Syndrome (e.g., fever, persistent swollen lymph nodes, unexplained weight loss, or night sweats).
- Description: A qualitative screening test for HIV-1 and HIV-2 antibodies is a laboratory or point-of-care procedure used to detect the presence of disease-fighting proteins produced by the immune system in response to an HIV infection.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): These tests are designed to provide a simple "Reactive" (Positive) or "Non-Reactive" (Negative) result rather than a numerical value.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 31201-7, 40733-8, 42768-2, 43009-0, 48345-3, 54086-4, 3905-2, 73906-0, 7918-6, 80387-4
  - Associated CPT:

6. HIV12_AB_SCR_QN (HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Screen method - quantitative)

   - Common Name: HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Screen method - quantitative
   - Long Name: HIV 1 and HIV 2 Antibody in Serum, Plasma or Blood by Screen method - quantitative
   - Test Type: HIV12_AB_SCR_QN
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: Indications for an HIV-1/2 antibody (and antigen) screen are generally divided into routine screening, risk-based screening, and diagnostic testing based on clinical symptoms.

- Description: In modern clinical practice, the HIV-1 and HIV-2 Antibody Screen is almost exclusively a qualitative test, meaning it reports a "Reactive" or "Nonreactive" result rather than a numerical quantity of antibodies. The standard screening method is a Fourth-Generation Immunoassay, which simultaneously detects HIV-1 Antibodies (Groups M and O)and HIV-2 Antibodies and HIV-1 p24 Antigen (a viral protein that appears before antibodies develop).
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Non-reactive (Negative): $<$ 1.00 signal-to-cutoff ratio
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 22357-8, 41290-8, 43008-2, 48346-1, 5223-3
  - Associated CPT:

7. HIV2_AB_CNF_QL (HIV 2 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative)
   - Common Name: HIV 2 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative
   - Long Name: HIV 2 Antibody in Serum, Plasma or Blood by Confirmatory method - qualitative
   - Test Type: HIV2_AB_CNF_QL
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The primary indication for a qualitative HIV-2 antibody confirmatory test is a reactive result on an initial HIV-1/2 screening assay (such as a 4th-generation Ag/Ab test). Because screening tests cannot distinguish between HIV-1 and HIV-2, this supplemental test is required to confirm the infection and identify HIV-2 virus type is present.

- Description: A qualitative HIV-2 antibody confirmatory test is a supplemental assay used to verify and differentiate HIV-2 infection in specimens that have already tested reactive on an initial screening test. Unlike screening tests that may give a combined "reactive" result for both HIV-1 and HIV-2, these confirmatory methods explicitly identify the presence of antibodies specific to HIV-2 proteins.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): HIV-2 "Reactive" confirms an HIV-2 infection.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 31073-0 , 5225-8
  - Associated CPT:

8. HIV2_AB_SCR_QL (HIV 2 Antibody in Serum, Plasma or Blood by Screen method - qualitative)

    - Common Name: HIV 2 Antibody in Serum, Plasma or Blood by Screen method - qualitative
    - Long Name: HIV 2 Antibody in Serum, Plasma or Blood by Screen method - qualitative
    - Test Type: HIV2_AB_SCR_QL
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The primary indications for an HIV-2 antibody screen include routine health screenings, specific clinical symptoms, or identified epidemiologic risk factors. In many modern healthcare settings, this test is part of a standard "opt-out" screening process for all adults.
    - Description: The HIV-2 Antibody Screen is a qualitative medical test used to detect the presence or absence of specific antibodies produced by the body in response to the Human Immunodeficiency Virus Type 2 (HIV-2).
    - Related Tests:

- Approximate Reference Range (Each site will have slight variations): The output is reported simply as Reactive (positive) or Non-reactive (negative).
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 30361-0, 33807-9, 7919-4, 81641-3
  - Associated CPT:

9. HIV2_AB_SCR_QN (HIV 2 Antibody in Serum, Plasma or Blood by Screen method - quantitative)

   - Common Name: HIV 2 Antibody in Serum, Plasma or Blood by Screen method - quantitative
   - Long Name: HIV 2 Antibody in Serum, Plasma or Blood by Screen method - quantitative
   - Test Type: HIV2_AB_SCR_QN
   - Panels:
   - Equivalent Tests:
   - Not Equivalent Tests:
   - Indications: The primary indication for this test is to screen for and diagnose HIV-2 infection, which is clinically distinct from HIV-1. Because HIV-1 and HIV-2 antigens are genetically different, a test sensitive only to HIV-1 may not reliably detect HIV-2.
   - Description: In modern clinical practice, a screening test for HIV-2 antibodies in serum, plasma, or blood is typically performed as part of a 4th-generation HIV-1/2 antigen-antibody combo assay.
   - Related Tests:
   - Approximate Reference Range (Each site will have slight variations): laboratory equipment calculates a numerical value known as the Signal-to-Cutoff (S/CO). Non-reactive (Negative): An S/CO < 1.0.
   - Comments:
   - Test Method Variance:
   - Other:
     - Associated LOINC: 22358-6, 51786-2, 5224-1

- Associated CPT:

10. HIV1_AG_QL (HIV 1 Antigen in Serum or Plasma - qualitative)
    - Common Name: HIV 1 Antigen in Serum or Plasma - qualitative
    - Long Name: HIV 1 Antigen in Serum or Plasma - qualitative
    - Test Type: HIV1_AG_QL
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The HIV-1 Antigen in Serum or Plasma qualitative test is indicated for detecting early infection, routine screening, and confirming exposure.
    - Description: An HIV-1 Antigen in Serum or Plasma qualitative lab test is used to detect the presence of the p24 antigen, a structural protein that makes up the viral core of HIV-1. This test is primarily used to identify acute or early-stage infection, as the p24 antigen typically appears in the blood before the body has produced detectable levels of HIV antibodies.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): Results are reported as Reactive (present) or Non-reactive (absent).
    - Comments:
    - Test Method Variance:
    - Other:
        - Associated LOINC: 18396-2, 24012-7, 33660-2, 5222-5, 9821-0
        - Associated CPT:

11. HIV1_AG_QN (HIV 1 Antigen in Serum or Plasma - quantitative)
    - Common Name: HIV 1 Antigen in Serum or Plasma - quantitative
    - Long Name: HIV 1 Antigen in Serum or Plasma - quantitative
    - Test Type: HIV1_AG_QN
    - Panels:
    - Equivalent Tests:

- Not Equivalent Tests:
- Indications: The HIV-1 Antigen in Serum or Plasma quantitative test is primarily indicated for clinical management and specialized diagnostic scenarios. While standard screening uses qualitative tests, this quantitative version is used when a numerical measurement of the virus is required.
- Description: The HIV-1 Antigen in Serum or Plasma quantitative test—often referred to as an HIV viral load test—measures the specific amount of HIV-1 genetic material (RNA) or the p24 antigen in a patient's blood sample. Unlike qualitative screening tests that only provide a "positive" or "negative" result, this test provides a numerical value, typically expressed as copies per milliliter (copies/mL).
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The range is used to assess how well antiretroviral therapy (ART) is working. A successful treatment goal is an "undetectable" viral load, usually defined as fewer than 20 to 50 copies/mL.
  - Detectable Viral Load: A numerical value indicating the concentration of the virus in the blood.
  - Undetectable: Does not mean the virus is gone; it means the level is below the lower limit of detection for that specific assay (e.g., <20 copies/mL).
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 16976-3, 42339-2, 53601-1, 9665-1
  - Associated CPT:

12. HIV1_RNA (HIV 1 RNA (viral load) in Serum or Plasma)

- Common Name: HIV 1 RNA (viral load) in Serum or Plasma
- Long Name: HIV 1 RNA (viral load) in Serum or Plasma
- Test Type: HIV1_RNA
- Panels:
- Equivalent Tests:

- Not Equivalent Tests:
- Indications: This test is primarily used as a surrogate marker to manage individuals already diagnosed with HIV-1.
- Description: An HIV-1 RNA (viral load) test is a quantitative molecular assay that measures the number of copies of HIV-1 genetic material in a patient's blood.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Clinicians use specific numerical milestones to guide treatment for patients on antiretroviral therapy (ART):

Result Label	Numerical Value (copies/mL)	Clinical Significance
Undetectable	<20 to <50	Indicates ART is highly effective; essentia
Viral Suppression	<200	The primary goal of treatment. Prevents
Low-Level Viremia	50–200	Often monitored closely; may include tem
Virologic Failure	>200	Confirmed results above this level may in
High Viral Load	>100,000	Associated with higher risk of rapid disea

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 10351-5, 10682-3, 20447-9, 21008-8, 21333-0, 23876-6, 41513-3,41515-8, 48511-0, 48551-6, 59419-2, 62469-2, 70241-5
  - Associated CPT:

13. HIV1_RNA_LOG (HIV 1 RNA (viral load - log) in Serum or Plasma)

- Common Name: HIV 1 RNA (viral load - log) in Serum or Plasma
- Long Name: HIV 1 RNA (viral load - log) in Serum or Plasma
- Test Type: HIV1_RNA_LOG
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:

- Indications: The primary indications for an HIV-1 RNA (viral load) test fall into three categories: managing an existing diagnosis, confirming a new or early infection, and testing infants.
- Description: An HIV-1 RNA (viral load - log) in Serum or Plasma test is a molecular assay that measures the amount of human immunodeficiency virus type 1 (HIV-1) genetic material in a patient's blood. This quantitative test, often performed using Nucleic Acid Amplification Testing (NAAT) or PCR, is primarily used to monitor the effectiveness of antiretroviral therapy (ART) and assess disease progression.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Clinicians use specific numerical milestones to guide treatment for patients on antiretroviral therapy (ART):

Result Label	Numerical Value	Clinical Significance
Undetectable	<Log l.3 : 20 copies/mL	Indicates ART is highly effective
Viral Suppression	<Log 2.3 : 200 copies/mL	The primary goal of treatment.
Low-Level Viremia	>Log 1.7 <Log 2.3 : 50-200 copies/mL	Often monitored closely; may in
Virologic Failure	>Log 2.3 : 200 copies/mL	Confirmed results above this lev
High Viral Load	>Log 5.0 : 100,000 copies/mL	Associated with higher risk of ra

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 29539-4, 29541-0, 41514-1, 41516-6, 48510-2, 48552-4, 51780-5
  - Associated CPT:

14. HIV2_RNA (HIV 2 RNA (viral load) in Serum or Plasma)
    - Common Name: HIV 2 RNA (viral load) in Serum or Plasma
    - Long Name: HIV 2 RNA (viral load) in Serum or Plasma
    - Test Type: HIV2_RNA
    - Panels:
    - Equivalent Tests:

- Not Equivalent Tests:
- Indications: The clinical indications for an HIV-2 RNA test generally fall into these four main categories:
  - Diagnostic Verification
  - Treatment Management (ART)
  - Disease Progression Monitoring
  - Prevention of Transmission
- Description: The HIV-2 RNA (viral load) test is a specialized molecular assay used to measure the amount of Human Immunodeficiency Virus Type 2 (HIV-2) genetic material in a patient's blood. Unlike the more common HIV-1, HIV-2 requires specific testing because standard HIV-1 viral load assays typically do not detect or accurately quantify HIV-2.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Clinicians use specific numerical milestones to guide treatment for patients on antiretroviral therapy (ART):

Result Label	Numerical Value (copies/mL)	Clinical Significance
Undetectable	<20 to <50	Indicates ART is highly effective; essentia
Viral Suppression	<200	The primary goal of treatment. Prevents
Low-Level Viremia	50–200	Often monitored closely; may include tem
Virologic Failure	>200	Confirmed results above this level may in
High Viral Load	>100,000	Associated with higher risk of rapid diseas

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 69354-9, 86548-5
  - Associated CPT:

15. HIV2_RNA_LOG (HIV 2 RNA (viral load - log) in Serum or Plasma)

- Common Name: HIV 2 RNA (viral load - log) in Serum or Plasma
- Long Name: HIV 2 RNA (viral load - log) in Serum or Plasma

- Test Type: HIV2_RNA_LOG
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The HIV-2 RNA (viral load) test is typically indicated for patients who have already been diagnosed with HIV-2 or for those where standard screening has produced inconclusive results. Because HIV-2 is genetically distinct from HIV-1, specific HIV-2 molecular tests are required for these scenarios.
- Description: An HIV-2 RNA (viral load - log) in Serum or Plasma test is a molecular diagnostic tool that measures the quantity of HIV-2 genetic material (RNA) in a person's blood. Unlike HIV-1, which is more common globally, HIV-2 is primarily found in West Africa and typically presents with lower viral loads.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Clinicians use specific numerical milestones to guide treatment for patients on antiretroviral therapy (ART):

Result Label	Numerical Value	Clinical Significance
Undetectable	<Log l.3 : 20 copies/mL	Indicates ART is highly effective
Viral Suppression	<Log 2.3 : 200 copies/mL	The primary goal of treatment.
Low-Level Viremia	>Log 1.7 <Log 2.3 : 50-200 copies/mL	Often monitored closely; may in
Virologic Failure	>Log 2.3 : 200 copies/mL	Confirmed results above this lev
High Viral Load	>Log 5.0 : 100,000 copies/mL	Associated with higher risk of ra

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 81652-0
  - Associated CPT:

16. HIV1_RNA_QL (HIV 1 RNA (detection) in Serum or Plasma - qualitative)

- Common Name: HIV 1 RNA (detection) in Serum or Plasma - qualitative

- Long Name: HIV 1 RNA (detection) in Serum or Plasma - qualitative
- Test Type: HIV1_RNA_QL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The HIV-1 RNA Qualitative test is used when a doctor needs to find the virus itself rather than the body's immune response. The primary indications for ordering it are:
  - Suspected Acute (Early) Infection
  - Confirming a "Reactive" Screen
  - Testing Newborns
  - Starting or Restarting PrEP/PEP

Inconclusive Standard Results

- Description: An HIV-1 RNA Qualitative test is a highly sensitive diagnostic tool used to detect the presence of the virus's genetic material in human serum or plasma. It typically uses Nucleic Acid Amplification Testing (NAAT), such as real-time PCR or transcription-mediated amplification (TMA), to amplify and identify specific RNA sequences of the HIV-1 virus.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Unlike quantitative "viral load" tests that count the number of copies, this test provides a simple "Detected" or "Not Detected" result.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 24013-5, 25835-0, 5017-9
  - Associated CPT:

17. HIV12_RNA_QL (HIV 1 and HIV 2 RNA (detection) in Serum or Plasma - qualitative)

- Common Name: HIV 1 and HIV 2 RNA (detection) in Serum or Plasma - qualitative

- Long Name: HIV 1 and HIV 2 RNA (detection) in Serum or Plasma - qualitative
- Test Type: HIV12_RNA_QL
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: The primary indication for an HIV-1 and HIV-2 RNA qualitative test is to diagnose infection when standard antibody-based tests are inconclusive or when infection is very recent.
  - Suspected Acute (Early) Infection
  - Confirming a "Reactive" Screen
  - Testing Newborns
  - Starting or Restarting PrEP/PEP
- Description: An HIV-1 and HIV-2 RNA qualitative lab test is a nucleic acid amplification test (NAAT) used to detect the presence of the genetic material (RNA) of the two main types of human immunodeficiency virus in serum or plasma.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Unlike quantitative "viral load" tests that count the number of virus particles, this test provides a simple "Detected" or "Undetected" result for each virus.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 85361-4, 85368-9, 96556-6, 69353-1
  - Associated CPT:

18. HIV1_AB_AG_QL (HIV 1 Antibody and Antigen)
    - Common Name: HIV 1 Antibody and Antigen
    - Long Name: HIV 1 Antibody and Antigen
    - Test Type: HIV1_AB_AG_QL
    - Panels:
    - Equivalent Tests:

- Not Equivalent Tests:
- Indications: The HIV-1 Antibody and Antigen (Ag/Ab) test is the preferred first-line test for both routine screening and diagnostic evaluation of suspected HIV infection.
- Description: The HIV-1 Antibody and Antigen (Ag/Ab) test, also known as a fourth-generation or combination test, is the standard laboratory method for screening for HIV. It is designed to detect both the virus itself and the body's response to it, allowing for earlier diagnosis than older tests.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):The reference range for an HIV-1 Antibody and Antigen (4th Generation) test is typically expressed as a Signal-to-Cutoff (S/CO) ratio or Index Value.

Value (S/CO or Index)	Interpretation	Meaning
Less than 0.90	Non-reactive (Negative)	No HIV-1 p24 antigen or HIV-1/2 a
0.90 to 0.99	Equivocal (Borderline)	The result is unclear. Doctors often
1.00 or Greater	Reactive (Preliminary Positive)	The test detected signs of HIV. Sup

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 51866-2
  - Associated CPT:

19. HIV12_AB_HIV1_AG_QL (HIV 1 and HIV 2 Antibody and HIV 1 Antigen - qualitative)
    - Common Name: HIV 1 and HIV 2 Antibody and HIV 1 Antigen - qualitative
    - Long Name: HIV 1 and HIV 2 Antibody and HIV 1 Antigen - qualitative
    - Test Type: HIV12_AB_HIV1_AG_QL
    - Panels:
    - Equivalent Tests:

- Not Equivalent Tests:
- Indications: The primary indication for this test is to screen for and diagnose HIV-1 and HIV-2 infections, including early "acute" infections.
- Description: This laboratory test, often called a 4th-generation HIV-1/2 Ag/Ab combo test, is a qualitative screen used for the simultaneous detection of both the HIV-1 p24 antigen and antibodies to HIV-1 and HIV-2. It is the current standard for initial HIV screening because it can identify infection earlier than older, antibody-only tests.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): The normal or "reference" result for a healthy, uninfected individual is Non-reactive (or Negative).
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 56888-1, 75666-8
  - Associated CPT:

20. HIV12_AB_HIV1_AG_QN (HIV 1 and HIV 2 Antibody and HIV 1 Antigen - quantitative)

    - Common Name: HIV 1 and HIV 2 Antibody and HIV 1 Antigen - quantitative
    - Long Name: HIV 1 and HIV 2 Antibody and HIV 1 Antigen - quantitative
    - Test Type: HIV12_AB_HIV1_AG_QN
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The indications for these tests depend on whether they are being used for routine screening, diagnosing a potential exposure, or monitoring an existing condition.
    - Description: A 4th Generation HIV-1/2 Antigen-Antibody test is the standard laboratory screening method used to detect both

the virus itself and the body's immune response. While the initial screen is qualitative (providing a "Reactive" or "Non-Reactive" result), it can lead to quantitative follow-up tests if a person is confirmed positive.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): measures number of viral copies

Result Label	Numerical Value (copies/mL)	Clinical Significance
Undetectable	<20 to <50	Indicates ART is highly effective; essentia
Viral Suppression	<200	The primary goal of treatment. Prevents
Low-Level Viremia	50–200	Often monitored closely; may include tem
Virologic Failure	>200	Confirmed results above this level may in
High Viral Load	>100,000	Associated with higher risk of rapid diseas

- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 58900-2
  - Associated CPT:

21. HIV1_QL (HIV 1 - qualitative)
    - Common Name: HIV 1 - qualitative
    - Long Name: HIV 1 - qualitative
    - Test Type: HIV1_QL
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: An HIV-1 Qualitative RNA test is indicated primarily in clinical situations where detecting the virus itself is more reliable than waiting for the body's immune response (antibodies) to appear.
    - Description: An HIV-1 qualitative lab test (RNA or DNA PCR) detects the presence or absence of the virus's genetic material in the blood, rather than antibodies. It is highly sensitive for diagnosing acute infections (within 1-2 weeks of exposure) or neonatal infection.

- Related Tests:
- Approximate Reference Range (Each site will have slight variations): provides a "detected" or "not detected" result.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 44607-0, 49483-1
  - Associated CPT:

22. HIV12_QL (HIV 1 and HIV 2 algorithm interpretation - qualitative)

    - Common Name: HIV 1 and HIV 2 algorithm interpretation - qualitative
    - Long Name: HIV 1 and HIV 2 algorithm interpretation - qualitative
    - Test Type: HIV12_QL
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The primary indication for this laboratory algorithm is to screen for and diagnose HIV-1 and HIV-2 infections, particularly in individuals where a differentiation between the two viruses is clinically necessary.
    - Description: The CDC-recommended HIV diagnostic algorithm is a multi-step process used to accurately diagnose and differentiate between HIV-1 and HIV-2 infections. It typically follows these three phases: initial screening, antibody differentiation, nucleic acid testing (NAT).
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): typically reported as Non-reactive or Undetected
    - Comments:
    - Test Method Variance:
    - Other:
      - Associated LOINC: 85380-4

- Associated CPT:

23. HIV1_AB_SCR_QL_S (HIV 1 Antibody in Saliva by Screen method - qualitative)

    - Common Name: HIV 1 Antibody in Saliva by Screen method - qualitative
    - Long Name: HIV 1 Antibody in Saliva by Screen method - qualitative
    - Test Type: HIV1_AB_SCR_QL_S
    - Panels:
    - Equivalent Tests:
    - Not Equivalent Tests:
    - Indications: The primary indications for this test include routine screening for adults and adolescents, and targeted screening for individuals with specific risk factors or symptoms of HIV infection.
    - Description: This laboratory test is a non-invasive screening method used to detect antibodies to the Human Immunodeficiency Virus Type 1 (HIV-1) in oral fluid.
    - Related Tests:
    - Approximate Reference Range (Each site will have slight variations): These antibodies are "present" or "absent" and does not measure the exact amount of the virus or stage of infection.
    - Comments:
    - Test Method Variance:
    - Other:
        - Associated LOINC: 35437-3, 35439-9, 41144-7
        - Associated CPT:

24. HIV1_AB_SCR_QN_S (HIV 1 Antibody in Saliva by Screen method - quantitative)

    - Common Name: HIV 1 Antibody in Saliva by Screen method - quantitative
    - Long Name: HIV 1 Antibody in Saliva by Screen method - quantitative

- Test Type: HIV1_AB_SCR_QN_S
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications: Indications for an HIV-1 antibody saliva-based screen include routine screening, known or suspected exposure, and symptoms of infection. This non-invasive method is often used when a blood draw is not feasible or desired.
- Description: An HIV-1 Antibody in Saliva by Screen method refers to a laboratory-based test that detects the presence of specific proteins (antibodies) produced by the immune system in response to an HIV-1 infection.
- Related Tests:
- Approximate Reference Range (Each site will have slight variations): Non-Reactive (Normal): S/CO < 1.0 reported as a signal-to-cutoff (S/CO) ratio or absorbance value.
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC: 35438-1, 41143-9
  - Associated CPT:

1.8.3 HPVDNA_CVX (Human Papilloma Virus DNA Test, Cervical Specimen)

- Common Name: HPV DNA, specimen from cervix
- Long Name: Human Papilloma virus (HPV) DNA test, cervical specimen, high/high + intermediate risk genotypes
- Test Type: HPVDNA_CVX
- Panels: In addition to single components, includes one LOINC for panel: human papilloma virus high + low risk DNA
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A

- Indications: This test is used in conjunction with Pap Tests in cervical cancer screening and prevention programs to test for the presence of the Human Papilloma Virus (HPV) which causes most cervical cancers. Its use varies with the presumed change in sexual partners in different age ranges. Routine use is often limited to women in the range from mid-teens until 30 years old. When Pap Test results for a woman in this age range are mildly abnormal, this test is used to determine the presence of "high risk" strains of HPV that are sufficient to warrant colposcopy and targeted biopsies to rule out serious cervical abnormalities. When women over age 30 have persistent mild cervical infections, as indicated by repeated abnormal Pap Tests, it is used to determine whether colposcopy and targeted biopsies or more frequent Pap Tests are warranted.

- Description: This test shows the presence or absence of DNA from "high risk" strains (also known as genotypes) of HPV. Currently, the only FDA approved HPV DNA test is the Digene Hybrid Capture 2 (HC2), though other tests may be used. The test is performed on a cervical specimen sample which may be residual specimen from a Pap Test performed using a liquidbased cytology collection method.

- Related Tests: N/A

- Approximate Reference Range (Each site will have slight variations): N/A

- Comments: Strains are classified as high risk relative to other HPV strains, not with respect to the absolute risk for developing serious cervical abnormalities. The prevalence of high risk HPV strains in the population of women of screening age is approximately 15%. As of 2010, high risk strains include 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, and 68 with 16 and 18 being most prevalent. Test results are unlikely to specify strain, only the presence or absence of 1+ high risk strains. For the purposes of the ABN_IND variable: abnormal = positive for 1+ high risk HPV strain(s); normal = no high risk HPV strains found. The HPV DNA tests came into widespread use in cervical cancer screening and prevention programs in the mid 2000s. HPV is also associated with vulvar, penile, and anal intraepithelial neoplasia and cancer. If you are able to distinguish specimen body source, only specimens taken from the cervix should be include in this

test type. A test does not need to test for every high risk strain to be included.

- Test Method Variance: N/A
- Other:
  - Associated LOINC: 11083-3, 14499-8, 14503-7, 21440-3, 30167-1, 38372-9, 42770-8, 44550-2 (updated 4/29/10)
  - Associated CPT:

1.8.4 HPVDNA_UN (Human Papilloma Virus DNA Test, Specimen Source/Site Unknown)

- Common Name: HPV DNA, specimen source/site unknown
- Long Name: Human Papilloma virus (HPV) DNA test, specimen from unknown site, high/high + intermediate risk genotypes
- Test Type: HPVDNA_UN
- Panels:
- Equivalent Tests: N/A
- Not Equivalent Tests: N/A
- Indications: This test can be used in conjunction with Pap Tests in cervical cancer screening and prevention programs to test for the presence of the Human Papilloma Virus (HPV) which causes most cervical cancers. However, for this specific test type, the body site from which the specimen was obtained is not known (i.e., could be cervical, anal, etc). The use of HPV DNA testing varies with the presumed change in sexual partners in different age ranges. Routine use is often limited to women in the range from mid-teens until 30 years old. When Pap Test results for a woman in this age range are mildly abnormal, this test is used to determine the presence of "high risk" strains of HPV that are sufficient to warrant colposcopy and targeted biopsies to rule out serious cervical abnormalities.
- Description: This test shows the presence or absence of DNA from "high risk" strains (also known as genotypes) of HPV.
- Related Tests: N/A

- Approximate Reference Range (Each site will have slight variations): N/A
- Comments: Strains are classified as high risk relative to other HPV strains, not with respect to the absolute risk for developing serious cervical abnormalities. The prevalence of high risk HPV strains in the population of women of screening age is approximately 15%. As of 2010, high risk strains include 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, and 68 with 16 and 18 being most prevalent. Test results are unlikely to specify strain, only the presence or absence of 1+ high risk strains. For the purposes of the ABN_IND variable: abnormal = positive for 1+ high risk HPV strain(s); normal = no high risk HPV strains found. The HPV DNA tests came into widespread use in cervical cancer screening and prevention programs in the mid 2000s. HPV is also associated with vulvar, penile, and anal intraepithelial neoplasia and cancer.
- Test Method Variance: N/A
- Other:
  - Associated LOINC: 11481-9, 16280-0, 17399-7, 17400-3, 17401-1, 17402-9, 17403-7, 17404-5, 17408-6, 17409-4, 17410-2, 44543-7, 44544-5, 44547-8, 48560-7, 49891-5, 49896-4 (updated 4/30/10)
  - Associated CPT:

1.9 Blank Template

1.9.1 TEST NAME—FOR FUTURE USE

- Common Name:
- Long Name:
- Test Type:
- Panels:
- Equivalent Tests:
- Not Equivalent Tests:
- Indications:

- Description:
- Related Tests:
- Approximate Reference Range (Each site will have slight variations):
- Comments:
- Test Method Variance:
- Other:
  - Associated LOINC:
  - Associated CPT: